

5G RAN

SA Mobility Management in Connected Mode Feature Parameter Description

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Vulnerability

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<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
Revised descriptions of QCI-specific cell-load-dependent measurement event configuration. For details, see [4.1.4.2.4 QCI-specific Cell-Load-Dependent Measurement Event Configuration](#).
Removed descriptions of the dependency of the RSRP threshold for event A2 related to service-based inter-frequency handover on service-based inter-frequency handover enhancement from [9.3.2.1 Prerequisite Functions](#).
Revised descriptions of the impact relationship between intra-RAT ANR and SSB SINR-based inter-frequency handover. For details, see [11.3.2.3 Function Impacts](#).
Revised descriptions of the impacts of optimized non-overlapping between gap and NR resources. For details, see [4.2.1 Benefits](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 03 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the service-based inter-frequency handover enhancement function. For details, see: • 3.3 SA Mobility Management in Connected Mode • 4.1.4.2.1 Measurement Event • 4.1.4.2.2 Triggering Quantity • 9.1.1 Handover Function Start Decision • 9.1.2 Measurement Configuration Delivery • 9.2.2 Impacts • 9.3.2.1 Prerequisite Functions • 9.4.1.1 Data Preparation • 9.4.1.2 Using MML Commands	Added the NRCellInterFHoMeaGrp.<i>InterFreqServHoA2ThldRsrp</i> parameter. Modified parameter: Added the SERV_BASED_ENH_HO option to the NRCellMobilityConfig.<i>InterFreqHoExtSwitch</i> parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the "optimized non-overlapping between gap and NR resources" function. For details, see: • 4.1.4.5 Measurement Gap Configuration • 4.2.1 Benefits • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the GAP_AVOID_NR_RES_OPT_SW option to the NRDUCellMeas.MeasAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the DRB session forwarding function. For details, see: • 2.3 Differences • 4.1.7.2 Handling Related to Handover Execution • 4.2.1 Benefits • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Modified parameter: Added the DRB_SESSION_FWD_SW option to the gNBMobilityCommParam.MobilityAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added high-load-specific handover in SA networking. For details, see the following sections as well as the "Data Preparation" and "Using MML Commands" sections of all chapters: • 4.1.4.2.4 QCI-specific Cell-Load-Dependent Measurement Event Configuration • 4.1.9 Anti-Ping-Pong Mechanism • 4.2.1 Benefits • 4.3.2.1 Prerequisite Functions • 8.1.4 Target Cell or Frequency Decision • 10.1.4 Target Cell or Frequency Decision • 11.1.4 Target Cell or Frequency Decision	Added the gNBScenarioDiffGrp MO. Added parameters: • NRCellQciBearer.ScenarioDiffGrpId • NRCellMobilityConfig.HoMeasAlgoSwitch • NRCellMobilityConfig.HighLoadDiffHoUeNumThld • NRCellMobilityConfig.HighLoadDiffHoDlPrbThld • NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added full configuration optimization. For details, see: • 2.3 Differences • 4.1.7.2 Handling Related to Handover Execution • 4.2.1 Benefits • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the NRCellFeatureSwitch.NrHoCompatibility-Switch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Optimized SMTC offset configuration. For details, see: • 4.1.4.4 SMTC Measurement Configuration • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the NRCellFreqRelation.SaSmcOffset parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs. For details, see: • 4.1.4.2.3 QCI-specific Measurement Event Configuration 4.1.4.2.4 QCI-specific Cell-Load-Dependent Measurement Event Configuration	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationship between RF module fast deep dormancy and voice-capability-based inter-frequency handover back to the HPLMN. For details, see 12.3.2.3 Function Impacts.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the impact relationship between energy-efficiency-based inter-frequency handover and frequency-priority-based inter-frequency handover. For details, see 7.3.2.3 Function Impacts.	None	Low-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between intra-frequency multi-carrier shutdown and voice-capability-based inter-frequency handover back to the HPLMN. For details, see 12.3.2.3 Function Impacts .	None	Low-frequency TDD	5900 series base stations (macro base stations)
Added support for configuring the offsets to the thresholds for events A4 and A5 related to RFSP-specific service-based inter-frequency measurements. For details, see: • 4.1.4.2.1 Measurement Event • 9.1.2 Measurement Configuration Delivery • 9.4.1.1 Data Preparation	Added the gNB RfSP Config.ServBasedInterFHoA4RsrpOfs parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for RSRP- and SINR-based joint handover. For details, see: • 4.1.6.2 Neighboring Cell Filtering Rules • 4.2.1 Benefits • 4.3.2.1 Prerequisite Functions • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added parameters: • NRCellInterFHoMeaGrp.RsrpSinnBasedNecSinnThld • NRCellInterFHoMeaGrp.RsrpSinnBasedUnecSinnThld Modified parameters: • Added the RSRP_SINR_HO_SW option to the NRCellMobilityConfig.InterFreqHoExtSwitch parameter. • Added the RSRP_SINR_HO_SW option to the gNBQciBearer.ServicceDiffAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for configurations of the offsets to the thresholds for events A4 and A5 related to RFSP-specific frequency-priority-based inter-frequency measurements. For details, see: • 4.1.4.2.1 Measurement Event • 7.1.5 Typical Parameter Configurations • 7.4.1.1 Data Preparation	Added the gNBRefspConfig.FreqPrilFHoA4RsrpThldOfs parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for "anti-ping-pong function based on the inter-frequency handover waiting time" and "QCI-specific suppression of unnecessary inter-frequency handover" by "downlink-service-based carrier selection in high-speed scenarios". For details, see 4.1.9 Anti-Ping-Pong Mechanism .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added the gap-assisted scheduling compatibility function. For details, see: 4.1.4.5 Measurement Gap Configuration 4.2.1 Benefits 4.4.1.1 Data Preparation 4.4.1.2 Using MML Commands	Modified parameter: Added the GAP_SCH_COMP_SW option to the gNodeBParam.CompatibilityAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Optimized the mechanism for selecting the target AMF in Xn-based handovers. For details, see: • 4.1.7.1 Handover Execution • 4.2.1 Benefits • 4.4.1.1 Data Preparation • 4.4.1.2 Using MML Commands	Added the gNodeBParam.NglnterfaceAlgoExtSw parameter and its options PATH_SWITCH_AMF_SEL_SW and PATH_SWITCH_BACKUP_AMF_SEL_SW.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

Added descriptions of the impact relationships between dedicated cell reselection priorities and the following functions: "inter-frequency handover policy for voice service-capable UEs" and "voice-capability-based inter-frequency handover back to the HPLMN". For details, see [12.3.2.3 Function Impacts](#).

Revised the MML-based configurations for "conflict handling between the PDU session release and NG-based handover procedures". For details, see [4.4.1.2 Using MML Commands](#).

Added descriptions of the scenarios where RSRP- and SINR-based joint handover takes effect. For details, see [4.1.6.2 Neighboring Cell Filtering Rules](#).

Revised the setting notes for the CIO specification compatibility switch. For details, see [4.4.1.1 Data Preparation](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-010014	Mobility Management	4 Basic Functions of Mobility Management in Connected Mode
FBFD-010014	Mobility Management	5 Coverage-based Intra-Frequency Handover
FBFD-010014	Mobility Management	6 Coverage-based Inter-Frequency Handover
FBFD-010014	Mobility Management	7 Frequency-Priority-based Inter-Frequency Handover
FBFD-010014	Mobility Management	8 Operator-specific-Priority-based Inter-Frequency Handover
FBFD-010014	Mobility Management	9 Service-based Inter-Frequency Handover
FBFD-010014	Mobility Management	10 Uplink-Interference-based Inter-Frequency Handover
FBFD-010014	Mobility Management	11 SSB SINR-based Inter-Frequency Handover (for LampSite)
FBFD-010014	Mobility Management	12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN

2.3 Differences

Differences Between NSA and SA

Only serving cell change is involved in SA networking. PSCell change and PCell change are involved in NSA networking. This document describes serving cell change in SA networking. For details about PSCell change and PCell change in NSA networking, see *NSA Mobility Management*.

Function Name	Difference	Chapter/Section
Basic functions of mobility management in connected mode	The basic principles and mechanisms of serving cell change in SA networking are similar to those of PSCell change in NSA networking. The differences are as follows: • The rule for determining the maximum number of non-serving NR frequencies that can be delivered differs between SA networking and NSA networking. For details, see Maximum Number of Non-Serving NR Frequencies That Can Be Delivered by the gNodeB in 4.1.4.1.2 Frequency Selection for Measurement. • In inter-frequency measurements, the method of QCI-specific measurement parameter configuration is different between SA UEs and NSA UEs. For details, see 4.1.4.2.3 QCI-specific Measurement Event Configuration. • Xn-based handover based on AMF set IDs is supported only in SA networking, not in NSA networking. For details, see 4.1.7.1 Handover Execution. • Non-overlapping between gap and NR resources is supported only in SA networking, not in NSA networking. For details, see Non-Overlapping Between Gap and NR Resources in 4.1.4.5 Measurement Gap Configuration. • Configuration of timer T312 is supported only in SA networking, not in NSA networking. For details, see Configuration of Timer T312 in 4.1.4.6 Other Configurations. • EPLMN-based intra-RAT inter-PLMN handover is supported only in SA networking, not in NSA networking. For details, see 4.1.6 Target Cell or Frequency Decision.	4 Basic Functions of Mobility Management in Connected Mode

Function Name	Difference	Chapter/Section
	• Full configurations during intra-base-station handovers are supported only in SA networking, not in NSA networking. For details, see 4.1.7.2 Handling Related to Handover Execution. • Full configuration optimization is supported only in SA networking, not in NSA networking. For details, see 4.1.7.2 Handling Related to Handover Execution. • DRB session forwarding is supported only in SA networking, not in NSA networking. For details, see 4.1.7.2 Handling Related to Handover Execution. • For other basic principles and mechanisms that are involved only in NSA networking, see <i>NSA Mobility Management</i>.	
Coverage-based intra-frequency handover	This function is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	5 Coverage-based Intra-Frequency Handover

Function Name	Difference	Chapter/Section
Coverage-based inter-frequency handover	This function is supported in both SA and NSA networking, with the same implementation and the following differences: In SA networking, measurement-based inter-frequency handover, measurement-based redirection, and blind redirection are supported. In NSA networking, measurement-based inter-frequency PSCell change and blind inter-frequency PSCell change are supported but redirection is not supported. Measurement-based inter-frequency PSCell change can be implemented between low frequency bands as well as between high and low frequency bands. Blind inter-frequency PSCell change can only be implemented from high frequency bands to low frequency bands; it cannot be implemented between high or low frequency bands as well as from low frequency bands to high frequency bands. Measurement-based inter-frequency handover is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	6 Coverage-based Inter-Frequency Handover
Frequency-priority-based inter-frequency handover	This function is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	7 Frequency-Priority-based Inter-Frequency Handover
Operator-specific-priority-based inter-frequency handover	This function is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	8 Operator-specific-Priority-based Inter-Frequency Handover
Service-based inter-frequency handover	Measurement-based mode is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	9 Service-based Inter-Frequency Handover

Function Name	Difference	Chapter/Section
Uplink-interference-based inter-frequency handover	This function is supported in both SA and NSA networking, with the same implementation. The only difference lies in function switches, activation verification, and performance monitoring.	10 Uplink-Interference-based Inter-Frequency Handover
SSB SINR-based inter-frequency handover (for LampSite)	This function is supported only in SA networking, not in NSA networking.	11 SSB SINR-based Inter-Frequency Handover (for LampSite)
Voice-capability-based inter-frequency handover back to the HPLMN	This function is supported in both SA and NSA networking, with the same implementation. The only difference lies in activation verification and performance monitoring.	12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic functions of mobility management in connected mode	Supported only in low frequency bands	4 Basic Functions of Mobility Management in Connected Mode
Coverage-based intra-frequency handover	Supported only in low frequency bands	5 Coverage-based Intra-Frequency Handover
Coverage-based inter-frequency handover	Supported only in low frequency bands	6 Coverage-based Inter-Frequency Handover
Frequency-priority-based inter-frequency handover	Supported only in low frequency bands	7 Frequency-Priority-based Inter-Frequency Handover

Function Name	Difference	Chapter/Section
Operator-specific-priority-based inter-frequency handover	Supported only in low frequency bands	8 Operator-specific-Priority-based Inter-Frequency Handover
Service-based inter-frequency handover	Supported only in low frequency bands	9 Service-based Inter-Frequency Handover
Uplink-interference-based inter-frequency handover	Supported only in low frequency bands	10 Uplink-Interference-based Inter-Frequency Handover
SSB SINR-based inter-frequency handover (for LampSite)	Supported only in low frequency bands	11 SSB SINR-based Inter-Frequency Handover (for LampSite)
Voice-capability-based inter-frequency handover back to the HPLMN	Supported only in low frequency bands	12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN

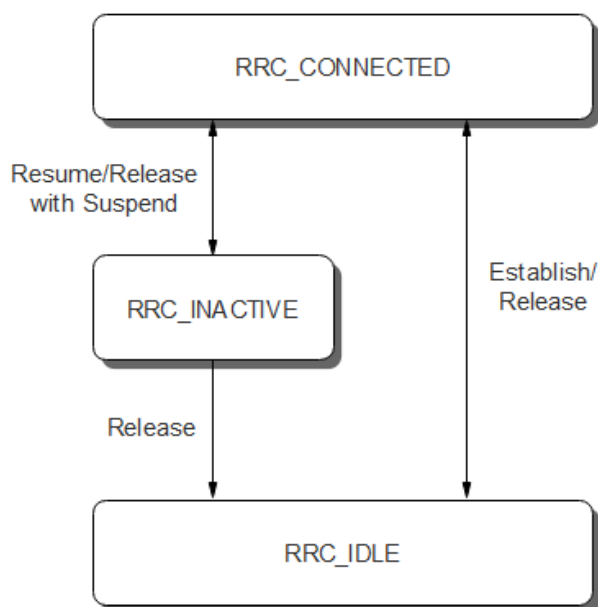
3 Overview

Mobility management is a basic function used to ensure that services stay uninterrupted as UEs move through mobile networks.

3.1 RRC State

In a 5G network, a UE at any time is in one of the following radio resource control (RRC) states: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states, as presented in [Figure 3-1](#).

Figure 3-1 RRC states



- RRC_CONNECTED: An RRC connection is established between the UE and the gNodeB.
- RRC_IDLE: No RRC connection is established between the UE and the gNodeB.
- RRC_INACTIVE: In this state, the UE suspends data processing. However, the gNodeB still maintains the UE's context information, and as a result, the core network considers that the UE remains in the CM-CONNECTED state (as if it

were still in the RRC_CONNECTED state). When data transmission is required, the UE can quickly transit to the RRC_CONNECTED state.

NOTE

RRC connection release is performed for transition from RRC_CONNECTED to RRC_IDLE or from RRC_CONNECTED to RRC_INACTIVE. The state (RRC_IDLE or RRC_INACTIVE) that a UE has entered can be checked based on whether *suspendConfig* is included in the RRCRelease message. If *suspendConfig* is included in the RRCRelease message, the UE has entered the RRC_INACTIVE state. If *suspendConfig* is not included in the RRCRelease message, the UE has entered the RRC_IDLE state.

For details about the RRC states, see section 4.2.1 "UE states and state transitions including inter RAT" in 3GPP TS 38.331 V15.5.1.

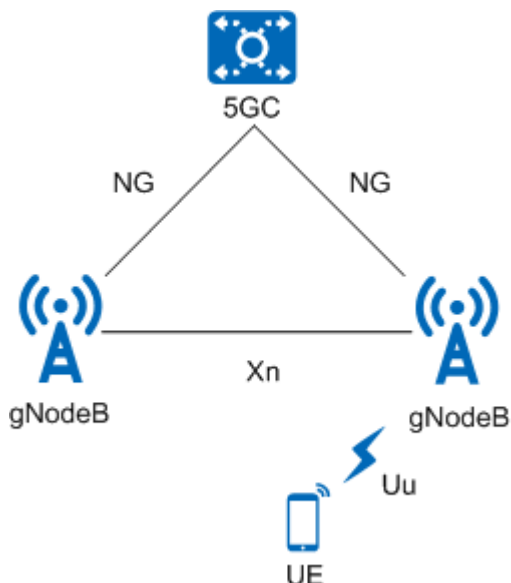
3.2 Mobility Management Scenarios

5G supports standalone (SA) networking and non-standalone (NSA) networking. Based on networking scenarios, mobility management is categorized into SA mobility management, NSA mobility management, and NSA and SA hybrid mobility management. **This document zooms in on functions related to SA mobility management in connected mode.**

SA Mobility Management

SA networking uses an end-to-end 5G network architecture, where the terminals, base stations, and core network comply with 5G specifications. [Figure 3-2](#) shows SA networking. For details about SA networking, see *5G Networking and Signaling*.

Figure 3-2 SA networking (Option 2 as an example)



In SA networking, mobility management is classified into intra-RAT mobility management and inter-RAT mobility management, depending on whether a serving cell changes between two networks of different radio access technologies (RATs). For details about SA inter-RAT mobility management, see *Interoperability*

Between E-UTRAN and NG-RAN in Connected Mode, Interoperability Between E-UTRAN and NG-RAN in Idle Mode, and NSA-SA Selection Based on User Experience.

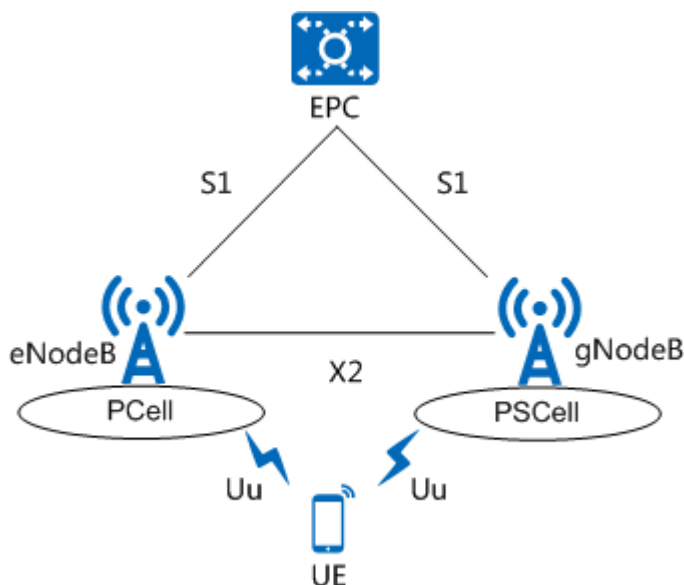
Based on the RRC states of UEs, intra-RAT mobility management in SA networking is categorized as follows:

- Mobility management in connected mode (RRC_CONNECTED state), as described in this document.
- Mobility management in idle mode (RRC_IDLE state). For details, see *SA Mobility Management in Idle Mode*.
- Mobility management in inactive mode (RRC_INACTIVE state). For details, see *SA Mobility Management in Inactive Mode*.

NSA Mobility Management

In NSA networking, terminals supporting E-UTRA-NR dual connectivity (EN-DC) connect to both an LTE base station and a 5G New Radio (NR) base station, using the radio resources of both of the base stations for transmission. The cell served by the LTE base station (eNodeB) is the primary cell (PCell), and the cell served by the 5G NR base station (gNodeB) is the primary SCG cell (PSCell). **Figure 3-3** shows the NSA EN-DC networking. For details about NSA networking, see *EPC-based NSA Basics*.

Figure 3-3 NSA EN-DC networking (Option 3x as an example)



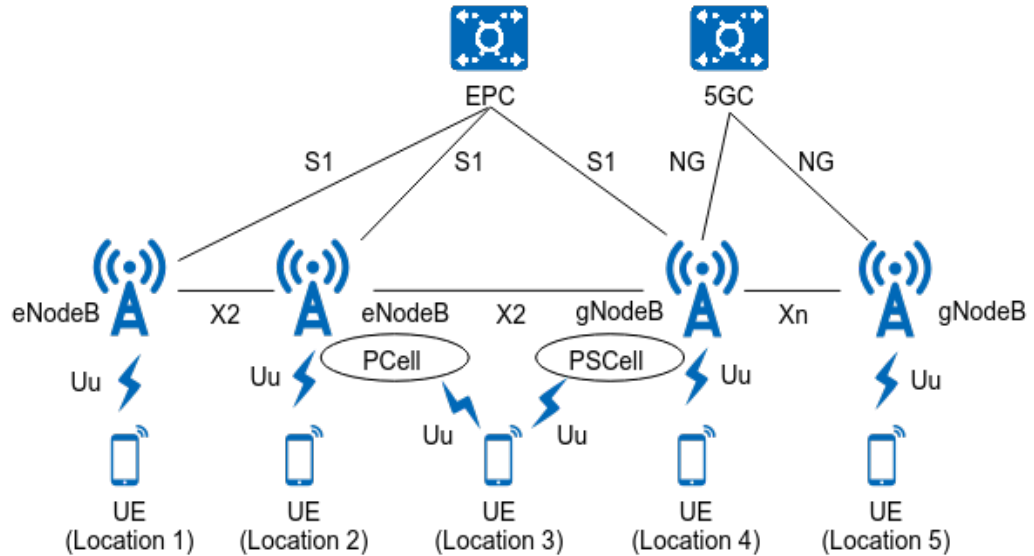
The RRC_INACTIVE state is not supported in NSA networking. Based on the RRC states of UEs, mobility management in NSA networking is categorized as follows:

- Mobility management in the RRC_CONNECTED state: UEs are connected to both the PCell and PSCell. Therefore, mobility management is categorized as PCell change and PSCell change. For details, see *NSA Mobility Management*.
- Mobility management in the RRC_IDLE state: UEs camp on LTE cells. For details, see *Idle Mode Management* in eRAN Feature Documentation.

NSA and SA Hybrid Mobility Management

NSA and SA hybrid networking marks a transition in the smooth evolution from NSA networking to SA networking, as shown in [Figure 3-4](#).

Figure 3-4 NSA and SA hybrid networking



As shown in [Figure 3-4](#), the following mobility scenarios occur in NSA and SA hybrid networking:

- When a UE moves from location 1 to location 2: The UE camps on an LTE cell at location 1 or location 2. In this scenario, intra-LTE mobility management is performed and the LTE mechanisms apply. For details, see *Idle Mode Management* and *Mobility Management in Connected Mode* in eRAN Feature Documentation.
- When a UE moves from location 2 to location 3: The UE camps on an LTE cell at location 2 and an NR cell in NSA networking at location 3. In this scenario, NSA mobility management is performed. For details, see *NSA Mobility Management*.
- When a UE moves from location 3 to location 4: The UE camps on an NR cell in NSA networking at location 3 and an NR cell in SA networking at location 4. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For details, see *NSA-SA Selection Based on User Experience*.
- When a UE moves from location 4 to location 5: The UE camps on an NR cell in SA networking at location 4 or location 5. In this scenario, SA mobility management is performed. If the UE is in connected mode during the movement, SA mobility management in connected mode is performed. For details, see this document. If the UE is in idle mode during the movement, SA mobility management in idle mode is performed. For details, see *SA Mobility Management in Idle Mode*. If the UE is in inactive mode during the movement, SA mobility management in inactive mode is performed. For details, see *SA Mobility Management in Inactive Mode*.
- When a UE directly moves from location 1 to location 5: The UE camps on an LTE cell at location 1 and an NR cell in SA networking at location 5. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For

details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

3.3 SA Mobility Management in Connected Mode

Mobility management in connected mode is a process during which a gNodeB monitors air interface status, determines serving cell change, and changes serving cells for RRC_CONNECTED UEs. Generally termed as handover, this process ensures that the UEs receive uninterrupted services from networks while on the move.

Overview of Handover Functions

5G supports multiple handover functions (listed in [Table 3-1](#)) in SA networking to guarantee UE mobility in various scenarios.

Table 3-1 Overview of handover functions

Handover Function	Application Scenario
5 Coverage-based Intra-Frequency Handover	On a network with continuous coverage, a coverage-based intra-frequency handover is triggered for a UE moving towards the cell edge, where it receives higher-quality signals from neighboring cells than from the serving cell. This prevents service drops caused by signal quality deterioration.
6 Coverage-based Inter-Frequency Handover	On a network with continuous coverage, a coverage-based inter-frequency handover is triggered for a UE moving towards the cell edge, where it receives higher-quality signals from neighboring cells than from the serving cell. This prevents service drops caused by signal quality deterioration.
7 Frequency-Priority-based Inter-Frequency Handover	In a network with multiple frequency bands having overlapping coverage, frequency-priority-based inter-frequency handover allows higher frequency bands to carry services as much as possible to spare lower frequency bands for continuous coverage.
8 Operator-specific-Priority-based Inter-Frequency Handover	In network sharing scenarios, operator-specific-priority-based inter-frequency handover allows the priority of a frequency to vary among operators, achieving networking flexibility.
9 Service-based Inter-Frequency Handover	In multi-frequency networking scenarios, service-based inter-frequency handover allows services with different QCI to be carried on different frequencies, achieving flexible networking and service steering.
10 Uplink-Interference-based Inter-Frequency Handover	In multi-frequency networking, if there is uplink interference on a frequency, the uplink user experience on this frequency will be affected. In this case, uplink-interference-based inter-frequency handover can be performed to hand over UEs from the serving cell under strong interference to inter-frequency neighboring cells, ensuring the uplink user experience.

Handover Function	Application Scenario
11 SSB SINR-based Inter-Frequency Handover (for LampSite)	For LampSite base stations in cell splitting heterogeneous networking, SSB SINR-based inter-frequency handover can be used to hand over UEs from the edge of a cell on a frequency to the center of another cell on a different frequency. This reduces interference between intra-frequency cells and improves user experience and system capacity.
12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN	Assume that a VoNR-incapable UE of operator A roams from a 5G cell of operator A to a 5G cell of operator B. Operators A and B are referred to as the home and visited operators, respectively. When the UE is paged in the visited 5G cell, the UE will experience an EPS fallback to a 4G cell of the home operator. In this scenario, if no N26 interface is configured between the home EPC and the visited 5GC, the home EPC cannot obtain the UE context from the visited 5GC over the N26 interface, resulting in a call failure. To avoid such a call failure, this function is used to proactively hand over the VoNR-incapable UE of the home operator that has roamed to the visited 5G cell back to the home 5G cell before the UE is paged in RRC_CONNECTED mode.
Note: Mobility management policies for special services or scenarios are not described in this document. Examples are as follows: • Inter-RAT handover: <i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>, <i>Interoperability Between E-UTRAN and NG-RAN in Idle Mode</i>, and <i>NSA-SA Selection Based on User Experience</i> • Load-based handover: <i>Mobility Load Balancing in Connected Mode</i> • Handover processing specific to VoNR: <i>VoNR</i> • Handover processing specific to ViNR: <i>ViNR</i> • Handover processing specific to designated UEs: <i>Flexible User Steering</i> • Handover enhancement for high speed mobility: <i>High Speed Mobility</i> • Handover processing in the case of carrier shutdown and energy-efficiency-based handover processing: <i>Energy Conservation and Emission Reduction</i> • Handover processing in the case of multi-operator sharing: <i>Multi-Operator Sharing</i> • Handover processing in the case of multi-carrier coordination: <i>Multi-Frequency Smart Selection</i> and <i>Multi-Frequency Smart Aggregation</i> • Mobility management in NSA networking: <i>NSA Mobility Management</i> • Mobility management in network slicing: <i>Network Slicing</i> and <i>Resource Control in NSA Networking</i>	

Handover functions have the same basic procedure, as described in [4 Basic Functions of Mobility Management in Connected Mode](#). In addition, [Table 3-2](#) summarizes the differences between these functions; for more detailed descriptions, see the sections corresponding to each function.

Table 3-2 Principles of handover functions

Handover Function	Processing Mode	Handover Policy	Starting Condition	Stopping Condition	Execution Condition
5 Coverage-based Intra-Frequency Handover	Measurement-based mode	Handover	This function is started by default. For coverage-based intra-frequency handover, the measurement configurations are delivered after the RRC connection is set up, not requiring handover function start decisions, and handover execution is controlled by the COVERAGE_BASED_HO option of the NRCellAlgoSwitch.IntraFreqHoSwitch parameter.	N/A	- Event A3 is received. - The serving cell is configured with neighboring cells.
6.1.1 Measurement-based Inter-Frequency Handover in 6 Coverage-based Inter-Frequency Handover	Measurement-based mode	Handover	- The COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - Event A2 is received.	Event A1 is received.	- Event A5 or A3 is received. - The serving cell is configured with neighboring cells.
6.1.2 Measurement-based Inter-Frequency Redirection in 6 Coverage-based Inter-Frequency Handover	Measurement-based mode	Redirection	- The REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - Event A2 is received.	Event A1 is received.	- Event A5 or A3 is received. - The serving cell is configured with neighboring cells.

Handover Function	Processing Mode	Handover Policy	Starting Condition	Stopping Condition	Execution Condition
6.1.3 Blind Inter-Frequency Redirection in 6 Coverage-based Inter-Frequency Handover	Blind mode	Redirection	- The REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - Event A2 is received.	N/A	The serving cell is configured with neighboring frequencies.
7 Frequency-Priority-based Inter-Frequency Handover	Measurement-based mode	Handover	- The FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - Event A1 or A2 is received.	Event A1 or A2 is received.	- Event A4 or A5 is received. - The serving cell is configured with neighboring cells.
8 Operator-specific-Priority-based Inter-Frequency Handover	Measurement-based mode	Handover	- The OP_DED_PRI_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - A UE accesses, is handed over to, or has its RRC connection reestablished in a cell whose NR frequency is not an operator-desired target frequency.	N/A	- Event A4 or A5 is received. - The serving cell is configured with neighboring cells.

Handover Function	Processing Mode	Handover Policy	Starting Condition	Stopping Condition	Execution Condition
9 Service-based Inter-Frequency Handover	Measurement-based mode	Handover	- A UE initiates a QCI-specific service that has the highest priority. - The SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - The base station checks the priority of a QoS flow when deciding whether to start service-based inter-frequency handover. - The service with this QCI is associated with a target frequency group. - The serving frequency of the UE is not within the target frequency range desired for this service. - Event A2 is received (required only when the SERV_BASED_ENH_HO option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter is selected).	N/A	- Event A4 or A5 is received. - The serving cell is configured with neighboring cells.
10 Uplink-Interference-based Inter-Frequency Handover	Measurement-based mode	Handover or redirection	- The UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected. - The strength of uplink interference in the serving cell is greater than the uplink interference threshold (specified by the NRDUCellIntrfIdent.UlIntrfThld parameter).	N/A	- Event A3 is received. - The serving cell is configured with neighboring cells.

Handover Function	Processing Mode	Handover Policy	Starting Condition	Stopping Condition	Execution Condition
11 SSB SINR-based Inter-Frequency Handover (for LampSite)	Measurement-based mode	Handover	- The SSB_SINR_BASED_HO option of the NRCellAlgoSwitch.<i>InterFreqHoSwitch</i> parameter is selected. - Event A2 is received.	Event A1 is received.	- Event A3 is received. - The serving cell is configured with neighboring cells.
12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN	Measurement-based mode	Handover or redirection	- The gNodeB identifies a UE as a roaming UE. - On the gNodeB, the VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.<i>HplmnAlgoSwitch</i> parameter is selected for the home operator of the roaming UE. - The roaming UE does not support VoNR. - The local operator and the roaming UE's home operator work in RAN sharing with common carrier mode, and the serving cell for the roaming UE is a shared cell.	N/A	- Event A5 is received. - The serving cell is configured with neighboring cells.

NOTE

For details about events, see [4.1.4.2.1 Measurement Event](#).

Handover policies consist of handover and redirection. For details, see [4.1.6 Target Cell or Frequency Decision](#).

Blind redirection does not have requirements for neighboring cell configurations. For details, see [6.1.3 Blind Inter-Frequency Redirection](#).

Inter-frequency handovers depend on inter-frequency measurements, which decrease UE throughput, especially for cell edge users (CEUs). In addition, an optimal carrier cannot be promptly selected for UEs due to inter-frequency measurement delay. Against this backdrop, virtual grid models are introduced to predict inter-frequency reference signal received power (RSRP), thus eliminating the need for inter-frequency measurements and improving user experience. For details, see *Virtual Grid-based Multi-Frequency Coordination*.

Handover Function Classification

In terms of necessity, intra-RAT handovers include:

- **Necessary handover:** When the serving cell is not able to provide services for a UE, a handover must be initiated for the UE to avoid service drops or interruptions. Such a handover is termed as a necessary handover. A necessary handover function can be:
 - Coverage-based handover: [5 Coverage-based Intra-Frequency Handover](#) and [6 Coverage-based Inter-Frequency Handover](#)
 - Voice-quality-based handover: *VoNR*
 - Voice-capability-based handover back to the HPLMN: [12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN](#)
- **Unnecessary handover:** A UE experiences a handover depending on network deployment policies. The serving cell is still able to provide services for the UE even if the handover ends up a failure. Such a handover is termed as an unnecessary handover. An unnecessary handover function can be any of those except the preceding necessary handover functions.

4 Basic Functions of Mobility Management in Connected Mode

4.1 Principles

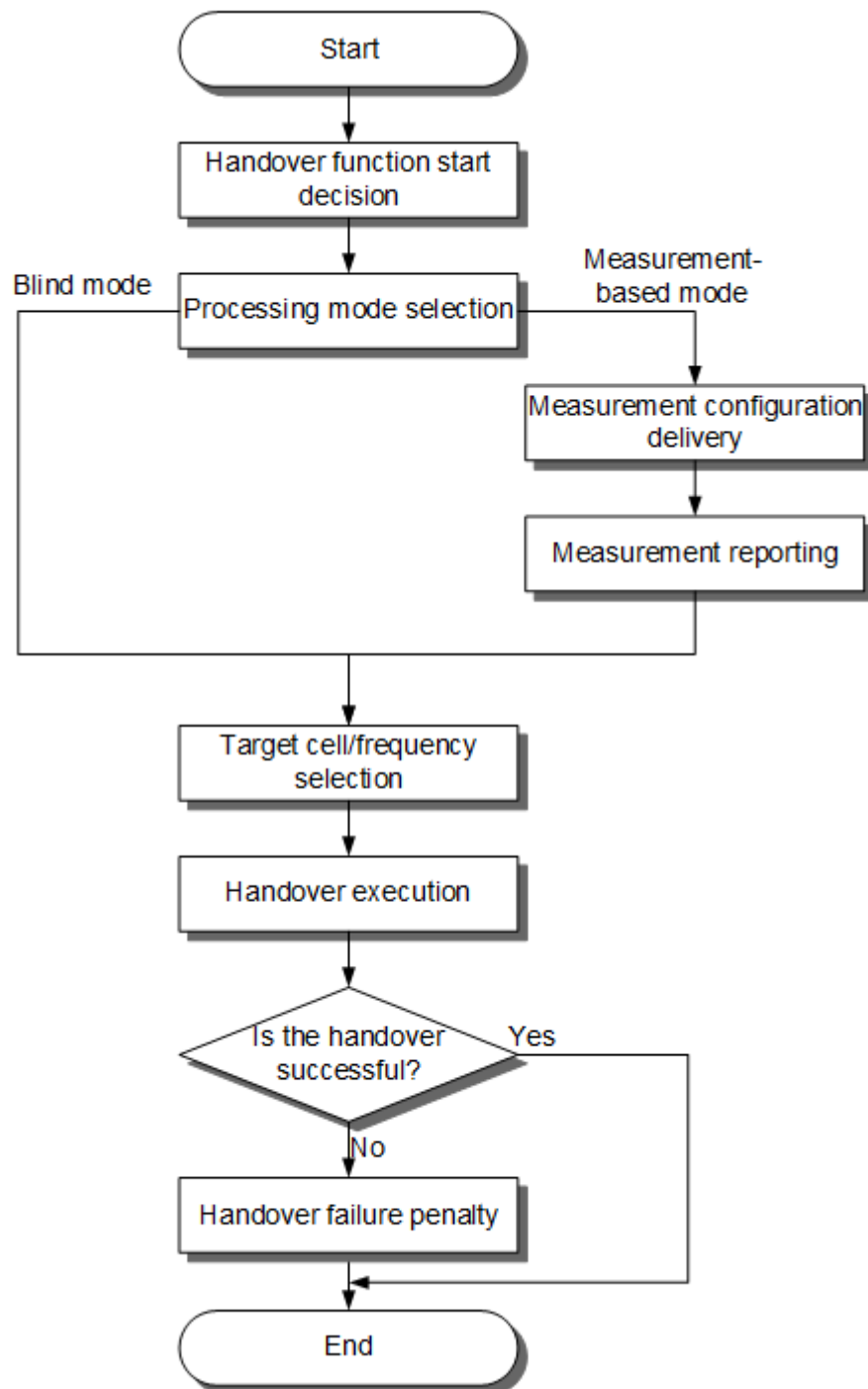
This section describes the basic principles and mechanisms of SA mobility management in connected mode. The detailed signaling procedure complies with the following 3GPP specifications. For details, see [13 Appendix](#).

- Section 9.2.3 "Mobility in RRC_CONNECTED" in 3GPP TS 38.300 V15.6.0
- Sections 5.3.5 "RRC reconfiguration" and 5.3.7 "RRC connection re-establishment" in 3GPP TS 38.331 V15.5.1
- Section 8.4 "UE Mobility Management Procedures" in 3GPP TS 38.413 V15.3.0

4.1.1 Basic Procedure

[Figure 4-1](#) shows the basic handover procedure in SA networking.

Figure 4-1 Basic handover procedure



1. **4.1.2 Handover Function Start Decision** describes the conditions for starting handover functions.
2. **4.1.3 Processing Mode Selection** describes processing mode selection (blind mode or measurement-based mode).
3. **4.1.4 Measurement Configuration Delivery** describes the method and contents of measurement configurations sent by the gNodeB to the UE.
4. **4.1.5 Measurement Reporting** describes the requirements for the UE to send measurement reports to the gNodeB.

5. **4.1.6 Target Cell or Frequency Decision** describes how the gNodeB selects a handover policy and generates target cells or frequencies for a handover.
6. **4.1.7 Handover Execution** describes how the gNodeB instructs the UE to execute a handover based on the handover policy.
7. **4.1.8 Handover Failure Penalty** describes the penalty mechanism for UE handover failures.

4.1.2 Handover Function Start Decision

A handover function start decision is a process of determining whether to start a handover function based on its starting conditions (provided in [Table 3-1](#)). The process depends on whether:

- The switch of the handover function is turned on.
- The signal quality of the serving cell meets the conditions.

4.1.3 Processing Mode Selection

After a handover function is started, the gNodeB selects a processing mode based on the signal quality of the serving cell and related events.

Depending on whether UEs measure the signal quality of candidate target cells before a handover, the processing mode is classified into measurement-based mode and blind mode.

- **Measurement-based mode: UEs measure the signal quality of candidate target cells and report the results to the gNodeB based on the measurement configurations delivered from the gNodeB. The gNodeB then uses the measurement report to generate a target cell list.**
- **Blind mode: The gNodeB generates a target cell or frequency list based on priority parameter configurations without instructing UEs to measure the signal quality of candidate target cells.**

NOTE

In blind mode, accessing a target cell has a high risk of failure due to the uncertain signal quality of the target cell. As such, blind mode is not recommended in most cases. This mode is only used when a quick handover is needed.

Blind mode is widely used in high-frequency scenarios. In these scenarios, mmWave signals attenuate quickly during transmission and provide a limited coverage range due to high frequencies and short wavelengths. When a UE moves at a high speed (for example, in high-speed railway scenarios), there is a possibility that the UE has moved out of the source cell before the measurement of the target cell or frequency is completed in measurement-based mode. Consequently, the service drop rate increases. To prevent this, blind mode can be used, in which case a handover is quickly implemented for the UE. However, blind mode has high requirements on neighboring cell planning. In the case of improper neighboring cell planning, the UE may experience an access failure.

4.1.4 Measurement Configuration Delivery

Measurement configuration delivery is a process during which the gNodeB sends measurement configurations to a UE.

The gNodeB delivers measurement configurations as follows:

- When a UE enters the RRC_CONNECTED state, the gNodeB includes measurement configurations in an RRCReconfiguration message and sends the message to the UE.
- After a UE enters the RRC_CONNECTED state or a handover is completed, in case of any updates to its measurement configurations, the gNodeB includes the updated measurement configurations in an RRCReconfiguration message and sends the message to the UE. This section describes measurement configuration delivery in this scenario.

Measurement configurations are configured by the gNodeB and then delivered to a UE. [Table 4-1](#) describes the information included in measurement configurations.

Table 4-1 Information included in measurement configurations

Measurement Configuration Information	Content	Function	Reference
Measurement object	It consists of attributes such as the RATs and frequencies (or cells) to be measured.	It specifies the frequencies or cells where a UE measures signal quality. The cells to be measured can be the serving cell or neighboring cells, for which the measurement procedures are the same. This section uses neighboring cell measurement as an example to describe measurement configuration delivery.	4.1.4.1 Measurement Object
Reporting configuration	It consists of attributes such as measurement event information and triggering quantities of the measurement reports.	It specifies the conditions and criteria for a UE to report measurement results.	4.1.4.2 Reporting Configuration
Measurement ID configuration	A measurement ID links a measurement object with a reporting configuration as a set.	Given a measurement ID, a UE measures the associated measurement object according to the reporting configuration.	4.1.4.3 Measurement ID Configuration
SSB-based measurement timing configuration (SMTC)	It consists of the SMTC period, SMTC duration, and SMTC offset.	It indicates the time window for a UE to perform measurements.	4.1.4.4 SMTC Measurement Configuration

Measurement Configuration Information	Content	Function	Reference
Measurement gap configuration	It consists of the measurement gap duration and measurement gap period.	It indicates a period of time during which a UE tunes its receiver towards a target frequency to perform inter-frequency measurements.	4.1.4.5 Measurement Gap Configuration
Other configurations	They include optimized configuration for deriveSSB-IndexFromCell, configuration of SSB beam measurement result selection and combination, and measurement filtering configuration.	They specify auxiliary information used for a UE to perform measurements and process measurement results.	4.1.4.6 Other Configurations

4.1.4.1 Measurement Object

The attributes of measurement objects include RATs and frequencies (or cells) to be measured. The key attributes of measurement objects vary with RATs. Inter-RAT measurement objects are described in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*. The key attributes of intra-RAT measurement objects include frequencies (or cells) to be measured.

The gNodeB determines an RAT for measurement first and then selects the frequencies from the cell-specific frequency relationships (or cells from the NRT) of this RAT.

4.1.4.1.1 RAT Selection for Measurement

The gNodeB selects a RAT for measurement based on measurement events.

- **For events A1 to A6, NR is selected.**
- **For events B1 and B2, a non-NR RAT (for example, E-UTRAN) is selected.**

For details about the events, see [4.1.4.2.1 Measurement Event](#).

4.1.4.1.2 Frequency Selection for Measurement

5G handovers are based on SSB measurements performed by UEs. Measurement is classified into intra-frequency measurement and inter-frequency measurement, in terms of the frequencies to be measured.

- Intra-frequency measurement refers to measuring cells that have the same SSB frequency and SSB frequency subcarrier spacing (SCS) as the serving cell, regardless of whether their bandwidths are the same or not.
- Inter-frequency measurement refers to measuring cells that have different SSB frequencies from the serving cell, or those that have the same SSB frequency as the serving cell but different SSB frequency SCS.

The gNodeB obtains the information about the SSB frequencies to be measured from the cell configurations or cell-specific frequency relationships of the selected RAT.

- For intra-frequency measurement, the gNodeB sends the NR frequencies determined by the **NRDUCell.SsbDescMethod** and **NRDUCell.SsbFreqPos** parameters for measurement.
- For inter-frequency measurement, the gNodeB selects frequencies for measurement as follows:

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Setting of the Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Selecting To-Be-Measured Frequencies
On	Any RFSP index is associated with a valid gNodeB frequency list group.	-	-	-	Method 1
On	No RFSP index is associated with a valid gNodeB frequency list group.	Not configured	-	-	Method 3
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	On	The operator index has been associated with a valid gNodeB frequency list group.	Method 2

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Setting of the Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Selecting To-Be-Measured Frequencies
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	On	The operator index is not associated with a valid gNodeB frequency list group.	Method 3
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	Off	-	Method 3
Off	-	Configured	On	The operator index has been associated with a valid gNodeB frequency list group.	Method 2
Off	-	Configured	On	The operator index is not associated with a valid gNodeB frequency list group.	Method 3

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Setting of the Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Selecting To-Be-Measured Frequencies
Off	-	Configured	Off	-	Method 3

Note:

- "-" in the table indicates that the corresponding factor does not need to be considered.
- RFSP-based NR inter-frequency handover is controlled by the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCelAlgoSwitch.RfspAlgoSwitch** parameter. For details, see *Flexible User Steering*.
- An RFSP index can be associated with a gNodeB frequency list group through a mapping between the **gNBRfspConfig.gNBFreqPriorityGroupPld** and **gNBFreqPriorityGroup.gNBFreqPriorityGroupPld** parameters. The **gNBRfspConfig.gNBFreqPriorityGroupPld** parameter is valid only when it is set to a value other than 255.
- The NR cell's operator-specific policy is configured using the **NRCelOpPolicy** MO. For details, see *Multi-Operator Sharing*.
- An operator index can be associated with a gNodeB frequency list group through a mapping between the **NRCelOpPolicy.gNBFreqPriorityGroupPld** and **gNBFreqPriorityGroup.gNBFreqPriorityGroupPld** parameters. The **NRCelOpPolicy.gNBFreqPriorityGroupPld** parameter is valid only when it is set to a value other than 255.
When configuring the **gNBFreqPriorityGroup** MO associated with the serving PLMN of a UE, ensure that all neighboring frequencies specific to this operator are configured. The neighboring frequencies that are not configured will not be delivered to the UE, and the UE will not be handed over to cells working on these frequencies.
- Operator-specific neighboring frequency configuration is controlled by the **OPERATOR_NR_FREQ_CFG_SW** option of the **NRCelOpPolicy.FreqConfigPolicySwitch** parameter.

- Method 1: The gNodeB filters NR frequencies in the intersection of the following NR frequencies using the filtering rules described in [Measurement Frequency Filtering](#), and then sends the filtered NR frequencies to the UE for measurement.
 - NR frequencies in the gNodeB frequency list group associated with the RFSP index

- NR frequencies determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters
- Method 2: The gNodeB filters NR frequencies in the intersection of the following NR frequencies using the filtering rules described in **Measurement Frequency Filtering**, and then sends the filtered NR frequencies to the UE for measurement.
 - NR frequencies in the gNodeB frequency list group associated with the operator index for the UE's serving PLMN
 - NR frequencies determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters
- Method 3: The gNodeB filters NR frequencies that are determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters using the filtering rules described in **Measurement Frequency Filtering**, and then sends the filtered frequencies to the UE for measurement.

Measurement Frequency Filtering

In measurement-based mode, the gNodeB filters non-serving NR frequencies using the following filtering rules, and then sends the filtered frequencies to the UE for measurement.

- **Filtering based on UE band capabilities**

For all UEs, the gNodeB filters out the frequencies of bands that are not supported by UEs.

The frequency bands supported by a UE are the intersection of frequency bands indicated by the *supportedBandListNR* and *supportedBandCombinationList* IEs in the UECapabilityInformation message signaled over the Uu interface.
- **Filtering based on UE measurement and handover capabilities**

For all UEs, the gNodeB filters non-serving frequencies based on UE measurement and handover capabilities.

 - If a UE does not support inter-frequency measurements (indicated by the absence of *intraAndInterF-MeasAndReport* and *eventA-MeasAndReport* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out all non-serving frequencies for the UE.
 - During an inter-frequency handover for which the handover policy does not support redirection, the gNodeB filters out the non-serving frequencies that are not supported by the UE. The details are as follows:
 - If the UE does not support inter-frequency handovers (indicated by the absence of *handoverInterF* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out all non-serving frequencies for the UE.
 - If the UE does not support inter-frequency FDD-TDD handovers (indicated by the absence of *handoverFDD-TDD* in the UECapabilityInformation message over the Uu interface), the

gNodeB filters out the corresponding TDD or FDD frequencies for the UE.

For example, if the UE is in a TDD cell, the gNodeB filters out all non-serving FDD frequencies for the UE; if the UE is in an FDD cell, the gNodeB filters out all non-serving TDD frequencies for the UE.

- If the UE does not support inter-frequency FR1-FR2 handovers (indicated by the absence of *handoverFR1-FR2* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out the corresponding FR1 or FR2 frequency for the UE.

For example, if the UE is in a cell on FR1, the gNodeB filters out all non-serving frequencies on FR2 for the UE; if the UE is in a cell on FR2, the gNodeB filters out all non-serving frequencies on FR1 for the UE.

- If the UE does not support network access from a cell on FR2 (indicated by the absence of *pCell-FR2* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out all FR2 frequencies for the UE. This means that the UE will not be handed over or redirected from a cell on FR1 to a cell on FR2.

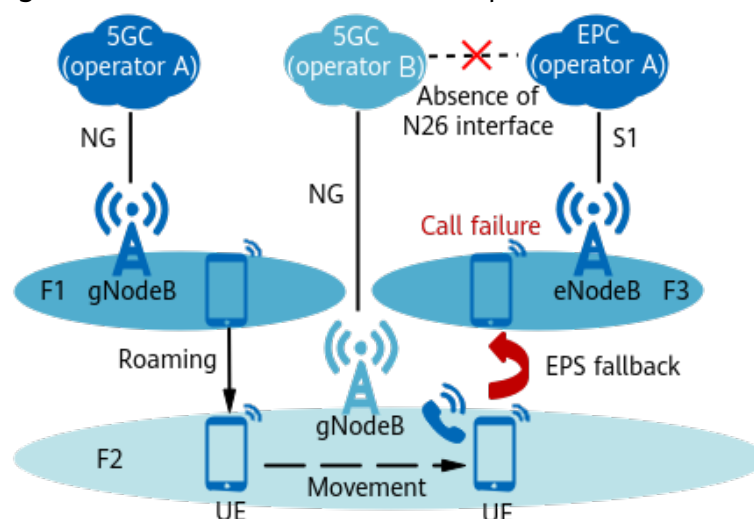
Note that, if the UE does not support network access from a cell on FR2 but supports inter-FR CA, then the UE can use a cell on FR1 as its PCell and a cell on FR2 as its SCell so that inter-FR CA can take effect for the UE. For details about inter-FR CA, see *Carrier Aggregation*.

- **Non-serving frequency filtering based on the inter-frequency handover policy for voice-service-capable UEs**

For VoNR-incapable UEs of the home operator in roaming scenarios, the gNodeB filters non-serving frequencies based on the inter-frequency handover policy for voice-service-capable UEs.

In the scenario shown in [Figure 4-2](#), a VoNR-incapable UE of operator A roams from a 5G cell of operator A to a 5G cell of operator B. Operators A and B are referred to as the home and visited operators, respectively. When the UE is paged in the visited 5G cell, the UE will experience an EPS fallback to a 4G cell of the home operator. In this scenario, if no N26 interface is configured between the home EPC and the visited 5GC, the home EPC cannot obtain the UE context from the visited 5GC over the N26 interface, resulting in a call failure. To prevent call failures in the preceding scenario, VoNR-incapable UEs of the home operator must be prohibited from being handed over from the home 5G cell to the visited 5G cell.

Figure 4-2 Call failure for a VoNR-incapable UE of the home operator



Note: The UE in the figure is a VoNR-incapable UE of operator A. Operator A provides the home network and operator B provides the visited network.

As such, Huawei introduces the inter-frequency handover policy for voice-service-capable UEs (controlled by the **VOICE_INTER_FREQ_HO_POLICY_SW** option of the **NRCellFreqRelation.AggregationAttribute** parameter). It is recommended that this option be selected for the NR frequency of the visited operator in the preceding scenario. This option applies only to VoNR-incapable UEs of the home operator and works only in low frequency bands. For VoNR-incapable UEs of the home operator, the gNodeB selects frequencies to be delivered for measurement based on the setting of this option.

- If this option is selected for a frequency, the gNodeB does not select the frequency to be delivered for measurement by the preceding UEs and prohibits them from being handed over to the frequency.
- If this option is deselected for a frequency, the gNodeB selects the frequency to be delivered for measurement by the preceding UEs and allows them to be handed over to the frequency.

When a UE meets all of the following conditions, the gNodeB identifies the UE as a VoNR-incapable UE of the home operator:

- The UE is registered with the home operator. The **gNBOperator.RoamingPlmnFlag** parameter is set to **FALSE** for the selected PLMN of the UE on the base station side. Here, the selected PLMN is a PLMN selected by the UE in the wireless network.
- The UE does not support VoNR. For details about whether a UE supports VoNR, see section 4.2.13 "IMS Parameters" in 3GPP TS 38.306 V15.5.0.

- **Non-serving frequency filtering based on the measurement activation flag**

For all UEs, the gNodeB filters out the frequencies for which the **NRCellFreqRelation.MeasActvFlag** parameter is set to **INACTIVE**.

- **Non-serving frequency filtering based on the inter-frequency periodic MR type**

For all UEs, the gNodeB filters out the frequencies for which the **NRCellFreqRelation.InterFreqPrdcMrType** parameter is set to

PERIODIC_MR_ONLY or **FORBID_MEAS** during inter-frequency handovers except RFSP-based inter-frequency handover back to the HPLMN.

- **Other non-serving frequency filtering rules**
For all UEs, the gNodeB filters out SDL frequencies that belong to band n75.
For all UEs, the gNodeB filters out frequencies for which the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter is selected during inter-frequency handovers where the high-speed UE return function is not involved.
For all UEs, the gNodeB filters out frequencies for which the **MOB_NEIGHBOR_FREQ_FILTER_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter is selected.

Maximum Number of Non-Serving NR Frequencies That Can Be Delivered by the gNodeB

Table 4-2 gives the maximum number of non-serving NR frequencies that can be delivered by the gNodeB.

Table 4-2 Maximum number of non-serving NR frequencies that can be delivered by the gNodeB

Networking Mode	Maximum Number of Non-serving NR Frequencies
SA networking	7
NSA networking	The eNodeB sends the maximum number of frequencies to the gNodeB through the <i>maxMeasFreqsSCG</i> IE in the SgNB Addition Request message over the X2 interface. If this IE does not contain the number, the maximum number of frequencies is 7.

NOTE

If the number of non-serving frequencies that meet the requirements exceeds the maximum, the gNodeB further selects frequencies for measurement in descending order of priorities until the selected frequencies reach the maximum. If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected. For details, see the section that describes measurement configuration delivery for each handover function.

Optimized UE Capability Indication for Inter-Frequency Handover

Evaluation on UE capabilities involves the *handoverInterF* IE, which can be carried in a range of IEs, including *measAndMobParametersXDD-Diff*, *measAndMobParametersFRX-Diff*, *fdd-Add-UE-NR-Capabilities*, *tdd-Add-UE-NR-Capabilities*, *fr1-Add-UE-NR-Capabilities*, and *fr2-Add-UE-NR-Capabilities*. As such, the inter-frequency handover capability of a UE needs to be collectively evaluated

based on the values of *handoverInterF* in all of the above IEs. For details about the *handoverInterF*, *handoverFDD-TDD*, and *handoverFR1-FR2* IEs, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V15.9.0.

When the values of *handoverInterF* in the aforementioned IEs differ from each other, the methods to evaluate whether UEs support inter-frequency handover vary among vendors. In this context, 3GPP TS 38.306 V16.1.0 updates the definition of the inter-frequency handover capability of UEs and defines the method of checking the capability based on the combination of *handoverInterF* values in different IEs. For details, see "Annex B: UE capability indication for UE capabilities with both FDD/TDD and FR1/FR2 differentiations" in 3GPP TS 38.306 V16.1.0.

To ensure UE compatibility and improve the accuracy of UE capability evaluation, compatibility processing optimization has been made in accordance with the latest definition in 3GPP specifications. The **INTER_FREQ_HO_CAPB_IND_OPT_SW** option of the **gNBMobilityCommParam.ProtocolCompatibilitySw** parameter determines whether to enable the optimization.

- If this option is selected:
During evaluation of UE support for inter-frequency handover, the gNodeB checks the *handoverInterF* IE in compliance with the new definition of checking the inter-frequency handover capability of UEs in 3GPP TS 38.306 V16.1.0.
During evaluation of UE support for inter-frequency FDD-TDD handover and inter-frequency FR1-FR2 handover, the gNodeB first checks the *handoverInterF* IE to determine whether outgoing handovers from the current cell are allowed. It then checks the *handoverFDD-TDD* IE to determine whether UEs support inter-frequency FDD-TDD handover and the *handoverFR1-FR2* IE to determine whether UEs support inter-frequency FR1-FR2 handover.
- If this option is deselected:
During evaluation of UE support for inter-frequency handover, the gNodeB checks the *handoverInterF* IE in compliance with the old definition of checking the inter-frequency handover capability of UEs in 3GPP TS 38.306 V16.1.0.
During evaluation of UE support for inter-frequency FDD-TDD handover and inter-frequency FR1-FR2 handover, the gNodeB directly checks the *handoverFDD-TDD* IE to determine whether UEs support inter-frequency FDD-TDD handover and the *handoverFR1-FR2* IE to determine whether UEs support inter-frequency FR1-FR2 handover, without first checking the *handoverInterF* IE.

It is recommended that this option be selected. If this option is selected, the number of NR inter-frequency handovers due to protocol incompatibility of UEs decreases, that is, the number of invalid inter-frequency handover attempts decreases.

4.1.4.1.3 Cell Selection for Measurement

Delivery of Cell CIO Information

The cell individual offset (CIO) of a cell is specified by the **NRCellRelation.CellIndividualOffset** parameter. For details about how this

parameter is used in measurement events, see the entering and leaving conditions of each measurement event in [4.1.4.2.1 Measurement Event](#). The gNodeB determines whether to deliver CIO information of a cell based on the setting of this parameter.

- If this parameter is set to a value other than **0**, CIO information of a cell can be included in a measurement object configuration.

CIO specification compatibility controls the maximum number of neighboring cells whose CIO information can be included in a measurement object configuration. CIO specification compatibility is controlled by the **CIO_SPEC_COMPAT_SW** option of the **gNB MobilityCommParam.ProtocolCompatibilitySw** parameter. [Table 4-3](#) lists the maximum number of neighboring cells whose CIO information can be included in a measurement object configuration when the function is enabled. [Table 4-4](#) lists the maximum number when the function is disabled.

Table 4-3 Maximum number of neighboring cells whose CIO information can be included in a measurement object configuration when CIO specification compatibility is enabled

Neighbor Relationship Type	Maximum Number of Neighboring Cells	IE Indicating the Maximum Number of Neighboring Cells
Intra-RAT intra-frequency neighboring cell	32	maxNrofCellMeas
Intra-RAT inter-frequency neighboring cell	32	maxNrofCellMeas
Neighboring E-UTRAN cell	32	maxCellMeasEUTRA

Table 4-4 Maximum number of neighboring cells whose CIO information can be included in a measurement object configuration when CIO specification compatibility is disabled

Neighbor Relationship Type	Maximum Number of Neighboring Cells	IE Indicating the Maximum Number of Neighboring Cells
Intra-RAT intra-frequency neighboring cell	8	maxNrofCellMeas
Intra-RAT inter-frequency neighboring cell	8	maxNrofCellMeas
Neighboring E-UTRAN cell	32	maxCellMeasEUTRA

 NOTE

If the number of cells that meet the requirements exceeds the aforementioned maximum number, the gNodeB preferentially selects the cells with larger CIO values until the selected cells reach the maximum number. If multiple cells have the same CIO value, the gNodeB preferentially selects the cells with larger PCI values.

3GPP specifications have stipulated the maximum number of neighboring cells whose information can be included in a measurement object configuration. For details, see section 6.4 "RRC multiplicity and type constraint values" in 3GPP TS 38.331 V16.5.0.

- If this parameter is set to **0**, CIO information of a cell cannot be included in a measurement object configuration.

Delivery of a Cell Blacklist

When the **NRCellMobilityConfig.MeasExcludedCellListSwitch** parameter is set to **ON** for the serving cell, the gNodeB includes a cell blacklist in the *excludedCellsToAddModList* IE of *measObjectNR* when delivering measurement configurations. If a UE receives measurement configurations in which *measObjectNR* contains the *excludedCellsToAddModList* IE, the UE does not measure cells on the blacklist in neighboring cell measurements. However, some UEs may still submit measurement reports on blacklisted cells to the gNodeB. If the measurement reports received by the gNodeB contain measurement results of blacklisted cells, the gNodeB filters out these cells when determining target cells and does not deliver the reportCGI measurement configurations of these cells to UEs.

Further details are as follows:

- Configuration of a cell blacklist
 - In blacklisted cell configuration, a cell blacklist is configured at the base station level and then is associated with the serving cell.
 - a. The **gNBCellBlacklist** MO is used to configure a cell blacklist at the base station level.
 - i. A cell blacklist is identified by **gNBCellBlacklist.CellBlacklistId**, with **gNBCellBlacklist.RatType** set to **NR**.
 - ii. A maximum of 16 blacklisted cell ranges can be configured in each cell blacklist. A blacklisted cell range is specified by the start physical cell ID and physical cell ID range. *N* consecutive PCI-specific cells starting from the one specified by the start physical cell ID are blacklisted.
where
 - The start physical cell ID is specified by **gNBCellBlackList.StartPhysicalCellId**.
 - The physical cell ID range is specified by **gNBCellBlackList.PhysicalCellIdRange**.
 - *N* is a positive integer to which the value of **gNBCellBlackList.PhysicalCellIdRange** is mapped.
 - b. A cell blacklist is associated with the serving cell.
 - Association with blacklisted intra-frequency cells
The intra-frequency cell blacklist identified by **gNBCellBlackList.CellBlacklistId** is associated with the serving cell

through **NRCellMobilityConfig.IntraFExcludedCellListId**. This way, the blacklisted intra-frequency cells on the list identified by **gNBCellBlackList.CellBlacklistId** are associated with the serving cell.

- Association with blacklisted inter-frequency cells

The inter-frequency cell blacklist identified by **gNBCellBlacklist.CellBlacklistId** is associated with the serving cell through **NRCellFreqRelation.InterFExcludedCellListId**. This way, the blacklisted inter-frequency cells on the list identified by **gNBCellBlacklist.CellBlacklistId** are associated with the serving cell.

NOTE

If **NRCellMobilityConfig.IntraFExcludedCellListId** is set to a value other than 65535, the value must be already configured in **gNBCellBlackList.CellBlacklistId** and the corresponding **gNBCellBlacklist.RatType** value must be NR.

If **NRCellFreqRelation.InterFExcludedCellListId** is set to a value other than 65535, the value must be already configured in **gNBCellBlackList.CellBlacklistId** and the corresponding **gNBCellBlacklist.RatType** value must be NR.

The values of **gNBCellBlacklist.RatType** in the records that have the same value of **gNBCellBlacklist.CellBlacklistId** must also be the same.

- Cell blacklist delivery specifications

One frequency can be associated with only one cell blacklist. A single cell blacklist can be configured with a maximum of 16 blacklisted cell ranges. According to 3GPP specifications, the *excludedCellsToAddModList* IE in each *measObjectNR* included in measurement configurations can contain a maximum of eight blacklisted cell ranges. If a frequency is associated with a cell blacklist that has more than eight blacklisted cell ranges, the gNodeB selects eight of the blacklisted cell ranges and adds them to the corresponding *excludedCellsToAddModList* IE based on the following principles:

- Blacklisted cell ranges with larger values of **gNBCellBlackList.PhysicalCellIdRange** are preferentially selected.
- If multiple blacklisted cell ranges have the same value of **gNBCellBlackList.PhysicalCellIdRange**, the ones with smaller values of **gNBCellBlackList.StartPhysicalCellId** are preferentially selected.

4.1.4.2 Reporting Configuration

Reporting configurations include:

- **Measurement event, which can be A1, A2, A3, A4, A5, A6, B1, or B2.** Handover functions determine the measurement events to be used. For details, see the sections corresponding to each handover function. For details about the definition, entering conditions, and leaving conditions for each event, see [4.1.4.2.1 Measurement Event](#).
- **Triggering quantity, which specifies the policy used to trigger event reporting and can be the reference signal received power (RSRP), reference signal received quality (RSRQ), or signal to interference plus noise ratio (SINR).** For details, see [4.1.4.2.2 Triggering Quantity](#).

4.1.4.2.1 Measurement Event

A measurement event indicates the signal quality of a cell. [Table 4-5](#) lists the definition of each measurement event.

Table 4-5 Event definition

Event Type	Event Definition
A1	The signal quality of the serving cell exceeds a specific threshold.
A2	The signal quality of the serving cell drops below a specific threshold.
A3	The signal quality of a neighboring cell is higher than that of the serving cell by a certain offset.
A4	The signal quality of a neighboring cell exceeds a specific threshold.
A5	The signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2.
A6	The signal quality of a neighboring cell is higher than that of the secondary cell by a certain offset.
B1	The signal quality of an inter-RAT neighboring cell exceeds a specific threshold.
B2	The signal quality of the serving cell drops below threshold 1 and the signal quality of an inter-RAT neighboring cell exceeds threshold 2.

NOTE

When an event uses different triggering quantities ([4.1.4.2.2 Triggering Quantity](#)), "signal quality" in the preceding event definitions needs to be adapted to the representation of the corresponding triggering quantity for more accurate event definitions. Specifically:

- In the case that an event uses RSRP as its triggering quantity, the "signal quality" in the preceding event definitions should be replaced with the "signal strength" because RSRP represents the signal strength. In this case, the UE measures the signal strength (RSRP) of the serving cell or a neighboring cell during the serving or neighboring cell measurements performed by the UE.
- In the case that an event uses RSRQ/SINR as its triggering quantity, the "signal quality" in the preceding event definitions should be retained because RSRQ/SINR represents the signal quality itself. In this case, the UE measures the signal quality (RSRQ/SINR) of the serving cell or a neighboring cell during the serving or neighboring cell measurements performed by the UE.

For details about RSRP, RSRQ, and SINR, see [4.1.4.2.2 Triggering Quantity](#).

Unless otherwise specified, this document uses "signal quality" in event definitions or UE measurements. The preceding explanations help you have a better understanding when reading this document.

The mechanisms for measurement configuration delivery and measurement reporting are the same for all these events.

- Events A1 and A2 are used to evaluate the signal quality of the serving cell in the handover function start decision phase and determine whether to start or stop a handover function.
- The other events (A3, A4, A5, A6, B1, and B2) are used to evaluate the signal quality of neighboring cells in the target cell or frequency decision phase and determine whether to initiate handover execution.

Table 4-6 lists the entering and leaving conditions for these events. For details, see section 5.5.4 "Measurement report triggering" in 3GPP TS 38.331 V15.5.1.

Table 4-6 Entering and leaving conditions for events

Event Type	Entering Condition	Leaving Condition	Setting Notes
A1	$(Ms - Hys > Thresh)$ throughout the time specified by <i>TimeToTrig</i>	$(Ms + Hys < Thresh)$ throughout the time specified by <i>TimeToTrig</i>	See Table 4-7 .
A2	$(Ms + Hys < Thresh)$ throughout the time specified by <i>TimeToTrig</i>	$(Ms - Hys > Thresh)$ throughout the time specified by <i>TimeToTrig</i>	See Table 4-8 .
A3	$(Mn + Ofn + Ocn - Hys > Ms + Ofs + Ocs + Off)$ throughout the time specified by <i>TimeToTrig</i>	$(Mn + Ofn + Ocn + Hys < Ms + Ofs + Ocs + Off)$ throughout the time specified by <i>TimeToTrig</i>	See Table 4-9 .
A4	$(Mn + Ofn + Ocn - Hys > Thresh)$ throughout the time specified by <i>TimeToTrig</i>	$(Mn + Ofn + Ocn + Hys < Thresh)$ throughout the time specified by <i>TimeToTrig</i>	See Table 4-10 .
A5	$(Ms + Hys < Thresh1)$ and $(Mn + Ofn + Ocn - Hys > Thresh2)$ throughout the time specified by <i>TimeToTrig</i>	$(Ms - Hys > Thresh1)$ or $(Mn + Ofn + Ocn + Hys < Thresh2)$ throughout the time specified by <i>TimeToTrig</i>	See Table 4-11 .
A6	$(Mn + Ocn - Hys > Ms + Ocs + Off)$ throughout the time specified by <i>TimeToTrig</i>	$(Mn + Ocn + Hys < Ms + Ocs + Off)$ throughout the time specified by <i>TimeToTrig</i>	Applicable to carrier aggregation (CA). For details, see <i>Carrier Aggregation</i> .

Event Type	Entering Condition	Leaving Condition	Setting Notes
B1	$(Mn + Ofn + Ocn - Hys > Thresh)$ throughout the time specified by <i>TimeToTrig</i>	$(Mn + Ofn + Ocn + Hys < Thresh)$ throughout the time specified by <i>TimeToTrig</i>	Applicable to inter-RAT procedures. For details, see <i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i> .
B2	$(Ms + Hys < Thresh1)$ and $(Mn + Ofn + Ocn - Hys > Thresh2)$ throughout the time specified by <i>TimeToTrig</i>	$(Ms - Hys > Thresh1)$ or $(Mn + Ofn + Ocn + Hys < Thresh2)$ throughout the time specified by <i>TimeToTrig</i>	Applicable to inter-RAT procedures. For details, see <i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i> .

The variables in the above formulas are defined as follows:

- *Ms* and *Mn* are the measurement results of the serving and neighboring cells, respectively.
- *Hys* is the hysteresis for the measurement result.
- *TimeToTrig* is the duration during which a condition is met before the event can be triggered.
- *Thresh*, *Thresh1*, and *Thresh2* are thresholds.
- *Ofs* and *Ofn* are the frequency-specific offsets for the serving cell and a neighboring cell, respectively.
- *Ocs* and *Ocn* are the CIOs for the serving cell and a neighboring cell, respectively.
- *Off* is the offset for the measurement result.

Variable configuration is event specific. The following describes the entering conditions and parameters for variables of each event.

NOTE

Unless otherwise specified, the following parameters for event configurations are applicable to all handover functions.

A1

Entering condition for event A1: $(Ms - Hys > Thresh)$ is true throughout the time specified by *TimeToTrig*. Figure 4-3 shows how event A1 is triggered and Table 4-7 describes the parameters for variables.

Figure 4-3 Triggering of event A1

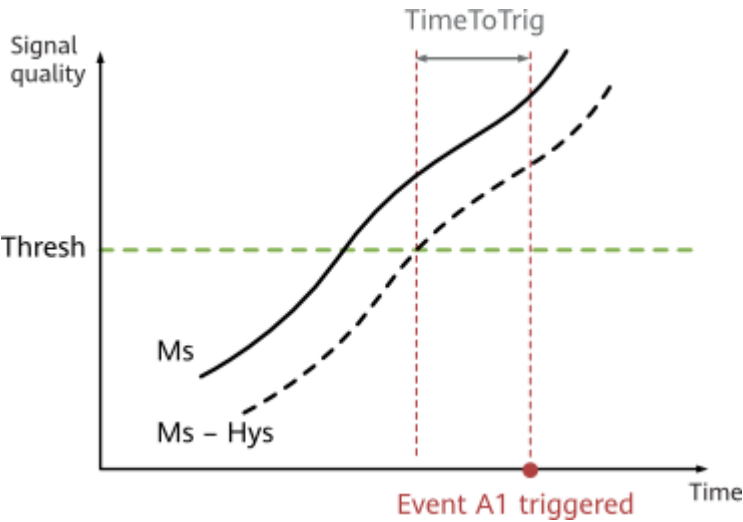


Table 4-7 Variable configuration of event A1

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-frequency A1A2 Hysteresis	NRCellInterFHoMeaGrp.InterFreqA1A2Hyst	This variable represents the hysteresis for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Cov Inter-freq Handover A1 RSRP Threshold	NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld	This variable represents the RSRP threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter. If this threshold needs to be frequency-specific, operators can set the NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs parameter to different values for different frequencies. The frequency-specific threshold will be equal to NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld plus NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs.
Thresh	Cov Inter-freq HO A1 RSRQ Threshold	NRCellInterFHoMeaGrp.CovInterFreqA1RsrqThld	This variable represents the RSRQ threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter.
Thresh	SSB SINR Inter-freq HO A1 Threshold	NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld	This variable represents the SINR threshold for event A1 related to coverage-based inter-frequency measurements and is configured through this parameter. In LampSite scenarios, this variable also represents the SINR threshold for event A1 related to SSB SINR-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Freq Priority Inter-freq A1 RSRP Threshold	NRCeIIInterFHoMeaGrp.FreqPriIInterFA1RsrpThld	- If the threshold for event A1 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the RSRP threshold for event A1 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriIInterFA1RsrpThld. - If the threshold for event A1 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the RSRP threshold for event A1 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriIInterFA1RsrpThld plus gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs.
Thresh	A3 Inter-freq Handover A1 RSRP Threshold	NRCeIIInterFHoMeaGrp.A3InterFreqHoA1RsrpThld	This variable represents the RSRP threshold for event A1 related to event A3-based coverage-based inter-frequency handover and is configured through this parameter.
TimeToTrig	Inter-frequency A1A2 Time to Trigger	NRCeIIInterFHoMeaGrp.InterFreqA1A2TimeToTrigger	This variable represents the time-to-trigger for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

A2

Entering condition for event A2: ($Ms + Hys < Thresh$) is true throughout the time specified by *TimeToTrig*. Figure 4-4 shows how event A2 is triggered and Table 4-8 describes the parameters for variables.

Figure 4-4 Triggering of event A2

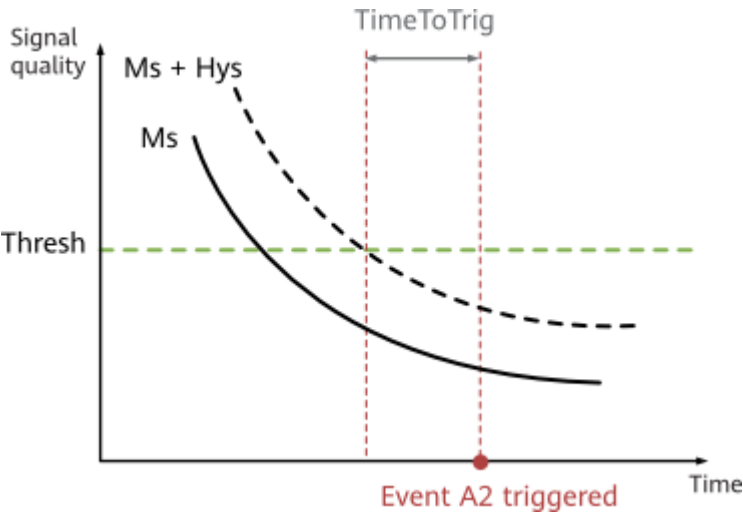


Table 4-8 Variable configuration of event A2

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-frequency A1A2 Hysteresis	NRCellInterFHoMeaGrp.InterFreqA1A2Hyst	This variable represents the hysteresis for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Cov Inter-freq Handover A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld	This variable represents the RSRP threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter. If this threshold needs to be frequency-specific, operators can set the NRCeIIFreqRelation.InterFreqHoA1A2A51FreqOfs parameter to different values for different frequencies. The frequency-specific threshold will be equal to NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld plus NRCeIIFreqRelation.InterFreqHoA1A2A51FreqOfs.
Thresh	Cov Inter-freq HO A2 RSRQ Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrqThld	This variable represents the RSRQ threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter.
Thresh	SSB SINR Inter-freq HO A2 Threshold	NRCeIIInterFHoMeaGrp.SsbSinrInterFreqHoA2Thld	This variable represents the SINR threshold for event A2 related to coverage-based inter-frequency measurements and is configured through this parameter. In LampSite scenarios, this variable also represents the SINR threshold for event A2 related to SSB SINR-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Cov HO to E-UTRAN Blind A2 RSRP Thld	NRCeIIInterRHoMeaGrp.CovHoToEutranBlindA2Thld	This variable represents the RSRP threshold for event A2 related to inter-frequency blind redirection and is configured through this parameter (also used for inter-RAT blind redirection).
Thresh	Cov HO to E-UTRAN Blind A2 RSRQ Thld	NRCeIIInterRHoMeaGrp.CovHoToLteBlindA2RsrqThld	This variable represents the RSRQ threshold for event A2 related to inter-frequency blind redirection and is configured through this parameter (also used for inter-RAT blind redirection).
Thresh	Inter-Freq Blind Redirect A2 SINR Thld	NRCeIIInterFHoMeaGrp.InterFBli ndRedirA2SinrThld	This variable represents the SINR threshold for event A2 related to inter-frequency blind redirection and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Freq Priority Inter-freq A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.FreqPriIInterFA2RsrpThld	- If the threshold for event A2 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the RSRP threshold for event A2 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriIInterFA2RsrpThld. - If the threshold for event A2 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the RSRP threshold for event A2 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriIInterFA2RsrpThld plus gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs.
Thresh	Service-based Inter-Freq HO A2 RSRP Thld	NRCeIIInterFHoMeaGrp.InterFreqServHoA2ThldRsrp	This variable represents the RSRP threshold for event A2 related to service-based inter-frequency handover and is configured through this parameter.
Thresh	A3 Inter-freq Handover A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.A3InterFreqHoA2RsrpThld	This variable represents the RSRP threshold for event A2 related to event A3-based coverage-based inter-frequency handover and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
TimeToTrig	Inter-frequency A1A2 Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA1A2TimeToTrigger	This variable represents the time-to-trigger for events A1/A2 related to inter-frequency measurements and is configured through this parameter.

A3

Entering condition for event A3: $(Mn + Ofn + Ocn - Hys > Ms + Ofc + Ocs + Off)$ throughout the time specified by *TimeToTrig*. Figure 4-5 shows how event A3 is triggered and Table 4-9 describes the parameters for variables.

Figure 4-5 Triggering of event A3

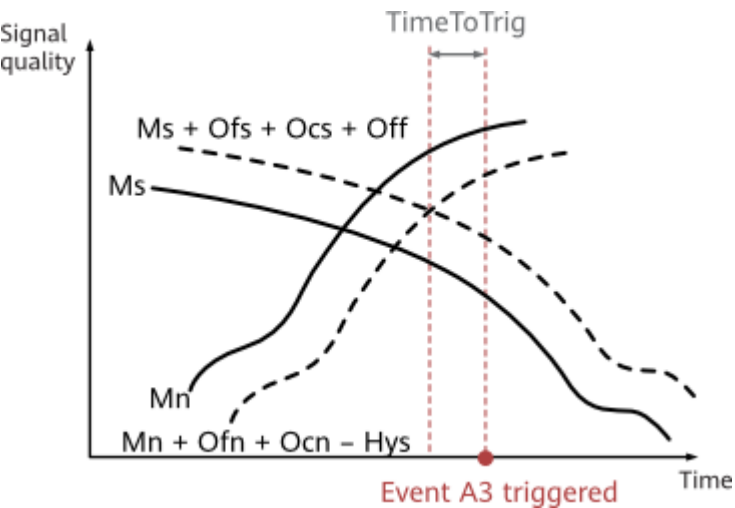


Table 4-9 Variable configuration of event A3

Variable	Parameter Name	Parameter ID	Remarks
Ofn	None	None	This variable is fixed at 0.

Variable	Parameter Name	Parameter ID	Remarks
Ocn	Cell Individual Offset	NRCellRelation.CellIndividualOffset	This variable represents the CIO of a neighboring cell. - If the value is not DB0, this parameter is included in measurement configurations. - If the value is DB0, this parameter is not included in measurement configurations and 0 is used for calculation by default.
Hys	Intra-frequency Handover A3 Hysteresis	NRCellIntraFHoMeaGrp.IntraFreqHoA3Hyst	This variable represents the hysteresis for event A3 related to intra-frequency measurements and is configured through this parameter.
Hys	Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Regarding to inter-frequency handovers except for uplink-interference-based inter-frequency handovers, this variable represents the hysteresis for event A3 related to inter-frequency handovers and is configured through this parameter.
Hys	None	None	In the case of uplink-interference-based inter-frequency handovers, the hysteresis for event A3 related to inter-frequency measurements is fixed at 1 dB.
Ofs	None	None	This variable is fixed at 0 .
Ocs	None	None	This variable is fixed at 0 .
Off	Intra-frequency Handover A3 Offset	NRCellIntraFHoMeaGrp.IntraFreqHoA3Offset	This variable represents the offset for event A3 related to intra-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Off	Inter-frequency Handover A3 Offset	NRCellInterFHoMeaGrp.InterFreqHoA3Offset	Regarding to inter-frequency handovers except for uplink-interference-based inter-frequency handovers, this variable represents the offset for event A3 related to inter-frequency handovers and is configured through this parameter.
Off	Intrf-based Inter-Freq HO A3 RSRP Offset	NRCellInterFHoMeaGrp.IntrfInterFreqHoA3RsrpOffset	This variable represents the RSRP offset for event A3 related to uplink-interference-based inter-frequency handover and is configured through this parameter.
TimeToTrig	Intra-frequency Handover A3 Time To Trigger	NRCellIntraFHoMeaGrp.IntraFreqHoA3TimeToTrigger	This variable represents the time-to-trigger for event A3 related to intra-frequency measurements and is configured through this parameter.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrigger	Regarding to inter-frequency handovers except for uplink-interference-based inter-frequency handovers, this variable represents the time-to-trigger for event A3 related to inter-frequency handovers and is configured through this parameter.
TimeToTrig	None	None	In the case of uplink-interference-based inter-frequency handovers, the time-to-trigger for event A3 related to inter-frequency measurements is fixed at 320 ms.

A4

Entering condition for event A4: $(Mn + Ofn + Ocn - Hys > Thresh)$ throughout the time specified by *TimeToTrig*. Figure 4-6 shows how event A4 is triggered and Table 4-10 describes the parameters for variables.

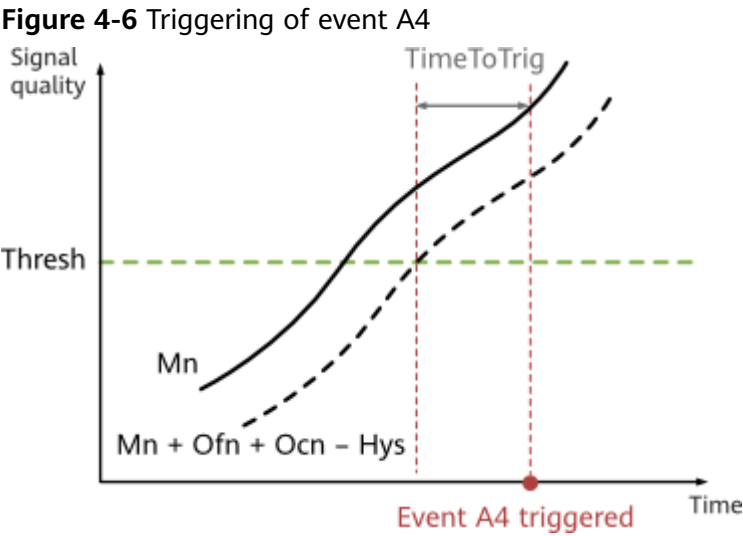


Table 4-10 Variable configuration of event A4

Variable	Parameter Name	Parameter ID	Remarks
Ofn	Frequency Offset for Connected Mode	NRCellFreqRelation.ConnFreqOffset	This variable represents the connected-mode frequency-specific offset of a non-serving frequency and is configured through this parameter.
Ocn	Cell Individual Offset	NRCellRelation.CellIndividualOffset	This variable represents the CIO of a neighboring cell. - If the value is not DB0, this parameter is included in measurement configurations. - If the value is DB0, this parameter is not included in measurement configurations and 0 is used for calculation by default.
Hys	Inter-freq Measurement Event Hysteresis	NRCellInterFreqMeaGrp.InterFreqA4A5Hyst	This variable represents the hysteresis for event A4 related to inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Freq Priority Inter-freq A4 RSRP Threshold	NRCeIIInterFHoMeaGrp.FreqPriIn terFA4RsrpThld	- If the threshold for event A4 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the RSRP threshold for event A4 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriInterFA4RsrpThld. - If the threshold for event A4 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the RSRP threshold for event A4 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPriInterFA4RsrpThld plus gNBRfspConfig.FreqPriIFHoA4RsrpThldOfs.
Thresh	Operator Ded Pri Inter-freq HO A4 RSRP Thld	NRCeIIInterFHoMeaGrp.OpDedPriHoA4RsrpThld	This variable represents the RSRP threshold for event A4 related to operator-specific-priority-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh	Srv-Based Inter-freq A4 RSRP Threshold	NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld	- If the threshold for event A4 related to RFSP-specific service-based inter-frequency measurements is not customized for a UE, the RSRP threshold for event A4 related to service-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld. - If the threshold for event A4 related to RFSP-specific service-based inter-frequency measurements is customized for a UE, the RSRP threshold for event A4 related to service-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld plus gNBRfspConfig.ServBasedInterFHoA4RsrpOfs.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	NRCeIIInterFHoMeaGrp.InterFreqA4A5TimeToTrigger	This variable represents the time-to-trigger for event A4 related to inter-frequency measurements and is configured through this parameter.

A5

Entering condition for event A5: $(Ms + Hys < Thresh1)$ and $(Mn + Ofn + Ocn - Hys > Thresh2)$ throughout the time specified by *TimeToTrig*. [Figure 4-7](#) shows how event A5 is triggered and [Table 4-11](#) describes the parameters for variables.

Figure 4-7 Triggering of event A5

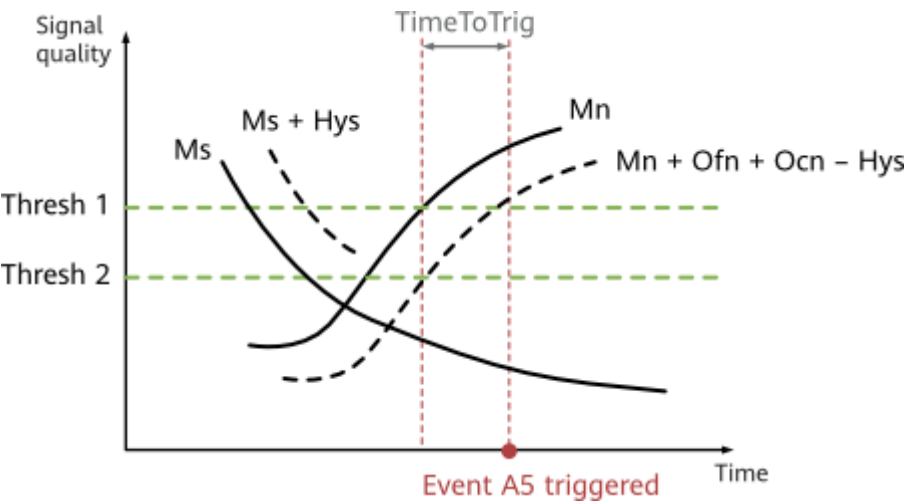


Table 4-11 Variable configuration of event A5

Variable	Parameter Name	Parameter ID	Remarks
Hys	Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Except for voice-capability-based inter-frequency handover back to the HPLMN, this variable represents the hysteresis for event A5 related to inter-frequency measurements and is configured through this parameter.
Hys	Inter-freq Measurement Event Hysteresis	None	This variable is fixed at 1 dB for voice-capability-based inter-frequency handover back to the HPLMN.

Variable	Parameter Name	Parameter ID	Remarks
Thresh1	Cov Inter-freq Handover A5 RSRP Threshold1	NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld1	- RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements is specified by this parameter when the NRCeIIIMobilityConfig.MeasTrigQuantity parameter is set to RSRP. If this threshold needs to be frequency-specific, operators can set the NRCeIIFreqRelation.InterFreqHoA1A2A5IFreqOfs parameter to different values for different frequencies. The frequency-specific threshold will be equal to NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld1 plus NRCeIIFreqRelation.InterFreqHoA1A2A5IFreqOfs. - RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements is fixed at –43 dBm when the NRCeIIIMobilityConfig.MeasTrigQuantity parameter is not set to RSRP. In this case, the value of this parameter is invalid.
Thresh1	RSRP threshold 1 for event A5 related to frequency-priority-based inter-frequency measurements	None	This variable is fixed at –43 dBm.

Variable	Parameter Name	Parameter ID	Remarks
Thresh1	RSRP threshold 1 for event A5 related to operator-specific-priority-based inter-frequency measurements	None	This variable is fixed at –43 dBm.
Thresh1	RSRP threshold 1 for event A5 related to service-based inter-frequency measurements	NRCeIIInterFHoMeaGrp.InterFreqServHoA2ThldRsrp	This variable represents RSRP threshold 1 for event A5 related to service-based inter-frequency handover and is configured through this parameter. This variable reuses the specific parameter of the RSRP threshold for event A2.
Thresh1	Voice-Capb-based Inter-freq HO Back to HPLMN A5 RSRP Threshold1	None	This variable is fixed at –43 dBm.
Thresh2	Cov Inter-freq Handover A5 RSRP Threshold2	NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld2	This variable represents RSRP threshold 2 for event A5 related to coverage-based inter-frequency measurements and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Thresh2	RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements	NRCeIIInterFHoMeaGrp.FreqPrilnterFA4RsrpThld	This variable reuses the specific parameter of the RSRP threshold for event A4. - If threshold 2 for event A5 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPrilnterFA4RsrpThld. - If threshold 2 for event A5 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.FreqPrilnterFA4RsrpThld plus gNBRfspConfig.FreqPrilFHoA4RsrpThldOfs.
Thresh2	RSRP threshold 2 for event A5 related to operator-specific-priority-based inter-frequency measurements	NRCeIIInterFHoMeaGrp.OpDedPriHoA4RsrpThld	This variable represents RSRP threshold 2 for event A5 related to operator-specific-priority-based inter-frequency measurements and is configured through this parameter. This variable reuses the specific parameter of the RSRP threshold for event A4.

Variable	Parameter Name	Parameter ID	Remarks
Thresh2	RSRP threshold 2 for event A5 related to service-based inter-frequency measurements	NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld	This variable reuses the specific parameter of the RSRP threshold for event A4. - If threshold 2 for event A5 related to RFSP-specific service-based inter-frequency measurements is not customized for a UE, RSRP threshold 2 for event A5 related to service-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld. - If threshold 2 for event A5 related to RFSP-specific service-based inter-frequency measurements is customized for a UE, RSRP threshold 2 for event A5 related to service-based inter-frequency measurements equals NRCeIIInterFHoMeaGrp.SrvInterFreqA4RsrpThld plus gNBRfspConfig.ServBasedInterFHoA4RsrpOfs.
Thresh2	Cov Inter-freq Handover A5 RSRP Threshold2	NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld2	Threshold 2 for event A5 related to voice-capability-based inter-frequency handover back to the HPLMN equals the sum of NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld2 and gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs.
	Inter-freq A5 RSRP Threshold2 Offset	gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs	
Ofn	Frequency Offset for Connected Mode	NRCeIIFreqRelation.ConnFreqOffset	This variable represents the connected-mode frequency-specific offset of a non-serving frequency and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Remarks
Ocn	Cell Individual Offset	NRCellRelation.CellIndividualOffset	This variable represents the CIO of a neighboring cell. - If the value is not DB0, this parameter is included in measurement configurations. - If the value is DB0, this parameter is not included in measurement configurations and 0 is used for calculation by default.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrigger	Except for voice-capability-based inter-frequency handover back to the HPLMN, this variable represents the time-to-trigger for event A5 related to inter-frequency measurements and is configured through this parameter.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	None	This variable is fixed at 320 ms for voice-capability-based inter-frequency handover back to the HPLMN.

4.1.4.2.2 Triggering Quantity

A triggering quantity refers to the policy that triggers event reporting. This policy uses concepts including RSRP, RSRQ, and SINR.

- **RSRP:** Indicates the strength of the received reference signals. RSRP refers to the linear average power of signals received by a UE on all resource elements (REs) that carry reference signals within the duration of an orthogonal frequency division multiplexing (OFDM) symbol across a measurement bandwidth. RSRP indicates the strength rather than the quality of downlink reference signals in a cell, excluding noise and interference.
- **Received signal strength indicator (RSSI):** Indicates the total signal strength of the current channel. RSSI refers to the linear average power of all signals (including reference signals, data signals, interference signals from a neighboring cell, and noise signals) received by a UE within the duration of an OFDM symbol across a measurement bandwidth. The signals include noise and interference.
- **RSRQ:** Indicates the quality of received reference signals and is similar to the result of RSRP/RSSI.

- SINR: Indicates a ratio of the strength of received wanted signals to the strength of received unwanted signals (noise and interference).

Table 4-12 lists the triggering quantities supported by the current version.

Table 4-12 Triggering quantities

Triggering Quantity	Definition	Description
Only RSRP	Measurement is triggered and stopped based only on RSRP.	1. The gNodeB sends only the RSRP-indicating measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP of a cell meets the entering conditions for the event, the UE sends a measurement report.
RSRP or RSRQ	Measurement is triggered and stopped based on both RSRP and RSRQ, and the RSRP-based process and the RSRQ-based process are independent of each other.	1. The gNodeB sends both RSRP- and RSRQ-indicating measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP or RSRQ of a cell meets the entering conditions for the event, the UE sends a measurement report.
RSRP or SINR	Measurement is triggered and stopped based on both RSRP and SINR, and the RSRP-based process and the SINR-based process are independent of each other.	1. The gNodeB sends both RSRP- and SINR-indicating measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured RSRP or SINR of a cell meets the entering conditions for the event, the UE sends a measurement report.
Only SINR	Measurement is triggered and stopped based only on SINR.	1. The gNodeB sends only the SINR-indicating measurement configurations related to an event to a UE. 2. The UE performs measurements based on the measurement configurations. When determining that the measured SINR of a cell meets the entering conditions for the event, the UE sends a measurement report.

Table 4-13 lists the triggering quantities specific to handover functions. For details, see the sections corresponding to each handover function.

Table 4-13 Event triggering quantities specific to handover functions

Handover Function	Event That Triggers the Start of a Handover Function	Event That Stops a Handover Function	Event That Triggers the Execution of a Handover
5 Coverage-based Intra-Frequency Handover	N/A	N/A	Event A3: only RSRP
6 Coverage-based Inter-Frequency Handover	Event A2: - Only RSRP - RSRP or RSRQ - RSRP or SINR	Event A1: - Only RSRP - RSRP or RSRQ - RSRP or SINR	Event A5: only RSRP Event A3: only RSRP
7 Frequency-Priority-based Inter-Frequency Handover	Event A1: only RSRP Event A2: only RSRP	Event A2: only RSRP Event A1: only RSRP	Event A4: only RSRP Event A5: only RSRP
8 Operator-specific-Priority-based Inter-Frequency Handover	N/A	N/A	Event A4: only RSRP Event A5: only RSRP
9 Service-based Inter-Frequency Handover	Event A2: only RSRP	N/A	Event A4: only RSRP Event A5: only RSRP
10 Uplink-Interference-based Inter-Frequency Handover	N/A	N/A	Event A3: only RSRP
11 SSB SINR-based Inter-Frequency Handover (for LampSite)	Event A2: only SINR	Event A1: only SINR	Event A3: only RSRP
12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN	N/A	N/A	Event A5: only RSRP

Handover Function	Event That Triggers the Start of a Handover Function	Event That Stops a Handover Function	Event That Triggers the Execution of a Handover
Note: The gNodeB sends a UE only the RSRP-indicating measurement configuration for event A5 or A3 related to coverage-based inter-frequency handover, based on which the UE performs measurements. When determining that the measured RSRP of a neighboring cell meets the entering conditions (as indicated in Table 4-6) for the event, the UE sends the gNodeB a measurement report that includes the measured RSRQ and SINR of the neighboring cell. • If NRCellFreqRelation.FreqIntrfFlag is set to NO_INTRF for a frequency, the gNodeB generates candidate cells based on the measured RSRPs included in the received measurement reports, and then filters candidate cells according to the rules described in Generation of the Target Cell or Frequency. In this case, the gNodeB does not filter candidate cells based on the measured RSRQs or SINRs. • If NRCellFreqRelation.FreqIntrfFlag is set to INTRF for a frequency, the gNodeB generates candidate cells based on the measured RSRPs included in the received measurement reports, and then filters candidate cells according to the rules described in Generation of the Target Cell or Frequency. In this case, the gNodeB filters candidate cells based on the measured RSRQs. • If NRCellFreqRelation.FreqIntrfFlag is set to INTRF_SINR for a frequency, the gNodeB generates candidate cells based on the measured RSRPs included in the received measurement reports, and then filters candidate cells according to the rules described in Generation of the Target Cell or Frequency. In this case, the gNodeB filters candidate cells based on the measured SINRs.			

NOTE

When a triggering quantity contains SINR, the UE must be capable of SSB SINR measurements, which is indicated by the *ss-SINR-Meas* field with the "supported" value in the UECapabilityInformation message sent over the Uu interface.

- If the *ss-SINR-Meas* field is included in *measAndMobParameters* and its value is "supported", the UE supports SINR measurements on FR1 and FR2.
- If the *ss-SINR-Meas* field is included in *fr1-Add-UE-NR-Capabilities* and its value is "supported", the UE supports SINR measurements on FR1.
- If the *ss-SINR-Meas* field is included in *fr2-Add-UE-NR-Capabilities* and its value is "supported", the UE supports SINR measurements on FR2.

SSB RSRP and SSB RSRQ measurements do not require UE capabilities.

Unless otherwise specified, only RSRP is used as the triggering quantity for an event mentioned in the following chapters and sections.

4.1.4.2.3 QCI-specific Measurement Event Configuration

Quality of service (QoS) defines the performance level of a service and indicates the level of customer satisfaction. The QoS class identifier (QCI) is used to identify the QoS class of a data radio bearer (DRB). **A measurement parameter group can be configured separately for each QCI** as follows:

- In the case of intra-frequency measurement, the same configuration method applies to SA UEs and NSA UEs.

The details are as follows: The **NRCellIntraFHoMeaGrp** MO is configured to specify a measurement parameter group (**NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId**) related to intra-frequency handovers for an NR cell.

Then, a QCI is associated with the group using the **NRCellQciBearer.IntraFreqHoMeasGroupId** and **NRCellQciBearer.Qci** parameters. Through such configuration, intra-frequency measurement parameters are configured separately for QCI-specific services.

- In the case of inter-frequency measurement, the configuration method is different between SA UEs and NSA UEs. The details are as follows:
 - SA UEs:
 - The **NRCellInterFHoMeaGrp** MO is configured to specify a measurement parameter group (**NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId**) related to inter-frequency handovers for an NR cell.
 - Then, a QCI is associated with the group using the **NRCellQciBearer.InterFreqHoMeasGroupId** and **NRCellQciBearer.Qci** parameters. Through such configuration, inter-frequency measurement parameters are configured separately for QCI-specific services.
 - NSA UEs:
 - The **NRCellInterFHoMeaGrp** MO is configured to specify a measurement parameter group (**NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId**) related to inter-frequency handovers for an NR cell.
 - Then, a QCI is associated with the group using the **NRCellQciBearer.NsaInterFreqHoMeasGroupId** and **NRCellQciBearer.Qci** parameters. Through such configuration, inter-frequency measurement parameters are configured separately for QCI-specific services.

NOTE

If the ID of a measurement parameter group related to inter-frequency handovers of NSA UEs (specified by the **NRCellQciBearer.NsaInterFreqHoMeasGroupId** parameter) is not configured in the **NRCellInterFHoMeaGrp** MO, the gNodeB does not deliver measurement configurations to the UEs with the QCI bound to the **NRCellQciBearer.NsaInterFreqHoMeasGroupId** parameter. As such, there is a possibility that service drops occur.

If a UE has multiple radio bearers (in the case of combined services), each of the bearers has a QCI and measurement parameters can be configured separately for each QCI. In the case of combined services, the gNodeB sends the UE a group of measurement parameters configured for the QCI that has the highest priority for handovers. For non-RedCap UEs, QCI priorities for handovers are determined as follows:

- If the **gNBQciBearer.QciPriorityForHo** parameter is set to a value other than **255**, then a smaller value of this parameter indicates a higher QCI priority for handovers.

- If the **gNBQciBearer.QciPriorityForHo** parameter is set to **255**, then:
 - QCI priorities are preferentially ranked based on the *PriorityLevelQos* IE carried in the *Non Dynamic 5QI Descriptor* IE included in the latest message among the INITIAL CONTEXT SETUP REQUEST, PDU SESSION RESOURCE SETUP REQUEST, and PDU SESSION RESOURCE MODIFY REQUEST messages sent from the core network. A smaller value of the *PriorityLevelQos* IE indicates a higher QCI priority.
 - If the *PriorityLevelQos* IE is not carried, QCI priorities are ranked based on the value of the **gNBQciBearer.PriorityLevel** parameter. A smaller parameter value indicates a higher QCI priority.

If the priorities of different QCIs for handovers obtained based on the preceding rules are the same, the gNodeB sorts the QCIs and sends the UE a group of measurement parameters configured for the smallest QCI. For details, see section 6.1.7 "Standardized QoS characteristics" in 3GPP TS 23.203 V15.4.0.

For details about how QCI priorities are determined during handovers of RedCap UEs, see the "QCI-specific Handover Priority Configuration" section in *RedCap*.

NOTE

The gNodeB supports RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs. This allows handover parameter settings to be differentiated between voice and data service UEs. In this context, voice service UEs refer to UEs performing VoNR services, while data service UEs refer to UEs not performing VoNR services. For details about flexible configuration of intra-RAT handover measurement parameters for voice and data service UEs, see *Flexible User Steering*.

When multiple differentiated measurement events are configured, the event configurations take effect on UEs in the following sequence:

1. RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs
2. QCI-specific cell-load-dependent measurement event configuration
3. QCI-specific measurement event configuration

4.1.4.2.4 QCI-specific Cell-Load-Dependent Measurement Event Configuration

In scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts), fluctuations in channel quality may lead to frequent variations in the measured signal quality of neighboring cells. This may trigger ping-pong handovers, which in turn degrade the user experience for services. To mitigate this, the high-load-specific handover function is introduced. This function binds different intra- or inter-frequency measurement parameter groups to UEs with different QCIs, meeting diverse user requirements and ensuring differentiated user experience.

This function is controlled by the **HIGH_LOAD_DIFF_HO_POL_SW** option of the **NRCellMobilityConfig.HoMeasAlgoSwitch** parameter. The function takes effect when the option is selected and all required conditions are met. The required conditions for the function to take effect and the parameter settings related to a bound measurement parameter group are as follows:

- Conditions:
 - Cell load requirements:
 - Number of RRC_CONNECTED UEs in the cell $\geq$ **NRCellMobilityConfig.HighLoadDiffHoUeNumThld**

- Downlink PRB usage of the cell $\geq$
NRCellMobilityConfig.HighLoadDiffHoDlPrbThld
- Uplink PRB usage of the cell $\geq$
NRCellMobilityConfig.HighLoadDiffHoUlPrbThld
- Cell configuration requirements: The **gNBScenarioDiffGrp** MO associated with the QCI specific to the UE's highest handover priority is present, and either the **NRCellIntraFHoMeaGrp** or **NRCellInterFHoMeaGrp** MO associated with the **gNBScenarioDiffGrp** MO is also present.
- **Table 4-14** describes the parameter settings related to a bound measurement parameter group.

Table 4-14 QCI-specific cell-load-dependent measurement event configuration

Parameter in the gNBScenarioDiffGrp MO	Description
gNBScenarioDiffGrp.OperatorId	Operator of the UE.
gNBScenarioDiffGrp.ScenarioType	Scenario type. Set this parameter to HIGH_LOAD .
gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter to the same value as the NRCellQciBearer.ScenarioDiffGrpId parameter. In this way, the QCI is associated with the gNBScenarioDiffGrp MO.
gNBScenarioDiffGrp.IntraFreqHoMeasGroupId	Set this parameter to the same value as the NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId parameter. In this way, the NRCellIntraFHoMeaGrp MO is associated with the gNBScenarioDiffGrp MO.
gNBScenarioDiffGrp.InterFreqHoMeasGroupId	Set this parameter to the same value as the NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId parameter. In this way, the NRCellInterFHoMeaGrp MO is associated with the gNBScenarioDiffGrp MO.
gNBScenarioDiffGrp.InterFreqHoWaitTime	Inter-frequency handover waiting timer, which is used to prevent ping-pong handovers. For details about the "anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time", see 4.1.9 Anti-Ping-Pong Mechanism .

 NOTE

The gNodeB supports RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs. This allows handover parameter settings to be differentiated between voice and data service UEs. In this context, voice service UEs refer to UEs performing VoNR services, while data service UEs refer to UEs not performing VoNR services. For details about flexible configuration of intra-RAT handover measurement parameters for voice and data service UEs, see *Flexible User Steering*.

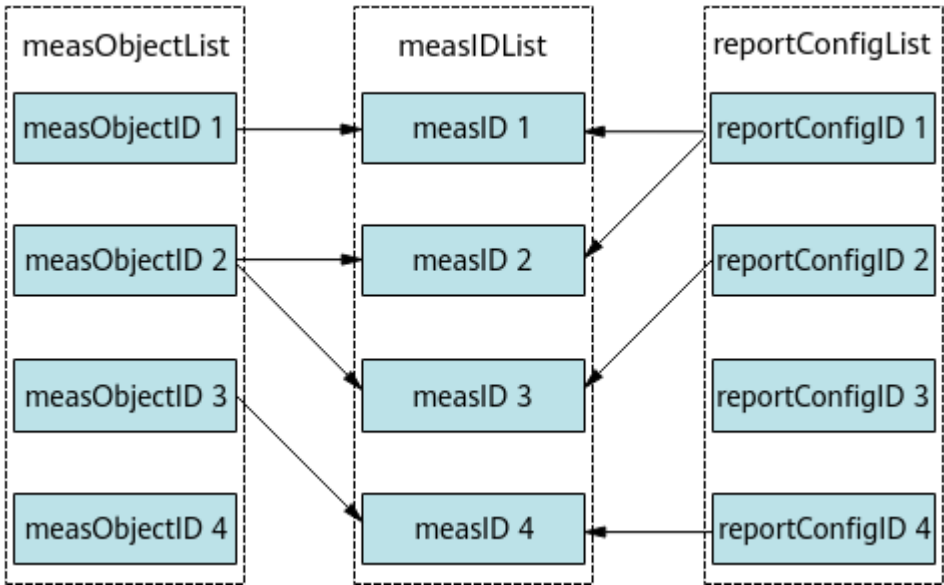
When multiple differentiated measurement events are configured, the event configurations take effect on UEs in the following sequence:

1. RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs
2. QCI-specific cell-load-dependent measurement event configuration
3. QCI-specific measurement event configuration

4.1.4.3 Measurement ID Configuration

A measurement ID links a measurement object with a reporting configuration as a set. Given a measurement ID, a UE measures the associated measurement object according to the reporting configuration. With multiple measurement IDs configured, it is possible to link multiple measurement objects to the same reporting configuration, as well as to link multiple reporting configurations to the same measurement object. **Figure 4-8** provides an example, in which both measObjectID 1 and measObjectID 2 are linked to reportConfigID 1, and measObjectID 2 is linked to both reportConfigID 1 and reportConfigID 2. A UE includes a measurement ID in a measurement report sent to the base station, which then identifies the corresponding measurement object and reporting configuration based on this measurement ID.

Figure 4-8 Measurement ID



The base station sends measurement configurations to UEs on a per measurement ID basis and in sequence based on the triggering time until the number of sent measurement IDs reaches the upper limit. Once this limit is reached, the base station cannot send new measurement configurations to UEs. Consequently, some

high-priority measurement configurations (such as those for coverage-based handovers) fail to be sent, leading to service drops or handover failures.

To prevent this, measurement ID preemption is introduced. This function is controlled by the **MEAS_ID_PREEMPTION_SW** option of the **gNBMobilityCommParam.MeasAlgoSwitch** parameter.

- If this option is selected, measurement ID preemption is enabled. If the number of sent measurement IDs has reached the upper limit, a UE's high-priority measurement IDs can preempt the measurement resources of its low-priority measurement IDs. This mechanism ensures that high-priority measurement configurations, such as mobility-related measurement configurations, can be successfully sent. If the number of sent measurement IDs has not reached the upper limit, measurement IDs are not preempted and measurement configurations are sent to UEs in sequence based on the triggering time.
- If this option is deselected, measurement ID preemption is disabled. Measurement IDs are not preempted and measurement configurations are sent to UEs in sequence based on the triggering time.

4.1.4.4 SMTC Measurement Configuration

5G handovers are based on SSB measurements performed by UEs. Measurement of all SSBs in a cell is preferred for a more accurate SSB measurement result. A cell sends SSBs in periodic scanning mode, and SSBs are not sent in all time-domain positions within a scanning period. If UEs search for and measure SSBs in all time-domain positions, a large amount of power is wasted. **To effectively indicate the time window for UEs to measure SSBs and reduce unnecessary measurement power consumption, 5G introduces SSB-based measurement timing configuration (SMTC), which is a time window configured by the gNodeB for UEs to measure SSBs. UEs measure SSBs only within the time window.**

SMTC includes an SMTC period, SMTC duration, and SMTC offset. For details about SMTC, see sections 5.5.2.10 "Reference signal measurement timing configuration" and 6.3.2 "Radio resource control information elements" in 3GPP TS 38.331 V15.5.1.

Table 4-15 SMTC

SMTC	Meaning	Parameter
SMTC period ^a	Interval at which the UE starts SSB measurements.	• The intra-frequency SMTC period is specified by the NRDUCell.SmtcPeriod parameter. • The inter-frequency SMTC period is specified by the NRCe11FreqRelation.SmtcPeriod parameter.

SMTC	Meaning	Parameter
SMTC duration	Time during which the UE performs SSB measurements in each SMTC period.	- The intra-frequency SMTC duration is specified by the NRDUCell.SmtcDuration parameter. - The inter-frequency SMTC duration is specified by the NRCellFreqRelation.SmtcDuration parameter.
SMTC offset ^b	Time offset for the UE to start SSB measurements in each SMTC period.	- The intra-frequency SMTC offset is calculated by the gNodeB to ensure accurate SSB measurements by the UE, with no parameters needed for configuration. - The inter-frequency SMTC offset is specified by the NRCellFreqRelation.SaSmtcOffset parameter. If this parameter is set to 255, the offset is calculated by the gNodeB.
a: SMTC is frequency-specific. If the cells on the same frequency have different SSB periods (configured by the NRDUCell.SsbPeriod parameter), there is a possibility that SSB measurements are inaccurate for the cells whose SSB periods are longer than the SMTC period specific to the frequency. To prevent this, ensure that the cells on the same frequency use the same SSB period (configured by the NRDUCell.SsbPeriod parameter) according to the network plan. b: If a parameter-defined SMTC offset, if the SMTC window cannot fully include the SSB of the target frequency due to inappropriate settings, inter-frequency measurement will be incomplete. This can make specific neighboring frequencies unavailable for handovers, thereby increasing the service drop rate.		

For accurate SMTC offset calculation by the gNodeB, the frame offsets of frequency bands must meet certain requirements.

Requirements for the Frame Offsets of Frequency Bands

The frame offsets of frequency bands affect the SMTC offset calculation. If they do not meet the corresponding requirements, the SSB measurement result will be inaccurate. The frame offsets of frequency bands must meet all of the following conditions:

- The frame offset of a frequency band must be the same for all base stations on the entire network.
If an operator configures different frame offsets for gNodeBs operating on the same frequency band, each gNodeB applies its own frame offset to calculate the SMTC offset. In this scenario, when both (1) UEs are located in the overlapping area of gNodeBs with different frame offsets, and (2) the serving gNodeB's frame offset exceeds that of a neighboring gNodeB, the intra- and inter-frequency measurements by UEs on the cells of neighboring gNodeBs will be incomplete, leading to an increased service drop rate. To address this

issue, an additional frame offset (specified by the **gNBFreqBandConfig.AdditionalFrameOffset** parameter) is introduced to the frequency band. It is recommended that the operator configure this additional frame offset for all gNodeBs in the overlapping area and set it to the minimum frame offset across all gNodeBs in that area. When calculating the SMTC offset, the gNodeB checks for its own valid additional frame offset for the frequency band. If there is a valid additional frame offset, the gNodeB uses the smaller value between the frame offset and additional frame offset as the input for the SMTC offset calculation. This ensures the complete intra- and inter-frequency measurements by UEs on the cells of neighboring gNodeBs operating on the frequency band.

Additionally, the cells served by gNodeBs with different frame offsets on the same frequency band have the same SSB sweeping period but different sweeping start times. As such, the SMTC duration must meet the following requirements to ensure that it can cover the SSB sweeping periods of all cells on the frequency band (see a configuration example in [4.4.1.2 Using MML Commands](#)):

- For all cells in the overlapping area served by gNodeBs operating on the frequency band: intra-frequency SMTC duration $\geq$ $\min\{\text{Roundup}\{\text{Maximum frame offset among all gNodeBs} + \text{SSB sweeping period (fixed at 2 ms) in low-frequency scenarios} - \text{Minimum frame offset among all gNodeBs}\}, 5\}$. Note that the frame offset should be expressed in units of ms.
- For all cells in the overlapping area served by gNodeBs not operating on the frequency band: inter-frequency SMTC duration $\geq$ $\min\{\text{Roundup}\{\text{Maximum frame offset among all gNodeBs} + \text{SSB sweeping period (fixed at 2 ms) in low-frequency scenarios} - \text{Minimum frame offset among all gNodeBs}\}, 5\}$. Note that the frame offset should be expressed in units of ms.
- If the frame offsets for the to-be-measured frequency bands differ, the maximum difference must not exceed 3 ms.

If the UE needs to measure multiple frequency bands with different frame offsets, the gNodeB selects the maximum gap to cover the SMTC durations of all the frequency bands to be measured. The protocol-defined maximum gap is 6 ms, but when the frame offsets for multiple to-be-measured frequency bands differ by up to more than 3 ms, the maximum gap may not cover the SMTC durations of all the to-be-measured frequency bands. As a result, inter-frequency measurement is incomplete, affecting the number of handover attempts.

- If the frame offsets for the serving and to-be-measured frequency bands differ, the maximum difference must not exceed 4 ms.

If it exceeds 4 ms, there is a possibility that inter-frequency measurement is incomplete, affecting the number of handover attempts.

- If the frame offset of a frequency band is configured on a base station on the network, the frame offset of this band must also be specified on neighboring base stations, even if these neighboring base stations do not serve any cell in this band. This ensures that this frequency band is accurately measured by UEs served by these neighboring base stations.

 NOTE

The effective frame offset of a frequency band configured on a base station is described as follows:

- When the frame offset of a frequency band is configured using **gNBFreqBandConfig.FrameOffset**, this configured value takes effect for the frequency band.
- When the frame offset of a frequency band is not configured, the frame offset of the base station (configured using **gNodeBParam.FrameOffset**) applies.

Subsequent frame offset-related parameter descriptions are based on **gNBFreqBandConfig.FrameOffset** (frame offset of a frequency band). For details about frame offsets of frequency bands, see *Standards Compliance*.

4.1.4.5 Measurement Gap Configuration

A simple approach to inter-frequency measurements for a UE is to configure two types of radio frequency receivers on the UE for separate measurements of the serving cell's frequency and a neighboring cell's frequency. However, this approach requires a high cost and results in strong interference among frequencies. As such, **3GPP specifications propose a measurement gap. It indicates a period of time, which is reserved for inter-frequency measurements only.** During a measurement gap, the UE does not send or receive any data but tunes its receiver towards a target frequency to perform inter-frequency measurements. When the measurement gap elapses, the UE tunes its receiver back to the serving frequency.

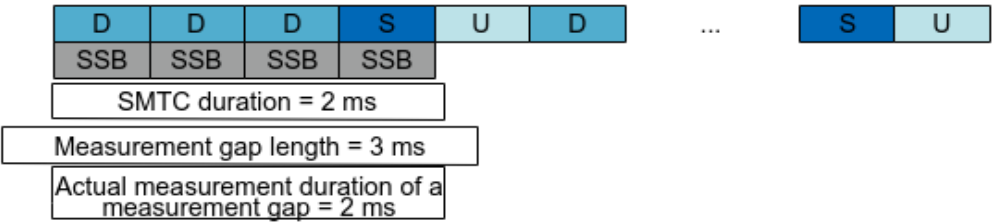
There are 24 protocol-defined gap patterns, with each specific to a measurement gap length and a measurement gap period. Measurement gap configurations are sent to UEs through measurement configurations. In the current version, the gNodeB supports gap pattern adaptation based on UE capabilities and SMTC to minimize the overhead of gap-assisted measurements. For details about measurement gaps, see section 9.1.2 "Measurement gap" in 3GPP TS 38.133 V15.5.0.

When an inter-frequency measurement is required, the gNodeB determines whether to send a measurement gap as follows:

- If a measurement gap has been configured, a new measurement gap is not sent.
- If no measurement gap has been configured, a measurement gap is sent.

To ensure the complete measurement of a non-serving frequency, the actual measurement duration within a measurement gap must cover the entire SMTC duration of this frequency. The actual measurement duration equals the measurement gap minus the time required for a frequency switch before and after the measurement. **Figure 4-9** shows an example for TDD.

Figure 4-9 Example of measurement gap and SMTC duration configuration



If a UE needs to start both intra-frequency and inter-frequency measurements and the SMTC period for intra-frequency measurements is greater than or equal to the measurement gap period, the SMTC duration for intra-frequency measurements is entirely included in the inter-frequency measurement gap. As a result, intra-frequency measurements cannot be performed. To avoid this problem, the gNodeB configures equal-opportunity intra-frequency and inter-frequency measurements, allowing the UE to alternatively perform such measurements through gap sharing.

If a UE needs to initiate inter-frequency measurements, frame offsets must be configured in accordance with the requirements in [Requirements for the Frame Offsets of Frequency Bands](#) for the serving and to-be-measured frequency bands.

Non-Overlapping Between Gap and NR Resources

The uplink and downlink throughput of UEs decrease in the measurement gaps for inter-frequency measurements because UEs cannot transmit uplink signals or receive downlink signals on the serving frequency throughout the measurement gaps. In addition, the following situations may occur:

- UEs experience service drops if PUCCH SRs and gap-assisted measurements conflict in the time domain due to the PUCCH SR period being greater than or equal to the gap period.
- The uplink IBLER and uplink RBLER increase if SRSs and gap-assisted measurements conflict at specific time-domain positions due to the SRS period being shorter than the gap period. This negatively affects the timeliness of measurement reporting, resulting in deteriorated service experience for moving UEs.

To address this issue, the "non-overlapping between gap and NR resources" and "optimized non-overlapping between gap and NR resources" functions are introduced.

- Non-overlapping between gap and NR resources takes effect when the NR resource allocation period is longer than or equal to the gap period.
- Optimized non-overlapping between gap and NR resources takes effect when the NR resource allocation period is shorter than the gap period.

The above two functions are designed to stagger the channel resources of specific signals (such as SRS, PUCCH SR, CSI-RS for CM, and CSI-RS for CM measurement reporting in low frequency bands) from measurement gaps in the time domain.

- Non-overlapping between gap and NR resources is controlled by the **GAP_AVOID_NR_RES_SW** option of the **gNB MobilityCommParam.MeasAlgoSwitch** parameter. This function is supported only in low frequency bands. This function is supported only in SA networking.
 - If this option is deselected, the channel resources for the aforesaid signals are not staggered from measurement gaps in the time domain. In this case, measurement gaps preferentially take effect on UEs, which do not transmit or receive signals throughout measurement gaps.
 - If this option is selected, the channel resources for the aforesaid signals are staggered from measurement gaps in the time domain. This setting reduces the loss in uplink and downlink UE throughput caused by gap-assisted measurements, reduces the rate of service drops caused by

conflicts between PUCCH SR and gap-assisted measurements, but prolongs the measurement duration.

- Optimized non-overlapping between gap and NR resources is controlled by the **GAP_AVOID_NR_RES_OPT_SW** option of the **NRDUCellMeas.MeasAlgoSwitch** parameter. This function is supported only in low frequency bands. It is recommended that this function be enabled when the channel quality fluctuates significantly.
 - If this option is deselected, the channel resources for the aforesaid signals are not staggered from measurement gaps in the time domain. In this case, measurement gaps preferentially take effect on UEs, which do not transmit or receive signals throughout measurement gaps.
 - If this option is selected, the channel resources for the aforesaid signals are staggered from measurement gaps in the time domain. This lowers the uplink IBLER and uplink RBLER, reduces the number of delayed measurement reports, but may increase the number of reconfiguration signaling messages. (IBLER: initial block error rate; RBLER: residual block error rate)

Non-Gap-Assisted Measurements for Active BWP (Trial)

The non-gap-assisted measurements for active BWP function is not yet ready for full commercial release in the current version for certain reasons such as industry chain (terminal) compatibility. However, this function can still be used for testing purposes or commercial network trials. As such, this function is used as a trial function in the current version.

The UEs capable of non-gap-assisted measurements support inter-frequency measurements (only within the frequency-domain range of the active BWP) without stopping downlink scheduling.

In this context, Huawei introduces non-gap-assisted measurements for active BWP. This function is developed to reduce data transmission interruptions during inter-frequency measurements for the preceding UEs in a cell. In addition, this function reduces the service drops caused by time-domain conflicts between PUCCH SRs and gap-assisted measurements, improving user experience.

The **SSB_MEAS_WITHOUT_GAP_SW** option of the **NRDUCellMeas.SsbMeasPolicySwitch** parameter specifies whether to enable non-gap-assisted measurements for active BWP, which is supported only in low frequency bands. When this function is enabled, it takes effect for the UE that meets all of the following conditions:

- The UE supports non-gap-assisted measurements. For details, see the *interFrequencyMeas-NoGap-r16* IE in the UECapabilityInformation message.
- At least one of the non-serving frequencies to be measured is within the frequency-domain range of the UE's active BWP.

After non-gap-assisted measurements for active BWP takes effect, the gNodeB sends an RRCReconfiguration message containing the *interFrequencyConfig-NoGap-r16* IE with the value **TRUE** to a UE that meets the preceding conditions. For non-serving frequencies included in the frequency-domain range of the UE's active BWP, the gNodeB determines whether to calculate or deliver measurement gaps as follows:

- If both the non-serving frequency to be measured and the serving frequency are TDD frequencies:
 - If an additional frame offset with a valid value is configured for a non-serving frequency to be measured, the gNodeB calculates and delivers the measurement gap for this frequency.
 - If an additional frame offset is not configured or is set to an invalid value for a non-serving frequency to be measured, then:
 - If the non-serving frequency to be measured has the same frame offset as the serving frequency, the gNodeB does not calculate and deliver the measurement gap for this non-serving frequency.
 - If the non-serving frequency to be measured has a different frame offset from the serving frequency, the gNodeB calculates and delivers the measurement gap for this non-serving frequency.

The additional frame offset is specified by the **gNBFreqBandConfig.AdditionalFrameOffset** parameter. A value of **4294967295** for this parameter indicates an invalid additional frame offset; a value other than **4294967295** indicates a valid one.

- In other scenarios, the gNodeB calculates and delivers measurement gaps for the corresponding non-serving frequencies.

NOTE

After non-gap-assisted measurements for active BWP takes effect, the gNodeB always calculates and delivers measurement gaps for the non-serving frequencies outside the frequency-domain range of the UE's active BWP.

Active BWP: the dedicated BWP activated for a UE in RRC_CONNECTED mode at a specified time. As defined in NR specifications, only one dedicated BWP can be activated for a UE in RRC_CONNECTED mode at a time. For details about the dedicated BWP and active BWP, see *UE Power Saving*.

In carrier aggregation scenarios, non-gap-assisted measurements for active BWP takes effect only for the non-serving frequencies within the frequency-domain range of the active BWP for the PCC of a UE enabled with carrier aggregation.

Gap-assisted Scheduling Compatibility

In low-frequency NR TDD scenarios, if the measurement gap timing advance (MGTA) value delivered by the gNodeB is not 0, UEs with compatibility issues fail to decode information in the first slot following the measurement gap. This triggers RLC retransmissions, resulting in increased delay for these UEs. To mitigate this, the gap-assisted scheduling compatibility function is introduced. This function is controlled by the **GAP_SCH_COMP_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter. When this option is selected, the function takes effect, and no service or signaling data is transmitted in the first slot following the measurement gap, improving the performance and experience of UEs with compatibility issues.

NOTE

For details about MGTA, see section 6.3.5 "Measurement information elements" in 3GPP TS 38.331 V16.7.0.

4.1.4.6 Other Configurations

Optimized Configuration for deriveSSB-IndexFromCell

The gNodeB includes the *deriveSSB-IndexFromCell* IE in measurement configurations.

- For intra-frequency neighboring cell measurements, the value of this IE (either set to **TRUE** or **FALSE**) indicates whether UEs derive the SSB indexes of all intra-frequency neighboring cells based on the timing information of the serving cell.
- For inter-frequency neighboring cell measurements, it indicates whether UEs derive the SSB indexes of all neighboring cells on the frequency of any detected inter-frequency cell based on the timing information of that cell.

If all cells on a frequency are time-synchronized, you can improve the efficiency and accuracy of detecting neighboring cells' SSB indexes by setting this IE to **TRUE**.

Optimized configuration for deriveSSB-IndexFromCell has been introduced, and is enabled if the **PROTOCOL_IE_OPT_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter is selected. When this function is enabled, the value of the *deriveSSB-IndexFromCell* IE in the measurement configurations sent from the gNodeB varies with the RATs corresponding to the frequencies that are to be measured.

If an NR TDD frequency is to be measured, the value of the *deriveSSB-IndexFromCell* IE is **TRUE**.

Configuration of SSB Beam Measurement Result Selection and Combination

A UE needs to combine the SSB beam-level measurement results of a cell when calculating the cell-level SSB measurement result. These beam-level results must meet the SSB combination threshold requirement. The SSB combination threshold is specified by the **NRCellMobilityConfig.SsbConsolidationThld** parameter. Upon detecting that multiple SSB beams in a cell meet the combination threshold requirement, the UE combines the measurement results of these SSB beams into a cell-level result. The maximum number of SSB beams whose measurement results can be combined is specified by the **NRCellMobilityConfig.SsbNumToAverage** parameter.

Measurement Filtering Configuration

Before evaluating the conditions for entering or leaving an event and sending measurement reports, the UE filters measurement results based on filtering coefficients as follows:

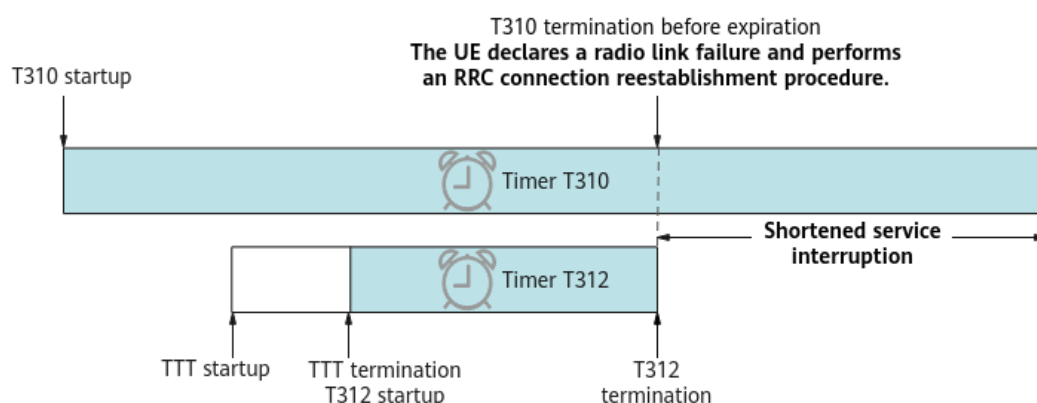
- Beam-level filtering, in which the filtering coefficient is specified by the **NRCellMobilityConfig.BeamRsrpFilterCoeff** parameter.
- Cell-level filtering, in which the filtering coefficient is specified by the **NRCellMobilityConfig.CellRsrpFilterCoeff** parameter.

Configuration of Timer T312

RRC connection reestablishment is a service processing procedure initiated by a UE to quickly reestablish the RRC connection. When a UE detects out-of-synchronization with the serving cell, the UE starts timer T310 (specified by **NRDUCellUeTimerConst. T310**). The UE declares a radio link failure when timer T310 expires, and then it performs an RRC connection reestablishment procedure to restore its service connection as soon as possible.

To reduce the time for RRC connection reestablishment due to out-of-synchronization at the physical layer during a handover, timer T312 (**NRCeUeTimer. T312**) has been introduced. As shown in [Figure 4-10](#), timer T312, which is shorter than timer T310, can trigger an RRC connection reestablishment procedure more quickly, so that recovery of a radio link failure takes a shorter time and the service interruption is shortened. It is recommended that **NRCeUeTimer. T312** be set to a value less than that of **NRDUCellUeTimerConst. T310**.

Figure 4-10 T312-based fast recovery of a radio link failure



Only in SA networking and when the **NRCeUeTimer. T312** parameter is not set to **NOT_CONFIG** for the serving cell, the gNodeB delivers the T312 setting to a UE that meets all of the following conditions:

- The UE supports T312. That is, the value of the **pcellT312-r16** field is "supported" in the UE Capability Information message sent by the UE over the Uu interface.
- The UE triggers a necessary handover, which means a coverage-based intra-frequency handover or coverage-based inter-frequency handover. In CA scenarios, the UE triggers a PCell change.

If a period of time (defined by *TimeToTrig*) elapses throughout which the measurement result meets the entering condition for an event, the UE will send a measurement report to the gNodeB and, if timer T310 has been started, timer T312 will be started. When timer T312 expires, the UE initiates an RRC connection reestablishment procedure.

For details about RRC connection reestablishment and timer T310, see *5G Networking and Signaling*.

4.1.5 Measurement Reporting

A UE performs measurements based on the measurement configuration received from the gNodeB, and filters the measurement results based on filtering coefficients. When determining that the entering conditions for the corresponding event are met, the UE sends measurement reports to the gNodeB.

- The maximum number of cells that can be reported by a UE is indicated by the *maxReportCells* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.
- The reporting type of a measurement report is indicated by the *reportType* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.
- The reporting interval of a UE is indicated by the *reportInterval* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.
- The number of reporting times allowed for a UE is indicated by the *reportAmount* IE (**MeasConfig** > **ReportConfigToAddModList**) in the RRCReconfiguration message.

4.1.6 Target Cell or Frequency Decision

The process includes:

- Measurement report processing (needed only in measurement-based mode)
- Determination of a handover policy
- Generation of the target cell or frequency

Measurement Report Processing

The gNodeB processes measurement reports in a first-come-first-handled manner. Based on the results, the gNodeB generates candidate cells or frequencies.

Determination of a Handover Policy

The gNodeB selects a handover policy for a UE based on information about candidate cells or frequencies, the UE's current service type, and UE capabilities. Handover policies include:

- Handover: When referred to as a handover policy, it means that the gNodeB, while retaining the RRC connections of a UE, transfers the UE from the source cell to the target cell to ensure service continuity.
- Redirection: The gNodeB releases the RRC connections of a UE and instructs the UE to access a cell working on a certain frequency.

 NOTE

- Unless otherwise stated, handover is used in this chapter as a generic term that covers procedures related to mobility management in connected mode.
- When referred to as a handover policy, handover is a specific term in contrast to redirection and describes a procedure in which the serving cell of a UE changes.
- To perform a medium-priority blocking for a cell, the RRC_CONNECTED UEs in the cell must be migrated out. The gNodeB selects redirection for transferring UEs out of the cell.

Generation of the Target Cell or Frequency

The target cell or frequency is generated as follows, depending on the processing modes and handover policies:

- Measurement-based mode
 - If the handover policy is handover, the gNodeB generates candidate cells based on measurement reports and then filters the candidate cells according to [4.1.6.2 Neighboring Cell Filtering Rules](#) to generate the target cell.
 - If the handover policy is redirection, the gNodeB generates a candidate frequency based on a measurement report. This candidate frequency is the frequency of cells in the measurement report. Then, the gNodeB releases the RRC connection of the UE and instructs the UE to access a cell working on this frequency.
- Blind mode
 - a. The gNodeB generates the target frequency for the UE as follows:

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Generating the Target Frequency
On	Any RFSP index is associated with a valid gNodeB frequency list group.	-	-	-	Method 1

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Generating the Target Frequency
On	No RFSP index is associated with a valid gNodeB frequency list group.	Not configured	-	-	Method 3
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	On	The operator index has been associated with a valid gNodeB frequency list group.	Method 2
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	On	The operator index is not associated with a valid gNodeB frequency list group.	Method 3
On	No RFSP index is associated with a valid gNodeB frequency list group.	Configured	Off	-	Method 3

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Generating the Target Frequency
Off	-	Configured	On	The operator index has been associated with a valid gNodeB frequency list group.	Method 2
Off	-	Configured	On	The operator index is not associated with a valid gNodeB frequency list group.	Method 3
Off	-	Configured	Off	-	Method 3

Setting of the RFSP-based NR Inter-Frequency Handover Switch	Configuration of RFSP Index 0 or the UE's RFSP Index	Whether the NR Cell's Operator-specific Policy Is Configured	Operator-specific Neighboring Frequency Configuration Switch	Configuration of the Operator Index for the UE's Serving PLMN	Method of Generating the Target Frequency
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Note:

- "-" in the table indicates that the corresponding factor does not need to be considered.
- RFSP-based NR inter-frequency handover is controlled by the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter. For details, see *Flexible User Steering*.
- An RFSP index can be associated with a gNodeB frequency list group through a mapping between the **gNB RfspConfig.gNBFreqPriorityGroupId** and **gNBFreqPriorityGroup.gNBFreqPriorityGroupId** parameters. The **gNB RfspConfig.gNBFreqPriorityGroupId** parameter is valid only when it is set to a value other than 255.
- The NR cell's operator-specific policy is configured using the **NRCellOpPolicy** MO. For details, see *Multi-Operator Sharing*.
- An operator index can be associated with a gNodeB frequency list group through a mapping between the **NRCellOpPolicy.gNBFreqPriorityGroupId** and **gNBFreqPriorityGroup.gNBFreqPriorityGroupId** parameters. The **NRCellOpPolicy.gNBFreqPriorityGroupId** parameter is valid only when it is set to a value other than 255.
When configuring the **gNBFreqPriorityGroup** MO associated with the serving PLMN of a UE, ensure that all neighboring frequencies specific to this operator are configured. The neighboring frequencies that are not configured will not be delivered to the UE, and the UE will not be handed over to cells working on these frequencies.
- Operator-specific neighboring frequency configuration is controlled by the **OPERATOR_NR_FREQ_CFG_SW** option of the **NRCellOpPolicy.FreqConfigPolicySwitch** parameter.

- Method 1: The gNodeB filters NR frequencies in the intersection of the following NR frequencies using the filtering rules described in [4.1.6.1 Frequency Filtering Rules](#). Then, the gNodeB generates a candidate frequency list based on frequency priorities. From the generated candidate frequency list, the gNodeB selects the frequency with the highest priority as the target frequency.
 - NR frequencies in the gNodeB frequency list group associated with the RFSP index

- NR frequencies determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters
- Method 2: The gNodeB filters NR frequencies in the intersection of the following NR frequencies using the filtering rules described in **4.1.6.1 Frequency Filtering Rules**. Then, the gNodeB generates a candidate frequency list based on frequency priorities. From the generated candidate frequency list, the gNodeB selects the frequency with the highest priority as the target frequency.
 - NR frequencies in the gNodeB frequency list group associated with the operator index for the UE's serving PLMN
 - NR frequencies determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters
- Method 3: The gNodeB filters NR frequencies that are determined by the **NRCellFreqRelation.SsbDescMethod** and **NRCellFreqRelation.SsbFreqPos** parameters using the filtering rules described in **4.1.6.1 Frequency Filtering Rules**. Then, the gNodeB generates a candidate frequency list based on frequency priorities. From the generated candidate frequency list, the gNodeB selects the frequency with the highest priority as the target frequency.
- b. The gNodeB instructs the UE to perform subsequent operations based on the specific handover policy.
 - If the handover policy is handover, the gNodeB filters cells working on the target frequency according to **4.1.6.2 Neighboring Cell Filtering Rules** to generate the target cell, and then instructs the UE to perform a handover to the target cell.
 - If the handover policy is redirection, the gNodeB releases the RRC connection of the UE and instructs the UE to access a cell working on the target frequency.

4.1.6.1 Frequency Filtering Rules

In blind mode, the gNodeB filters non-serving NR frequencies based on the following filtering rules, and then generates a candidate frequency list using the filtered non-serving NR frequencies.

- **Filtering based on UE band capabilities**

For all UEs, the gNodeB filters out the frequencies of bands that are not supported by UEs.

The frequency bands supported by a UE are the intersection of frequency bands indicated by the *supportedBandListNR* and *supportedBandCombinationList* IEs in the UECapabilityInformation message signaled over the Uu interface.

- **Filtering based on UE handover capabilities**

For all UEs, the gNodeB filters out the non-serving frequencies that are not supported by the UEs during inter-frequency handovers for which the handover policy does not support redirection. The details are as follows:

- If the UE does not support inter-frequency handovers (indicated by the absence of *handoverInterF* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out all non-serving frequencies for the UE.
- If the UE does not support inter-frequency FDD-TDD handovers (indicated by the absence of *handoverFDD-TDD* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out the corresponding TDD or FDD frequencies for the UE.
For example, if the UE is in a TDD cell, the gNodeB filters out all non-serving FDD frequencies for the UE; if the UE is in an FDD cell, the gNodeB filters out all non-serving TDD frequencies for the UE.
- If the UE does not support inter-frequency FR1-FR2 handovers (indicated by the absence of *handoverFR1-FR2* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out the corresponding FR1 or FR2 frequency for the UE.
For example, if the UE is in a cell on FR1, the gNodeB filters out all non-serving frequencies on FR2 for the UE; if the UE is in a cell on FR2, the gNodeB filters out all non-serving frequencies on FR1 for the UE.
- If the UE does not support network access from a cell on FR2 (indicated by the absence of *pCell-FR2* in the UECapabilityInformation message over the Uu interface), the gNodeB filters out all FR2 frequencies for the UE.
This means that the UE will not be handed over or redirected from a cell on FR1 to a cell on FR2.
- **Non-serving frequency filtering based on the frequency interference flag**
For all UEs involved in unnecessary blind inter-frequency handovers, the gNodeB also selects candidate frequencies based on the frequency interference flag. If the **NRCellFreqRelation.FreqIntrfrFlag** parameter is set to **NO_INTRF** for a frequency, the frequency can be selected as a candidate frequency. Otherwise, the frequency cannot be selected as a candidate frequency.
- **Non-serving frequency filtering based on the inter-frequency handover policy for voice-service-capable UEs**
For VoNR-incapable UEs of the home operator in roaming scenarios, the gNodeB filters frequencies based on the setting of the **VOICE_INTER_FREQ_HO_POLICY_SW** option of the **NRCellFreqRelation.AggregationAttribute** parameter. This option takes effect only for VoNR-incapable UEs of the home operator and applies only to low frequency bands.
 - If this option is selected for a frequency, the gNodeB filters out the frequency for the preceding UEs and prohibits them from being handed over to the frequency.
 - If this option is deselected for a frequency, the gNodeB does not filter out the frequency for the preceding UEs and allows them to be handed over to the frequency.

For details about how to identify VoNR-incapable UEs of the home operator and the scenarios where the voice-capability-based inter-frequency handover policy is recommended, see [Measurement Frequency Filtering](#) in [4.1.4.1.2 Frequency Selection for Measurement](#).
- **Non-serving frequency filtering based on the measurement activation flag**

For all UEs, the gNodeB filters out frequencies for which the **NRCellFreqRelation.MeasActvFlag** parameter is set to **INACTIVE**.

- **Non-serving frequency filtering based on the inter-frequency periodic MR type**

For all UEs, the gNodeB filters out frequencies for which the **NRCellFreqRelation.InterFreqPrdcMrType** parameter is set to **PERIODIC_MR_ONLY** or **FORBID_MEAS** during inter-frequency handovers except RFSP-based inter-frequency handover back to the HPLMN.

- **Other non-serving frequency filtering rules**

For all UEs, the gNodeB filters out SDL frequencies that belong to band n75.

For all UEs, the gNodeB filters out frequencies for which the **HIGH_SPEED_UE_REDIRECT_ONLY_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter is selected during inter-frequency handovers where the high-speed UE return function is not involved.

For all UEs, the gNodeB filters out frequencies for which the **MOB_NEIGHBOR_FREQ_FILTER_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter is selected.

4.1.6.2 Neighboring Cell Filtering Rules

For measurement-based inter-frequency handovers, the gNodeB filters candidate cells based on the following neighboring cell filtering rules to generate the target cell. For blind inter-frequency handovers, the gNodeB filters cells on the target frequency based on the following neighboring cell filtering rules to generate the target cell.

- The gNodeB filters out cells that do not have NR neighbor relationships (configured by the **NRCellRelation.MO**) with the serving cell.
- The gNodeB filters out the cells to which handovers are forbidden. That is, the gNodeB filters cells based on the value of the **NRCellRelation.NoHoFlag** parameter. For example, when this parameter is set to **FORBID_HO**, the corresponding cells are filtered out for all UEs; when this parameter is set to **SA_FORBID_NSA_PERMIT_HO**, the corresponding cells are filtered out for SA UEs but not for NSA UEs.
- The gNodeB filters out the cells that are not supported by the current UE (SA or NSA UE).

For an SA UE, candidate neighboring cells in the **NSA** networking option are filtered out, while candidate neighboring cells in the **SA** or **SA_NSA** networking option are retained.

Networking option of a neighboring cell:

- The networking option of a neighboring cell is specified by **NRExternalNCell.NrNetworkingOption** (applicable in scenarios other than multi-operator sharing) or **NRExternalNCellPlmn.NrNetworkingOption** (applicable in multi-operator sharing scenarios).
- The networking options of neighboring cells can be automatically maintained through ANR when the **NR_NR_ANR_SW** option of the **NRCellAlgoSwitch.AnrSwitch** parameter is selected. For details about ANR, see *ANR*.

- In the case of coverage-based inter-frequency handover (**6 Coverage-based Inter-Frequency Handover**), the gNodeB filters neighboring cells based on their RSRQs or SINRs.
 - If **NRCellFreqRelation.FreqIntrfFlag** is set to **INTRF** for a frequency, the gNodeB filters the cells working on this frequency based on the following rules:
 The gNodeB filters candidate cells based on the measured RSRQs. That is, the gNodeB determines whether the RSRQ of a cell meets the following condition. If it is met, the gNodeB filters out the cell. Otherwise, the gNodeB does not filter out the cell.

$$Mn + Ofn + Ocn + Hys < Thresh2$$
 where
 - *Mn* is the measured RSRQ of the neighboring cell.
 - *Ofn* is the frequency-specific offset for RRC_CONNECTED UEs, and is specified by the **NRCellFreqRelation.ConnFreqOffset** parameter.
 - *Ocn* is the CIO, and is specified by the **NRCellRelation.CellIndividualOffset** parameter. If the value is not **DB0**, this parameter is included in measurement configurations. If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.
 - *Hys* is the hysteresis for events related to inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.InterFreqA4A5Hyst** parameter.
 - *Thresh2* is the RSRQ threshold for events related to coverage-based inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2** parameter.
 - If **NRCellFreqRelation.FreqIntrfFlag** is set to **INTRF_SINR** for a frequency, the gNodeB filters the cells working on this frequency based on the following rules:
 The gNodeB filters candidate cells based on the measured SINRs. That is, the gNodeB determines whether the SINR of a cell meets the following condition. If the condition is met, the gNodeB filters out the cell. Otherwise, the gNodeB does not filter out the cell.

$$Mn + Ofn + Ocn + Hys < Thresh2$$
 where
 - *Mn* is the measured SINR of the neighboring cell.
 - *Ofn* is the frequency-specific offset for RRC_CONNECTED UEs, and is specified by the **NRCellFreqRelation.ConnFreqOffset** parameter.
 - *Ocn* is the CIO, and is specified by the **NRCellRelation.CellIndividualOffset** parameter. If the value is not **DB0**, this parameter is included in measurement configurations. If the value is **DB0**, this parameter is not included in measurement configurations and **0** is used for calculation by default.

- *Hys* is the hysteresis for events related to inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.InterFreqA4A5Hyst** parameter.
- *Thresh2* is the SINR threshold for coverage-based inter-frequency measurements, and is specified by the **NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2** parameter.
- **In low-frequency scenarios, the gNodeB filters out the ground neighboring cells with SINRs below a specific threshold for UEs using specific bearers.**

In specific scenarios, if (1) the cell-specific switch for RSRP- and SINR-based joint handover is turned on, and (2) the bearer-specific switch for RSRP- and SINR-based joint handover is turned on for any bearer of a UE, RSRP- and SINR-based joint handover will take effect for the UE. With this function, the gNodeB uses UE-reported SINR measurements to evaluate ground neighboring cells and, as a result, filters out those whose SINRs fall below a specific threshold. The specific scenarios and related parameters are explained as follows:

 - Specific scenarios refer to handover or redirection scenarios other than the following:
 - Blind handovers
 - Blind redirections
 - Handovers in which the gNodeB performs evaluations based on measurement reports—for example, in a handover scenario where experience-based smart carrier selection is applied, the gNodeB evaluates user experience after receiving the reports.
 - Grid-based handovers
 - Related parameters:
 - Cell-specific RSRP- and SINR-based joint handover switch: **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter
 - Bearer-specific RSRP- and SINR-based joint handover switch: **RSRP_SINR_HO_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch** parameter
 - Ground neighboring cell flag: **NRExternalNCell.AerialCellFlag** set to **GROUND_CELL**
 - Threshold: Indicates the threshold corresponding to the bearer with the highest handover priority on a UE. For details about bearer-specific handover priorities, refer to the descriptions of QCI priorities for handovers in [4.1.4.2.3 QCI-specific Measurement Event Configuration](#).
 - In the case of necessary handovers, the threshold is specified by the **NRCellInterFHoMeaGrp.RsrpSinrBasedNecSinrThld** parameter.

- In the case of unnecessary handovers, the threshold is specified by the **NRCellInterFHoMeaGrp.RsrpSinrBasedUnecSinrThld** parameter.

- **The gNodeB filters out the neighboring cells that handover restrictions apply to.**

When the **MOBILITY_RESTRICTION_SW** option of the **gNBMobilityCommParam.MobilityAlgoSwitch** parameter is selected, the gNodeB filters out the cells that handover restrictions apply to according to the *Mobility Restriction List* IE in the INITIAL CONTEXT SETUP REQUEST message sent from the AMF.

- **The gNodeB filters out neighboring cells with PLMNs not included in the list of those allowed for UE handovers.**

If the target cell supports multiple PLMNs (including the current serving PLMN), the source gNodeB preferentially selects the current serving PLMN as that used after the handover.

Table 4-16 lists the PLMNs allowed for intra-RAT handovers in SA networking given different settings of the **INTRA_RAT_HO_WITH_GNB_EPLMN_SW** option (specifying whether to enable EPLMN-based intra-RAT inter-PLMN handover) of the **gNodeBParam.EqvPlmnAlgoSwitch** parameter.

Table 4-16 PLMNs allowed for UE handovers

EPLMN-based Intra-RAT Inter-PLMN Handover Switch	PLMN Allowed for UE Handovers
Deselected	Current serving PLMN
Selected	● Current serving PLMN ● EPLMNs obtained by the gNodeB from the <i>Mobility Restriction List</i> IE ● EPLMNs configured using the gNBEqvPlmn MO
Note: ● When configuring the gNBEqvPlmn MO, ensure that there is a roaming agreement between the PLMN corresponding to gNBEqvPlmn.OperatorId and the EPLMN collectively determined by gNBEqvPlmn.EquivalentMcc and gNBEqvPlmn.EquivalentMnc. ● The PLMNs that are graylisted or blacklisted using the PLMN ID management function are excluded from those allowed for UE handovers. For details about the PLMN ID management function, see <i>ANR</i>.	

- **The gNodeB filters out the neighboring cells whose frequency bands do not match UE frequency band capabilities and whose uplink and downlink bandwidths do not match UE bandwidth capabilities. The details are as follows:**
 - For intra-base-station handovers, the gNodeB filters out certain neighboring cells as follows:

- If an intra-base-station neighboring cell is configured with a frequency band that does not match the UE-supported frequency bands, the gNodeB filters out the neighboring cell.
- If the minimum bandwidth supported by a UE is greater than the downlink bandwidth and uplink bandwidth of an intra-base-station neighboring cell, the gNodeB filters out the neighboring cell.
- For inter-base-station handovers, the gNodeB filters out certain neighboring cells that have been configured with frequency bands as follows:
 - If an inter-base-station neighboring cell is configured with a frequency band that does not match the UE-supported frequency bands, the gNodeB filters out the neighboring cell.
 - If an Xn interface is set up between the gNodeB and a neighboring base station, the gNodeB can obtain the uplink and downlink bandwidths of inter-base-station neighboring cells. In this scenario, if the minimum bandwidth supported by a UE is greater than the downlink bandwidth and uplink bandwidth of an inter-base-station neighboring cell, the gNodeB filters out the neighboring cell. If an Xn interface is not set up between the gNodeB and a neighboring base station, the gNodeB cannot obtain the uplink and downlink bandwidths of inter-base-station neighboring cells. In this scenario, the gNodeB does not filter out such neighboring cells based on UE bandwidth capabilities.
- For inter-base-station handovers, the gNodeB does not filter out inter-base-station neighboring cells that are not configured with frequency bands based on UE frequency band capabilities or UE bandwidth capabilities.

Where

- The frequency band of an intra-base-station neighboring cell is specified by the **NRCell.FrequencyBand** parameter, which is mandatory.
- The frequency band of an inter-base-station neighboring cell is specified by the **NRExternalNCell.FrequencyBand** parameter, which is optional.
- The frequency bands supported by a UE are the intersection of frequency bands indicated by the *supportedBandListNR* and *supportedBandCombinationList* IEs in the UECapabilityInformation message signaled over the Uu interface.
- The minimum bandwidth supported by a UE can be queried from the *channelBWs-UL*, *channelBWs-DL*, *channelBWs-UL-v1590*, and *channelBWs-DL-v1590* IEs in the UECapabilityInformation message signaled over the Uu interface.

For details, see section 4.2 "UE Capability Parameters" in 3GPP TS 38.306 V16.5.0.

- **The gNodeB filters out the NR cells served by blacklisted gNodeBs.**

After other gNodeBs are added to the blacklist using the **gNBNRBlacklist** MO, the gNodeB filters out the NR cells served by blacklisted gNodeBs. For details about principles and configurations of a gNodeB blacklist, see *ANR*.

4.1.7 Handover Execution

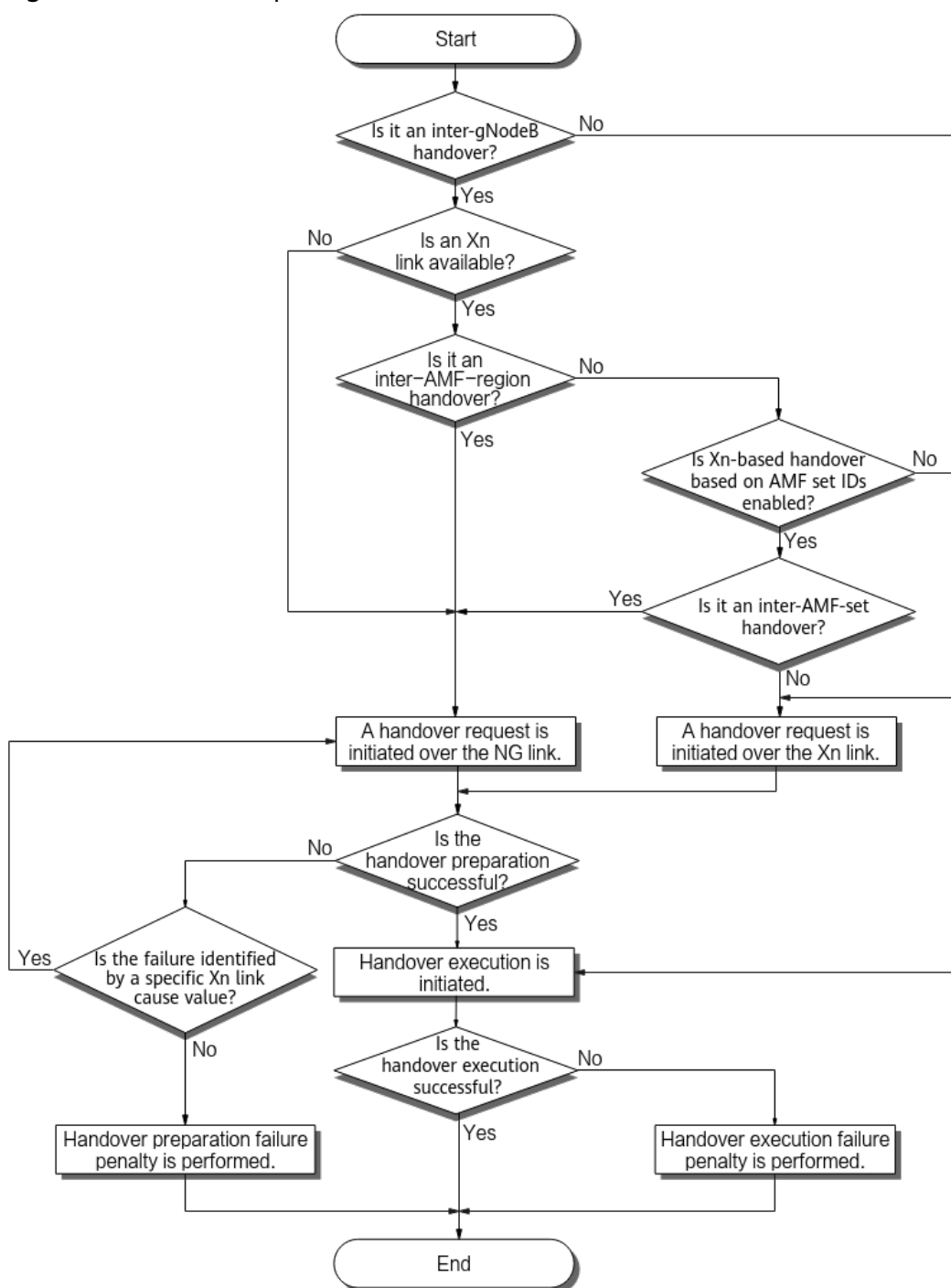
4.1.7.1 Handover Execution

After determining the target cell or frequency list, the gNodeB executes a handover based on the selected handover policy, which can be handover or redirection.

Handover

Figure 4-11 shows the procedure when the handover policy is handover.

Figure 4-11 Handover procedure



1. Inter-gNodeB determination: The source gNodeB selects the cell with the best quality from the target cell or frequency list and then starts inter-gNodeB determination.
 - If the source cell and the target cell are served by the same gNodeB, handover execution is initiated. In this case, the procedure goes to [5](#).
 - If the source cell and the target cell are served by different gNodeBs, the procedure goes to [2](#).

2. Link determination: The source gNodeB checks whether it has an Xn link with the target gNodeB.
 - If an Xn link exists between the gNodeB serving the source cell and the gNodeB serving the target cell, the procedure goes to 3.
 - If no Xn link is present between the gNodeB serving the source cell and the gNodeB serving the target cell, the source gNodeB initiates an NG-based handover request. In this case, the procedure goes to 4.
3. Inter-AMF-region or inter-AMF-set determination: The source gNodeB checks whether it belongs to the same AMF region or AMF set as the target gNodeB. In an Xn self-setup procedure (detailed in *NG and Xn Self-Management*), the target gNodeB and source gNodeB exchange AMF region ID information through an Xn Setup Request message.
 - If the source and target gNodeBs belong to different AMF regions, an NG-based handover request is initiated. In this case, the procedure goes to 4.
 - If the source and target gNodeBs belong to the same AMF region:

The target gNodeB sends the AMF set ID to the source gNodeB through a proprietary message, namely Private Vendor Configuration Transfer, over the Xn interface. Upon receipt of the message, the source gNodeB determines the subsequent processing, which depends on the enabled/disabled status of Xn-based handover based on AMF set IDs.

 - If Xn-based handover based on AMF set IDs is disabled for the source gNodeB, the source gNodeB does not check whether it belongs to the same AMF set as the target gNodeB based on the received AMF set ID. Instead, the source gNodeB initiates an Xn-based handover request and the procedure goes to 4.

In this scenario, the source gNodeB includes its globally unique AMF identifier (GUAMI) in the handover request that is sent to the target gNodeB. Based on the GUAMI sent from the source gNodeB, the target gNodeB then checks whether it belongs to the same AMF set as the source gNodeB. If the target gNodeB belongs to a different AMF set, the target gNodeB responds to the source gNodeB with a Handover Preparation Failure message with the cause value "Unknown GUAMI ID". After that, the source gNodeB attempts to initiate an NG-based handover request. As such, the Xn-based handover preparation success rate of the source cell is low in this scenario.
 - If Xn-based handover based on AMF set IDs is enabled for the source gNodeB, the source gNodeB checks whether it belongs to the same AMF set as the target gNodeB based on the received AMF set ID.
 - If the source and target gNodeBs belong to different AMF sets, the source gNodeB initiates an NG-based handover request and the procedure goes to 4.
 - If the source and target gNodeBs belong to the same AMF set, the source gNodeB initiates an Xn-based handover request and the procedure goes to 4.

If the source and target gNodeBs belong to different AMF sets and the Xn-based handover preparation success rate of the source cell is low, enabling Xn-based handover based on AMF set IDs for the

source gNodeB increases the Xn-based handover preparation success rate of the source cell.

The **AMF_SET_ID_BASED_XN_HO_SW** option of the **gNB MobilityCommParam.ProtocolCompatibilitySw** parameter specifies whether to enable Xn-based handover based on AMF set IDs.

4. Handover preparation: After receiving the handover request, the target gNodeB prepares for the handover and responds with the preparation result.
 - If the admission to the target gNodeB succeeds, the target gNodeB sends a Handover Request Acknowledge or Handover Command message to the source gNodeB. As such, the source gNodeB considers the handover preparation to have succeeded and initiates handover execution. In this case, the procedure goes to 5.
 - If the admission to the target gNodeB fails, the target gNodeB sends a Handover Preparation Failure message to the source gNodeB.
 - If the failure cause value is "Invalid AMF Set ID", "Unknown GUAMI ID", or "Transport Resource Unavailable", the source gNodeB re-initiates a handover request over the NG link.
 - If any other failure cause value is received, the source gNodeB considers the handover preparation to have failed and applies a penalty. For details, see [4.1.8 Handover Failure Penalty](#).

During handover preparation, if a handover request is initiated over the Xn interface, the target AMF selection process proceeds as follows:

- The target gNodeB preferentially selects the UE's serving AMF as the target AMF. However, an AMF with the *Relative AMF Capacity* IE set to 0 cannot be selected. To allow a UE to select the serving AMF with the *Relative AMF Capacity* IE set to 0 as the target AMF for a handover, you can enable the path switch AMF selection function (by selecting the **PATH_SWITCH_AMF_SEL_SW** option of the **gNodeBParam.NgInterfaceAlgoExtSw** parameter). Path switch AMF selection is enabled by default. For details about the *Relative AMF Capacity* IE, see section 9.3.1.32 "Relative AMF Capacity" in 3GPP TS 38.413 V17.6.0.
 - If (1) the target gNodeB cannot select the UE's serving AMF and instead selects a backup AMF, and (2) the target gNodeB and the selected AMF are supplied by different vendors, the UE may encounter a handover failure, resulting in a service drop. To mitigate this risk, the path switch backup AMF selection function can be enabled to prevent selection of a backup AMF. This function is controlled by the **PATH_SWITCH_BACKUP_AMF_SEL_SW** option of the **gNodeBParam.NgInterfaceAlgoExtSw** parameter. When this option is selected, the target gNodeB prevents selection of a backup AMF. Consequently, if handover preparation over the Xn interface fails, the UE can perform handover preparation over the NG interface instead. When this option is deselected, the target gNodeB can select a backup AMF. Path switch backup AMF selection is enabled by default.
5. Handover execution: The source gNodeB sends the handover command to the UE, instructing it to execute the handover.

- If the handover is successfully executed, the source gNodeB releases resources after receiving the Release Resource message from the target gNodeB.

In the case of an Xn-based handover, the target gNodeB forwards the uplink data of the UE to the core network after the handover is completed. As such, this process interrupts uplink data transmission. To reduce the interruption duration, the function of Xn-based handover interruption duration optimization has been introduced. This function is controlled by the **XN_HO_INTERRUPT_DUR_OPT_SW** option of the **gNBMobilityCommParam.ProtocolCompatibilitySw** parameter.

- If this option is deselected, the target gNodeB starts to forward uplink data to the core network after receiving a PATH SWITCH REQUEST ACKNOWLEDGE message from the AMF.
 - If this option is selected, the target gNodeB starts to forward uplink data to the core network after receiving an RRCReconfigurationComplete message from the UE. This shortens the duration of uplink data transmission interruption during an Xn-based handover.
- If the handover execution fails, the handover execution failure penalty begins. For details, see [4.1.8 Handover Failure Penalty](#).

If the UE experiences a service drop during the handover, the UE re-accesses the network through RRC connection reestablishment. For details about RRC connection reestablishment, see *5G Networking and Signaling*.

NOTE

For details about AMF regions, see *NG-flex*.

Upon receiving a handover request message from the source gNodeB, the target gNodeB automatically determines whether it complies with a protocol version earlier than that of the source gNodeB. If so, the target gNodeB replies with a handover request response message that carries full configurations and the *fullConfig* IE. The source gNodeB then sends the UE a handover command that carries full configurations and the *fullConfig* IE.

Redirection

If the handover policy is redirection, the gNodeB selects the cell of the highest-priority frequency from the target cell or frequency list, and sends it to the UE through an RRC Release message. The UE then performs redirection.

4.1.7.2 Handling Related to Handover Execution

Conflict Handling Between the Voice Fallback and Handover Procedures

Assume that the gNodeB is preparing for an intra-RAT outgoing handover. During the preparation, if the core network initiates a procedure for setting up a bearer with 5QI 1 for voice services and the gNodeB decides to perform voice fallback, the handover procedure conflicts with the voice fallback procedure. Alternatively, if the core network initiates an emergency call fallback procedure during the preparation, the handover procedure conflicts with the emergency call fallback procedure. The **VOICE_FB_FIRST_SWITCH** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter determines how to handle the preceding conflict between handovers and PDU sessions.

- If the **VOICE_FB_FIRST_SWITCH** option of this parameter is selected, the gNodeB preferentially handles voice fallback or emergency call fallback and sends a handover cancellation request.
- If the **VOICE_FB_FIRST_SWITCH** option of this parameter is deselected, the gNodeB preferentially handles the handover procedure and responds with a message, notifying the core network that the voice fallback or emergency call fallback fails as the gNodeB is performing an intra-RAT handover. After the handover is completed, the core network re-initiates a voice fallback or emergency call fallback procedure, as specified in 3GPP TS 23.501. If the core network does not support re-initiation of the said procedure, the voice fallback or emergency call fallback procedure will fail.

Conflict Handling Between the QoS and Handover Procedures

Assume that the gNodeB is preparing for an intra-RAT outgoing handover. During the preparation, if the gNodeB receives PDU session setup or modification signaling indicating the addition or modification of a QoS flow, the handover procedure conflicts with the QoS procedure. The **QOS_FLOW_FIRST_SW** option of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter determines how to handle the preceding conflict between the QoS and handover procedures.

- When the **QOS_FLOW_FIRST_SW** option of this parameter is selected, the gNodeB preferentially handles the QoS procedure and initiates a handover cancellation request. It is recommended that this option be selected when the core network does not support PDU session setup or modification in the target cell during a handover procedure.
- When the **QOS_FLOW_FIRST_SW** option of this parameter is deselected, the gNodeB preferentially handles the handover procedure and notifies the core network that the gNodeB is performing an intra-RAT handover. After the handover is complete, the core network re-initiates a QoS procedure, as specified in 3GPP TS 23.501. If the core network does not support the re-initiation of the said procedure, the QoS procedure will fail.

NOTE

This function applies to all 5QI-specific QoS flow procedures. It can be enabled for a specific 5QI by binding the 5QI to the **NRCellQciBearer.Qci** parameter. Note that among the bearers with 5QI 1 (with the **NRCellQciBearer.Qci** parameter set to 1), this function is only applicable to the bearers carrying VoNR services. For details about the handling of conflicts between the 5QI-1-specific voice fallback and handover procedures, see [Conflict Handling Between the Voice Fallback and Handover Procedures](#).

Conflict Handling Between the PDU Session Release and NG-based Handover Procedures

The signaling procedures on the core network experience exceptions when the delay between the core network and NEs is abnormal. In this situation, if (1) the intra-RAT NG-based outgoing handover preparation phase is in progress on the gNodeB, and (2) the gNodeB receives a PDU session release message, then the NG-based handover procedure will conflict with the PDU session release procedure. Consequently, the handover fails and the UE fails to be handed over or released.

To address this issue, a new function is introduced: conflict handling between the PDU session release and NG-based handover procedures. This function is

controlled by the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter. With this function, the gNodeB handles the conflict between the PDU session release and NG-based handover procedures as follows:

- If the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter is selected, the gNodeB preferentially handles the PDU session release procedure and initiates a handover cancellation request.
- If the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter is deselected, the gNodeB preferentially handles the NG-based handover procedure and notifies the core network that an intra-RAT NG-based handover procedure is in progress on the gNodeB. In this situation, the core network starts a timer if it can re-initiate a PDU session release procedure (according to section 4.9.1 "Handover procedures in 3GPP access" in 3GPP TS 23.502 V18.6.0). When the core network detects a handover completion or failure message or when the timer expires, the core network attempts to re-initiate a PDU session release procedure.

 NOTE

This function is applicable only to NG-based handover requests that are irrelevant to voice fallback procedures. For details about conflict handling between the voice fallback and handover procedures, see [Conflict Handling Between the Voice Fallback and Handover Procedures](#).

Handling of Oversized Inter-Base-Station NG-based Handover Request Messages

During an inter-base-station NG-based handover, if an oversized handover request message (HANDOVER REQUIRED) exceeds the specifications of the core network, the message will be truncated or rejected by the core network, leading to a handover failure. To address this issue, Huawei introduces the handling of oversized inter-base-station NG-based handover request messages. That is, the size of the UE capability field (specified by **gNBMobilityCommParam.IntraRatHoReqUeCapbSize**) in an intra-RAT handover request message can be configured on the gNodeB side.

To use this function, the size of the UE capability field in an intra-RAT handover request message needs to be properly configured on the gNodeB side, based on the core network's capability to process handover request messages. After the size is configured, the gNodeB checks whether the size of the UE capability field (indicating NR, E-UTRA, and EN-DC capabilities of the UE, detailed in section 11.2.2 "Message definitions" in 3GPP TS 38.331 V17.4.0) reported by the UE exceeds the configured value, and then determines UE capability information to be carried in the handover request message sent to the core network.

- When the size of the UE capability field in an intra-RAT handover request message is set to a value other than 0, handling of oversized inter-base-station NG-based handover request messages is enabled. Specifically,
 - If the size of the UE capability field reported by the UE exceeds the configured value, the NG-based handover request message carries only the NR capabilities reported by the UE.

- If the size of the UE capability field reported by the UE does not exceed the configured value, the NG-based handover request message carries the complete capabilities reported by the UE.
- When the size of the UE capability field in an intra-RAT handover request message is set to **0**, handling of oversized inter-base-station NG-based handover request messages is disabled, and the NG-based handover request message carries the complete capabilities reported by the UE.

Handling in the Case of Abnormal UEs or Interconnection Across Multiple Vendors

The handover command carries only incremental configurations during a handover that involves abnormal UEs or interconnection across multiple vendors. This triggers interconnection across multiple vendors or compatibility issues due to abnormal UEs in such a handover. As a result, the service drop rate increases and the traffic volume drops, affecting user experience. To address this issue, full configurations during inter-base-station and intra-base-station handovers are introduced. When "full configurations during inter-base-station handovers" is enabled, the source gNodeB does not forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB. As a result, these packets are lost, negatively impacting user experience. To address this, the DRB session forwarding function is introduced.

- Full configurations during inter-base-station handovers: For inter-base-station handovers, the **INTER_GNB_HO_FULL_CONFIG_SW** option of the **NRCCellAlgoSwitch.HoCompatibilitySwitch** parameter needs to be selected if the source and target gNodeBs are provided by different vendors or there are abnormal UEs on live networks. When this option is selected, upon receiving a handover request message from the source gNodeB, the target gNodeB replies with a handover request response message that carries full configurations and the *fullConfig* IE. The source gNodeB then sends the UE a handover command that carries full configurations and the *fullConfig* IE. When this option is deselected, the source gNodeB sends the UE a handover command that carries only incremental configurations. This function applies to inter-base-station handovers in NSA or SA networking.

Specially, "full configurations during inter-base-station handovers" needs to be disabled by selecting the **FULL_CONFIG_OPT_SW** option of the **NRCCellFeatureSwitch.NrHoCompatibilitySwitch** parameter when all of the following conditions are met in the same scenario: (i) an Xn-based handover occurs in SA networking, (ii) the target and source gNodeBs are provided by the same vendor, and (iii) "full configurations during inter-base-station handovers" is enabled. Disabling the function under these conditions prevents the loss of specific packets for the UE involved in the handover, thereby ensuring user experience.

- DRB session forwarding: This function is controlled by the **DRB_SESSION_FWD_SW** option of the **gNBMMobilityCommParam.MobilityAlgoSwitch** parameter. When this function takes effect, the source gNodeB can forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB, thereby reducing packet loss and enhancing user experience. It is recommended that this function be enabled when "full configurations during inter-base-station handovers" is enabled. This function is supported only in SA networking.

- Full configurations during intra-base-station handovers: For intra-base-station handovers, the **SA_INTRA_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter needs to be selected if there are abnormal UEs on live networks. When this option is selected, the gNodeB sends the UE a handover command that carries full configurations and the *fullConfig* IE. When this option is deselected, the gNodeB sends the UE a handover command that carries only incremental configurations. This function applies only to intra-base-station handovers in SA networking.

Full configurations during inter-base-station or intra-base-station handovers can take effect only for specified UEs. For details, see [Table 4-17](#) and [Table 4-18](#).

Table 4-17 Function effectiveness of full configurations during inter-base-station handovers

Setting of the HO_FULL_CONFIG_SW_ ON Option of the gNBUECompatSw. <i>Whit elistCtrlSwitch</i> Parameter	Setting of the INTER_GNB_HO_FULL_C ONFIG_SW Option of the NRCellAlgoSwitch. <i>HoCo mpatibilitySwitch</i> Parameter	Function Effectiveness for UEs
Deselected	Deselected	Full configurations during inter-base-station handovers do not take effect for any UE.
Deselected	Selected	Full configurations during inter-base-station handovers take effect for all UEs.
Selected	Deselected	Full configurations during inter-base-station handovers do not take effect for any UE.

Setting of the HO_FULL_CONFIG_SW_ ON Option of the gNBUECompatSw. <i>WhitelistCtrlSwitch</i> Parameter	Setting of the INTER_GNB_HO_FULL_CONFIG_SW Option of the NRCellAlgoSwitch. <i>HoCompatibilitySwitch</i> Parameter	Function Effectiveness for UEs
Selected	Selected	• Full configurations during inter-base-station handovers take effect only for specific UEs that correspond to the UE information index for which the HO_FULL_CONFIG_SW_ON option of the gNBUECompatSw.<i>WhitelistCtrlSwitch</i> parameter is selected. • Full configurations during inter-base-station handovers do not take effect for other UEs.

Table 4-18 Function effectiveness of full configurations during intra-base-station handovers

Setting of the HO_FULL_CONFIG_SW_ ON Option of the gNBUECompatSw. <i>WhitelistCtrlSwitch</i> Parameter	Setting of the SA_INTRA_GNB_HO_FULL_CONFIG_SW Option of the NRCellAlgoSwitch. <i>HoCompatibilitySwitch</i> Parameter	Function Effectiveness for UEs
Deselected	Deselected	Full configurations during intra-base-station handovers do not take effect for any UE.
Deselected	Selected	Full configurations during intra-base-station handovers take effect for all UEs.

Setting of the HO_FULL_CONFIG_SW_ON Option of the gNBUECompatSw.WhitelistCtrlSwitch Parameter	Setting of the SA_INTRA_GNB_HO_FULL_CONFIG_SW Option of the NRCCellAlgoSwitch.HoCompatibilitySwitch Parameter	Function Effectiveness for UEs
Selected	Deselected	Full configurations during intra-base-station handovers do not take effect for any UE.
Selected	Selected	- Full configurations during intra-base-station handovers take effect only for specific UEs that correspond to the UE information index for which the HO_FULL_CONFIG_SW_ON option of the gNBUECompatSw.WhitelistCtrlSwitch parameter is selected. - Full configurations during intra-base-station handovers do not take effect for other UEs.

NOTE

The **HO_FULL_CONFIG_SW_ON** option of the **gNBUECompatSw.WhitelistCtrlSwitch** parameter takes effect only when the **INTER_GNB_HO_FULL_CONFIG_SW** or **SA_INTRA_GNB_HO_FULL_CONFIG_SW** option of the **NRCCellAlgoSwitch.HoCompatibilitySwitch** parameter is selected.

4.1.8 Handover Failure Penalty

In the case of a handover preparation failure:

- If handover preparation of the cell with the best signal quality fails in the process of a necessary handover, the gNodeB sorts the cells in the measurement report in descending order of signal quality and instructs the UE to initiate a handover attempt to the cell with the second-best signal quality. If the UE has tried all cells in the candidate cell list but handover preparation still fails, the gNodeB waits for another measurement report from the UE.
- If handover preparation of the cell with the best signal quality fails in the process of an unnecessary handover, the gNodeB waits for another

measurement report from the UE because the gNodeB instructs the UE to initiate a handover attempt only to the cell with the best signal quality in this scenario.

If handover preparation fails in the process of a blind handover, the gNodeB terminates the handover procedure.

The handover failure penalty mechanism aims to reduce the number of ineffective handover retries that follow a handover failure. Depending on the execution phase in which the handover failure occurs, handover failure penalty is classified into handover preparation failure penalty and handover execution failure penalty.

Handover Preparation Failure Penalty

Depending on the causes, handover preparation failures are categorized as resource-caused handover preparation failures and non-resource-caused handover preparation failures.

- **For resource-caused handover preparation failures, a penalty timer is configured to impose penalties on the target cell. A handover retry cannot be initiated to the target cell for the UE until the penalty timer expires.**
 - For coverage-based intra-frequency and inter-frequency handovers, the penalty timer is specified by the **gNBMobilityCommParam.NecHoPrepFailPunishTimer** parameter.
 - For other handover functions, the penalty timer is specified by the **gNBMobilityCommParam.OptHoPrepFailPunishTimer** parameter.

NOTE

For handover preparation failures with the cause value "Transport resource unavailable", the following applies within the penalty time:

- If the UE involved is an NSA DC UE, the NSA DC UEs that are connected to the same eNodeB and served by the same PLMN as this NSA DC UE cannot initiate handover attempts to the target cell.
- If the UE involved is an SA UE, the SA UEs that are served by the same PLMN as this SA UE cannot initiate handover attempts to the target cell.
- **For non-resource-caused handover preparation failures, the number of penalty times is configured to impose penalties on the target cell. A handover retry cannot be initiated to the target cell for the UE until the number of penalty times is reached.** If the handover preparation still fails after a retry, the UE can no longer initiate a handover to the target cell until the UE re-camps on a cell. The number of penalty times is specified by the **gNBMobilityCommParam.NonResHoPrepFailPunishTimes** parameter.

For a serving cell, the total number of the following types of punished neighboring cells cannot exceed the predefined limit of punished neighboring cells. Otherwise, the handover preparation failure penalty cannot be imposed on other neighboring cells. Specifically, if the total number of punished neighboring cells has reached the limit, the UE can continue to initiate a handover attempt to another neighboring cell, but the gNodeB does not impose penalty on the neighboring cell even when it experiences a handover preparation failure. That is, the number of penalty times (specified by the **gNBMobilityCommParam.NonResHoPrepFailPunishTimes** parameter) is invalid for the neighboring cell.

- Punished intra-frequency neighboring NR cells, due to handover preparation failures from the serving NR cell to intra-frequency neighboring NR cells
- Punished inter-frequency neighboring NR cells, due to handover preparation failures from the serving NR cell to inter-frequency neighboring NR cells
- Punished neighboring LTE cells, due to handover preparation failures from the serving NR cell to neighboring LTE cells

In this context, the **NON_RES_INTRA_NR_HO_PENALTY_SW** and **NON_RES_EUTRAN_HO_PENALTY_SW** options of the **NRCCellAlgoSwitch.HoCompatibilitySwitch** parameter are introduced for low frequency band scenarios. The two options are used as follows:

- In the case that the total number of punished neighboring cells of the preceding types has exceeded the limit of punished neighboring cells, the **NON_RES_INTRA_NR_HO_PENALTY_SW** option of the **NRCCellAlgoSwitch.HoCompatibilitySwitch** parameter can be used to control whether to prohibit intra- or inter-frequency handovers within NR.
 - When the **NON_RES_INTRA_NR_HO_PENALTY_SW** option is deselected, the UE can continue to initiate a handover attempt to another intra- or inter-frequency neighboring NR cell, and the gNodeB does not impose penalty on the neighboring cell even when it experiences a handover preparation failure.
 - When the **NON_RES_INTRA_NR_HO_PENALTY_SW** option is selected, the UE is no longer allowed to initiate a handover attempt to any other intra- or inter-frequency neighboring NR cell.
- In the case that the total number of punished neighboring cells of the preceding types has exceeded the limit of punished neighboring cells, the **NON_RES_EUTRAN_HO_PENALTY_SW** option of the **NRCCellAlgoSwitch.HoCompatibilitySwitch** parameter can be used to control whether to prohibit handovers from NR to LTE.
 - When the **NON_RES_EUTRAN_HO_PENALTY_SW** option is deselected, the UE can continue to initiate a handover attempt to another neighboring LTE cell, and the gNodeB does not impose penalty on the neighboring cell even when it experiences a handover preparation failure.
 - When the **NON_RES_EUTRAN_HO_PENALTY_SW** option is selected, the UE is no longer allowed to initiate a handover attempt to any other neighboring LTE cell.

 NOTE

- After cell A enters the penalty state, the number of penalty times is incremented by one each time the measurement result indicates that cell A meets the handover condition.
- Retry means that a request for a handover to cell A is sent again when this cell meets the handover condition after the specified penalties imposed on it have ended.
- Resource-caused handover preparation failures are only those whose failure causes are as follows:
 - No radio resources available in target cell
 - Transport resource unavailable
 - Not enough User Plane Processing Resources
 - Radio resources not available
 - Control Processing Overload

Handover Execution Failure Penalty

If a UE fails to be handed over to a target cell, its RRC connection will be reestablished in the source cell. When a handover attempt is initiated to the target cell again and the handover preparation attempt succeeds, the source gNodeB instructs the UE to initiate a handover retry to the target cell. If the retries to the target cell continue to fail after 10 attempts, the source gNodeB no longer instructs the UE to send requests for a handover to the target cell to prevent a higher service drop rate due to exceptions.

4.1.9 Anti-Ping-Pong Mechanism

If an unnecessary handover function is enabled for multiple cells, a UE will be handed over back to the original source cell, due to inappropriate network planning, through unnecessary handovers of the same or different types. That is, the UE experiences ping-pong handovers. Following are examples:

- After a UE is handed over from cell 1 to cell 2 through a frequency-priority-based handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.
- After a UE is handed over from cell 1 to cell 2 through a frequency-priority-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a service-based inter-frequency handover.

To minimize the probability of ping-pong handovers, Huawei introduces a range of anti-ping-pong functions, as described in [Table 4-19](#).

 NOTE

The anti-ping-pong functions described in this section apply to inter-frequency handovers. Unless otherwise specified, "unnecessary handover function" in this section refers to any intra-RAT unnecessary inter-frequency handover function, including those related to special services or specific scenarios not covered in this document. For details, see [Table 3-1](#).

Table 4-19 Anti-ping-pong functions

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Subscription to uplink interference information about inter-frequency neighboring cells	INTER_FREQ_UL_INTRF_SW option of the NRCellAlgoSwitch.NCellInfoCtrlSwitch parameter	Target cell or frequency decision	In the target cell or frequency decision phase of an unnecessary handover function, the serving cell obtains uplink interference information about intra- or inter-base-station neighboring cells. If the serving cell determines that the target cell is exposed to low uplink interference, the gNodeB instructs the UE to perform a handover to the target cell. Otherwise, the UE is not instructed to perform a handover. For details about how the serving cell determines whether a neighboring cell is exposed to low uplink interference, see 10.1.4 Target Cell or Frequency Decision .	When an unnecessary handover function is enabled in the serving cell and uplink-interference-based inter-frequency handover is enabled in the target cell, the probability of ping-pong handovers decreases if the serving cell is enabled with subscription to uplink interference information about inter-frequency neighboring cells.

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Inter-frequency handover collaboration	INTER_FREQ_HO_COLLABORATION_SWITCH option of the NRCellAlgoSwitch . InterFreqHoSwitch parameter	Target cell or frequency decision	In the target cell or frequency decision phase of an unnecessary handover function, if the serving cell identifies that the target cell has entered the high load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell.	When an unnecessary handover function is enabled in the serving cell and inter-frequency mobility load balancing (MLB) by transferring UEs in connected mode is enabled in the target cell, the probability of ping-pong handovers decreases if inter-frequency handover collaboration is enabled in the serving cell. For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see <i>Mobility Load Balancing in Connected Mode</i> .
Inter-frequency handover collaboration	INTER_FREQ_HO_COLLABORATION_SWITCH option of the NRCellAlgoSwitch . InterFreqHoSwitch parameter	Target cell or frequency decision	In the target cell or frequency decision phase of an unnecessary handover function, the serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.	When an unnecessary handover function is enabled in the serving cell and multi-frequency beam coordination is enabled in the target cell, the probability of ping-pong handovers decreases if inter-frequency handover collaboration is enabled in the serving cell. For details about high-load beams and multi-frequency beam coordination, see <i>Multi-Frequency Smart Selection</i> .

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Inter-frequency handover collaboration	INTER_FREQ_HO_COLLABORATION_SWITCH option of the NRCellAlgoSwitch . <i>InterFreqHoSwitch</i> parameter	Target cell or frequency decision	In the target cell or frequency decision phase of frequency-priority-based inter-frequency handover, the serving cell checks whether the UE has a voice bearer. If so, the gNodeB does not instruct the UE to perform a handover to the target cell. If not, the gNodeB instructs the UE to perform a handover to the target cell.	When frequency-priority-based inter-frequency handover is enabled in the serving cell and voice-quality-based inter-frequency handover is enabled in the target cell, the probability of ping-pong handovers decreases if inter-frequency handover collaboration is enabled in the serving cell. For details about voice-quality-based inter-frequency handover, see <i>VoNR</i> .

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Anti-ping-pong function based on the inter-frequency handover waiting time	NRCellMobilityConfig.InterFreqHoWaitTime set to a value other than 0	Target cell or frequency decision	For a UE that is handed over to a cell (the current serving cell) through any handover function, a timer (specified by NRCellMobilityConfig.InterFreqHoWaitTime) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an unnecessary handover function to which the anti-ping-pong function is applied before the timer expires. If the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time preferentially takes effect.	If the anti-ping-pong function based on the inter-frequency handover waiting time is enabled for the current serving cell, the UE cannot be handed over back to the original serving cell within a short period of time through an unnecessary handover function to which the anti-ping-pong function is applied. In this case, an unnecessary handover function to which the anti-ping-pong function is applied can be: • Operator-specific-priority-based inter-frequency handover • Uplink-interference-based inter-frequency handover • SSB SINR-based inter-frequency handover (for LampSite) • RFSP-based inter-frequency handover back to the HPLMN • Experience-based smart carrier selection • Uplink multi-frequency coordination • MLB by transferring UEs in connected mode • Uplink-service-based carrier selection in high-speed scenarios • Downlink-service-based carrier selection in high-speed scenarios For details about operator-specific-priority-based inter-frequency handover, see 8 Operator-specific-Priority-

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
				based Inter-Frequency Handover. For details about uplink-interference-based inter-frequency handover, see 10 Uplink-Interference-based Inter-Frequency Handover. For details about SSB SINR-based inter-frequency handover (for LampSite), see 11 SSB SINR-based Inter-Frequency Handover (for LampSite). For details about RFSP-based inter-frequency handover back to the HPLMN, see <i>Flexible User Steering.</i> For details about experience-based smart carrier selection and uplink multi-frequency coordination, see <i>Multi-Frequency Smart Selection.</i> For details about MLB by transferring UEs in connected mode, see <i>Mobility Load Balancing in Connected Mode.</i> For details about uplink- and downlink-service-based carrier selection in high-speed scenarios, see <i>High Speed Mobility.</i>

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time	gNBScenarioDiffGrp.InterFreqHoWaitTime	Target cell or frequency decision	For a UE with "QCI-specific cell-load-dependent inter-frequency handover" in effect and handed over to a cell (the current serving cell) through any handover function, a timer (specified by gNBScenarioDiffGrp.InterFreqHoWaitTime) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an unnecessary handover function to which the anti-ping-pong function is applied before the timer expires. If the anti-ping-pong function based on the inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific load-dependent inter-frequency handover waiting time takes precedence.	If the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time is enabled for the current serving cell, the UE with "QCI-specific cell-load-dependent inter-frequency handover" in effect cannot be handed over back to the original serving cell within a short period of time through an unnecessary handover function to which the anti-ping-pong function is applied. In this case, an unnecessary handover function to which the anti-ping-pong function is applied can be: • Operator-specific-priority-based inter-frequency handover • Uplink-interference-based inter-frequency handover • SSB SINR-based inter-frequency handover (for LampSite) • RFSP-based inter-frequency handover back to the HPLMN • Experience-based smart carrier selection • Uplink multi-frequency coordination • MLB by transferring UEs in connected mode • Uplink-service-based carrier selection in high-speed scenarios • Downlink-service-based carrier selection in high-speed scenarios For details about operator-specific-priority-based inter-

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
				frequency handover, see 8 Operator-specific-Priority-based Inter-Frequency Handover. For details about uplink-interference-based inter-frequency handover, see 10 Uplink-Interference-based Inter-Frequency Handover. For details about SSB SINR-based inter-frequency handover (for LampSite), see 11 SSB SINR-based Inter-Frequency Handover (for LampSite). For details about RFSP-based inter-frequency handover back to the HPLMN, see <i>Flexible User Steering</i>. For details about experience-based smart carrier selection and uplink multi-frequency coordination, see <i>Multi-Frequency Smart Selection</i>. For details about MLB by transferring UEs in connected mode, see <i>Mobility Load Balancing in Connected Mode</i>. For details about uplink- and downlink-service-based carrier selection in high-speed scenarios, see <i>High Speed Mobility</i>.

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Anti-ping-pong function with delay for frequency-priority-based inter-frequency handover	NRCellMobilityConfig.InterFreqMeasDelay set to a value other than 0	Measurement configuration delivery	For a UE that is handed over to the current serving cell through any unnecessary handover, a timer (specified by NRCellMobilityConfig.InterFreqMeasDelay) is started when the incoming handover is complete. The gNodeB does not deliver frequency-priority-based inter-frequency measurement configurations to the UE before the timer expires.	If (1) both "frequency-priority-based inter-frequency handover" and "anti-ping-pong function with delay for frequency-priority-based inter-frequency handover" are enabled in the current serving cell, and (2) a UE is handed over to the current serving cell through any unnecessary handover, then the UE will not be handed over back to the original serving cell through "frequency-priority-based inter-frequency handover" within a short period of time.

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
QCI-specific suppression of unnecessary inter-frequency handover	INTER_FREQ_HO_SUPPRESS_SW option of the gNBQciBearer.MobilityFunInd parameter	Measurement configuration delivery	After an unnecessary handover function to which the anti-ping-pong function is applied is started, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. Specifically: • If the UE has a QCI-specific bearer for which the INTER_FREQ_HO_SUPPRESS_SW option of the gNBQciBearer.MobilityFunInd parameter is selected, the gNodeB does not deliver measurement configurations to the UE. • If the UE does not have a QCI-specific bearer for which the INTER_FREQ_HO_SUPPRESS_SW option of the gNBQciBearer.MobilityFunInd parameter is selected, the gNodeB delivers measurement configurations to the UE.	When an unnecessary handover function to which the anti-ping-pong function is applied is enabled in the serving cell and service-based inter-frequency handover is enabled in the target cell, the probability of ping-pong handovers decreases if the INTER_FREQ_HO_SUPPRESS_SW option of the gNBQciBearer.MobilityFunInd parameter is selected for the QCI-specific bearers associated with the target frequency group. In this case, an unnecessary handover function to which the anti-ping-pong function is applied can be: • Frequency-priority-based inter-frequency handover • Operator-specific-priority-based inter-frequency handover • SSB SINR-based inter-frequency handover (for LampSite) • MLB by transferring UEs in connected mode • Energy-efficiency-based inter-frequency handover • Multi-frequency beam coordination • Uplink-service-based carrier selection in high-speed scenarios • Downlink-service-based carrier selection in high-speed scenarios For details about frequency-priority-based inter-frequency handover, see 7

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
				Frequency-Priority-based Inter-Frequency Handover. For details about operator-specific-priority-based inter-frequency handover, see 8 Operator-specific-Priority-based Inter-Frequency Handover. For details about SSB SINR-based inter-frequency handover (for LampSite), see 11 SSB SINR-based Inter-Frequency Handover (for LampSite). For details about MLB by transferring UEs in connected mode, see <i>Mobility Load Balancing in Connected Mode.</i> For details about energy-efficiency-based inter-frequency handover, see <i>Energy Conservation and Emission Reduction.</i> For details about multi-frequency beam coordination, see <i>Multi-Frequency Smart Selection.</i> For details about uplink- and downlink-service-based carrier selection in high-speed scenarios, see <i>High Speed Mobility.</i>

Anti-Ping-Pong Function	Function Switch	Function Applied To	Function Description	Application Scenario
Anti-ping-pong function between service-based inter-frequency handover and voice-quality-based inter-frequency handover	VOICE_INTER_FREQ_HO_COLLAB_SW option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	Before the handover function start phase	For a UE that is handed over to a cell (the serving cell) through a voice-quality-based inter-frequency handover, a 10s timer is started when the incoming handover is complete. Initiating a service-based inter-frequency handover for the UE is allowed only when the timer expires. If the UE accesses the serving cell in other ways, relevant processing is not performed.	When service-based inter-frequency handover is enabled in the serving cell and voice-quality-based inter-frequency handover is enabled in the target cell, the probability of ping-pong handovers decreases if the serving cell is enabled with the anti-ping-pong function between service-based inter-frequency handover and voice-quality-based inter-frequency handover. For details about voice-quality-based inter-frequency handover, see <i>VoNR</i> .

 NOTE

After a preceding anti-ping-pong function takes effect, the number of inter-frequency handovers caused by the unnecessary handover function to which the anti-ping-pong function is applied decreases, but the probability of ping-pong handovers decreases and user experience improves.

An unnecessary handover function enables a UE to be handed over only to the neighboring cell with the strongest signal strength indicated by the measurement report. If such a neighboring cell happens to be the previous serving cell, and that cell has any of the functions enabled—"subscription to uplink interference information about inter-frequency neighboring cells", "inter-frequency handover collaboration", or "anti-ping-pong function based on the inter-frequency handover waiting time"—the UE cannot be handed over back to the previous serving cell. In this case, if the UE keeps performing inter-frequency measurements, the measurement gap will be widened, resulting in decreased throughput.

4.2 Network Analysis

4.2.1 Benefits

The basic function of mobility management in connected mode aims to select the cell with the best quality for UEs in RRC_CONNECTED mode to transmit data, ensuring continuous coverage and providing users with seamless service experience.

Other sub-functions have the following impacts on the network:

- **Configuration of the inter-frequency handover policy for voice service-capable UEs on non-serving frequencies**

After this function is enabled in the recommended scenario (described in [4.1.4.1 Measurement Object](#)), the number of call failures of VoNR-incapable UEs of the home operator decreases, the number of handover attempts in the 4G cell of the home operator decreases, and the number of outgoing EPS fallback preparation attempts in the 5G cell of the visited operator decreases.

- **Measurement ID preemption**

After this function is enabled in the case of measurement resource insufficiency, the rate of service drops due to such insufficiency decreases and voice service experience improves. If measurement resources are sufficient, this function has no impact on network performance.

- **Non-overlapping between gap and NR resources**

This function reduces the loss in uplink and downlink UE throughput caused by gap-assisted measurement and the rate of service drops caused by conflicts between PUCCH SR and gap-assisted measurement but prolongs the measurement duration.

Due to cell-level measurement, the values of the following indicators may change after this function is enabled. Specifically, the values of **N.ThpTime.UL.Cell** (duration of uplink data transmission at the MAC layer in a cell) and **N.ThpTime.DL.Cell** (duration of downlink data transmission at the MAC layer in a cell) increase; the values of **Cell Uplink Average Throughput (DU)** (average uplink cell throughput) and **Cell Downlink Average Throughput (DU)** (average downlink cell throughput) decrease.

- **Optimized non-overlapping between gap and NR resources**

This function lowers the uplink IBLER and uplink RBLER, reduces the loss in uplink and downlink UE throughput caused by gap-assisted measurements, reduces the number of delayed measurement reports, but may increase the number of reconfiguration signaling messages.

Due to cell-level measurement, the values of the following indicators may change after this function is enabled. Specifically, the values of **N.ThpTime.UL.Cell** (duration of uplink data transmission at the MAC layer in a cell) and **N.ThpTime.DL.Cell** (duration of downlink data transmission at the MAC layer in a cell) increase; the values of **Cell Uplink Average Throughput (DU)** (average uplink cell throughput) and **Cell Downlink Average Throughput (DU)** (average downlink cell throughput) decrease.

- **Non-gap-assisted measurements for active BWP**

This function improves user experience. This function reduces data transmission interruptions during inter-frequency measurements for UEs capable of non-gap-assisted measurements in a cell. In addition, this function reduces the service drops caused by time-domain conflicts between PUCCH SRs and gap-assisted measurements.

- **Gap-assisted scheduling compatibility**

This function reduces RLC retransmissions triggered by decoding failures experienced by UEs with compatibility issues, thereby minimizing delay performance degradation caused by retransmissions. This function reduces the **User Uplink Average Throughput (DU)** and **User Downlink Average Throughput (DU)** for UEs without compatibility issues.

- **Optimized configuration for deriveSSB-IndexFromCell**
In TDD scenarios, this function improves the efficiency and accuracy in UEs' detection of neighboring cells' SSB indexes because cells working on TDD frequencies are time-synchronized.
- **Configuration of timer T312**
Enabling this function (by setting **NRCeUeTimer.T312** to a value other than **NOT_CONFIG**) has the following impacts on the network:
 - On the UE side: After this function is enabled, the RRC connection reestablishment procedure due to out-of-synchronization at the physical layer takes a shorter time in the case of handovers, reducing service loss. When this function takes effect, a smaller value of **NRCeUeTimer.T312** results in faster UE service recovery and a larger value results in slower UE service recovery.
 - On the gNodeB side: After this function is enabled, downlink out-of-synchronization detection takes a shorter time, the number of RRC connection reestablishments may increase, and the RRC connection reestablishment ratio ($\frac{N.RRC.ReEst.Att}{(N.RRC.SetupReq.Succ + N.RRC.ResumeReq.Succ + N.HO.InterRAT.E2N.ExecSuccIn)} \times 100\%$) may increase. When this function takes effect, a smaller value of **NRCeUeTimer.T312** results in a higher RRC connection reestablishment ratio and a larger value results in a lower RRC connection reestablishment ratio.
- **Xn-based handover based on AMF set IDs**
If the source and target gNodeBs for a handover belong to different AMF sets, it is recommended that this function be enabled on the source gNodeB. In this case, the Xn-based handover preparation success rate of the source cell increases.
- **Xn-based handover interruption duration optimization**
This function shortens the duration of uplink data transmission interruption during Xn-based handovers, but can cause service drops due to incompatibility with the core network.
- **Conflict handling between the voice fallback and handover procedures**
In the case that a handover procedure conflicts with a voice fallback procedure or an emergency call voice fallback procedure, this function has the following impact:
 - The values of the following counters (related to canceled handovers) will increase when the **VOICE_FB_FIRST_SWITCH** option of the **gNB MobilityCommParam.HoPduSessConflHandlingSw** parameter is selected. It is recommended that the values of these counters be deducted during the calculation of the outgoing handover success rate.
 - **N.HO.IntraFreq.Prep.FailOut.HOCancel**
 - **N.HO.IntraFreq.FailOut.HOCancel**
 - **N.HO.InterFreq.Prep.FailOut.HOCancel**
 - **N.HO.InterFreq.FailOut.HOCancel**
 - When the **VOICE_FB_FIRST_SWITCH** option of the **gNB MobilityCommParam.HoPduSessConflHandlingSw** parameter is

deselected, the bearer establishment success rate for 5QI 1 and other 5QIs involved in the PDU session request for 5QI 1 establishment (indicated by the result of **N.QosFlow.Est.Succ.Cell5QI** divided by **N.QosFlow.Est.Att.Cell5QI**) will decline.

- **Conflict handling between the QoS and handover procedures**

In the case that the QoS and handover procedures conflict with each other, this function has the following impact:

- The values of the following counters (related to canceled handovers) will increase when the **QOS_FLOW_FIRST_SW** option of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter is selected. It is recommended that the values of these counters be deducted during the calculation of the outgoing handover success rate.
 - **N.HO.IntraFreq.Prep.FailOut.HOCancel**
 - **N.HO.IntraFreq.FailOut.HOCancel**
 - **N.HO.InterFreq.Prep.FailOut.HOCancel**
 - **N.HO.InterFreq.FailOut.HOCancel**
- When the **QOS_FLOW_FIRST_SW** option of the **NRCellQciBearer.HoQosFlowConflProcStrat** parameter is deselected, the bearer setup success rate for a specific 5QI (indicated by the result of **N.QosFlow.Est.Succ.Cell5QI** divided by **N.QosFlow.Est.Att.Cell5QI**) will decrease.

- **Conflict handling between the PDU session release and NG-based handover procedures**

In the case that the PDU session release and NG-based handover procedures conflict with each other, enabling the **PDU_SESS_REL_NG_HO_REQ_SEL_SW** option of the **gNBMobilityCommParam.HoPduSessConflHandlingSw** parameter has the following impacts:

- If the signaling procedure is abnormal on the core network side, the NG-based outgoing handover success rate increases. NG-based outgoing handover success rate: $(\text{N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut} + \text{N.HO.InterFreq.Ng.IntergNB.ExecSuccOut}) / (\text{N.HO.IntraFreq.Ng.IntergNB.PrepAttOut} + \text{N.HO.InterFreq.Ng.IntergNB.PrepAttOut}) \times 100\%$
- If the signaling procedure is normal on the core network side, the NG-based outgoing handover success rate decreases.

- **Handling of oversized inter-base-station NG-based handover request messages**

When handling of oversized inter-base-station NG-based handover request messages is enabled (by setting the **gNBMobilityCommParam.IntraRatHoReqUeCapbSize** parameter to a value other than 0), this function has the following impact on the network:

- A smaller value of this parameter results in a higher probability that an intra-RAT handover request carries only the NR capability of a UE, a lower probability that the handover request is rejected by the core network, a higher NG-based handover success rate, a higher probability that the target base station needs to query the UE's capability of other RATs, and higher air interface overhead of the target base station.

- A larger value of this parameter results in a lower probability that an intra-RAT handover request carries only the NR capability of a UE, a higher probability that the handover request is rejected by the core network, a lower NG-based handover success rate, a lower probability that the target base station needs to query the UE's capability of other RATs, and lower air interface overhead of the target base station.
- **Handling in the case of abnormal UEs or interconnection across multiple vendors**
For the function of full configurations during inter-base-station handovers in NSA or SA networking (controlled by the **INTER_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter):
 - When this function is disabled, the inter-base-station handover command carries only incremental configurations during inter-base-station handovers that involve abnormal UEs or interconnection across multiple vendors. This triggers interconnection across multiple vendors or compatibility issues due to abnormal UEs in such handovers. As a result, the service drop rate increases and the traffic drops sharply, affecting user experience. The preceding issues do not occur for inter-base-station handovers that do not involve abnormal UEs or interconnection across multiple vendors.
 - When this function is enabled, the inter-base-station handover command carries full configurations during inter-base-station handovers that involve abnormal UEs or interconnection across multiple vendors. In this way, the preceding issues can be avoided. For inter-base-station handovers that do not involve abnormal UEs or interconnection across multiple vendors, the size of the RRC reconfiguration message containing a handover command with full configurations increases. This increases the resources consumed by the base station and leads to complex UE processing. As a result, the handover success rate decreases. To address this issue in SA networking, full configuration optimization (controlled by the **FULL_CONFIG_OPT_SW** option of the **NRCellFeatureSwitch.NrHoCompatibilitySwitch** parameter) can be enabled. With this optimization, the "full configurations during inter-base-station handovers" function does not apply to Xn-based handovers between base stations of the same vendor. This reduces packet loss associated with the "full configurations during inter-base-station handovers" function and ensures user experience.

For the DRB session forwarding function (controlled by the **DRB_SESSION_FWD_SW** option of the **gNBMobilityCommParam.MobilityAlgoSwitch** parameter) in SA networking:

- When this function is disabled, the source gNodeB does not forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB. As a result, these packets are lost, negatively impacting user experience.
- When this function is enabled, the source gNodeB can forward the packets that have been sent but not acknowledged over the air interface to the target gNodeB in an inter-base-station handover, where the handover command carries full configurations and the *fullConfig* IE. This reduces packet loss and enhances user experience.

For the function of full configurations during intra-base-station handovers in SA networking (controlled by the **SA_INTRA_GNB_HO_FULL_CONFIG_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter):

- When this function is disabled, the intra-base-station handover command carries only incremental configurations during intra-base-station handovers that involve abnormal UEs. This triggers compatibility issues due to abnormal UEs in such handovers. As a result, the service drop rate increases and the traffic drops sharply, affecting user experience. The preceding issues do not occur for intra-base-station handovers that do not involve abnormal UEs.
- When this function is enabled, the intra-base-station handover command carries full configurations during intra-base-station handovers that involve abnormal UEs. In this way, the preceding issues can be avoided. For intra-base-station handovers that do not involve abnormal UEs, the size of the RRC reconfiguration message containing a handover command with full configurations increases. This increases the resources consumed by the base station and leads to complex UE processing. As a result, the handover success rate decreases.

- **Non-serving frequency filtering**

UEs do not perform measurements on the non-serving frequencies for which this function is enabled. Consequently, the number of UEs handed over or redirected to these non-serving frequencies decreases.

The gNodeB delivers less frequency information due to non-serving frequency filtering. As such, the cells enabled with this function may experience the following impacts:

- The value of **Inter-Frequency Handover Out Success Rate (CU)** fluctuates.
- The values of the counters in the **Number of Intra-RAT Redirections** function subset decrease.
- The value of **Service Call Drop Rate (CU)** increases.
- The values of **User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)** increase, as the number of gap-assisted measurements decreases.

- **CIO specification compatibility**

The function allows CIO information of more cells to be delivered, which may increase handover attempts. As such, the values of the following counters may fluctuate for the NR cells enabled with the function:

- **N.HO.IntraFreq.IntragNB.PrepAttOut**
- **N.HO.IntraFreq.Xn.IntergNB.PrepAttOut**
- **N.HO.IntraFreq.Ng.IntergNB.PrepAttOut**
- **N.HO.InterFreq.IntragNB.PrepAttOut**
- **N.HO.InterFreq.Xn.IntergNB.PrepAttOut**
- **N.HO.InterFreq.Ng.IntergNB.PrepAttOut**

- **RSRP- and SINR-based joint handover**

When (1) the **RSRP_SINR_HO_SW** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter is selected, and (2) the **RSRP_SINR_HO_SW** option of the **gNBQciBearer.ServiceDiffAlgoSwitch**

parameter is selected for any bearer of a UE, the ground neighboring cells with SINRs below a specific threshold will be filtered out. In this way, the UE can select a ground neighboring cell with a better SINR as the target cell, thereby improving user experience after it is handed over to the target cell. Note that, if the SINR threshold is too high, there will be no available ground neighboring cell for handovers, increasing the service drop rate.

- **High-load-specific handover**

This function causes the number of handovers to change in a high-load cell, as the intra- or inter-frequency measurement parameter groups bound to the gNodeB scenario differentiation group apply to UEs in the cell. Any larger value of the following thresholds results in a lower probability that high-load-specific handover takes effect and a smaller change in the number of handovers in the high-load cell. Any smaller value of the following thresholds produces the opposite effects.

- **NRCellMobilityConfig.HighLoadDiffHoUeNumThld** (indicating the UE number threshold for high-load-specific handover)
- **NRCellMobilityConfig.HighLoadDiffHoDlPrbThld** (indicating the downlink PRB usage threshold for high-load-specific handover)
- **NRCellMobilityConfig.HighLoadDiffHoUlPrbThld** (indicating the uplink PRB usage threshold for high-load-specific handover)

- **Path switch AMF selection**

This function may increase the Xn-based handover success rate.

- **Path switch backup AMF selection**

This function may increase the Xn-based handover success rate and may reduce the service drop rate.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

The basic functions of mobility management in connected mode have no impact on the network.

The network impacts produced by other sub-functions are described in benefit analysis. For details, see [4.2.2 Impacts](#). If the network impacts are not described in [4.2.1 Benefits](#), the corresponding sub-function has no impact on the network.

4.3 Requirements

4.3.1 Licenses

There are no license requirements.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RSRP- and SINR-based joint handover	RSRP_SINR_HO_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Bearer-specific RSRP- and SINR-based joint handover (controlled by the RSRP_SINR_HO_SW option of the gNBQciBearer.ServiceDiffAgoSwitch parameter) can take effect only when cell-specific RSRP- and SINR-based joint handover has been enabled (by selecting the RSRP_SINR_HO_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QCI-specific cell-load-dependent measurement event configuration	HIGH_LOAD_DIFF_HO_POL_SW option of the NRCellMobilityConfig.HoMeasAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	The following parameters take effect only when QCI-specific cell-load-dependent measurement event configuration is enabled (by selecting the HIGH_LOAD_DIFF_HO_POL_SW option of the NRCellMobilityConfig.HoMeasAlgoSwitch parameter). • NRCellMobilityConfig.HighLoadDiffHoUeNumThld (indicating the UE number threshold for high-load-specific handover) • NRCellMobilityConfig.HighLoadDiffHoDLPrbThld (indicating the downlink PRB usage threshold for high-load-specific handover) • NRCellMobilityConfig.HighLoadDiffHoULPrbThld (indicating the uplink PRB usage threshold for high-load-specific handover) • NRCellQciBearer.ScenarioDiffGrpId (indicating the ID of a scenario differentiation parameter group)

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	VG_MODEL_A LLOW_BUILD_ FLAG option of the NRCellFreqRelation.AggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	When Virtual Grid-based Multi-Frequency Coordination is enabled, the NRCellFreqRelation.InterFreqPrdcMrType parameter cannot be set to PERIODIC_MR_ONLY .

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellAlgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Inter-frequency measurements require the use of measurement gaps, causing high reliability performance to deteriorate during inter-frequency measurements performed by UEs. UEs perform inter-frequency measurements for a variety of handovers, including coverage-based inter-frequency handovers, frequency-priority-based inter-frequency handovers, and service-based inter-frequency handovers.
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SELECTION_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When both experience-based smart carrier selection (controlled by the MULTI_FREQ_SMART_SELECTION_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter) and EPLMN-based intra-RAT inter-PLMN handover are enabled, the list of PLMNs to which a UE can be handed over includes only the serving PLMN.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Latency-based optimal carrier selection	DELAY_BASED_CARR_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When both EPLMN-based intra-RAT inter-PLMN handover and latency-based optimal carrier selection are enabled, the list of PLMNs to which a UE can be handed over contains only the serving PLMN.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSwitch parameter	<i>UE Power Saving</i>	When both "non-gap-assisted measurements for active BWP" and "power saving BWP" are enabled, the frequency-domain range of BWP2 is used to calculate the frequency-domain range of the active BWP enabled with non-gap-assisted measurements. In this case, "non-gap-assisted measurements for active BWP" takes effect in fewer scenarios.
Low-frequency TDD	LTE TDD and NR spectrum sharing	LTE_NR_TDD_SPCT_SHR_SW option of the NRDUCellAlgoSwitch.SpectrumCloudSwitch parameter	None	When both "non-gap-assisted measurements for active BWP" and "LTE TDD and NR spectrum sharing" are enabled, the NR-exclusive bandwidth is used to calculate the frequency-domain range of the active BWP enabled with non-gap-assisted measurements.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Virtual Grid-based Multi-Frequency Coordination	VG_MODEL_ALLOWED_BUILD_FLAG option of the NRCellFrequencyRelationAggregationAttribute parameter	<i>Virtual Grid-based Multi-Frequency Coordination</i>	When both "non-gap-assisted measurements for active BWP" and "virtual grid-based multi-frequency coordination" are enabled, the latter function preferentially takes effect. That is, virtual grid-based prediction is preferentially performed to select the optimal inter-frequency cell for an inter-frequency handover. If virtual grid-based prediction fails, whether "non-gap-assisted measurements for active BWP" takes effect is determined when measurement configurations are delivered.
High-frequency TDD	None	None	None	None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- The configuration requirements for external cells (through the **MO NRExternalNCell** MO) are as follows:
 - If the **NRExternalNCell.NrNetworkingOption** parameter is set to **NSA**, no tracking area code (TAC) needs to be planned. In this case, you are advised to set the TAC to an invalid value (4294967295).
 - If the **NRExternalNCell.NrNetworkingOption** parameter is set to **SA** or **SA_NSA**, a TAC equal to the actual TAC of the external cell must be configured. Otherwise, handovers may fail.
- If there are multiple AMFs, the core network must support inter-AMF handovers.
- If an AMF region is planned on the core network, the gNodeB must support at least one network slice in the AMF region and connect to all AMFs in the AMF set corresponding to the network slice.
- If the source cell and target cell for an intra-base-station handover are in the same PLMN, they cannot belong to different AMFs in the same AMF region. Otherwise, the UE registration update fails after the intra-base-station handover is complete, resulting in a service drop. As such, during network planning, cells in the same PLMN under a base station need to be mapped to the same AMF through the TACs.
- Xn-based handovers have the following requirements for the gNodeB:
 - The target gNodeB must comply with 3GPP TS 38.423 V15.5.0 or later. Otherwise, Xn-based handovers will fail as the source gNodeB complies with 3GPP TS 38.423 V15.5.0 or later. As such, the function of protocol compatibility processing for Service Area Information exchanged between base stations has been introduced. This function is controlled by the **XN_SERVICE_AREA_INFO_PROC_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter. It is recommended that this option be deselected when the source and target gNodeBs comply with different protocol versions.
 - If this option is selected, Huawei base stations send and receive the *Service Area Information* IE in compliance with 3GPP TS 38.423 V15.5.0 or later.
 - If this option is deselected, Huawei base stations do not send the *Service Area Information* IE and they receive this IE in compliance with 3GPP TS 38.423 V15.4.0 or earlier. With this setting, Xn-based handovers can succeed, but UEs may be handed over to the forbidden cells indicated by the *Mobility Restriction List* IE in the INITIAL CONTEXT SETUP REQUEST message sent from the AMF.
 - Compared with 3GPP TS 38.423 V16.6.0, 3GPP TS 38.423 V16.7.0 introduced an incompatibility change in the definition of the *UE Security Capabilities* IE. For details, see section 9.2.3.49 "UE Security Capabilities"

in 3GPP TS 38.423 V16.7.0. During an Xn-based handover, the target gNodeB processes the *UE Security Capabilities* IE in the Handover Request message sent by the source gNodeB to select a ciphering algorithm. If the two base stations comply with different protocols during the processing, a protocol compatibility issue occurs. To prevent handover failures due to protocol inconsistency between the source and target gNodeBs, Huawei gNodeBs have optimized the compatibility processing in accordance with 3GPP TS 38.423 V16.7.0. This optimization is controlled by the **XN_UE_SECURITY_CAP_PROC_SW** option of the **gNodeBParam.CompatibilityAlgoSwitch** parameter. It is recommended that this option be selected when a Huawei gNodeB is connected to a non-Huawei gNodeB over the Xn interface and the non-Huawei gNodeB sends and receives the Handover Request message containing the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.7.0.

- When this option is selected, Huawei gNodeBs send and receive the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.7.0.
- When this option is deselected, Huawei gNodeBs send and receive the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.6.0.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Parameter settings vary among handover functions. This section describes parameter settings related to common configurations. Set parameters related to common configurations, and then set parameters to activate a specific handover function.

Mobility management in connected mode depends on neighboring cell configurations. Before using this function, ensure that the parameters related to neighboring cells have been configured. [Table 4-20](#) lists the parameters related to neighboring cells involved in mobility management in connected mode. [Table 4-21](#) lists the parameters related to cell-specific frequency relationships involved in inter-frequency handovers. If ANR is enabled, neighboring cells do not require manual configuration. For details, see *ANR*.

Table 4-20 Parameters used to configure neighboring cells

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCell.Mcc	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobile Network Code	NRExternalNCell.Mnc	Set this parameter based on the network plan.
gNodeB ID	NRExternalNCell.gNBId	Set this parameter based on the network plan.
Cell ID	NRExternalNCell.CellId	Set this parameter based on the network plan.
Physical Cell ID	NRExternalNCell.PhysicalCellId	Set this parameter based on the network plan.
Cell Name	NRExternalNCell.CellName	Set this parameter based on the network plan.
Tracking Area Code	NRExternalNCell.Tac	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRExternalNCell.SsbDescMethod	Use the recommended value.
SSB Frequency Position	NRExternalNCell.SsbFreqPos	Set this parameter based on the network plan.
Frequency Band	NRExternalNCell.FrequencyBand	Set this parameter based on the network plan.
NR Networking Option	NRExternalNCell.NrNetworkingOption	Set this parameter based on the network plan.
NR Cell ID	NRCellRelation.NrCellId	Set this parameter based on the network plan.
Mobile Network Code	NRCellRelation.Mnc	Set this parameter based on the network plan.
Mobile Country Code	NRCellRelation.Mcc	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NRCellRelation.gNBId	Set this parameter based on the network plan.
Cell ID	NRCellRelation.CellId	Set this parameter based on the network plan.
Cell Individual Offset	NRCellRelation.CellIndividualOffset	Use the recommended value.

 NOTE

- Maximum number of external NR cells that can be configured for a gNodeB: Maximum number of cells supported by the gNodeB x 32
- Maximum number of neighboring NR cells that can be configured for an NR cell: 768 when the **INTRA_RAT_NRT_NUM_PERCELL_ENH** option of the **NRCellAlgoSwitch.NeighborCellOptSwitch** parameter is selected to enable increased intra-RAT neighbor relationship capacity per cell; 384 when this option is deselected to disable increased intra-RAT neighbor relationship capacity per cell
- Maximum number of neighboring NR cells that can be configured for a gNodeB: Maximum number of cells supported by the gNodeB x 256

The **NRExternalNCell** MO defines external NR cells, and the **NRCellRelation** MO defines neighboring NR cells. For details about the maximum number of cells supported by a gNodeB, see "Capacity Specifications" in the corresponding BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Table 4-21 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Aggregation Attribute	NRCellFreqRelation.AggregationAttribute	The VOICE_INTER_FREQ_HO_POLICY_SW option of this parameter specifies the inter-frequency handover policy for voice-service-capable UEs. For details about the scenarios where selecting this option is recommended, see 4.1.4.1.2 Frequency Selection for Measurement .

Table 4-22 Common parameters used for optimization

Parameter Name	Parameter ID	Option	Setting Notes
Measurement Excluded Cell List Switch	NRCellMobilityConfig.MeasurementExcludedCellListSwitch	None	Set this parameter based on the network plan.
Cell Blacklist ID	gNBCellBlacklist.CellBlacklistId	None	Set this parameter based on the network plan.
Start Physical Cell ID	gNBCellBlacklist.StartPhysicalCellId	None	Set this parameter based on the network plan.
Physical Cell ID Range	gNBCellBlacklist.PhysicalCellIdRange	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
RAT Type	gNBCellBlacklist.RatType	None	Set this parameter based on the network plan. The values of gNBCellBlacklist.RatType in the records that have the same value of gNBCellBlacklist.CellBlacklistId must also be the same.
Intra-Frequency Excluded Cell List ID	NRCellMobilityConfig.IntraFExcludedCellListId	None	Set this parameter based on the network plan. If NRCellMobilityConfig.IntraFExcludedCellListId is set to a value other than 65535 , the value must have been configured in gNBCellBlackList.CellBlacklistId and the corresponding gNBCellBlacklist.RatType must be set to NR .
Inter-Frequency Excluded Cell List ID	NRCellFreqRelation.InterFExcludedCellListId	None	Set this parameter based on the network plan. If NRCellFreqRelation.InterFExcludedCellListId is set to a value other than 65535 , the value must have been configured in gNBCellBlackList.CellBlacklistId and the corresponding gNBCellBlacklist.RatType must be set to NR .
Necessary HO Prepare Fail Punish Timer	gNBMobilityCommParam.NecHoPrepFailPunishTimer	None	Use the recommended value.
Optimized HO Prepare Fail Punish Timer	gNBMobilityCommParam.OptHoPrepFailPunishTimer	None	Use the recommended value.
Non-Res HO Prepare Fail Punish Times	gNBMobilityCommParam.NonResHoPrepFailPunishTimes	None	Use the recommended value.

Parameter Name	Parameter ID	Option	Setting Notes
SMTC Period	NRDUCell.SmtcPeriod	None	Use the recommended value.
SMTC Duration	NRDUCell.SmtcDuration	None	Use the recommended value.
SMTC Period	NRCellFreqRelation.SmtcPeriod	None	Use the recommended value.
SMTC Duration	NRCellFreqRelation.SmtcDuration	None	Use the recommended value.
SA SMTC Offset	NRCellFreqRelation.SaSmtcOffset	None	Set this parameter based on the network plan. If this parameter is set to a value other than 255, the value must be less than the value of the SMTC period.
Frame Offset	gNBFreqBandConfig.FrameOffset	None	Set this parameter based on the network plan.
Additional Frame Offset	gNBFreqBandConfig.AdditionalFrameOffset	None	Use the recommended value.
Measurement Algorithm Switch	gNBMobilityCommParam.MeasAlgoSwitch	GAP_AVOID_N R_RES_SW	It is recommended that this option be selected for inter-frequency measurements in low-frequency SA networking.
Measurement Algorithm Switch	NRDUCellMeas.MeasAlgoSwitch	GAP_AVOID_N R_RES_OPT_SW	It is recommended that this option be selected for inter-frequency measurements in low-frequency scenarios where the channel quality fluctuates significantly.
Compatibility Algorithm Switch	gNodeBParam.CompatibilityAlgoSwitch	GAP_SCH_CO MP_SW	It is recommended that this option be selected when UEs encounter compatibility issues.

Parameter Name	Parameter ID	Option	Setting Notes
SSB Measurement Policy Switch	NRDUCellMeas.SsbMeasPolicySwitch	SSB_MEAS_WITHOUT_GAP_SW	It is recommended that this option be selected in multi-frequency coverage scenarios where there is a large proportion of CA UEs capable of non-gap-assisted measurements.
Measurement Algorithm Switch	gNBMobilityCommParam.MeasAlgoSwitch	MEAS_ID_PREFERENCE_SW	It is recommended that this option be selected.
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibilitySwitch	PROTOCOL_IE_OPT_SW	It is recommended that this option be selected.
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibilitySwitch	INTER_GNB_HO_FULL_CONFIG_SW	For inter-base-station handovers, it is recommended that this option be selected to avoid interconnections between the source and target gNodeBs provided by different vendors or compatibility issues due to abnormal UEs on live networks.
NR Handover Compatibility Switch	NRCellFeatureSwitch.NrHoCompatibilitySwitch	FULL_CONFIG_OPT_SW	It is recommended that this option be selected when the "full configurations during inter-base-station handovers" function is enabled for Xn-based handovers between base stations of the same vendor.
Mobility Algorithm Switch	gNBMobilityCommParam.MobilityAlgoSwitch	DRB_SESSION_FWD_SW	It is recommended that this option be selected when the "full configurations during inter-base-station handovers" function is enabled.
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibilitySwitch	SA_INTRA_GNB_HO_FULL_CONFIG_SW	For intra-base-station handovers, it is recommended that this option be selected to avoid compatibility issues due to abnormal UEs on live networks.

Parameter Name	Parameter ID	Option	Setting Notes
Whitelist Control Switch	gNBUECompantSw . <i>WhitelistCtrlSwitch</i>	HO_FULL_CONFIG_SW_ON	It is recommended that this option be selected if full configurations during inter-base-station or intra-base-station handovers are enabled for specified UEs.
Compatibility Algorithm Switch	gNodeBParam . <i>CompatibilityAlgoSwitch</i>	XN_SERVICE_AREA_INFO_PROC_SW	It is recommended that this option be deselected when the source and target gNodeBs comply with different versions of 3GPP TS 38.423 (one complying with V15.5.0 or later whereas the other complying with V15.4.0 or earlier). In other scenarios, it is recommended that this option be selected.
Compatibility Algorithm Switch	gNodeBParam . <i>CompatibilityAlgoSwitch</i>	XN_UE_SECURITY_CAP_PROC_SW	It is recommended that this option be selected when a Huawei gNodeB is connected to a non-Huawei gNodeB over the Xn interface and the non-Huawei gNodeB signals the Handover Request message containing the <i>UE Security Capabilities</i> IE in compliance with 3GPP TS 38.423 V16.7.0. In other scenarios, it is recommended that this option be deselected.
Protocol Compatibility Switch	gNBMobilityCommParam . <i>ProtocolCompatibilitySw</i>	INTER_FREQ_HO_CAPB_IND_OPT_SW	It is recommended that this option be selected.
Protocol Compatibility Switch	gNBMobilityCommParam . <i>ProtocolCompatibilitySw</i>	XN_HO_INTERRUPT_DURATION_OPT_SW	It is recommended that this option be selected when the duration of uplink data transmission interruption needs to be reduced for UEs during Xn-based handovers.

Parameter Name	Parameter ID	Option	Setting Notes
Protocol Compatibility Switch	gNBMobilityCommParam.ProtocolCompatibilitySw	AMF_SET_ID_B ASED_XN_HO_SW	It is recommended that this option be selected for the source gNodeB when the source and target gNodeBs involved in an Xn-based handover belong to different AMF sets.
Equivalent PLMN Algorithm Switch	gNodeBParam.EqvPlmnAlgorithmSwitch	INTRA_RAT_HO_WITH_GNB_EPLMN_SW	Set this parameter based on the network plan. It is recommended that the INTRA_RAT_HO_WITH_GNB_EPLMN_SW option be selected if EPLMN-based intra-RAT inter-PLMN handover is required in SA networking. For details about the PLMNs allowed for UE handovers given different option settings, see Table 4-16 .
SSB Consolidation Threshold	NRCellMobilityConfig.SsbConsolidationThreshold	None	Use the recommended value.
SSB Number To Average	NRCellMobilityConfig.SsbNumToAverage	None	Use the recommended value.
Beam RSRP Filter Coefficient	NRCellMobilityConfig.BeamRsrpFilterCoefficient	None	Use the recommended value.
Cell RSRP Filter Coefficient	NRCellMobilityConfig.CellRsrpFilterCoefficient	None	Use the recommended value.
T312	NRCellUeTimer.T312	None	It is recommended that NRCellUeTimer.T312 be set to a value less than that of NRDUCellUeTimerConst.T310 . Set this parameter based on the network plan.
Operator ID ^a	gNBEqvPlmn.OperatorId	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Equivalent Mobile Country Code	gNBEqvPlmn.EquivalentMcc	None	Set this parameter based on the network plan.
Equivalent Mobile Network Code	gNBEqvPlmn.EquivalentMnc	None	Set this parameter based on the network plan.
HO and PDU Session Conflict Handling Sw	gNBMobilityCommParam.HoPduSessConflHandlingSw	VOICE_FB_FIR ST_SWITCH	It is recommended that this option be selected if voice fallback needs to be preferentially handled when a voice fallback request is received during outgoing handover preparation.
HO and QoS Flow Conflict Proc Strategy	NRCellQciBearer.HoQosFlowConflProcStrat	QOS_FLOW_FI RST_SW	It is recommended that this option be selected if the QoS flow procedure needs to be preferentially handled when PDU session setup or modification signaling indicating the addition or modification of a QoS flow is received during outgoing handover preparation.
HO and PDU Session Conflict Handling Sw	gNBMobilityCommParam.HoPduSessConflHandlingSw	PDU_SESS_REL _NG_HO_REQ_ SEL_SW	It is recommended that this option be selected if both of the following conditions apply: (1) the signaling procedure is abnormal on the core network side; (2) the NG-based handover procedure conflicts with the PDU session procedure, resulting in UE handover failures or UE release failures.
Mobility Algorithm Switch	gNBMobilityCommParam.MobilityAlgoSwitch	MOBILITY_RES TRITION_SW	It is recommended that this option be selected if the INITIAL CONTEXT SETUP REQUEST message sent from the core network includes the <i>Mobility Restriction List</i> IE.
Frequency Interference Flag	NRCellFreqRelatation.FreqIntrfFlag	None	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Configuration Policy Switch	NRCellOpPolicy.FreqConfigPolicySwitch	OPERATOR_NR_FREQ_CFG_SW	The policy on frequency delivery for inter-frequency measurement is configured as follows: Select OPERATOR_NR_FREQ_CFG_SW if NR frequencies in the intersection of NR frequencies configured in the gNBFreqPriorityGroup MO and those configured in the NRCellFreqRelation MO are to be delivered. Deselect OPERATOR_NR_FREQ_CFG_SW if NR frequencies configured in the NRCellFreqRelation MO are to be delivered. Frequency filtering in blind mode is configured as follows: Select OPERATOR_NR_FREQ_CFG_SW if frequencies outside the intersection of NR frequencies configured in the gNBFreqPriorityGroup MO and those configured in the NRCellFreqRelation MO need to be filtered out. Deselect OPERATOR_NR_FREQ_CFG_SW if frequencies that do not belong to NR frequencies configured in the NRCellFreqRelation MO need to be filtered out. Use the recommended value.
Roaming PLMN Flag	gNBOperator.RoamingPlmnFlag	None	Set this parameter based on the network plan.
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibilitySwitch	NON_RES_INTRA_NR_HO_PENALTY_SW	This option indicates the non-resource-induced intra-NR handover penalty switch. It is recommended that this option be selected if the base station and the core network receive inconsistent information about the inter-frequency handover capability reported by a UE.

Parameter Name	Parameter ID	Option	Setting Notes
Handover Compatibility Switch	NRCellAlgoSwitch.HoCompatibilitySwitch	NON_RES_EUTRAN_HO_PENALTY_SW	This option indicates the non-resource-induced E-UTRAN handover penalty switch. It is recommended that this option be selected if the base station and the core network receive inconsistent information about the inter-RAT handover capability reported by a UE.
Intra-RAT Handover Request UE Capability Size	gNBMMobilityCommParam.IntraRatHoReqUECapbSize	None	Set this parameter based on the network plan. Before modifying the value of this parameter, check the core network's capability in processing handover request messages. If the parameter value exceeds the processing capability of the core network, there is a possibility that NG-based handover requests fail.
Frequency Control Switch	NRCellFreqRelation.FrequencyCtrlSwitch	MOB_NEIGHBOR_FREQ_FILTER_SW	Set this parameter based on the network plan. If this option is selected for a frequency, the gNodeB does not deliver information about the frequency to a UE involved in a handover or redirection procedure.
Protocol Compatibility Switch	gNBMMobilityCommParam.ProtocolCompatibilitySw	CIO_SPEC_COMPAT_SW	For a cell with the NRCellRelation.CellIndividualOffset parameter set to a value other than 0, it is recommended that this option be selected if a measurement object contains more than eight intra-RAT intra-frequency or inter-frequency neighboring cells whose CIO information needs to be delivered.

Parameter Name	Parameter ID	Option	Setting Notes
Inter-Frequency Handover Extend Switch	NRCellMobilityConfig.InterFreqHoExtSwitch	RSRP_SINR_HO_SW	In specific scenarios, to enable the gNodeB to filter out the ground neighboring cells with SINRs below a specific threshold for a specific UE, you are advised to select the RSRP_SINR_HO_SW option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter, and also select the RSRP_SINR_HO_SW option of the gNBQciBearer.ServiceDiffAlgorithmSwitch parameter for any bearer of the UE.
Service Differentiation Algorithm Switch	gNBQciBearer.ServiceDiffAlgorithmSwitch	RSRP_SINR_HO_SW	
RSRP and SINR-based Nec HO SINR Thld	NRCellInterFHoMeaGrp.RsrpSinrBasedNecSinrThld	None	Set this parameter based on the network plan. For necessary handovers, a larger value of this parameter results in a lower probability of initiating handovers; a smaller value of this parameter results in a higher probability of initiating handovers.
RSRP and SINR-based Unec HO SINR Thld	NRCellInterFHoMeaGrp.RsrpSinrBasedUnecSinrThld	None	Set this parameter based on the network plan. For unnecessary handovers, a larger value of this parameter results in a lower probability of initiating handovers; a smaller value of this parameter results in a higher probability of initiating handovers.
Ng Interface Algorithm Extension Switch	gNodeBParam.NgInterfaceAlgorithmExtSw	PATH_SWITCH_AMF_SEL_SW	It is recommended that this option be selected to enable path switch AMF selection.
Ng Interface Algorithm Extension Switch	gNodeBParam.NgInterfaceAlgorithmExtSw	PATH_SWITCH_BACKUP_AMF_SEL_SW	It is recommended that this option be selected to enable path switch backup AMF selection.

Parameter Name	Parameter ID	Option	Setting Notes
a: When configuring the gNBEqvPlmn MO, ensure that there is a roaming agreement between the PLMN corresponding to gNBEqvPlmn.OperatorId and the EPLMN collectively determined by gNBEqvPlmn.EquivalentMcc and gNBEqvPlmn.EquivalentMnc .			

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Configuring a frequency relationship with a neighboring frequency
//Adding a frequency relationship for an NR cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1,
AggregationAttribute=VOICE_INTER_FREQ_HO_POLICY_SW-0;
//Configuring a neighbor relationship if it is not configured
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
```

Optimization Command Examples

TDD configuration examples

```
//Modifying common gNodeB mobility parameters
MOD GNBMOBILITYCOMMPARAM: NecHoPrepFailPunishTimer=0, OptHoPrepFailPunishTimer=0,
NonResHoPrepFailPunishTimes=10;
//Configuring the policy on frequency delivery for inter-frequency measurement with NR frequencies in the
intersection of NR frequencies configured in the gNBFreqPriorityGroup MO and those configured in the
NRCellFreqRelation MO to be sent as well as frequency filtering in blind mode with frequencies outside the
intersection of NR frequencies configured in the gNBFreqPriorityGroup MO and those configured in the
NRCellFreqRelation MO to be filtered out
MOD NRCELLLOPPOLICY: NrCellId=0, OperatorId=0,
FreqConfigPolicySwitch=OPERATOR_NR_FREQ_CFG_SW-1;
//Configuring the roaming PLMN flag
MOD GNBOPERATOR: OperatorId=0, Mcc="302", Mnc="220", OperatorName="5G",
RoamingPlmnFlag=FALSE;
//Turning on the excluded cell list switch for measurement
MOD NRCELLMOBILITYCONFIG: NrCellId=0, MeasExcludedCellListSwitch=ON;
//Adding cell blacklists for the gNodeB
ADD GNBCELLBLACKLIST: CellBlacklistId=1, StartPhysicalCellId=231, PhysicalCellIdRange=N4, RatType=NR;
```

```

ADD GNBCELLBLACKLIST: CellBlacklistId=2, StartPhysicalCellId=1, PhysicalCellIdRange=N8, RatType=NR;
//Associating the intra-frequency excluded cell list ID
MOD NRCELLMOBILITYCONFIG: NrCellId=0, IntraFExcludedCellListId=1;
//Associating the inter-frequency excluded cell list ID
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77,
InterFExcludedCellListId=2;
//Modifying the SMTC settings (assuming the SMTC offset is set to 0)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, SmtcPeriod=MS20, SmtcDuration=MS2;
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=7853, FrequencyBand=N78, SmtcPeriod=MS20,
SmtcDuration=MS3, SaSmtcOffset=0;
//Modifying the configuration for SSB beam measurement result selection and combination
MOD NRCELLMOBILITYCONFIG: NrCellId=0, SsbConsolidationThld=70, SsbNumToAverage=8;
//Modifying the measurement filtering configuration
MOD NRCELLMOBILITYCONFIG: NrCellId=0, BeamRsrpFilterCoeff=FC4, CellRsrpFilterCoeff=FC4;
//Modifying timer T312
MOD NRCELLUETIMER: NrCellId=0, T312=MS200;
//Enabling the function of non-overlapping between gap and NR resources for inter-frequency
measurements in low-frequency SA networking
MOD GNBMOBILITYCOMMPARAM: MeasAlgoSwitch=GAP_AVOID_NR_RES_SW-1;
//((Recommended) Enabling the "optimized non-overlapping between gap and NR resources" function for
inter-frequency measurements in low-frequency scenarios where the channel quality fluctuates significantly
MOD NRDUCELLMEAS: NrDuCellId=0, MeasAlgoSwitch=GAP_AVOID_NR_RES_OPT_SW-1;
//Enabling non-gap-assisted measurements for active BWP
MOD NRDUCELLMEAS: NrDuCellId=0, SsbMeasPolicySwitch=SSB_MEAS_WITHOUT_GAP_SW-1;
//Enabling gap-assisted scheduling compatibility
MOD GNODEBPARAM: CompatibilityAlgoSwitch=GAP_SCH_COMP_SW-1;
//Selecting the INTER_FREQ_HO_CAPB_IND_OPT_SW option
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=INTER_FREQ_HO_CAPB_IND_OPT_SW-1;
//Turning on the measurement ID preemption switch
MOD GNBMOBILITYCOMMPARAM: MeasAlgoSwitch=MEAS_ID_PREEMPTION_SW-1;
//Turning on the protocol IE optimization switch
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=PROTOCOL_IE_OPT_SW-1;
//Turning on the Xn service area information processing switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=XN_SERVICE_AREA_INFO_PROC_SW-1;
//Turning on the Xn UE security capability processing switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=XN_UE_SECURITY_CAP_PROC_SW-1;
//Enabling the sending of full configurations during inter-base-station handovers
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=INTER_GNB_HO_FULL_CONFIG_SW-1;
//((Recommended) Enabling full configuration optimization when the "full configurations during inter-base-
station handovers" function is enabled for Xn-based handovers between base stations of the same vendor
MOD NRCELLFEATURESWITCH: NrCellId=0, NrHoCompatibilitySwitch=FULL_CONFIG_OPT_SW-1;
//((Recommended) Enabling DRB session forwarding when the "full configurations during inter-base-station
handovers" function is enabled
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=DRB_SESSION_FWD_SW-1;
//Enabling the sending of full configurations during intra-base-station handovers
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=SA_INTRA_GNB_HO_FULL_CONFIG_SW-1;
//Enabling the sending of full configurations during inter-base-station and intra-base-station handovers for
specified UEs, given an assumption that UeInfoIndex (UE information index) with the value of 1 has been
added
MOD GNBUECOMPATSW: UeInfoIndex=1, ParamCtrlType=ALL,
WhitelistCtrlSwitch=HO_FULL_CONFIG_SW_ON-1;
//Enabling preferential handling of voice fallback
MOD GNBMOBILITYCOMMPARAM: HoPduSessConflHandlingSw=VOICE_FB_FIRST_SWITCH-1;
//Enabling preferential handling of QoS procedures (QCI 1 used as an example)
MOD NRCELLQCI BEARER: NrCellId=0, Qci=1, HoQosFlowConflProcStrat=QOS_FLOW_FIRST_SW-1;
//Turning on the switch for conflict handling between the PDU session release and NG-based handover
procedures
MOD GNBMOBILITYCOMMPARAM: HoPduSessConflHandlingSw=PDU_SESS_REL_NG_HO_REQ_SEL_SW-1;
//Turning on the mobility restriction switch
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=MOBILITY_RESTRICTION_SW-1;
//Turning on the EPLMN-based intra-RAT inter-PLMN handover switch
MOD GNODEBPARAM: EqvPlmnAlgoSwitch=INTRA_RAT_HO_WITH_GNB_EPLMN_SW-1;
//Adding an EPLMN for an operator of the gNodeB (In this example, OperatorId is set to 0, EquivalentMcc
is set to 520, and EquivalentMnc is set to 15. The configuration depends on the actual requirements.)
ADD GNBEQVPLMN: OperatorId=0, EquivalentMcc="520", EquivalentMnc="15";
//Setting the frequency interference flag (given an assumption that candidate cells are filtered based on the
measured RSRQs of neighboring cells)
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=7853, FrequencyBand=N78, FreqIntrfFlag=INTRF;
//Turning on the switch for Xn-based handover based on AMF set IDs

```

```
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=AMF_SET_ID_BASED_XN_HO_SW-1;
//Selecting XN_HO_INTERRUPT_DUR_OPT_SW
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=XN_HO_INTERRUPT_DUR_OPT_SW-1;
//Turning on the non-resource-induced intra-NR handover penalty switch and the non-resource-induced E-
UTRAN handover penalty switch
MOD NRCELLALGOSWITCH: NrCellId=0,
HoCompatibilitySwitch=NON_RES_INTRA_NR_HO_PENALTY_SW-1&NON_RES_EUTRAN_HO_PENALTY_SW-1;
//Setting the size of the UE capability field in an intra-RAT handover request
MOD GNBMOBILITYCOMMPARAM: IntraRatHoReqUeCapbSize=10;
//Enabling non-serving frequency filtering
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77,
FrequencyCtrlSwitch=MOB_NEIGHBOR_FREQ_FILTER_SW-1;
//Enabling increased intra-RAT neighbor relationship capacity per cell
MOD NRCELLALGOSWITCH: NrCellId=0, NeighborCellOptSwitch=INTRA_RAT_NRT_NUM_PERCELL_ENH-1;
//Enabling CIO specification compatibility
MOD GNBMOBILITYCOMMPARAM: ProtocolCompatibilitySw=CIO_SPEC_COMPAT_SW-1;
//Enabling RSRP- and SINR-based joint handover
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=RSRP_SINR_HO_SW-1;
MOD GNBQCIBEARER: Qci=9, ServiceDiffAlgoSwitch=RSRP_SINR_HO_SW-1;
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, RsrpSinrBasedNecSinrThld=-6,
RsrpSinrBasedUnecSinrThld=-6;
//Enabling path switch AMF selection and path switch backup AMF selection
MOD GNODEBPARAM:
NgInterfaceAlgoExtSw=PATH_SWITCH_AMF_SEL_SW-1&PATH_SWITCH_BACKUP_AMF_SEL_SW-1;
```

In the preceding MML command examples, the same frame offset is configured for the same frequency band by default. (For details about to configure a frame offset, see *Standards Compliance*.)

To configure different frame offsets for the same frequency band, an operator is required to set an additional frame offset for all base stations in the overlapping area. The following example assumes that: (i) the n28 frequency band is used in NR FDD, (ii) base stations A and B are in the overlapping area, and (iii) base station A's frame offset is 3 ms and base station B's frame offset is 2.3 ms. In the following example, it is recommended that the additional frame offset be set to 2.3 ms for both base stations A and B.

- Settings of base station A

```
//(For base station A) Modifying the frame offset
MOD GNBFBREQBANDCONFIG: FrequencyBand=N28, FrameOffset=92160;
//(For base station A) Modifying the intra-frequency SMTC duration for the cell operating on the n28
band and the inter-frequency SMTC duration for the cell operating on a non-n28 band, both of which
are required to be greater than or equal to 3 ms (assuming that base station A has Cell 1 operating
on the n28 band and Cell 2 operating on a non-n28 band)
MOD NRDUCELL: NrDuCellId=1, DuplexMode=CELL_FDD, SmtcPeriod=MS20, SmtcDuration=MS3;
MOD NRCELLFREORELATION: NrCellId=2, SsbFreqPos=2000, FrequencyBand=N28, SmtcPeriod=MS20,
SmtcDuration=MS3;
//(For base station A) Modifying the additional frame offset
MOD GNBFBREQBANDCONFIG: FrequencyBand=N28, AdditionalFrameOffset=70656;
```

- Settings of base station B

```
//(For base station B) Modifying the frame offset
MOD GNBFBREQBANDCONFIG: FrequencyBand=N28, FrameOffset=70656;
//(For base station B) Modifying the intra-frequency SMTC duration for the cell operating on the n28
band and the inter-frequency SMTC duration for the cell operating on a non-n28 band, both of which
are required to be greater than or equal to 3 ms (assuming that base station B has Cell 1 operating
on the n28 band and Cell 2 operating on a non-n28 band)
MOD NRDUCELL: NrDuCellId=1, DuplexMode=CELL_FDD, SmtcPeriod=MS20, SmtcDuration=MS3;
MOD NRCELLFREORELATION: NrCellId=2, SsbFreqPos=2000, FrequencyBand=N28, SmtcPeriod=MS20,
SmtcDuration=MS3;
//(For base station B) Modifying the additional frame offset
MOD GNBFBREQBANDCONFIG: FrequencyBand=N28, AdditionalFrameOffset=70656;
```

Deactivation Command Examples

MML commands are not required for deactivating this function. You can determine whether to restore the settings of other parameters based on actual network conditions.

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

To verify the activation of a handover function, see the "Activation Verification" section of the corresponding handover function.

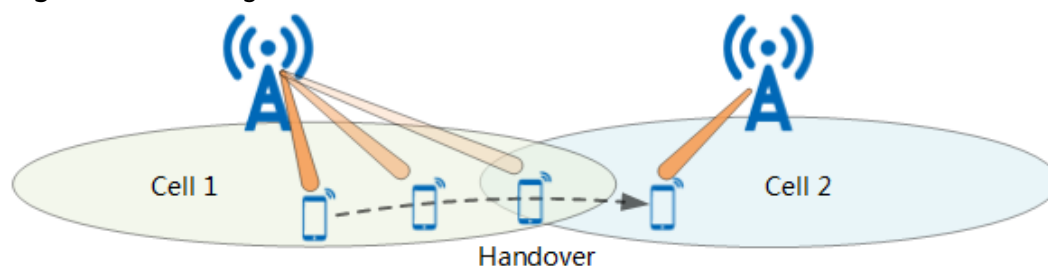
4.4.3 Network Monitoring

To perform network monitoring for a handover function, see the "Network Monitoring" section of the corresponding handover function.

5 Coverage-based Intra-Frequency Handover

On a network with continuous coverage, a coverage-based handover is triggered for a UE moving towards the cell edge, where it receives higher-quality signals from neighboring cells than from the serving cell. This prevents service drops or deteriorated experience caused by signal quality deterioration. [Figure 5-1](#) illustrates a coverage-based handover.

Figure 5-1 Coverage-based handover

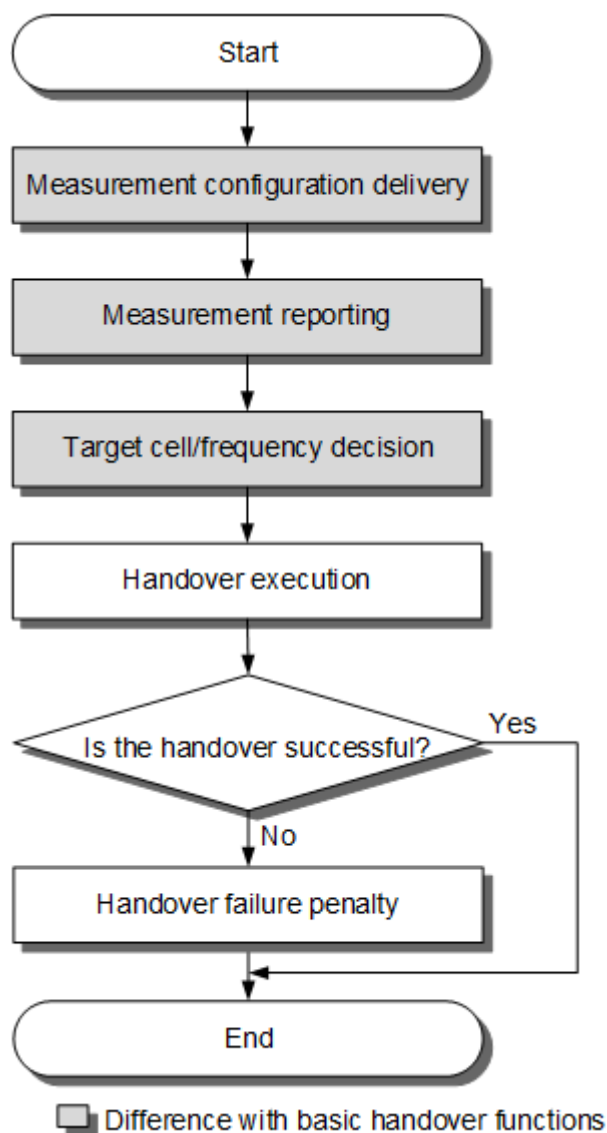


Coverage-based handovers are classified into intra-frequency handovers and inter-frequency handovers. This section describes coverage-based intra-frequency handovers. For details about coverage-based inter-frequency handovers, see [6 Coverage-based Inter-Frequency Handover](#).

5.1 Principles

Coverage-based intra-frequency handover is a basic function that ensures continuous coverage. Measurement configurations are sent after RRC connections are set up. The handover function start decision is not required, but whether a handover is executed depends on the setting of the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch.IntraFreqHoSwitch** parameter. Coverage-based intra-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports only handover but does not support redirection. [Figure 5-2](#) shows the procedure of coverage-based intra-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 5-2 Procedure of coverage-based intra-frequency handover



This function differs from the basic handover functions as follows:

- **Measurement configuration delivery:** The measurement event in the reporting configuration of this function is event A3 (indicating the signal quality of a neighboring cell is higher than that of the serving cell by a certain threshold).
- **Measurement reporting:** Upon determining that the measurement results meet the criteria for reporting event A3, UEs send the measurement reports to the gNodeB.

The measurement reporting mode is controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch.IntraFreqHoSwitch** parameter. After a UE detects a neighboring cell that meets certain conditions, the gNodeB notifies the UE of the measurement reporting mode through the *ReportConfigToAddModList* IE in MeasConfig of the RRCReconfiguration message.

- If the option is selected, the UE periodically sends measurement reports.

- If the option is deselected, the UE sends the measurement report only once.

Note that any change in the status of the option (from selected to deselected, or from deselected to selected) controlling the measurement reporting mode takes effect only for UEs that newly access the network but does not take effect for UEs that have accessed the network. For example, if the status of the option is changed from selected (its default status) to deselected, the UE that accesses the network after the option is deselected will send only one measurement report. If the option is selected again, the UE that has sent a measurement report before the option is selected will not send any measurement report again.

- Target cell or frequency decision: In the target cell or frequency decision phase, whether the gNodeB instructs the UE to perform a handover is also controlled by the **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch.IntraFreqHoSwitch** parameter.
 - If this option is selected, the gNodeB instructs the UE to initiate handover attempts towards the filtered cells one by one, which are sorted in the order that event A3 measurement reports are submitted as well as in descending order of RSRPs of the cells per event A3 measurement report.
 - If this option is deselected, the gNodeB does not instruct the UE to perform a handover.

For details about the definition, entering conditions, and leaving conditions for event A3, see [4.1.4.2.1 Measurement Event](#). For the variable configurations for event A3, see [Table 4-9](#).

5.2 Network Analysis

5.2.1 Benefits

Coverage-based intra-frequency handovers reduce interference caused by intra-frequency neighboring cells in SA intra-frequency networking, thereby reducing the service drop rate and providing users with seamless service experience.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

After the coverage-based intra-frequency handover function takes effect, the UE synchronizes with the target cell and initiates random access during a handover. This process introduces a delay due to service interruption, affecting data rates.

The status of the switch for coverage-based intra-frequency handover (**COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch.IntraFreqHoSwitch** parameter) has the following impacts:

- If this option is deselected, intra-frequency handovers cannot be performed for UEs in the cell. As a result, service drops may occur, leading to an increased service drop rate in the cell; in addition, there is a possibility that intra-frequency interference increases, leading to deteriorated user experience in the cell.

- If the status of this option is changed from selected to deselected, air interface resources are wasted. This is because the UEs that have accessed the network before this option is deselected periodically send measurement reports about neighboring cells that meet certain conditions but these UEs do not perform intra-frequency handovers.
- If the status of this option is changed from selected to deselected, UEs that access the network after this option is deselected send measurement reports only once. If this option is selected again, the preceding UEs that have sent measurement reports once before this option is selected will not send measurement reports again. This may increase the service drop rate in the cell, because these UEs cannot perform intra-frequency handovers and may experience service drops.

5.3 Requirements

5.3.1 Licenses

There are no license requirements.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-frequency MRO under cell-level MRO	MRO_SCENARIO_IDENT_SWITCH option of the gNB-MRO-ScenarioIdentifySwitch parameter and the INTRA_FREQ_MRO_SWITCH option of the NRCellMRO-MROSwitch parameter	<i>MRO</i>	When both coverage-based intra-frequency handover and cell-level intra-frequency MRO are enabled, CIO adjustment by MRO affects the number of intra-frequency handovers and the RRC connection reestablishment rate.
Low-frequency TDD	UE-level MRO	UE_MRO_SWITCH option of the NRCellMRO-MROSwitch parameter	<i>MRO</i>	When both coverage-based intra-frequency handover and UE-level MRO are enabled, CIO adjustment by MRO affects the number of intra-frequency handovers.
High-frequency TDD	None	None	None	None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

None

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

The effectiveness of this function depends on neighbor relationships. For details about neighbor relationship configurations, see [4.4.1 Data Configuration](#).

[Table 5-1](#) describes the parameters used for function activation.

[Table 5-2](#) describes the parameters used for function optimization.

Table 5-1 Intra-frequency handover algorithm switch

Parameter Name	Parameter ID	Setting Notes
Intra-Frequency Handover Switch	NRCellAlgoSwitch.IntraFreqHoSwitch	The COVERAGE_BASED_HO option of this parameter specifies whether to enable coverage-based intra-frequency handover. It is recommended that this option be selected.

Table 5-2 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellIntraFHoMeaGrp.NrCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Intra-freq Handover Measurement Group ID	NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId	Set this parameter based on the network plan.
Intra-frequency Handover A3 Time To Trigger	NRCellIntraFHoMeaGrp.IntraFreqHoA3TimeToTrig	Set this parameter to its recommended value.
Intra-frequency Handover A3 Hysteresis	NRCellIntraFHoMeaGrp.IntraFreqHoA3Hyst	Set this parameter to its recommended value.
Intra-frequency Handover A3 Offset	NRCellIntraFHoMeaGrp.IntraFreqHoA3Offset	Set this parameter to its recommended value.
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Intra-freq Handover Measurement Group ID	NRCellQciBearer.IntraFreqHoMeasGroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).

Parameter Name	Parameter ID	Setting Notes
High Load Differentiated HO UE Num Threshold	NRCellMobilityCon-fig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityCon-fig.HighLoadDiffHoDLPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityCon-fig.HighLoadDiffHoULPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpld	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.Operatorld	Set this parameter based on the network plan.
Intra-freq Handover Measurement Group ID	gNBScenarioDiffGrp.IntraFreqHoMeasureGroupld	Set this parameter based on the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Configuring a neighbor relationship if it is not configured
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6300, FrequencyBand=N41,
NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Enabling coverage-based intra-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, IntraFreqHoSwitch=COVERAGE_BASED_HO-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring measurement parameters for intra-frequency handovers
MOD NRCELLINTRAHOMEAGRP: NrCellId=0, IntraFreqHoMeasGroupId=0, IntraFreqHoA3Offset=2,
IntraFreqHoA3Hyst=2, IntraFreqHoA3TimeToTrig=320MS;
ADD NRCELLINTRAHOMEAGRP: NrCellId=0, IntraFreqHoMeasGroupId=1, IntraFreqHoA3Offset=2,
IntraFreqHoA3Hyst=2, IntraFreqHoA3TimeToTrig=320MS;
//((Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, IntraFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, IntraFreqHoMeasGroupId=1;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
IntraFreqHoMeasGroupId=1;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating coverage-based intra-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, IntraFreqHoSwitch=COVERAGE_BASED_HO-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

If any of the following counters have a value other than 0, coverage-based intra-frequency handover has taken effect:

- **N.HO.IntraFreq.Ng.IntergNB.PrepAttOut**
- **N.HO.IntraFreq.Xn.IntergNB.PrepAttOut**
- **N.HO.IntraFreq.IntragNB.PrepAttOut**

5.4.3 Network Monitoring

This function provides basic assurance for intra-frequency handovers and can be monitored using the **Intra-Frequency Handover Out Success Rate (CU)** KPI.

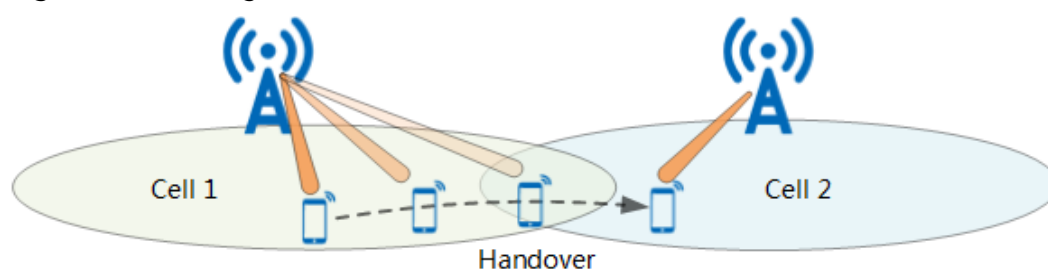
6 Coverage-based Inter-Frequency Handover

6.1 Principles

Application Scenario

On a network with continuous coverage, a coverage-based handover is triggered for a UE moving towards the cell edge, where it receives higher-quality signals from neighboring cells than from the serving cell. This prevents service drops or deteriorated experience caused by signal quality deterioration. [Figure 6-1](#) illustrates a coverage-based handover.

Figure 6-1 Coverage-based handover

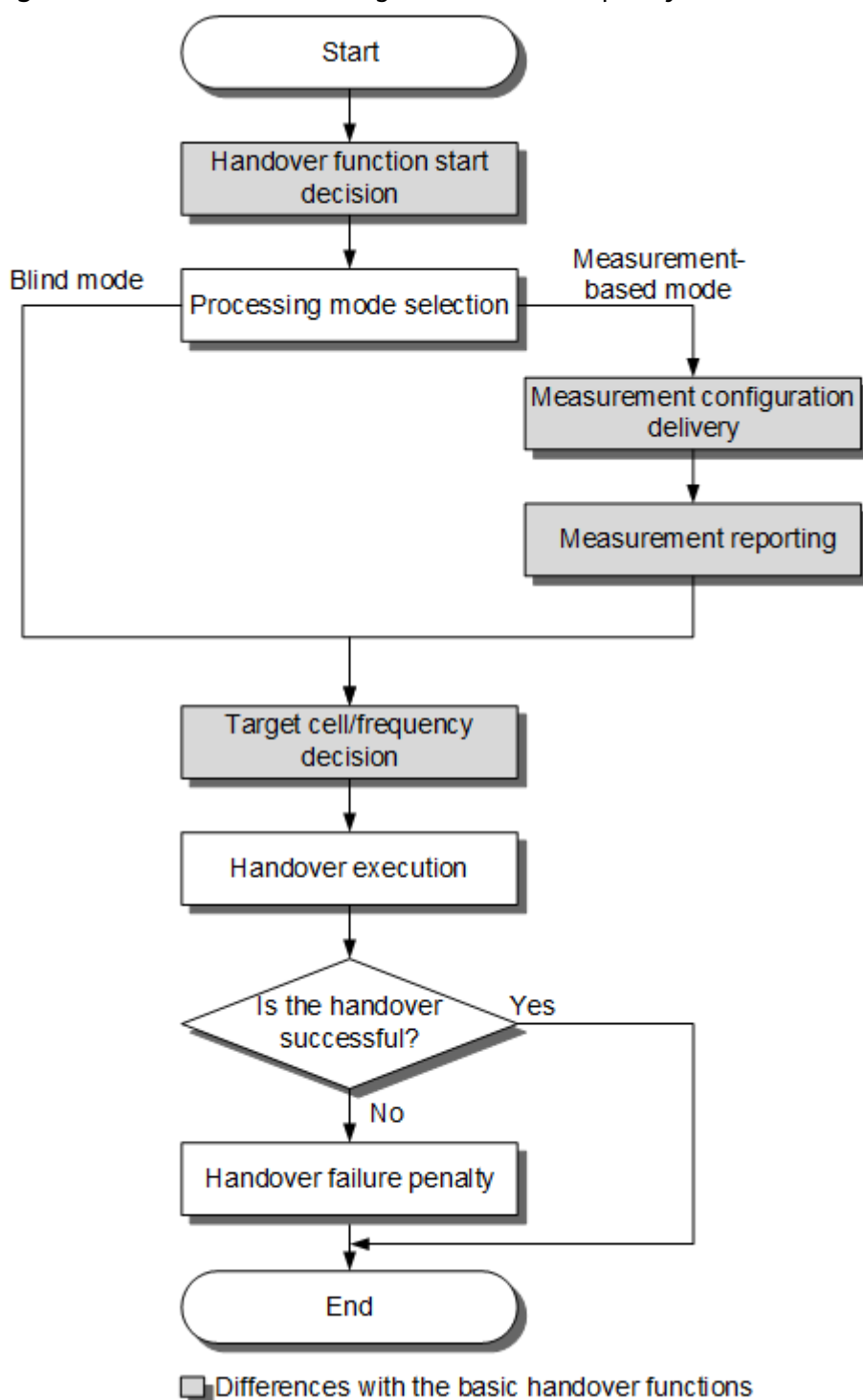


Coverage-based handovers are classified into intra-frequency handovers and inter-frequency handovers. This section describes coverage-based inter-frequency handovers. For details about coverage-based intra-frequency handovers, see [5 Coverage-based Intra-Frequency Handover](#).

Procedure of Coverage-based Inter-Frequency Handover

[Figure 6-2](#) shows the procedure of coverage-based inter-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 6-2 Procedure of coverage-based inter-frequency handover



Depending on the processing mode (measurement-based mode or blind mode) and handover policy (handover or redirection), coverage-based inter-frequency handover includes the following sub-functions:

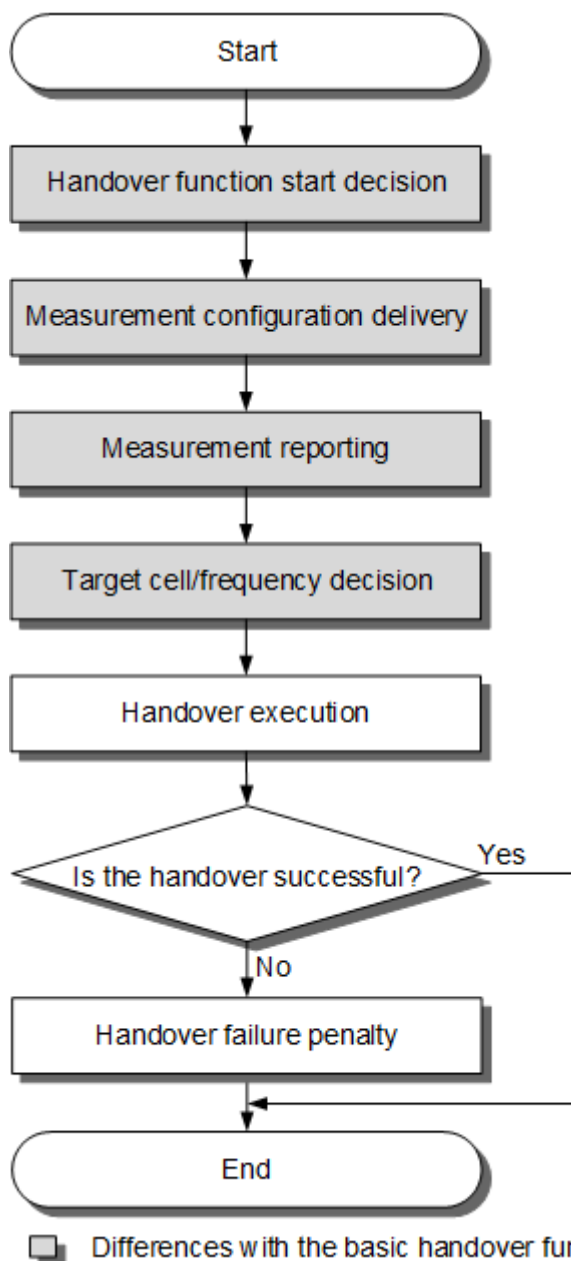
- Measurement-based inter-frequency handover, which is described in [6.1.1 Measurement-based Inter-Frequency Handover](#)

- Measurement-based inter-frequency redirection, which is described in [6.1.2 Measurement-based Inter-Frequency Redirection](#)
- Blind inter-frequency redirection, which is described in [6.1.3 Blind Inter-Frequency Redirection](#)

6.1.1 Measurement-based Inter-Frequency Handover

[Figure 6-3](#) shows the procedure of measurement-based inter-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 6-3 Procedure of a measurement-based inter-frequency handover



 NOTE

In low frequency bands, event A5 or A3 can be used to trigger execution of measurement-based inter-frequency handovers, depending on the value **EVENT_A5** or **EVENT_A3** configured for the **NRCellFreqRelation.InterFreqHoEventType** parameter. Assume that spectrums overlap between cells and the configured SSB frequency-domain positions are different. If event A5 is used to trigger handover execution, there is a possibility that handovers are not executed promptly due to severe interference between the cells, causing service drops. Therefore, it is recommended that event A3 be used to quickly trigger handover execution in such scenarios. In other scenarios, it is recommended that event A5 be used to trigger handover execution.

6.1.1.1 Handover Function Start Decision

The function is started when all of the following conditions are met:

- The **COVERAGE_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell.
- The gNodeB receives a measurement report indicating that the serving cell meets the entering conditions for event A2.
- The UE supports inter-frequency measurements and inter-frequency handovers. The UE supports the frequency bands containing the non-serving frequencies to be measured.

The triggering quantity for event A2 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A2 is related to its triggering quantity, as described in [Table 6-1](#). For details about the configuration of other variables for event A2, see [Table 4-8](#).

Table 6-1 Threshold configuration for event A2

Triggering Quantity for Event A2	To-Be-Configured Threshold for Event A2	Threshold Configuration for Event A2
Only RSRP	RSRP threshold	- Scenario 1: Whether there is a non-serving frequency for which the NRCellFreqRelation.InterFreqHoEventType parameter is set to EVENT_A5 is evaluated. - If there is such a frequency, the RSRP threshold for event A2 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld parameter. If this threshold needs to be frequency-specific, operators can set the NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs parameter to different values for different frequencies. The frequency-specific threshold will be equal to NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld plus NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs. - If there is no such frequency, the RSRP threshold for event A2 is not specified by the preceding parameter. - Scenario 2: Whether there is a non-serving frequency for which the NRCellFreqRelation.InterFreqHoEventType parameter is set to EVENT_A3 is evaluated. - If there is such a frequency, the RSRP threshold for event A2 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld parameter. - If there is no such frequency, the RSRP threshold for event A2 is not specified by the preceding parameter. If both of the preceding scenarios exist, the two RSRP thresholds for event A2 must be configured at the same time.
RSRP or RSRQ	RSRP and RSRQ thresholds	- For event A2, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. - The RSRQ threshold for event A2 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA2RsrqThld parameter.

Triggering Quantity for Event A2	To-Be-Configured Threshold for Event A2	Threshold Configuration for Event A2
RSRP or SINR	RSRP and SINR thresholds	- For event A2, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. - The SINR threshold for event A2 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreq-HoA2Thld parameter.

For details about the definition, entering conditions, and leaving conditions for event A2, see [Table 4-6](#). For details about triggering quantities, see [4.1.4.2.2 Triggering Quantity](#).

6.1.1.2 Measurement Configuration Delivery

After measurement-based inter-frequency handover is started, the gNodeB selects frequencies and delivers related measurement configurations.

- The gNodeB selects frequencies based on the following rules:
 - If the gNodeB receives event A2 in [Scenario 1](#) in [Table 6-1](#) during the handover function start decision, the gNodeB selects frequencies from the non-serving frequencies for which **NRCellFreqRelation.InterFreqHoEventType** is set to **EVENT_A5** and delivers the measurement configurations related to the selected frequencies.
 - If the gNodeB receives event A2 in [Scenario 2](#) in [Table 6-1](#) during the handover function start decision, the gNodeB selects frequencies from the non-serving frequencies for which **NRCellFreqRelation.InterFreqHoEventType** is set to **EVENT_A3** and delivers the measurement configurations related to the selected frequencies.
 - If the thresholds (**NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld** and **NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld**) for event A2 in [Scenario 1](#) and [Scenario 2](#) are set to the same value, the gNodeB cannot determine whether the received event A2 is generated in [Scenario 1](#) or [Scenario 2](#) during the handover function start decision. In this case, the gNodeB selects frequencies from all of the non-serving frequencies configured in frequency relationships without distinguishing the setting of **NRCellFreqRelation.InterFreqHoEventType**, and then delivers the measurement configurations related to the selected frequencies.

If the number of selected non-serving frequencies exceeds the maximum, the gNodeB further selects frequencies for measurement in descending order of priorities indicated by the **NRCellFreqRelation.ConnFreqPriority** parameter until the selected frequencies reach the maximum. If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected. For details about the maximum number of to-be-

measured frequencies in measurement configurations, see [4.1.4.1.2 Frequency Selection for Measurement](#).

- The gNodeB delivers measurement configurations related to events based on the following rules:

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A5** for a frequency, the gNodeB delivers the measurement configurations related to events A5 and A1.

Event A5 can be triggered only by RSRP. For details about its variable configuration, see [Table 4-11](#).

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the

NRCellMobilityConfig.MeasTrigQuantity parameter. The threshold configuration for event A1 is related to its triggering quantity, as described in [Table 6-2](#). For details about the configuration of other variables for event A1, see [Table 4-7](#).

Table 6-2 Threshold configuration for event A1

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
Only RSRP	RSRP threshold	The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld parameter. If this threshold needs to be frequency-specific, operators can set the NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs parameter to different values for different frequencies. The frequency-specific threshold will be equal to NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld plus NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs .
RSRP or RSRQ	RSRP and RSRQ thresholds	• For event A1, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. • The RSRQ threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrqThld parameter.

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
RSRP or SINR	RSRP and SINR thresholds	- For event A1, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. - The SINR threshold for event A1 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld parameter.

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A3** for a frequency, the gNodeB delivers the measurement configurations related to events A3 and A1.

Event A3 can be triggered only by RSRP. For details about its variable configuration, see [Table 4-9](#).

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A1 is related to its triggering quantity, as described in [Table 6-3](#). For details about the configuration of other variables for event A1, see [Table 4-7](#).

Table 6-3 Threshold configuration for event A1

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
Only RSRP	RSRP threshold	The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter.
RSRP or RSRQ	RSRP and RSRQ thresholds	- The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter. - The RSRQ threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrqThld parameter.

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
RSRP or SINR	RSRP and SINR thresholds	- The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter. - The SINR threshold for event A1 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld parameter.

Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2. Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset. Event A1 indicates that the signal quality of the serving cell exceeds a specific threshold.

6.1.1.3 Measurement Reporting

After measurement-based inter-frequency handover is started, a UE performs measurements based on the event configuration included in the measurement configurations received from the gNodeB, and sends measurement reports to the gNodeB based on the measurement results.

6.1.1.4 Target Cell or Frequency Decision

The gNodeB performs target cell or frequency decision based on the received measurement reports.

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A5** for the frequency in the received measurement report, target cell or frequency decision follows the mechanism in [Event A5 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report](#).
- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A3** for the frequency in the received measurement report, target cell or frequency decision follows the mechanism in [Event A3 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report](#).

Event A5 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report

- If the gNodeB receives an event A5 measurement report indicating that there is an NR neighbor relationship (configured by the **NRCellRelation MO**) between the serving cell and the cell with the best signal quality, then:
The gNodeB filters cells indicated in event A5 measurement reports by using neighboring cell filtering rules and then instructs the UE to perform an inter-frequency handover or redirection based on the UE capabilities.

- In the case that the UE supports inter-frequency handover:
The gNodeB instructs the UE to initiate handover attempts towards the filtered cells one by one, which are sorted in the order that event A5 measurement reports are submitted as well as in descending order of RSRPs of the cells per event A5 measurement report.
- In the case that the UE does not support inter-frequency handover:
 - If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell, the gNodeB instructs the UE to perform a redirection to the frequency of the cell indicated in the event A5 measurement report.
 - If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is deselected for the serving cell, the procedure ends. The gNodeB waits for the next event A5 measurement report from the UE.
- If the gNodeB receives an event A5 measurement report indicating that there is no NR neighbor relationship (configured by the **NRCellRelation MO**) between the serving cell and the cell with the best signal quality, then:
 - In the case that automatic detection and addition of missing neighboring cells in intra-RAT ANR is enabled for the serving cell and the UE supports reportCGI measurements on neighboring NR cells (nr-CGI-Reporting):
The gNodeB sends the reportCGI measurement configuration of the optimal cell to the UE, then obtains neighboring cell information from the reportCGI measurement report, and finally instructs the UE to perform an inter-frequency handover or redirection.
 - If the gNodeB obtains neighboring cell information from the reportCGI measurement report:
 - In the case that the UE supports inter-frequency handover, the gNodeB instructs the UE to initiate a handover attempt to the cell indicated in the reportCGI measurement report.
 - In the case that the UE does not support inter-frequency handover:
If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell, the gNodeB instructs the UE to perform a redirection to the frequency of the cell indicated in the reportCGI measurement report.
If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is deselected for the serving cell, the procedure ends. The gNodeB waits for the next event A5 measurement report from the UE.
 - If the gNodeB does not receive the reportCGI measurement report or fails to obtain neighboring cell information from the reportCGI measurement report:
 - In the case that both the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter and the **MR_BASED_REDIRECT_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter are

- selected for the serving cell, the gNodeB instructs the UE to perform a redirection to the frequency of the optimal cell.
 - Otherwise, the procedure ends. The gNodeB waits for the next event A5 measurement report from the UE.
- In the case that automatic detection and addition of missing neighboring cells in intra-RAT ANR is disabled for the serving cell or the UE does not support reportCGI measurements on neighboring NR cells (nr-CGI-Reporting):
 - In the case that both the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter and the **MR_BASED_REDIRECT_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter are selected for the serving cell, the gNodeB instructs the UE to perform a redirection to the frequency of the optimal cell.
 - Otherwise:

The gNodeB filters cells indicated in event A5 measurement reports by using neighboring cell filtering rules and then instructs the UE to perform an inter-frequency handover or redirection based on the UE capabilities.

 - In the case that the UE supports inter-frequency handover:

The gNodeB instructs the UE to initiate handover attempts towards the filtered cells one by one, which are sorted in the order that event A5 measurement reports are submitted as well as in descending order of RSRPs of the cells per event A5 measurement report.
 - In the case that the UE does not support inter-frequency handover:

If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell, the gNodeB instructs the UE to perform a redirection to the frequency of the cell indicated in the event A5 measurement report.

If the **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is deselected for the serving cell, the procedure ends. The gNodeB waits for the next event A5 measurement report from the UE.

For details about the neighboring cell filtering rules, see [Generation of the Target Cell or Frequency](#) in [4.1.6 Target Cell or Frequency Decision](#).

For details about intra-RAT ANR, see *ANR*.

- If the gNodeB receives an event A1 measurement report, it instructs the UE to stop event A5 measurements. The function is stopped.

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter.

 - When RSRP or RSRQ is used as the triggering quantity, the UE is instructed to stop event A5 measurements only after RSRP- and RSRQ-triggered event A1 reports that respectively correspond to previous RSRP-

and RSRQ-triggered event A2 reports are all received. In this case, the function is stopped.

- When RSRP or SINR is used as the triggering quantity, the UE is instructed to stop event A5 measurements only after RSRP- and SINR-triggered event A1 reports that respectively correspond to previous RSRP- and SINR-triggered event A2 reports are all received. In this case, the function is stopped.

Event A3 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report

- After receiving an event A3 measurement report, the gNodeB performs in the same way as it does after receiving an event A5 measurement report in two scenarios: **an NR neighbor relationship** and **no NR neighbor relationship** between the serving cell and the cell with the best signal quality. For details, see **Event A5 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report**.
- If the gNodeB receives an event A1 measurement report, it instructs the UE to stop event A3 measurements. The function is stopped.

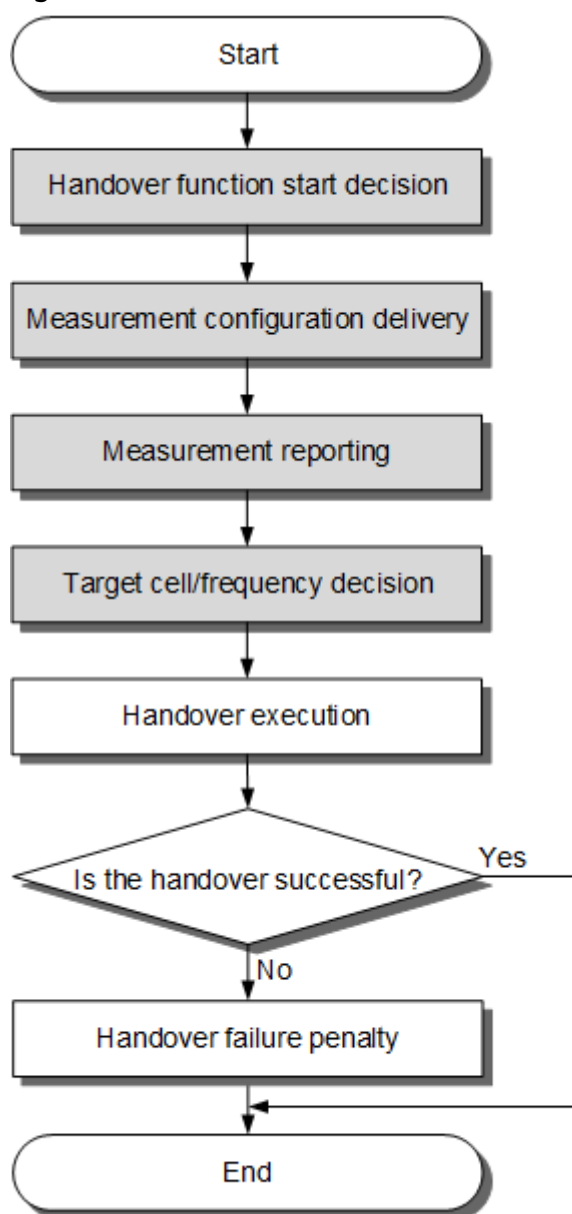
The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter.

- When RSRP or RSRQ is used as the triggering quantity, the UE is instructed to stop event A3 measurements only after RSRP- and RSRQ-triggered event A1 reports that respectively correspond to previous RSRP- and RSRQ-triggered event A2 reports are all received. In this case, the function is stopped.
- When RSRP or SINR is used as the triggering quantity, the UE is instructed to stop event A3 measurements only after RSRP- and SINR-triggered event A1 reports that respectively correspond to previous RSRP- and SINR-triggered event A2 reports are all received. In this case, the function is stopped.

6.1.2 Measurement-based Inter-Frequency Redirection

Figure 6-4 shows the procedure of measurement-based inter-frequency redirection. This section describes the differences between this function and the basic handover functions presented in **4 Basic Functions of Mobility Management in Connected Mode**.

Figure 6-4 Procedure of measurement-based inter-frequency redirection



 Differences with the basic handover functions

NOTE

In low frequency bands, event A5 or A3 can be used to trigger execution of measurement-based inter-frequency redirection, depending on the value **EVENT_A5** or **EVENT_A3** configured for the **NRCellFreqRelation.InterFreqHoEventType** parameter. Assume that spectrums overlap between cells and the configured SSB frequency-domain positions are different. If event A5 is used to trigger redirection execution, there is a possibility that redirections are not executed promptly due to severe interference between the cells, causing service drops. Therefore, it is recommended that event A3 be used to quickly trigger redirection execution in such scenarios. In other scenarios, it is recommended that event A5 be used to trigger redirection execution.

6.1.2.1 Handover Function Start Decision

The function is started when all of the following conditions are met:

- The **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell.
- The gNodeB receives a measurement report indicating that the serving cell meets the entering conditions for event A2.
- The UE supports inter-frequency measurement. The UE supports the frequency bands containing the non-serving frequencies to be measured.

The triggering quantity for event A2 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A2 is related to its triggering quantity, as described in [Table 6-4](#). For details about the configuration of other variables for event A2, see [Table 4-8](#).

Table 6-4 Threshold configuration for event A2

Triggering Quantity for Event A2	To-Be-Configured Threshold for Event A2	Threshold Configuration for Event A2
Only RSRP	RSRP threshold	• Scenario 1: Whether there is a non-serving frequency for which the NRCellFreqRelation.InterFreqHoEventType parameter is set to EVENT_A5 is evaluated. – If there is such a frequency, the RSRP threshold for event A2 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld parameter. – If there is not such a frequency, the RSRP threshold for event A2 is not specified by the preceding parameter. • Scenario 2: Whether there is a non-serving frequency for which the NRCellFreqRelation.InterFreqHoEventType parameter is set to EVENT_A3 is evaluated. – If there is such a frequency, the RSRP threshold for event A2 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld parameter. – If there is not such a frequency, the RSRP threshold for event A2 is not specified by the preceding parameter. If both of the preceding scenarios exist, the two RSRP thresholds for event A2 must be configured at the same time.

Triggering Quantity for Event A2	To-Be-Configured Threshold for Event A2	Threshold Configuration for Event A2
RSRP or RSRQ	RSRP and RSRQ thresholds	- For event A2, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. - The RSRQ threshold for event A2 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA2RsrqThld parameter.
RSRP or SINR	RSRP and SINR thresholds	- For event A2, the RSRP threshold configuration mode is the same as that when only RSRP is used as the triggering quantity. - The SINR threshold for event A2 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreq-HoA2Thld parameter.

For details about the definition, entering conditions, and leaving conditions for event A2, see [Table 4-6](#). For details about triggering quantities, see [4.1.4.2.2 Triggering Quantity](#).

6.1.2.2 Measurement Configuration Delivery

After measurement-based inter-frequency redirection is started, the gNodeB selects frequencies and delivers related measurement configurations.

- The gNodeB selects frequencies based on the following principles:
 - If the gNodeB receives event A2 in [Scenario 1](#) in [Table 6-4](#) during the handover function start decision, the gNodeB selects frequencies from the non-serving frequencies for which **NRCellFreqRelation.InterFreqHoEventType** is set to **EVENT_A5** and delivers related measurement configurations.
 - If the gNodeB receives event A2 in [Scenario 2](#) in [Table 6-4](#) during the handover function start decision, the gNodeB selects frequencies from the non-serving frequencies for which **NRCellFreqRelation.InterFreqHoEventType** is set to **EVENT_A3** and delivers related measurement configurations.
 - If the thresholds for event A2 in [Scenario 1](#) and [Scenario 2](#) (**NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld** and **NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld**) are set to the same value, the gNodeB cannot determine whether the received event A2 is in [Scenario 1](#) or [Scenario 2](#) during the handover function start decision. In this case, the gNodeB does not distinguish the value of **NRCellFreqRelation.InterFreqHoEventType**, and the gNodeB selects frequencies from all of the non-serving frequencies configured in frequency relationships and delivers related measurement configurations.

If the number of selected non-serving frequencies exceeds the maximum, the gNodeB further selects frequencies for measurement in descending order of priorities indicated by the **NRCellFreqRelation.ConnFreqPriority** parameter until the selected frequencies reach the maximum. If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#).

- The gNodeB delivers measurement configurations related to events based on the following principles:

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A5** for a frequency, the gNodeB delivers the measurement configurations related to events A5 and A1.

Event A5 can be triggered only by RSRP. For details about its variable configuration, see [Table 4-11](#).

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A1 is related to its triggering quantity, as described in [Table 6-5](#). For details about the configuration of other variables for event A1, see [Table 4-7](#).

Table 6-5 Threshold configuration for event A1

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
Only RSRP	RSRP threshold	The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld parameter.
RSRP or RSRQ	RSRP and RSRQ thresholds	• The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld parameter. • The RSRQ threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrqThld parameter.
RSRP or SINR	RSRP and SINR thresholds	• The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld parameter. • The SINR threshold for event A1 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld parameter.

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A3** for a frequency, the gNodeB delivers the measurement configurations related to events A3 and A1.

Event A3 can be triggered only by RSRP. For details about its variable configuration, see [Table 4-9](#).

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A1 is related to its triggering quantity, as described in [Table 6-6](#). For details about the configuration of other variables for event A1, see [Table 4-7](#).

Table 6-6 Threshold configuration for event A1

Triggering Quantity for Event A1	To-Be-Configured Threshold for Event A1	Threshold Configuration for Event A1
Only RSRP	RSRP threshold	The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter.
RSRP or RSRQ	RSRP and RSRQ thresholds	• The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter. • The RSRQ threshold for event A1 is specified by the NRCellInterFHoMeaGrp.CovInterFreqA1RsrqThld parameter.
RSRP or SINR	RSRP and SINR thresholds	• The RSRP threshold for event A1 is specified by the NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld parameter. • The SINR threshold for event A1 is specified by the NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld parameter.

Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2. Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset. Event A1 indicates that the signal quality of the serving cell exceeds a specific threshold.

6.1.2.3 Measurement Reporting

After measurement-based inter-frequency redirection is started, a UE performs measurements based on the event configuration included in the measurement configurations received from the gNodeB, and sends measurement reports to the gNodeB based on the measurement results.

6.1.2.4 Target Cell or Frequency Decision

The gNodeB performs target cell or frequency decision based on the received measurement reports.

- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A5** for the frequency in the received measurement report, target cell or frequency decision follows the mechanism in [Event A5 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report](#).
- If the **NRCellFreqRelation.InterFreqHoEventType** parameter is set to **EVENT_A3** for the frequency in the received measurement report, target cell or frequency decision follows the mechanism in [Event A3 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report](#).

Event A5 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report

- If the gNodeB receives an event A5 measurement report:
 - In the case that the UE does not support inter-frequency handover, the gNodeB instructs the UE to perform a redirection to the frequency of the cell with the best signal quality indicated in the measurement result.
 - In the case that the UE supports inter-frequency handover and measurement-based inter-frequency handover has been enabled for the serving cell, the gNodeB triggers a measurement-based redirection if there is no NR neighbor relationship (configured by the **NRCellRelation MO**) between the serving cell and the cell with the best signal quality indicated in the measurement report sent by the UE. For details, see [6.1.1.4 Target Cell or Frequency Decision](#).
- If the gNodeB receives an event A1 measurement report, it instructs the UE to stop event A5 measurements. The function is stopped.

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter.

- When RSRP or RSRQ is used as the triggering quantity, the UE is instructed to stop event A5 measurements only after RSRP- and RSRQ-triggered event A1 reports that respectively correspond to previous RSRP- and RSRQ-triggered event A2 reports are all received. In this case, the function is stopped.
- When RSRP or SINR is used as the triggering quantity, the UE is instructed to stop event A5 measurements only after RSRP- and SINR-triggered event A1 reports that respectively correspond to previous RSRP- and SINR-triggered event A2 reports are all received. In this case, the function is stopped.

Event A3 Triggering Inter-Frequency Handover for the Frequency in the Measurement Report

- If the gNodeB receives an event A3 measurement report:
 - In the case that the UE does not support inter-frequency handover, the gNodeB instructs the UE to perform a redirection to the frequency of the cell with the best signal quality indicated in the measurement result.
 - In the case that the UE supports inter-frequency handover and measurement-based inter-frequency handover has been enabled for the serving cell, the gNodeB triggers a measurement-based redirection if there is no NR neighbor relationship (configured by the **NRCellRelation** MO) between the serving cell and the cell with the best signal quality indicated in the measurement report sent by the UE. For details, see [6.1.1.4 Target Cell or Frequency Decision](#).
- If the gNodeB receives an event A1 measurement report, it instructs the UE to stop event A3 measurements. The function is stopped.

The triggering quantity for event A1 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter.

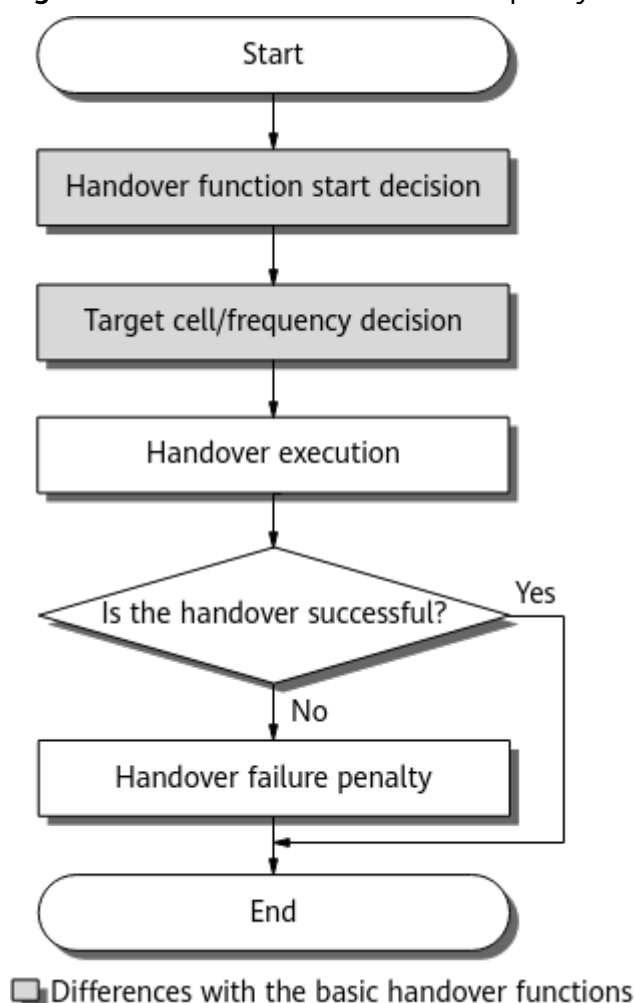
- When RSRP or RSRQ is used as the triggering quantity, the UE is instructed to stop event A3 measurements only after RSRP- and RSRQ-triggered event A1 reports that respectively correspond to previous RSRP- and RSRQ-triggered event A2 reports are all received. In this case, the function is stopped.
- When RSRP or SINR is used as the triggering quantity, the UE is instructed to stop event A3 measurements only after RSRP- and SINR-triggered event A1 reports that respectively correspond to previous RSRP- and SINR-triggered event A2 reports are all received. In this case, the function is stopped.

6.1.3 Blind Inter-Frequency Redirection

6.1.3.1 Basic Procedure

[Figure 6-5](#) shows the procedure of blind inter-frequency redirection. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 6-5 Procedure of blind inter-frequency redirection



6.1.3.1.1 Handover Function Start Decision

The function is started when all of the following conditions are met:

- The **REDIRECTION** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the serving cell.
- The gNodeB receives a measurement report indicating that the serving cell meets the entering conditions for event A2.

The triggering quantity for event A2 can be only RSRP, RSRP or RSRQ, or RSRP or SINR, which is specified by the **NRCellMobilityConfig.MeasTrigQuantity** parameter. The threshold configuration for event A2 is related to its triggering quantity, as described in [Table 6-7](#). For details about the configuration of other variables for event A2, see [Table 4-8](#).

Table 6-7 Threshold configuration for event A2

Triggering Quantity for Event A2	To-Be-Configured Threshold for Event A2	Threshold Configuration for Event A2
Only RSRP	RSRP threshold	The RSRP threshold for event A2 is specified by the NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld parameter.
RSRP or RSRQ	RSRP and RSRQ thresholds	- The RSRP threshold for event A2 is specified by the NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld parameter. - The RSRQ threshold for event A2 is specified by the NRCellInterRHoMeaGrp.CovHoToLteBlindA2RsrqThld parameter.
RSRP or SINR	RSRP and SINR thresholds	- The RSRP threshold for event A2 is specified by the NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld parameter. - The SINR threshold for event A2 is specified by the NRCellInterFHoMeaGrp.InterFBlindRedirA2SinnrThld parameter.

For details about the definition, entering conditions, and leaving conditions for event A2, see [Table 4-6](#). For details about triggering quantities, see [4.1.4.2.2 Triggering Quantity](#).

6.1.3.1.2 Target Cell or Frequency Decision

When blind inter-frequency redirection is enabled, the gNodeB filters the neighboring NR frequencies of the serving cell and instructs the UE to select a high-priority frequency for blind redirection.

- Filtering candidate frequencies
 - Filtering rules exclusive to data services: The **NRCellFreqRelation.BlindRedirectPriority** parameter specifies whether a non-serving frequency can be selected as the target frequency for blind redirection of data services.
 - If the parameter is not set to **0** for a non-serving frequency, this frequency can be selected as the target frequency for blind redirection of data services.
 - If the parameter is set to **0** for a non-serving frequency, this frequency cannot be selected as the target frequency for blind redirection of data services.

If the parameter is set to **0** for all neighboring NR frequencies (configured by the **NRCellFreqRelation** MO) of the serving cell, blind inter-frequency redirection for data services is prohibited. In this case, the gNodeB selects a neighboring E-UTRAN frequency of the serving cell as the target frequency for blind redirection of data services on the UE. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

- Filtering rules exclusive to voice services: The **VOICE_BLIND_REDIRECT_SW** option of the **NRCellFreqRelation.FrequencyCtrlSwitch** parameter specifies whether a non-serving frequency can be selected as the target frequency for blind redirection of voice services. For details about optimized blind redirection for voice services, see *VoNR*.
 - If the option is selected for a non-serving frequency, this frequency can be selected as the target frequency for blind redirection of voice services.
 - If the option is deselected for a non-serving frequency, this frequency cannot be selected as the target frequency for blind redirection of voice services.
- Common filtering rules for all services: The measurement-based redirection function (enabled by selecting the **MR_BASED_REDIRECT_SW** option of the **NRCellAlgoSwitch.HoCompatibilitySwitch** parameter) controls whether to filter the neighboring NR frequencies of the serving cell based on historical measurement reports before the target frequency is selected for blind redirection.
 - If the option is selected for the serving cell, the gNodeB selects the intersection of the following NR frequencies as candidate frequencies for blind redirection:
 - Neighboring NR frequencies of the serving cell (NR cell frequency relationships configured by the **NRCellFreqRelation** MO)
 - NR frequencies of cells in the historical measurement reports recorded by the gNodeB
 - If the option is deselected for the serving cell, the gNodeB selects the neighboring NR frequencies of the serving cell as candidate frequencies for blind redirection. The NR cell frequency relationships are configured by the **NRCellFreqRelation** MO.

2. Performing a blind redirection

The gNodeB instructs the UE to sort the non-serving frequencies that are retained after filtering based on their blind redirection priorities and select the frequency with the highest priority as the target frequency for blind redirection. Note that, if multiple non-serving frequencies have the same priority, the frequency with the smallest value of *SsbFreqPos* is selected for blind redirection.

The blind redirection priority of a frequency is specified by the **NRCellFreqRelation.BlindRedirectPriority** parameter.

- This parameter is applicable to both voice and data services.

- This parameter is useful in prioritizing non-serving frequencies only when it is set to a value other than 0.

6.1.3.2 Preferential Triggering of Blind Redirection

When the A2-based blind redirection control switch is turned on (by selecting the **A2_BLIND_REDIRECT_CTRL_SW** option of the **NRCellAlgoSwitch.ProcessSwitch** parameter), the gNodeB preferentially triggers a blind redirection procedure in the following scenarios:

- After a UE sends a blind event A2 report, if the gNodeB initiates an RRC connection release procedure due to an air interface-related exception (for example, RRC reconfiguration timeout), the gNodeB performs subsequent operations based on the setting of the A2-based blind redirection control switch.
 - If this switch is turned off, the gNodeB triggers an RRC connection release procedure to release the UE. As a result, the UE experiences a service drop.
 - If this switch is turned on, the gNodeB triggers an RRC connection release procedure for blind redirection.
- After the gNodeB sends an RRCReconfiguration message to a UE, if the gNodeB receives blind event A2 and measurement-based event A2 reports before receiving an RRCReconfigurationComplete message from the UE, upon receipt of the RRCReconfigurationComplete message, the gNodeB performs subsequent operations based on the setting of the A2-based blind redirection control switch.
 - If this switch is turned off, the gNodeB processes the measurement reports according to the time sequence in which they are received.
 - If this switch is turned on, the gNodeB preferentially processes the blind event A2 report.
- After receiving a measurement-based event A2 report from a UE, the gNodeB performs subsequent operations based on the setting of the A2-based blind redirection control switch.
 - If this switch is turned off, the gNodeB directly processes the measurement-based event A2 report.
 - If this switch is turned on, the gNodeB checks whether the signal quality of the serving cell is lower than the value of **NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld**. If so, the gNodeB triggers a blind redirection procedure. If not, the gNodeB processes the measurement-based event A2 report.

6.2 Network Analysis

6.2.1 Benefits

Coverage-based inter-frequency handovers ensure that coverage is continuous and service experience is consistent for moving UEs.

The service drop rate decreases after the blind redirection optimization function is enabled. The service drop rate is indicated by counters such as **Service Call Drop**

Rate (CU), Service Call Drop Rate (CU, Inactive), and Call Drop Rate (CU, VoNR).

6.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

Coverage-based inter-frequency handover has the following impacts on the network:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.

6.3 Requirements

6.3.1 Licenses

There are no license requirements.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB SINR-based inter-frequency handover	SSB_SINR_BASSED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When both coverage-based inter-frequency handover and SSB SINR-based inter-frequency handover are enabled, the NRCellMobilityConfig.MeasTrigQuantity parameter cannot be set to RSRP_OR_SINR .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB SINR-based inter-frequency handover	SSB_SINR_BASSED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When both inter-frequency redirection and SSB SINR-based inter-frequency handover are enabled, the NRCellMobilityConfig.MeasTrigQuantity parameter cannot be set to RSRP_OR_SINR .
Low-frequency TDD	User-experience-based coverage expansion	COV_THLD_ADAPT_SW option of the gNBMobilityCommonParam.MobilityAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	When the NRCellMobilityConfig.MeasTrigQuantity parameter is set to RSRP_OR_RSRQ or RSRP_OR_SINR , user-experience-based coverage expansion cannot be enabled.
High-frequency TDD	None	None	None	None

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	If measurement-based inter-frequency handover is disabled, intra-RAT ANR will not be triggered by measurement-based inter-frequency handover because such handover does not take effect. If redirection is disabled, intra-RAT ANR will not be triggered by measurement-based redirection because such redirection does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	If measurement-based inter-frequency handover is enabled and the NRCellFreqRelation.FreqIntrfFlag parameter is set to INTRF or INTRF_SINR, the gNodeB delivers measurement configurations with the report type of reportCGI to the UE involving only the neighboring cells whose measured signal quality is higher than the value of the NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2 or NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2 parameter. If redirection is enabled and the NRCellFreqRelation.FreqIntrfFlag parameter is set to INTRF or INTRF_SINR, the gNodeB delivers measurement configurations with the report type of reportCGI to the UE involving only the neighboring cells whose measured signal quality is higher than the value of the NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2 or NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2 parameter.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both coverage-based inter-frequency handover and connected mode MLB are enabled, (2) coverage-based inter-frequency handovers are triggered by event A5, and (3) the value of NRCellInterFHoMeaGrp.InterFreqMlbA4RsrpThld is less than or equal to the value of NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 , then ping-pong between coverage-based handovers and MLB-based handovers will occur.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both coverage-based inter-frequency handover and connected mode MLB are enabled, (2) coverage-based inter-frequency handovers are triggered by event A5, and (3) the value of NRCellInterFHoMeaGrp.InterFreqMlbA4RsrpThld is less than or equal to the value of NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 , then ping-pong between coverage-based handovers and MLB-based handovers will occur.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	5QI-specific UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to SPECIFIED_5QI_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both coverage-based inter-frequency handover and connected mode MLB are enabled, (2) coverage-based inter-frequency handovers are triggered by event A5, and (3) the value of NRCellInterFHoMeaGrp.InterFreqMlbA4RsrpThld is less than or equal to the value of NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 , then ping-pong between coverage-based handovers and MLB-based handovers will occur.
Low-frequency TDD	User-experience-based coverage expansion	COV_THLD_ADAPT_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	User-experience-based coverage expansion modifies the thresholds for events A1 and A2 and threshold 2 for event A5 related to coverage-based inter-frequency handover only when measurement-based inter-frequency handover (controlled by the COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter) or measurement-based inter-frequency redirection (controlled by the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter) is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When event A2 is used (that is, NRCellMobilityConfig.FreqPriHoTriggerMode is set to A2_BASED) in the handover function start decision for frequency-priority-based inter-frequency handover, it is recommended that the threshold for frequency-priority-based inter-frequency event A2 (specified by NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld) be higher than the threshold for coverage-based inter-frequency event A2 (specified by NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld). Otherwise, event A2 cannot be used to trigger frequency-priority-based inter-frequency handover.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for coverage-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency list group bound to the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers: If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency.

RAT	Function Name	Function Switch	Reference	Description
				If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is deselected, coverage-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target cell's frequency in the frequency list group bound to the roaming UE, the following mechanism is applied: - If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency. - If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is deselected, coverage-based inter-frequency handover does not take effect for the roaming UE. In the decision-making phase for coverage-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-

RAT	Function Name	Function Switch	Reference	Description
				incapable roaming UE, and (2) the COVERAGE_BASED_HO and REDIRECTION options of the NRCellAlgoSwitch.InterFrequencyHoSwitch parameter are respectively deselected and selected for the serving cell, then the gNodeB will instruct the roaming UE to immediately perform a redirection to the target cell's frequency.
Low-frequency TDD	Blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN	• MOBILITY_TO_EUTRAN_SW option of the NRCellAlgoSwitch.InterRATServiceMobilitySwitch parameter • EUTRAN_REDIRECT_SWITCH option of the NRInterRatHoParam.HoModeSwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	If both "intra-RAT blind inter-frequency redirection" and "blind redirection in downlink-coverage-based inter-RAT mobility from NG-RAN to E-UTRAN" are enabled, "intra-RAT blind inter-frequency redirection" preferentially takes effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-experience-based inter-frequency handover	UL_EXP_BASED_INTER_FREQ_HO_SW option of the NRCellRedCap.RedCapAlgorithmSwitch parameter	<i>RedCap</i>	If (1) both uplink-experience-based inter-frequency handover and coverage-based inter-frequency handover are enabled, and (2) the value of NRCellRedCap.RedCapUplinkExperienceBasedInterFreqA5RsrpTh2 is less than or equal to that of NRCellInterFHoMeaGrp.CoverageBasedInterFreqA2RsrpThld , then ping-pong between uplink-experience-based inter-frequency handovers and coverage-based inter-frequency handovers will occur.
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SW option of the NRCellAlgorithmSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	If event A2 (NRCellMobilityConfig.FreqPriHoTriggerMode set to A2_BASED) is used in the handover function start decision for CA-UE-specific frequency-priority-based inter-frequency handover, it is recommended that the threshold for event A2 (NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld + NRCellUserMeasCfg.RsrpThldOffset) related to CA-UE-specific frequency-priority-based inter-frequency handover be higher than the threshold for event A2 (NRCellInterFHoMeaGrp.CoverageBasedInterFreqA2RsrpThld) related to coverage-based inter-frequency handover. Otherwise, event A2 cannot be used to trigger CA-UE-specific frequency-priority-based inter-frequency handover.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	None	None	None	None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Others

When a UE simultaneously measures multiple non-serving frequencies, it is recommended that the frame offset difference between any two frequencies be less than or equal to 3 ms so that they can be correctly measured. The frame offset is specified by the **gNBFreqBandConfig.FrameOffset** parameter.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

The NR cell's frequency relationships and neighbor relationships must be configured before this function is activated. For details about the NR cell's

frequency relationship configurations, see [Table 6-8](#). For details about neighbor relationship configurations, see [Table 4-20](#).

[Table 6-9](#) describes the function switches.

[Table 6-10](#) describes other parameters used for function optimization.

Table 6-8 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Frequency Priority for Connected Mode	NRCellFreqRelation.ConnFreqPriority	It is recommended that this parameter be set to a non-zero value for frequencies providing contiguous coverage, with a larger value set for a frequency in a lower frequency band. It is recommended that this parameter be set to 0 for frequencies providing discontinuous coverage, which means that such a frequency will not be selected as the target frequency.
Blind Redirection Priority	NRCellFreqRelation.BlindRedirectPriority	Set this parameter based on the network plan.
Inter-Freq HO Event Type	NRCellFreqRelation.InterFreqHoEventType	In low frequency bands, it is recommended that this parameter be set to EVENT_A3 if spectrums overlap between cells and the configured SSB frequency-domain positions are different and to EVENT_A5 in other scenarios.

Parameter Name	Parameter ID	Setting Notes
Frequency Interference Flag	NRCellFreqRelation.FreqIntrfFlag	Set this parameter based on the network plan.
Inter-Freq HO A1/A2/A5-1 Freq Offset	NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs	Set this parameter based on the network plan. A smaller value of this parameter results in a lower probability of UE-triggered coverage-based inter-frequency handovers to the frequency. A larger value of this parameter results in a higher probability.

Table 6-9 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgorithmSwitch.InterFreqHoSwitch	Select the COVERAGE_BASED_HO option if coverage-based measurement-based inter-frequency handover needs to be enabled. Select the REDIRECTION option if inter-frequency redirection needs to be enabled.
Process Switch	NRCellAlgorithmSwitch.ProcessSwitch	Select the A2_BLIND_REDIRECT_CTRL_SW option if blind redirection optimization needs to be enabled.
Handover Compatibility Switch	NRCellAlgorithmSwitch.HoCompatibilitySwitch	It is recommended that the MR_BASED_REDIRECT_SW option be selected in the case of many missing neighboring cells of the serving cell or weak coverage by the neighboring NR frequencies of the serving cell. When the option is selected, it is recommended that the "automatic detection and addition of missing neighboring cells" subfunction of intra-RAT ANR be enabled. For details about related operations and maintenance, see <i>ANR</i> .

Table 6-10 Parameters related to inter-frequency handover measurement configurations of NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCeIIInterFHoMeaGrp.NrCellId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	NRCeIIInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
Inter-frequency A1A2 Time to Trigger	NRCeIIInterFHoMeaGrp.InterFreqA1A2TimeToTrigger	Use the recommended value.
Inter-frequency A1A2 Hysteresis	NRCeIIInterFHoMeaGrp.InterFreqA1A2Hyst	Use the recommended value.
Cov Inter-freq Handover A1 RSRP Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA1RsrpThld	Use the recommended value. This parameter must be set to a value greater than NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld .
A3 Inter-freq Handover A1 RSRP Threshold	NRCeIIInterFHoMeaGrp.A3InterFreqHoA1RsrpThld	Use the recommended value. This parameter must be set to a value greater than NRCeIIInterFHoMeaGrp.A3InterFreqHoA2RsrpThld .
Cov Inter-freq HO A1 RSRQ Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA1RsrqThld	Use the recommended value.
SSB SINR Inter-freq HO A1 Threshold	NRCeIIInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld	Use the recommended value.
Cov Inter-freq Handover A2 RSRP Threshold	NRCeIIInterFHoMeaGrp.CovInterFreqA2RsrpThld	Use the recommended value. This parameter must be set to a value less than NRCeIIInterFHoMeaGrp.CovInterFreqA1RsrpThld .

Parameter Name	Parameter ID	Setting Notes
A3 Inter-freq Handover A2 RSRP Threshold	NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld	Use the recommended value. This parameter must be set to a value less than NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld .
Cov Inter-freq HO A2 RSRQ Threshold	NRCellInterFHoMeaGrp.CovInterFreqA2RsrqThld	Use the recommended value.
SSB SINR Inter-freq HO A2 Threshold	NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA2Thld	Use the recommended value.
Cov HO to E-UTRAN Blind A2 RSRP Thld	NRCellInterRHoMeaGrp.CovHoToEutranBlindA2Thld	Use the recommended value.
Cov HO to E-UTRAN Blind A2 RSRQ Thld	NRCellInterRHoMeaGrp.CovHoToLteBlindA2RsrqThld	Use the recommended value.
Inter-Freq Blind Redirect A2 SINR Thld	NRCellInterFHoMeaGrp.InterFBli ndRedirA2SinrThld	Use the recommended value.
Cov Inter-freq Handover A5 RSRP Threshold1	NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld1	Use the recommended value. - RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements is specified by this parameter when the NRCellMobilityConfig.MeasTrigQuantity parameter is set to RSRP. - RSRP threshold 1 for event A5 related to coverage-based inter-frequency measurements is fixed at -43 dBm when the NRCellMobilityConfig.MeasTrigQuantity parameter is not set to RSRP. In this case, the value of this parameter is invalid.
Cov Inter-freq Handover A5 RSRP Threshold2	NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Cov Inter-freq HO RSRQ Threshold	NRCellInterFHoMeaGrp.CovInterFreqA5RsrqThld2	Use the recommended value.
Cov Inter-Freq HO A5 SINR Threshold2	NRCellInterFHoMeaGrp.CovInterFreqA5SinrThld2	Use the recommended value.
Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrigger	Use the recommended value.
Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Use the recommended value.
Inter-frequency Handover A3 Offset	NRCellInterFHoMeaGrp.InterFreqHoA3Offset	Use the recommended value.
Measurement Trigger Quantity	NRCellMobilityConfig.MeasTriggerQuantity	Set this parameter based on the network plan.
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	NRCellQciBearer.NsaInterFreqHoMeasGroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgorithmSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).

Parameter Name	Parameter ID	Setting Notes
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpld	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

- Configuration of NR cell frequency relationships, external cells, and NR cell relationships

The following examples are specific to low frequency bands in common scenarios where event A5 is used to trigger handover execution.

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1, BlindRedirectPriority=2,
InterFreqHoEventType=EVENT_A5;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
CellIndividualOffset=DB0;
```

The following examples are specific to low frequency band scenarios where the serving cell and the neighboring cell have an overlapping spectrum but different SSB frequency-domain positions and event A3 is used to trigger handover execution.

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREQRELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1, BlindRedirectPriority=2,
InterFreqHoEventType=EVENT_A3;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0,
CellIndividualOffset=DB0;
```

- Enabling the functions

```
//Activating coverage-based measurement-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=COVERAGE_BASED_HO-1;
//Activating coverage-based inter-frequency redirection
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=REDIRECTION-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring a group of measurement parameters for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, CovInterFreqA5RsrpThld1=-110, CovInterFreqA5RsrpThld2=-106,
CovInterFreqA2RsrpThld=-110, CovInterFreqA2RsrqThld=-30, SsbSinrInterFreqHoA2Thld=-10,
CovInterFreqA1RsrpThld=-106, CovInterFreqA1RsrqThld=-26, CovInterFreqA5RsrqThld2=-24,
CovInterFreqA5SinrThld2=-4, SsbSinrInterFreqHoA1Thld=-6, A3InterFreqHoA1RsrpThld=-96,
A3InterFreqHoA2RsrpThld=-100, InterFreqHoA3Offset=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, CovInterFreqA5RsrpThld1=-110, CovInterFreqA5RsrpThld2=-106,
CovInterFreqA2RsrpThld=-110, CovInterFreqA2RsrqThld=-30, SsbSinrInterFreqHoA2Thld=-10,
CovInterFreqA1RsrpThld=-106, CovInterFreqA1RsrqThld=-26, CovInterFreqA5RsrqThld2=-24,
CovInterFreqA5SinrThld2=-4, SsbSinrInterFreqHoA1Thld=-6, A3InterFreqHoA1RsrpThld=-96,
A3InterFreqHoA2RsrpThld=-100, InterFreqHoA3Offset=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=2,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, CovInterFreqA5RsrpThld1=-105, CovInterFreqA5RsrpThld2=-101,
CovInterFreqA2RsrpThld=-105, CovInterFreqA2RsrqThld=-25, SsbSinrInterFreqHoA2Thld=-10,
CovInterFreqA1RsrpThld=-100, CovInterFreqA1RsrqThld=-21, CovInterFreqA5RsrqThld2=-19,
CovInterFreqA5SinrThld2=-4, SsbSinrInterFreqHoA1Thld=-6, A3InterFreqHoA1RsrpThld=-96,
A3InterFreqHoA2RsrpThld=-100, InterFreqHoA3Offset=2;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=3,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, CovInterFreqA5RsrpThld1=-105, CovInterFreqA5RsrpThld2=-101,
```

```
CovInterFreqA2RsrpThld=-105, CovInterFreqA2RsrqThld=-25, SsbSinrInterFreqHoA2Thld=-10,
CovInterFreqA1RsrpThld=-100, CovInterFreqA1RsrqThld=-21, CovInterFreqA5RsrqThld2=-19,
CovInterFreqA5SinrThld2=-4, SsbSinrInterFreqHoA1Thld=-6, A3InterFreqHoA1RsrpThld=-96,
A3InterFreqHoA2RsrpThld=-100, InterFreqHoA3Offset=2;
//Configuring CovHoToEutranBlindA2Thld in the measurement parameter group related to inter-RAT
handover for an NR cell (CovHoToEutranBlindA2Thld is shared between inter-frequency and inter-RAT blind
redirection.)
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=0, CovHoToEutranBlindA2Thld=-126;
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=1, CovHoToEutranBlindA2Thld=-126;
//Configuring CovHoToLteBlindA2RsrqThld in the measurement parameter group related to inter-RAT
handover for an NR cell (CovHoToLteBlindA2RsrqThld is shared between inter-frequency and inter-RAT
blind redirection.)
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=0, CovHoToLteBlindA2RsrqThld=-38;
MOD NRCELLINTERRHOMEAGRP: NrCellId=0, InterRatHoMeasGroupId=1, CovHoToLteBlindA2RsrqThld=-38;
//Configuring the SINR threshold for a coverage-based blind event A2 in the measurement parameter group
related to inter-frequency handover for an NR cell
MOD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFBlindRedirA2SinrThld=-16;
MOD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=1, InterFBlindRedirA2SinrThld=-16;
//Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement for SA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//Optional) Configuring QCI-specific measurement parameters for NSA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, NsaInterFreqHoMeasGroupId=2;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, NsaInterFreqHoMeasGroupId=3;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Setting the triggering quantity for events A1 and A2 given an assumption that RSRP or RSRQ is used as
the triggering quantity
MOD NRCELLMOBILITYCONFIG: NrCellId=0, MeasTrigQuantity=RSRP_OR_RSRQ;
//Setting the frequency interference flag (given an assumption that candidate cells are filtered based on the
measured RSRQs of neighboring cells)
MOD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77, FreqIntrfFlag=INTRF;
//Turning on the A2-based blind redirection control switch if blind redirection optimization needs to be
enabled
MOD NRCELLALGOSWITCH: NrCellId=0, ProcessSwitch=A2_BLIND_REDIRECT_CTRL_SW-1;
//Turning on the measurement-based redirection switch in the case of many missing neighboring cells for
the serving cell or weak coverage by the neighboring NR frequencies of the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, HoCompatibilitySwitch=MR_BASED_REDIRECT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating coverage-based measurement-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=COVERAGE_BASED_HO-0;
//Deactivating coverage-based inter-frequency redirection
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=REDIRECTION-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

If any of the following counters has a value other than 0, coverage-based inter-frequency handover has taken effect:

- **N.HO.InterFreq.Coverage.PrepAttOut**
- **N.HO.InterFreq.Coverage.ExecAttOut**
- **N.HO.InterFreq.Coverage.ExecSuccOut**

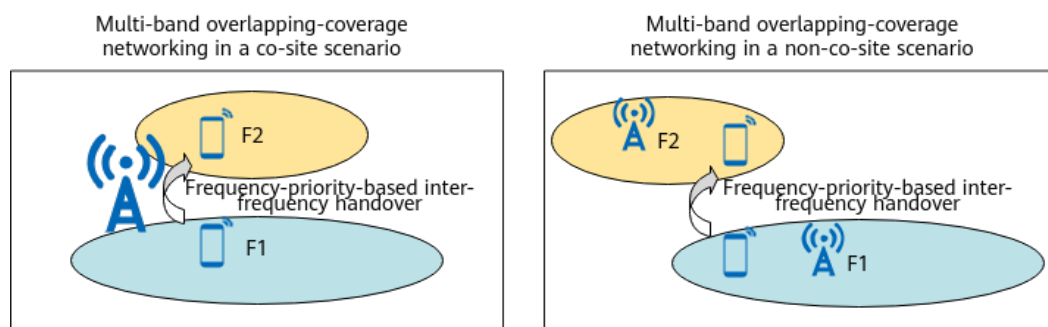
6.4.3 Network Monitoring

This function provides basic assurance for inter-frequency handovers and can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

7 Frequency-Priority-based Inter-Frequency Handover

In a network with multiple frequency bands having overlapping coverage, frequency-priority-based inter-frequency handover allows higher frequency bands to carry services as much as possible to spare lower frequency bands for contiguous coverage, as shown in [Figure 7-1](#).

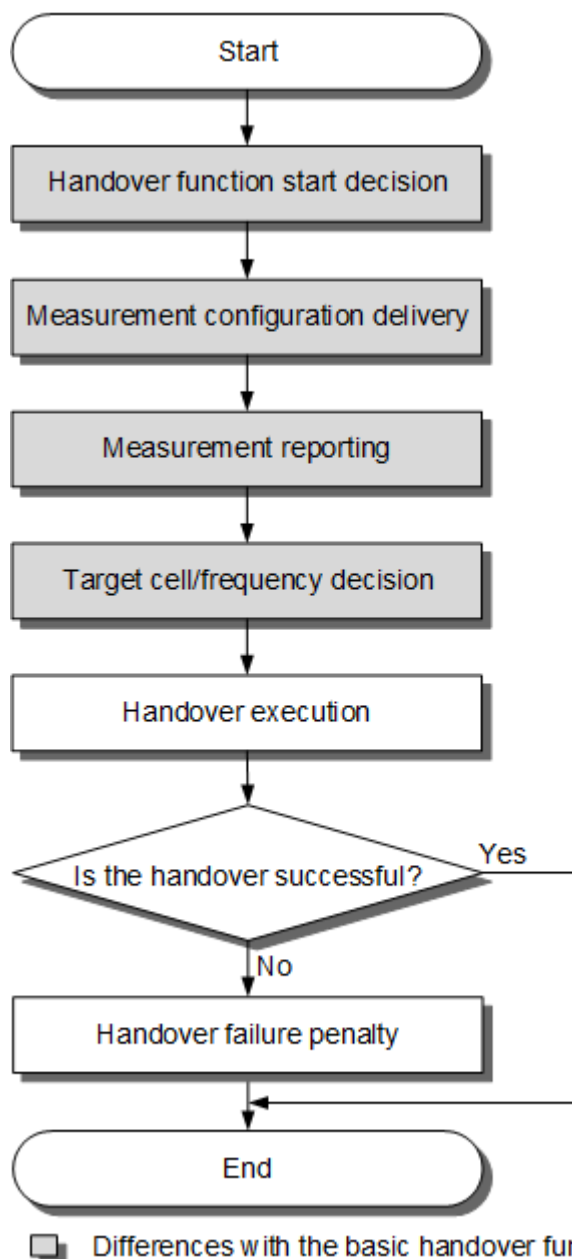
Figure 7-1 Frequency-priority-based inter-frequency handover in multi-band overlapping-coverage networking



7.1 Principles

Frequency-priority-based inter-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports only handover but does not support redirection. [Figure 7-2](#) shows the procedure of frequency-priority-based inter-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 7-2 Procedure of frequency-priority-based inter-frequency handover



In addition to the basic function of frequency-priority-based inter-frequency handover, the gNodeB also supports the following functions specific to different services and UE types:

- QCI-specific frequency-priority-based inter-frequency handover: Traffic priorities and frequency lists can be QCI-customized for UEs. This function is controlled by the **QCI_BASED_FREQ_PRIORITY_HO_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter.
- RFSP-specific frequency-priority-based inter-frequency handover: Traffic priorities, frequency lists, and thresholds for events A1 and A2 can be RFSP-customized for UEs. This function is controlled by the **FREQ_PRIORITY_BASED_HO_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

- CA-UE-specific frequency-priority-based inter-frequency handover: Traffic priorities, frequency lists, and thresholds for events A1 and A2 can be customized for CA UEs. This function supports non-gap-assisted measurement. This function is controlled by the **CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter.

This chapter describes the basic function of frequency-priority-based inter-frequency handover and QCI-specific frequency-priority-based inter-frequency handover. For RFSP-specific frequency-priority-based inter-frequency handover, see this chapter and *Flexible User Steering*. For CA-UE-specific frequency-priority-based inter-frequency handover, see *CA Experience Improvement*. Any of the aforesaid functions can take effect separately. When multiple of them are enabled and a UE meets all of the conditions for the enabled functions to take effect, the gNodeB makes these functions take effect on the UE in the following sequence: RFSP-specific frequency-priority-based inter-frequency handover, CA-UE-specific frequency-priority-based inter-frequency handover, QCI-specific frequency-priority-based inter-frequency handover, and the basic function of frequency-priority-based inter-frequency handover.

7.1.1 Handover Function Start Decision

Frequency-priority-based inter-frequency handover is started when all of the following conditions are met:

- The **FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected.
- The gNodeB receives a measurement report indicating that the serving cell meets the entering conditions for event A1 or A2.

For details about whether event A1 or A2 is used in the handover function start decision for frequency-priority-based inter-frequency handover, see [Triggering Mode of Frequency-Priority-based Handover](#).

Event A1 indicates that the signal quality of the serving cell exceeds a specific threshold, and event A2 indicates that the signal quality of the serving cell drops below a specific threshold.

- The UE supports inter-frequency measurements and inter-frequency handovers. The UE supports the frequency bands containing the non-serving frequencies to be measured.

When the switch for "QCI-specific frequency-priority-based inter-frequency handover" or "RFSP-specific frequency-priority-based inter-frequency handover" replaces the function switch in the aforesaid conditions with other conditions remaining unchanged, "QCI-specific frequency-priority-based inter-frequency handover" or "RFSP-specific frequency-priority-based inter-frequency handover" is started.

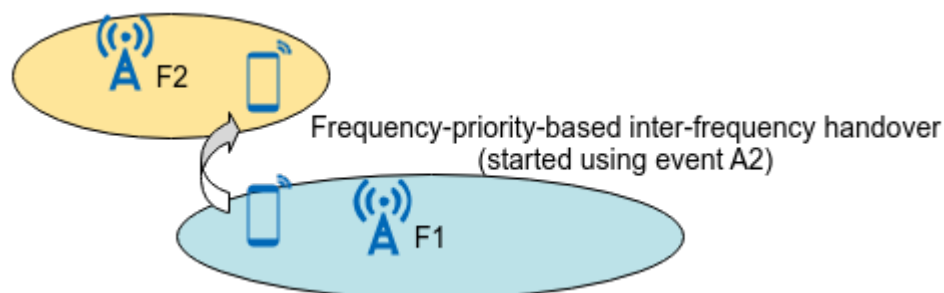
Triggering Mode of Frequency-Priority-based Handover

Frequency-priority-based inter-frequency handover supports the event A1- or A2-based handover function start decision, which is specified by **A1_BASED** or **A2_BASED** of the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter.

In multi-band overlapping-coverage networking among multiple sites, event A2 is recommended for use in the handover function start decision for a cell operating

on a low-priority frequency, and it is recommended that the threshold for frequency-priority-based inter-frequency event A2 be higher than the threshold for coverage-based inter-frequency event A2. Take the scenario shown in [Figure 7-3](#) as an example, where there is overlapping coverage between frequencies F1 and F2, with a given assumption that F1 is a lower-priority frequency compared with F2. As such, it is recommended that event A2 be used in the handover function start decision for frequency-priority-based inter-frequency handover in the cell operating on F1. In other scenarios except for the aforesaid scenario, it is recommended that event A1 be used in the handover function start decision.

Figure 7-3 Handover function start decision using event A2



For details about coverage-based inter-frequency handover, see [6 Coverage-based Inter-Frequency Handover](#).

7.1.2 Measurement Configuration Delivery

The gNodeB selects frequencies and delivers related measurement configurations according to the following rules: Frequency selection is affected by CA UE frequency steering, as detailed in [CA UE Frequency Steering](#). Whether measurement configurations are delivered is affected by "anti-ping-pong function with delay for frequency-priority-based inter-frequency handover" and "QCI-specific suppression of unnecessary inter-frequency handover", as detailed in [Anti-Ping-Pong Function with Delay for Frequency-Priority-based Inter-Frequency Handover](#) and [QCI-specific Suppression of Unnecessary Inter-Frequency Handover](#), respectively.

- The gNodeB selects frequencies based on the following rules:
 - If the value of the **NRCellFreqRelation.TrafficPriority** parameter is used as the traffic priority, the gNodeB selects frequencies whose traffic priorities are not 0 and delivers related measurement configurations.
 - If the value of the **gNBFreqPriorityGroup.TrafficPriority** parameter is used as the traffic priority, the gNodeB selects the UE's RFSP-index- or QCI-specific frequencies whose traffic priorities are not 0 and higher than the serving frequency. Then, the gNodeB delivers related measurement configurations.
 - When CA UE frequency steering is enabled, the gNodeB filters out the frequencies that meet specific conditions and then delivers the frequencies that are not filtered out to UEs, with frequency priorities taken into account.

For details about which parameter value is used as the traffic priority in a specific scenario, see [7.1.5 Typical Parameter Configurations](#).

The gNodeB selects frequencies in descending order of traffic priorities until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.

- The gNodeB delivers measurement configurations related to events based on the following rules:
The event (A1 or A2) selected in the delivered measurement configurations is used to determine whether to stop a handover function and therefore is opposite to the event (A2 or A1) for determining whether to start the handover function. For example, if the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A1_BASED**, event A1 is used to determine whether to start a handover function whereas event A2 is used to determine whether to stop the handover function.
 - When the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter (indicating the triggering mode of frequency-priority-based handover) is set to **A1_BASED**:
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A4**, the gNodeB delivers measurement configurations related to events A4 and A2.
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A5**, the gNodeB delivers measurement configurations related to events A5 and A2.
 - When the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter (indicating the triggering mode of frequency-priority-based handover) is set to **A2_BASED**:
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A4**, the gNodeB delivers measurement configurations related to events A4 and A1.
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A5**, the gNodeB delivers measurement configurations related to events A5 and A1.

Event A4 indicates that the signal quality of a neighboring cell exceeds a specific threshold. Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2.

Anti-Ping-Pong Function with Delay for Frequency-Priority-based Inter-Frequency Handover

If (1) "frequency-priority-based inter-frequency handover" is started, (2) "anti-ping-pong function with delay for frequency-priority-based inter-frequency handover" is enabled (by setting the **NRCellMobilityConfig.InterFreqMeasDelay** parameter to a value other than 0), and (3) a UE is handed over to the current serving cell through any unnecessary inter-frequency handover, then a timer specified by the **NRCellMobilityConfig.InterFreqMeasDelay** parameter will start at the moment the handover is complete. The gNodeB does not deliver frequency-priority-based inter-frequency measurement configurations to the UE before the timer expires, thereby preventing ping-pong handovers.

QCI-specific Suppression of Unnecessary Inter-Frequency Handover

After frequency-priority-based inter-frequency handover is started, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. This process is referred to as QCI-specific suppression of unnecessary inter-frequency handover. The details are as follows:

- If the UE has a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB does not deliver measurement configurations to the UE.
- If the UE does not have a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB delivers measurement configurations to the UE.

NOTE

When frequency-priority-based inter-frequency handover is enabled for the serving cell, you are advised to select the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for each QCI-specific bearer associated with the target frequency group to avoid ping-pong handovers if service-based inter-frequency handover (specified by the **SERVICE_BASED_HO** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter) is enabled for the target cell.

CA UE Frequency Steering

After frequency-priority-based inter-frequency handover takes effect and CA UE frequency steering is enabled (by selecting the **CA_UE_FREQ_STEER_SW** option of the **NRCCellAlgoSwitch.MultiFreqAlgoSwitch** parameter), the gNodeB determines whether to filter out a frequency based on (a) CA capabilities of the current UE, (b) frequency of the serving cell on which the UE camps, and (c) network continuity. Information about the frequencies that are not filtered out is included in measurement configurations and delivered to the CA-capable UE, with frequency priorities taken into account. [Table 7-1](#) describes the scenarios where the function takes effect in non-serving frequency filtering. The function does not take effect if the combinations of criteria in this table do not apply.

Table 7-1 Scenarios where the function takes effect in non-serving frequency filtering

PCC Anchor Flag for Frequency A of the Serving Cell	PCC Anchor Flag for Frequency B of a Neighboring Cell	UE Capability	Result
TRUE	FALSE	All of the following criteria must be met: - A UE supports CA. - The UE supports a frequency combination of frequency A as the PCC and frequency B as an SCC.	Frequency B of the neighboring cell is filtered out.
TRUE	TRUE	All of the following criteria must be met: - A UE supports CA. - The UE supports a frequency combination of frequency A as the PCC and frequency B as an SCC. - The UE does not support a frequency combination of frequency B as the PCC and frequency A as an SCC.	Frequency B of the neighboring cell is filtered out.

 NOTE

The PCC anchor flag (specified by **NRCellMobilityConfig.PccAnchorFlag** for the serving cell and by **NRCellFreqRelation.PccAnchorFlag** for a neighboring cell) is usually configured depending on networking continuity. If the PCC anchor flag is set to **TRUE**, it indicates that the frequency of the corresponding cell provides contiguous coverage. If the PCC anchor flag is set to **FALSE**, it indicates that the frequency of the corresponding cell provides discontinuous coverage. It is recommended that this function be disabled in discontinuous network coverage scenarios.

7.1.3 Measurement Reporting

A UE performs measurements based on the event measurement configuration and sends the measurement report based on measurement results.

- If event A4 or A5 is reported, the gNodeB determines the target cell or frequency based on the measurement report.

- When the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter (indicating the triggering mode of frequency-priority-based handover) is set to **A1_BASED** and an event A2 report is received, the gNodeB instructs the UE to stop event A4 or A5 measurements. The function stops.
- When the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter (indicating the triggering mode of frequency-priority-based handover) is set to **A2_BASED** and an event A1 report is received, the gNodeB instructs the UE to stop event A4 or A5 measurements. The function stops.

If no suitable neighboring cell is reported after a UE has performed inter-frequency A4 or A5 measurements for a period specified by **NRCellMobilityConfig.InterFreqMeasWaitTime** (indicating the inter-frequency measurement waiting time), the gNodeB releases the measurement configurations, enabling the UE to stop the measurements for this event A4 or A5. The gNodeB then periodically delivers measurement configurations for event A4 or A5 to the UE.

The interval for delivery of measurement configurations is calculated as follows:

Interval = **NRCellMobilityConfig.FreqPriInterFHoRetryIntvl** (retry period for frequency-priority-based inter-frequency handover) + **NRCellMobilityConfig.InterFreqMeasWaitTime** (inter-frequency measurement waiting time)

- A smaller retry period for frequency-priority-based inter-frequency handover results in a higher probability of triggering frequency-priority-based inter-frequency handover. When the signal quality of the target cell is poor, the UE frequently performs inter-frequency measurements, decreasing user-perceived throughput.
- A larger retry period for frequency-priority-based inter-frequency handover results in a lower probability of triggering frequency-priority-based inter-frequency handover. When the signal quality of the target cell is poor, the number of unnecessary inter-frequency measurements decreases, increasing user-perceived throughput.

NOTE

If the source gNodeB receives a measurement report from the UE and sends a Handover Request message to the target gNodeB but does not receive a response from the target gNodeB during a period specified by **NRCellMobilityConfig.InterFreqMeasWaitTime**, the source gNodeB will again send measurement configurations for event A4 or A5 to the UE.

7.1.4 Target Cell or Frequency Decision

After receiving an event A4 or A5 report, the gNodeB obtains frequency information from the event A4 or A5 report and determines the target frequency as follows:

- If the frequency has the highest traffic priority, the gNodeB chooses this frequency as the target frequency and instructs the UE to perform a handover to the target cell (that is, the optimal cell running on the frequency with the highest traffic priority in the measurement report). Note that whether such an instruction is finally delivered is also affected by subscription to uplink interference information about inter-frequency neighboring cells and inter-frequency handover collaboration, as detailed in [Subscription to Uplink](#)

Interference Information About Inter-Frequency Neighboring Cells and Inter-Frequency Handover Collaboration.

- If the frequency does not have the highest traffic priority, the gNodeB starts a timer (fixed at 2s) and continues to receive measurement reports. After the timer expires, the gNodeB chooses the inter-frequency neighboring cell with the highest traffic priority among those reported in this period for handover.

Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells

When the function of subscription to uplink interference information about inter-frequency neighboring cells is enabled (by selecting the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch.NCellInfoCtrlSwitch** parameter) for the serving cell, the serving cell obtains uplink interference information about intra- or inter-base-station neighboring cells, based on which the serving cell determines whether to instruct the UE to initiate a handover. Specifically,

- The UE is instructed to perform a handover if the serving cell determines that the target cell is exposed to low uplink interference.
- Otherwise, the UE is not instructed to perform a handover.

NOTE

For details about how the serving cell determines whether a neighboring cell is exposed to low uplink interference, see [10.1.4 Target Cell or Frequency Decision](#).

If frequency-priority-based inter-frequency handover is enabled for the serving cell and uplink-interference-based inter-frequency handover is enabled for the target cell, you are advised to enable the function of subscription to uplink interference information about inter-frequency neighboring cells for the serving cell, avoiding ping-pong handovers.

Inter-Frequency Handover Collaboration

When the inter-frequency handover collaboration function is enabled (by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter) for the serving cell, the serving cell further determines whether to instruct the UE to initiate a handover as follows:

- If the serving cell identifies that the target cell has entered the high-load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.
- The serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.
- The serving cell checks whether the UE has a voice bearer.
 - If so, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.
 - If not, the gNodeB proceeds to instruct the UE to perform a handover to the target cell.

NOTE

When frequency-priority-based inter-frequency handover is enabled for the serving cell, you are advised to enable the inter-frequency handover collaboration function for the serving cell to avoid ping-pong handovers if any of the following functions is enabled for the target cell:

- Inter-frequency MLB by transferring UEs in connected mode (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter)
- Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter)
- Voice-quality-based inter-frequency handover (controlled by the **QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp.QciAlgoSwitch** parameter)

For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see *Mobility Load Balancing in Connected Mode*. For details about high-load beams and multi-frequency beam coordination, see *Multi-Frequency Smart Selection*. For details about voice-quality-based inter-frequency handover, see *VoNR*.

7.1.5 Typical Parameter Configurations

This section describes typical parameters related to frequency-priority-based inter-frequency handover.

Traffic Priority

The gNodeB selects the parameters for configuring the traffic priority based on the enabling status of RFSP-specific frequency-priority-based inter-frequency handover (controlled by the **FREQ_PRIORITY_BASED_HO_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter) and QCI-specific frequency-priority-based inter-frequency handover (controlled by the **QCI_BASED_FREQ_PRIORITY_HO_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter).

- If both RFSP-specific frequency-priority-based inter-frequency handover and QCI-specific frequency-priority-based inter-frequency handover are disabled, the traffic priority is specified by the **NRCellFreqRelation.TrafficPriority** parameter.
- If RFSP-specific frequency-priority-based inter-frequency handover is enabled, the parameters for configuring the traffic priority are scenario-specific, as described in [Table 7-2](#).

Table 7-2 Parameters for configuring the traffic priority when RFSP-specific frequency-priority-based inter-frequency handover is enabled

Scenario	Scenario Description	Traffic Priority Parameter
Scenario 1	All of the following conditions are met: - The gNodeB obtains the UE's RFSP index from the core network. - The gNBRfspConfig.RfspIndex parameter corresponding to the RFSP index is configured on the gNodeB. - The gNBRfspConfig.RfspIndex parameter is associated with the gNBRfspConfig.gNBFreqPriorityGroupId parameter that has a valid value (not equal to 255).	The traffic priority is specified by the gNBFreqPriorityGroup.TrafficPriority parameter, which is associated with the gNBRfspConfig.RfspIndex parameter corresponding to the UE's RFSP index.
Scenario 2	Any of the following cases: Case 1: All of the following conditions are met: - The gNodeB obtains the UE's RFSP index from the core network. - The gNBRfspConfig.RfspIndex parameter corresponding to the RFSP index is not configured on the gNodeB. - The gNBRfspConfig.RfspIndex parameter with a value of 0 is configured on the gNodeB, and this parameter is associated with the gNBRfspConfig.gNBFreqPriorityGroupId parameter that has a valid value (not equal to 255). Case 2: All of the following conditions are met: - The gNodeB does not obtain the UE's RFSP index from the core network. - The gNBRfspConfig.RfspIndex parameter with a value of 0 is configured on the gNodeB, and this parameter is associated with the gNBRfspConfig.gNBFreqPriorityGroupId parameter that has a valid value (not equal to 255).	The traffic priority is specified by the gNBFreqPriorityGroup.TrafficPriority parameter, which is associated with the gNBRfspConfig.RfspIndex parameter set to 0.
Scenario 3	Scenarios other than scenarios 1 and 2	The traffic priority is specified by the NRCellFreqRelation.TrafficPriority parameter.

Scenario	Scenario Description	Traffic Priority Parameter
	Note: - In scenarios 1 and 2, the gNodeB uses RFSP to customize the traffic priority for a UE. - In scenarios 1 and 2, the gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs parameter can be used to customize the thresholds of events A1 and A2 and the gNBRfspConfig.FreqPriFHoA4RsrpThldOfs parameter can be used to customize the thresholds of events A4 and A5 for a UE. For details, see Thresholds for Events A1, A2, A4, and A5. - In scenarios 1 and 2, in the RFSP-specific dedicated frequency list group with which the gNBRfspConfig.RfspIndex parameter is associated, there must be at least one non-serving frequency with the gNBFreqPriorityGroup.TrafficPriority parameter set to a non-0 value. Otherwise, RFSP-specific frequency-priority-based inter-frequency handover does not take effect. If RFSP-specific frequency-priority-based inter-frequency handover is enabled but does not take effect, cell-level frequency-priority-based inter-frequency handover takes effect (that is, the traffic priority is specified by the NRCellFreqRelation.TrafficPriority parameter). - In NSA networking, the gNodeB uses the SPID to customize the traffic priority, threshold for event A1, and threshold for event A2 for the UE. After receiving the SPID from the eNodeB, the gNodeB processes the SPID in the same way as in SA networking.	

- If QCI-specific frequency-priority-based inter-frequency handover is enabled, the parameters for configuring the traffic priority are scenario-specific, as described in [Table 7-3](#).

Table 7-3 Parameters for configuring the traffic priority when QCI-specific frequency-priority-based inter-frequency handover is enabled

Scenario	Scenario Description	Traffic Priority Parameter
Scenario 1	All of the following conditions are met: - The NRCellQciBearer.gNBFreqPriorityGroupId parameter is set to a value other than 255 for the QCI that has the highest handover priority. - There is at least one non-serving frequency with the gNBFreqPriorityGroup.TrafficPriority parameter set to a value other than 0 in the frequency priority group (specified by the gNBFreqPriorityGroup.gNBFreqPriorityGroupId parameter) bound to the QCI that has the highest handover priority.	The traffic priority is specified by the gNBFreqPriorityGroup.TrafficPriority parameter, which is bound to the NRCellQciBearer.Qci parameter for the QCI that has the highest handover priority on the UE.

Scenario	Scenario Description	Traffic Priority Parameter
Scenario 2	Scenarios other than scenario 1	The traffic priority is specified by the NRCellFreqRelation.TrafficPriority parameter.
Note: The priority of a QCI for handovers is specified by the gNBQciBearer.QciPriorityForHo parameter and a smaller value of this parameter indicates a higher QCI priority for handovers. For details about the QCI priority for handovers, see 4.1.4.2.3 QCI-specific Measurement Event Configuration. If different QCIs on a UE have the same priority for handovers, the gNodeB sorts the QCIs based on their values, selects the frequency priority group specific to the QCI with the smallest value, and then delivers the corresponding measurement configurations.		

Triggering Mode of Frequency-Priority-based Handover

- If the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A1_BASED**:
 - Frequency-priority-based inter-frequency handover is triggered when the UE's RSRP in the serving cell exceeds the threshold for event A1.
 - Frequency-priority-based inter-frequency handover is stopped when the UE's RSRP in the serving cell drops below the threshold for event A2.
- If the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A2_BASED**:
 - Frequency-priority-based inter-frequency handover is triggered when the UE's RSRP in the serving cell drops below the threshold for event A2.
 - Frequency-priority-based inter-frequency handover is stopped when the UE's RSRP in the serving cell exceeds the threshold for event A1.

Thresholds for Events A1, A2, A4, and A5

- Threshold for event A1
 - If the threshold for event A1 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the threshold for event A1 equals **NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld**.
 - If the threshold for event A1 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the threshold for event A1 equals **NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld** plus **gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs**.

For more information, see [Table 4-7](#).
- Threshold for event A2
 - If the threshold for event A2 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the

threshold for event A2 equals

NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld.

- If the threshold for event A2 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the threshold for event A2 equals

NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld plus
gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs.

For more information, see [Table 4-8](#).

- Threshold for event A4

- If the threshold for event A4 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, the threshold for event A4 equals

NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld.

- If the threshold for event A4 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, the threshold for event A4 equals

NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld plus
gNBRfspConfig.FreqPriFHoA4RsrpThldOfs.

For more information, see [Table 4-10](#).

- Thresholds for event A5

Threshold 1 for event A5 uses a fixed value of –43 dBm.

Threshold 2 for event A5 is set as follows:

- If threshold 2 for event A5 related to RFSP-specific frequency-priority-based inter-frequency measurements is not customized for a UE, threshold 2 for event A5 equals

NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld.

- If threshold 2 for event A5 related to RFSP-specific frequency-priority-based inter-frequency measurements is customized for a UE, threshold 2 for event A5 equals ***NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld*** plus ***gNBRfspConfig.FreqPriFHoA4RsrpThldOfs.***

For more information, see [Table 4-11](#).

7.2 Network Analysis

7.2.1 Benefits

This function provides a method to hand over UEs and allows for flexible networking to help operators implement service steering.

- QCI-specific frequency-priority-based inter-frequency handover can provide different frequency priority policies for different services.
- After CA UE frequency steering is enabled, the number of inter-frequency handovers for CA UEs may decrease and the average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) in a cell may increase.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

This function results in the following network impacts:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- Networking of the live network and the impact relationships between frequency-priority-based inter-frequency handover and other handover functions (such as service-based inter-frequency handover) must be considered to prevent ping-pong handovers and ensure user experience. For details about networking requirements, see [7.3.4 Networking](#).

7.3 Requirements

7.3.1 Licenses

There are no license requirements.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RFSP-based NR inter-frequency handover policy	INTER_FREQ_LIST_CTRL_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter	<i>Flexible User Steering</i>	If RFSP-specific frequency-priority-based inter-frequency handover is enabled (by selecting the FREQ_PRIORITY_BASED_HO_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter), the RFSP-based NR inter-frequency handover policy must also be enabled.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	None	None	None	None

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	If frequency-priority-based inter-frequency handover is disabled, intra-RAT ANR will not be triggered by frequency-priority-based inter-frequency handover because such handover does not take effect.
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCCellAlgoSwitch . <i>MlbAlgoSwitch</i> parameter, and the NRCCellMlb.MlbTriggerMode parameter set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both UE-number-based connected mode MLB and frequency-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both spectral-efficiency-based connected mode MLB and frequency-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both radio-resource-based connected mode MLB and frequency-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch . <i>MultiFreqAlgoSwitch</i> parameter	<i>Multi-Frequency Smart Selection</i>	- When the smart carrier selection policy (specified by the NRCellCaMgmtConfig.<i>SmartCarrSelPolicy</i> parameter) is set to EXPERIENCE_FIRST, you are not advised to enable both experience-based smart carrier selection (controlled by the MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.<i>MultiFreqAlgoSwitch</i> parameter) and frequency-priority-based inter-frequency handover (controlled by the FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.<i>InterFreqHoSwitch</i> parameter). If they are both enabled, ping-pong handovers may occur. - It is not recommended that event A2 be used in the handover function start decision for frequency-priority-based inter-frequency handover (NRCellMobilityConfig.<i>FreqPriHoTriggerMode</i> set to A2_BASED) when experience-based smart carrier selection is enabled. Otherwise, user experience of UEs located at the cell edge or at a medium distance from the cell center may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When event A2 is used in the handover function start decision for frequency-priority-based inter-frequency handover (that is, NRCellMobilityConfig.FreqPriHoTriggerMode is set to A2_BASED), it is recommended that the threshold for frequency-priority-based inter-frequency event A2 (specified by NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld) be higher than the threshold for coverage-based inter-frequency event A2 (specified by NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld). Otherwise, event A2 cannot be used to trigger frequency-priority-based inter-frequency handover.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for frequency-priority-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, frequency-priority-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, frequency-priority-based inter-frequency handover does not take effect for the roaming UE.
Low-frequency TDD	Multi-RF-module coordinated energy saving	ENERGY_EFFICIENCY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	It is not recommended that frequency-priority-based inter-frequency handover and energy-efficiency-based inter-frequency handover be both enabled, as enabling both of them will cause energy saving gains to decrease.
Low-frequency TDD	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs	3CC_FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	After frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs takes effect, frequency-priority-based inter-frequency handover does not take effect for non-3CC UEs or non-massive-CA UEs.
Low-frequency TDD	Latency-based optimal carrier selection	DELAY_BASED_CARR_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	- After a UE is handed over to a target cell for latency-based optimal carrier selection, frequency-priority-based inter-frequency handover does not take effect for the UE in the target cell. - For a UE where latency-based optimal carrier selection has not taken effect, frequency-priority-based inter-frequency handover takes effect when the relevant handover conditions are met.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-experience-based inter-frequency handover	UL_EXP_BASED_INTER_FREQ_HO_SW option of the NRDUCellRedCap.RedCapAlgorithmSwitch parameter	<i>RedCap</i>	Frequency-priority-based inter-frequency handovers will not be triggered for RedCap UEs if they have completed uplink-experience-based inter-frequency handovers, thereby preventing ping-pong handovers.
High-frequency TDD	None	None	None	None

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-

based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through a frequency-priority-based handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through a frequency-priority-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a service-based inter-frequency handover.

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

The NR cell's frequency relationships and neighbor relationships must be configured before this function is activated. For details about the NR cell's frequency relationship configurations, see [Table 7-4](#). For details about neighbor relationship configurations, see [Table 4-20](#).

[Table 7-5](#) describes the function switches.

[Table 7-6](#) describes other parameters used for function optimization.

Table 7-4 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Traffic Priority	NRCellFreqRelation.TrafficPriority	Set this parameter based on the network plan.
PCC Anchor Flag	NRCellFreqRelation.PccAnchorFlag	Set this parameter based on the network plan.

Table 7-5 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	Select the FREQ_PRIORITY_BASED_HO option.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	When frequency-priority-based inter-frequency handover is enabled, it is recommended that the INTER_FREQ_HO_COOPERATION_SW option be selected if either inter-frequency MLB by transferring UEs in connected mode or voice-quality-based inter-frequency handover is also enabled.

Table 7-6 Parameters related to inter-frequency handover measurement configurations of NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellInterFHoMeaGrp.NrCellId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
Inter-frequency A1A2 Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA1A2TimeToTrig	Use the recommended value.
Inter-frequency A1A2 Hysteresis	NRCellInterFHoMeaGrp.InterFreqA1A2Hyst	Use the recommended value.
Freq Priority Inter-freq A1 RSRP Threshold	NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld	Use the recommended value.
Freq Priority Inter-freq A2 RSRP Threshold	NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld	Use the recommended value.
Frequency Priority HO Trigger Mode	NRCellMobilityConfig.FreqPriHoTriggerMode	For details about the setting notes, see 7.1.1 Handover Function Start Decision . After frequency-priority-based inter-frequency handover is enabled, the parameter modification immediately takes effect for newly accessed UEs but does not immediately take effect for UEs that have accessed the network.

Parameter Name	Parameter ID	Setting Notes
Inter-Freq Measurement Event Configuration	NRCellMobilityConfig.InterFreqMeasEventConfig	Use the recommended value.
Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrig	Use the recommended value.
Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Use the recommended value.
Freq Priority Inter-freq A4 RSRP Threshold	NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld	This parameter indicates the RSRP threshold for event A4 related to frequency-priority-based inter-frequency handover and RSRP threshold 2 for event A5 related to frequency-priority-based inter-frequency handover. Use the recommended value.
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	NRCellQciBearer.NsaInterFreqHoMeasGroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobility Function Indication	gNBQciBearer.MobilityFunInd	The INTER_FREQ_HO_SU PPRESS_SW option specifies whether to enable QCI-specific suppression of unnecessary inter-frequency handover. When frequency-priority-based inter-frequency handover is enabled for the serving cell, you are advised to select this option for each QCI-specific bearer associated with the target frequency group if service-based inter-frequency handover is enabled for the target cell.
Freq Priority Inter-Freq HO Retry Intvl	NRCellMobilityConfig.FreqPriInterFHoRetryIntvl	Use the recommended value.
Inter-Freq Measurement Wait Time	NRCellMobilityConfig.InterFreqMeasWaitTime	Use the recommended value.
Inter-Freq Measurement Delay	NRCellMobilityConfig.InterFreqMeasDelay	Use the recommended value.
Traffic Priority	gNBFreqPriorityGroup.TrafficPriority	Set this parameter based on the network plan.
Freq Priority HO A1A2 RSRP Thld Offset	gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs	Set this parameter based on the network plan.
Freq Pri Inter-Freq HO A4 RSRP Thld Offset	gNBRfspConfig.FreqPriIFHoA4RsrpThldOfs	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
RFSP-based Algorithm Switch	NRCellAlgoSwitch.RfspAlgoSwitch	Select the FREQ_PRIORITY_BASED_HO_SW option if RFSP-specific frequency-priority-based inter-frequency handover needs to be enabled.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	Select the QCI_BASED_FREQ_PRIORITY_HO_SW option if QCI-specific frequency-priority-based inter-frequency handover needs to be enabled.
Neighboring Cell Information Control Switch	NRCellAlgoSwitch.NCellInfoCtrlSwitch	The INTER_FREQ_UL_INTF_SW option specifies whether to enable subscription to uplink interference information about inter-frequency neighboring cells. Use the recommended value.
PCC Anchor Flag	NRCellMobilityConfig.PccAnchorFlag	Set this parameter based on the network plan.
Multi-Frequency Algorithm Switch	NRCellAlgoSwitch.MultiFreqAlgoSwitch	CA UE frequency steering is controlled by the CA_UE_FREQ_STEER_SW option. It is recommended that this option be deselected in discontinuous network coverage scenarios.

Parameter Name	Parameter ID	Setting Notes
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDLPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoULPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77,
SubcarrierSpacing=30KHZ, TrafficPriority=1, SsbDescMethod=SSB_DESC_TYPE_GSCN;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=SA;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Activating frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=FREQ_PRIORITY_BASED_HO-1;
//Activating RFSP-specific frequency-priority-based inter-frequency handover (Note that the RFSP-based NR
inter-frequency handover policy must be already activated.)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=INTER_FREQ_LIST_CTRL_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=FREQ_PRIORITY_BASED_HO_SW-1;
//Activating QCI-specific frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=QCI_BASED_FREQ_PRIORITY_HO_SW-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring a group of measurement parameters for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, FreqPriInterFA2RsrpThld=-89, FreqPriInterFA1RsrpThld=-86,
FreqPriInterFA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, FreqPriInterFA2RsrpThld=-89, FreqPriInterFA1RsrpThld=-86,
FreqPriInterFA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=2,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, FreqPriInterFA2RsrpThld=-89, FreqPriInterFA1RsrpThld=-86,
FreqPriInterFA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=3,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, FreqPriInterFA2RsrpThld=-89, FreqPriInterFA1RsrpThld=-86,
FreqPriInterFA4RsrpThld=-104;
//((Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement for SA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//((Optional) Configuring QCI-specific measurement parameters for NSA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, NsaInterFreqHoMeasGroupId=2;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, NsaInterFreqHoMeasGroupId=3;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
```

```

MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Enabling QCI-specific suppression of unnecessary inter-frequency handover for the QCI-specific bearers
associated with the target frequency group
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
MOD GNBQCIBEARER: Qci=9, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Setting the triggering mode for frequency-priority-based inter-frequency handovers (A1-based triggering
mode is used as an example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, FreqPriHoTriggerMode=A1_BASED;
//Configuring the type of measurement event for unnecessary inter-frequency handovers (event A5 as an
example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A5;
//Modifying the retry period for frequency-priority-based inter-frequency handover
MOD NRCELLMOBILITYCONFIG: NrCellId=0, FreqPriInterFHoRetryIntvl=S60;
//Modifying the inter-frequency measurement waiting time
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasWaitTime=6;
//Modifying the inter-frequency handover delay
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasDelay=6;
//Recommended when frequency-priority-based inter-frequency handover is enabled for the serving cell
while inter-frequency MLB by transferring UEs in connected mode, multi-frequency beam coordination, or
voice-quality-based inter-frequency handover is enabled for the target cell) Turning on the inter-frequency
handover collaboration switch for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Recommended when frequency-priority-based inter-frequency handover is enabled for the serving cell
and uplink-interference-based inter-frequency handover is enabled for the target cell) Turning on the switch
for subscription to uplink interference information about inter-frequency neighboring cells for the serving
cell
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
//Recommended when the serving cell is enabled with frequency-priority-based inter-frequency handover
and the target cell is deployed in a continuous network coverage scenario) Turning on the switch of CA UE
frequency steering for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, MultiFreqAlgoSwitch=CA_UE_FREQ_STEER_SW-1;
//Recommended when the serving cell is enabled with frequency-priority-based inter-frequency handover
and the target cell is deployed in a continuous network coverage scenario) Setting the PCC anchor flag to
TRUE for the target frequency
MOD NRCELLFREQLRELATION: NrCellId=0, SsbFreqPos=8000, PccAnchorFlag=TRUE;
//Recommended when the serving cell is enabled with frequency-priority-based inter-frequency handover
in a discontinuous network coverage scenario) Setting the PCC anchor flag to FALSE for the serving cell
MOD NRCELLMOBILITYCONFIG: NrCellId=0, PccAnchorFlag=FALSE;
//Configuring parameters related to RFSP-specific frequency-priority-based inter-frequency handover
//Configuring a frequency priority group and the traffic priority of an NR frequency in the frequency priority
group
ADD GNBFBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR, SsbFreqPos=26639,
TrafficPriority=1;
ADD GNBFBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR, SsbFreqPos=26638,
TrafficPriority=2;
//Binding an RFSP index to the configured frequency priority group
ADD GNBFBFSPCONFIG: OperatorId=0, RfspIndex=2, gNBFreqPriorityGroupId=0;
//Modifying the A1 or A2 RSRP threshold offset for RFSP-specific frequency-priority-based handover
MOD GNBFBFSPCONFIG: OperatorId=0, RfspIndex=2, FreqPriHoA1A2RsrpThldOfs=-10;
//Configuring parameters related to QCI-specific frequency-priority-based inter-frequency handover
//Configuring a frequency priority group and the traffic priority of an NR frequency in the frequency priority
group
ADD GNBFBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR, SsbFreqPos=26639,
TrafficPriority=1;
ADD GNBFBFREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR, SsbFreqPos=26638,
TrafficPriority=2;
//Binding a QCI to the configured frequency priority group
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, gNBFreqPriorityGroupId=0;

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating frequency-priority-based inter-frequency handover  
MOD NRCELLALGOSWITCH: NrCellId=0,InterFreqHoSwitch=FREQ_PRIORITY_BASED_HO-0;  
//Deactivating RFSP-specific frequency-priority-based inter-frequency handover  
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=FREQ_PRIORITY_BASED_HO_SW-0;  
//Deactivating QCI-specific frequency-priority-based inter-frequency handover  
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=QCI_BASED_FREQ_PRIORITY_HO_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

If any of the following counters have a value other than 0, frequency-priority-based inter-frequency handover has taken effect:

- **N.HO.InterFreq.FreqPri.PrepAttOut**
- **N.HO.InterFreq.FreqPri.ExecAttOut**
- **N.HO.InterFreq.FreqPri.ExecSuccOut**

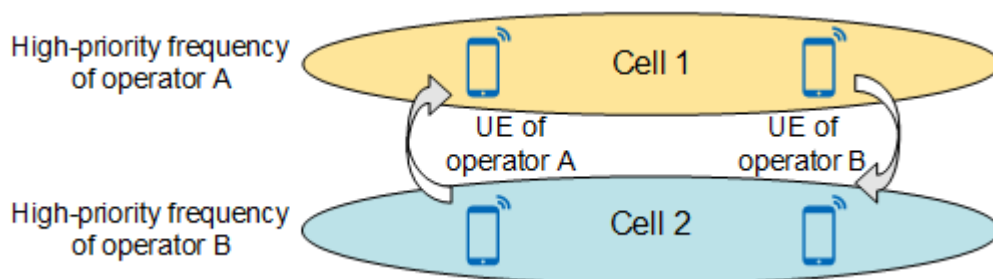
7.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

8 Operator-specific-Priority-based Inter-Frequency Handover

In network sharing scenarios, operator-specific-priority-based inter-frequency handover allows the priority of a frequency to vary among operators, achieving networking flexibility. As shown in [Figure 8-1](#), cells 1 and 2 are shared by operators. The frequency of cell 1 has a higher priority for operator A, whereas the frequency of cell 2 has a higher priority for operator B. This function transfers each operator's UEs to cells on the higher-priority frequency of that operator.

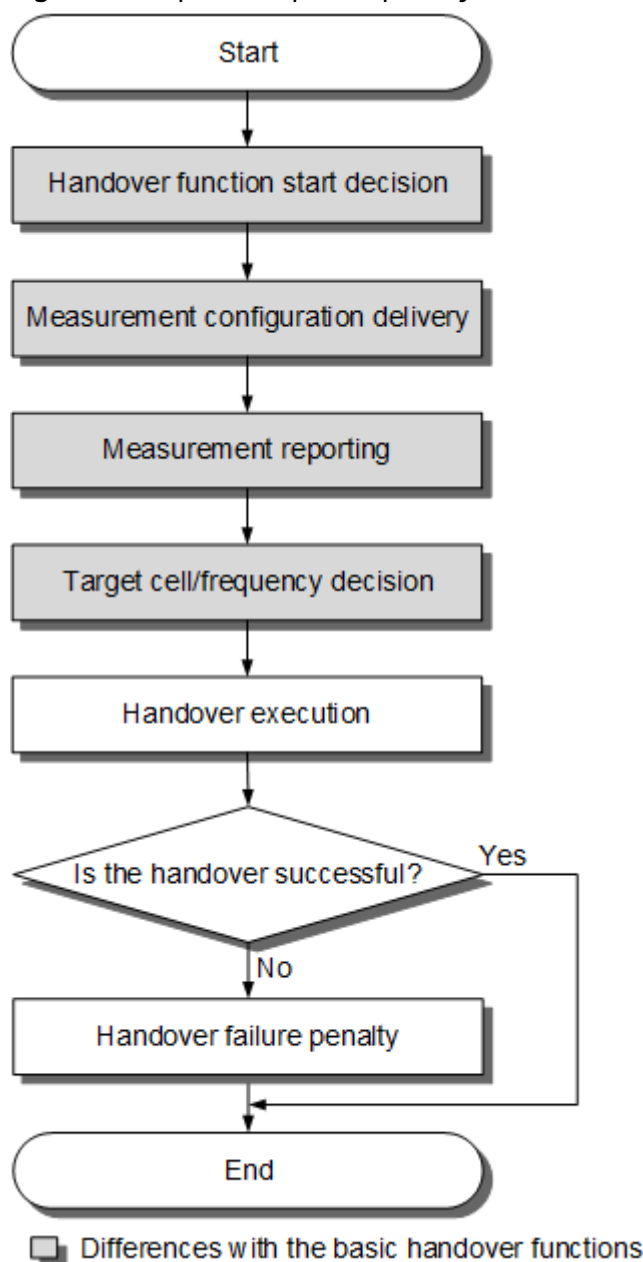
Figure 8-1 Operator-specific-priority-based inter-frequency handover in network sharing scenarios



8.1 Principles

Operator-specific-priority-based inter-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports only handover but does not support redirection. [Figure 8-2](#) shows the procedure of operator-specific-priority-based inter-frequency handover. This section describes only the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 8-2 Operator-specific-priority-based inter-frequency handover



8.1.1 Handover Function Start Decision

Operator-specific-priority-based inter-frequency handover is started when all of the following conditions are met:

- The **OP_DED_PRI_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter or the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter is selected.
- A UE accesses, is handed over to, or has its RRC connection reestablished in a cell whose NR frequency is not an operator-desired target frequency.

Procedure for selecting an operator-desired target frequency: The gNodeB obtains the to-be-measured frequencies that can be delivered to the UE based

on the method of selecting to-be-measured frequencies, and then filters out the to-be-measured frequencies for which the **gNBFreqPriorityGroup.ConnDedFreqPriority** parameter is set to **0**. For details about how the to-be-measured frequencies are selected, see "Method of Selecting To-Be-Measured Frequencies" in [4.1.4.1.2 Frequency Selection for Measurement](#).

Frequency priority policies can vary among operators. The base station determines which operator's frequency priority policy is applied to a UE based on the UE's PLMN information.

- The UE supports inter-frequency measurement and inter-frequency handover. The UE supports the frequency bands containing the non-serving frequencies to be measured.

8.1.2 Measurement Configuration Delivery

After operator-specific-priority-based inter-frequency handover is started, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. This process is referred to as QCI-specific suppression of unnecessary inter-frequency handover. The details are as follows:

- If the UE has a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB does not deliver measurement configurations to the UE.
- If the UE does not have a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB delivers measurement configurations to the UE.

NOTE

When operator-specific-priority-based inter-frequency handover is enabled for the serving cell, you are advised to select the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for each QCI-specific bearer associated with the target frequency group in the target cell to avoid ping-pong handovers, if service-based inter-frequency handover (specified by the **SERVICE_BASED_HO** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter) is enabled for the target cell.

The gNodeB selects frequencies and delivers related measurement configurations based on the following principles:

- The gNodeB selects frequencies based on the following principles:
The gNodeB selects frequencies whose connected-mode dedicated frequency priorities are not 0 and delivers related measurement configurations.
The gNodeB selects frequencies in descending order of connected-mode dedicated frequency priorities until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.
The connected-mode dedicated frequency priority is specified by the **gNBFreqPriorityGroup.ConnDedFreqPriority** parameter that is associated with the **NRCCellOPPolicy** MO.

- The gNodeB delivers measurement configurations related to events based on the following principles:
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A4**, the gNodeB delivers measurement configurations related to event A4.
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A5**, the gNodeB delivers measurement configurations related to event A5.

Event A4 indicates that the signal quality of a neighboring cell exceeds a specific threshold. Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2. The thresholds for events A4 and A5 are specified by the following parameters:

- The threshold for event A4 is specified by the **NRCellInterFHoMeaGrp.OpDedPriHoA4RsrpThld** parameter. For more information, see [Table 4-10](#).
- Thresholds for event A5
 - Threshold 1 for event A5 uses a fixed value of –43 dBm.
 - Threshold 2 for event A5 is specified by the **NRCellInterFHoMeaGrp.OpDedPriHoA4RsrpThld** parameter.

For more information, see [Table 4-11](#).

8.1.3 Measurement Reporting

A UE performs measurements based on the event measurement configuration and sends the measurement report based on measurement results. If event A4 or A5 is reported, the gNodeB determines the target cells or frequencies based on the measurement report.

If no suitable neighboring cell is reported 6s after the measurement of event A4 or A5 has started, the gNodeB releases the measurement configurations, enabling the UE to stop the measurement for this event A4 or A5. In this scenario, if the UE is always in connected mode, the gNodeB no longer sends the same measurement configurations.

8.1.4 Target Cell or Frequency Decision

After the gNodeB receives a measurement report of event A4 or A5, the gNodeB selects the optimal cell from the reported cells as the target cell for handover execution. Note that whether the gNodeB instructs a UE to perform a handover is also affected by the following functions: subscription to uplink interference information about inter-frequency neighboring cells, inter-frequency handover collaboration, anti-ping-pong function based on the inter-frequency handover waiting time, and anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time. For details about these functions, see [Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells](#), [Inter-Frequency Handover Collaboration](#), [Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time](#), and [Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time](#).

Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells

When the function of subscription to uplink interference information about inter-frequency neighboring cells is enabled (by selecting the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch.NCellInfoCtrlSwitch** parameter) for the serving cell, the serving cell obtains uplink interference information about intra- or inter-base-station neighboring cells, based on which the serving cell determines whether to instruct the UE to initiate a handover. Specifically,

- The UE is instructed to perform a handover if the serving cell determines that the target cell is exposed to low uplink interference.
- Otherwise, the UE is not instructed to perform a handover.

NOTE

For details about how the serving cell determines whether a neighboring cell is exposed to low uplink interference, see [10.1.4 Target Cell or Frequency Decision](#).

If operator-specific-priority-based inter-frequency handover is enabled for the serving cell and uplink-interference-based inter-frequency handover is enabled for the target cell, you are advised to enable the function of subscription to uplink interference information about inter-frequency neighboring cells for the serving cell, avoiding ping-pong handovers.

Inter-Frequency Handover Collaboration

When the inter-frequency handover collaboration function is enabled (by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter) for the serving cell, the serving cell further determines whether to instruct the UE to initiate a handover as follows:

If the serving cell identifies that the target cell has entered the high load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.

The serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.

NOTE

When operator-specific-priority-based inter-frequency handover is enabled for the serving cell, you are advised to enable the inter-frequency handover collaboration function for the serving cell to avoid ping-pong handovers if any of the following functions is enabled for the target cell:

Inter-frequency MLB by transferring UEs in connected mode (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter)

Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter)

For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see *Mobility Load Balancing in Connected Mode*. For details about high-load beams and multi-frequency beam coordination, see *Multi-Frequency Smart Selection*.

Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time

For a UE that is handed over to a cell (the current serving cell) through any handover function, a timer (specified by **NRCellMobilityConfig.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an operator-specific-priority-based inter-frequency handover before the timer expires.

If the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time preferentially takes effect.

Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time

For a UE with "QCI-specific cell-load-dependent inter-frequency handover" in effect and handed over to a cell (the current serving cell) through any handover function, a timer (specified by **gNBScenarioDiffGrp.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an operator-specific-priority-based inter-frequency handover before the timer expires.

If the anti-ping-pong function based on the inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific load-dependent inter-frequency handover waiting time takes precedence.

8.2 Network Analysis

8.2.1 Benefits

In network sharing scenarios, this function facilitates the transfer of each operator's UEs to cells on the higher-priority frequency of that operator, providing flexible networking for operators.

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

This function results in the following network impacts:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- Networking of the live network and the impact relationship between operator-specific-priority-based inter-frequency handover and other handover functions (such as frequency-priority-based inter-frequency handover) must

be considered to prevent ping-pong handovers and ensure user experience.
For details about networking requirements, see [8.3.4 Networking](#).

8.3 Requirements

8.3.1 Licenses

There are no license requirements.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

None

8.3.2.2 Mutually Exclusive Functions

None

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	If operator-specific-priority-based inter-frequency handover is disabled, intra-RAT ANR will not be triggered by operator-specific-priority-based inter-frequency handover.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both UE-number-based connected mode MLB and operator-specific-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both spectral-efficiency-based connected mode MLB and operator-specific-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both radio-resource-based connected mode MLB and operator-specific-priority-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Policy of RFSP-based inter-frequency handover back to the HPLMN	BACK_TO_HPLMN_INTER_FREQ_HO_SWITCH option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter	<i>Flexible User Steering</i>	When both the "policy of RFSP-based inter-frequency handover back to the HPLMN" and "operator-specific-priority-based inter-frequency handover" functions are enabled, only the former, not the latter, can be triggered for a roaming UE.
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SWITCH option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the OP_DED_PRI_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then voice-capability-based inter-frequency handover back to the HPLMN takes effect while operator-specific-priority-based inter-frequency handover does not.
High-frequency TDD	None	None	None	None

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an operator-specific-priority-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through an operator-specific-priority-based inter-frequency handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an operator-specific-priority-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

The NR cell's frequency relationships ([Table 8-1](#)), neighbor relationships ([Table 4-20](#)), frequency priorities ([Table 8-2](#)), and the NR cell's operator-specific policies ([Table 8-3](#)) must be configured before this function is activated.

[Table 8-4](#) describes the function switches.

[Table 8-5](#) describes other parameters used for function optimization.

Table 8-1 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.

Table 8-2 Parameters used to configure frequency priority groups of the gNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	Set this parameter based on the network plan.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	Set this parameter based on the network plan.
RAT Type	gNBFreqPriorityGroup.RatType	Set this parameter based on the network plan.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	Set this parameter based on the network plan.
Dedicated Freq Priority for Connected Mode	gNBFreqPriorityGroup.ConnDedFreqPriority	Set this parameter based on the network plan.

Table 8-3 Parameters used to configure operator-specific policies for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellOpPolicy.NrCellId	Set this parameter based on the network plan.
Operator ID	NRCellOpPolicy.OperatorId	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	NRCellOpPolicy.gNBFreqPriority-GroupId	Set this parameter based on the network plan.

Table 8-4 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	- The OP_DED_PRI_BASED_HO option determines whether to enable operator-specific-priority-based inter-frequency handover. Set this parameter based on the network plan. - When operator-specific-priority-based inter-frequency handover is enabled, it is recommended that the INTER_FREQ_HO_COLLABORATION_SW option be selected if inter-frequency MLB by transferring UEs in connected mode is also enabled.

Table 8-5 Other related parameters

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellInterFHoMeaGrp.NrCellId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
Inter-Freq Measurement Event Configuration	NRCellMobilityConfig.InterFreqMeasEventConfig	Use the recommended value.
Inter-Frequency Handover Wait Time	NRCellMobilityConfig.InterFreqHoWaitTime	Use the recommended value.
Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrig	Use the recommended value.
Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Use the recommended value.
Operator Ded Pri Inter-freq HO A4 RSRP Thld	NRCellInterFHoMeaGrp.OpDedPriHoA4RsrpThld	This parameter indicates the RSRP threshold for event A4 related to operator-specific-priority-based inter-frequency handover and RSRP threshold 2 for event A5 related to operator-specific-priority-based inter-frequency handover. Use the recommended value.
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	NRCellQciBearer.NsaInterFreqHoMeasGroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.
Mobility Function Indication	gNBQciBearer.MobilityFunInd	The INTER_FREQ_HO_SUPPRESS_SW option specifies whether to enable QCI-specific suppression of unnecessary inter-frequency handover. When operator-specific-priority-based inter-frequency handover is enabled for the serving cell, you are advised to select this option for each QCI-specific bearer associated with the target frequency group if service-based inter-frequency handover is enabled for the target cell.

Parameter Name	Parameter ID	Setting Notes
Neighboring Cell Information Control Switch	NRCellAlgoSwitch.NCellInfoCtrlSwitch	The INTER_FREQ_UL_INTERF_SW option specifies whether to enable subscription to uplink interference information about inter-frequency neighboring cells. Use the recommended value.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDLPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoULPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.
Inter-Frequency Handover Wait Time	gNBScenarioDiffGrp.InterFreqHoWaitTime	Set this parameter based on the network plan.

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREQUENCYRELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ;
ADD NRCELLFREQUENCYRELATION: NrCellId=0, SsbFreqPos=8580, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N79, SubcarrierSpacing=30KHZ;
//Adding external cells
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=UNLIMITED;
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=1, CellName="B", PhysicalCellId=1,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8580, FrequencyBand=N79,
NrNetworkingOption=UNLIMITED;
//Adding a record to the NR external-cell PLMN list
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=0, SharedMcc="460",
SharedMnc="03", Tac=0, NrNetworkingOption=SA;
ADD NREXTERNALNCELLPLMN: Mcc="302", Mnc="220", gNBId=1, CellId=1, SharedMcc="460",
SharedMnc="03", Tac=0, NrNetworkingOption=SA;
//Adding a relationship with the NR cell
```

```
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, CellIndividualOffset=DB0;
//Adding a gNodeB frequency priority group
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=0, RatType=NR, SsbFreqPos=8000,
CellReselPri=1, CellReselSubPri=0DOT2, ConnDedFreqPriority=1;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1, RatType=NR, SsbFreqPos=8580,
CellReselPri=1, CellReselSubPri=0DOT2, ConnDedFreqPriority=2;
//Configuring an operator-specific policy for the NR cell
ADD NRCELLPOLICY: NrCellId=0, OperatorId=1, gNBFreqPriorityGroupId=1;
//Activating operator-specific-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=OP_DED_PRI_BASED_HO-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring measurement parameter groups related to inter-frequency handover
MOD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, OpDedPriHoA4RsrpThld=-104;
ADD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=1,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, OpDedPriHoA4RsrpThld=-104;
ADD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=2,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, OpDedPriHoA4RsrpThld=-104;
ADD NRCELLINTERFHOMEGRP: NrCellId=0, InterFreqHoMeasGroupId=3,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, OpDedPriHoA4RsrpThld=-104;
//Optional Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement for SA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//Optional Configuring QCI-specific measurement parameters for NSA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, NsaInterFreqHoMeasGroupId=2;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, NsaInterFreqHoMeasGroupId=3;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1, InterFreqHoWaitTime=10;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Enabling QCI-specific suppression of unnecessary inter-frequency handover for the QCI-specific bearers
associated with the target frequency group
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
MOD GNBQCIBEARER: Qci=9, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Configuring the type of measurement event for unnecessary inter-frequency handovers (event A5 as an
example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A5;
//Setting the inter-frequency handover waiting time
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoWaitTime=10;
//Recommended when operator-specific-priority-based inter-frequency handover is enabled for the serving
cell while inter-frequency MLB by transferring UEs in connected mode or multi-frequency beam
coordination is enabled for the target cell) Turning on the inter-frequency handover collaboration switch for
the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Recommended when operator-specific-priority-based inter-frequency handover is enabled for the serving
cell and uplink-interference-based inter-frequency handover is enabled for the target cell) Turning on the
switch for subscription to uplink interference information about inter-frequency neighboring cells for the
serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating operator-specific-priority-based inter-frequency handover  
MOD NRCELLALGOSWITCH: NrCellId=0,InterFreqHoSwitch=OP_DED_PRI_BASED_HO-0;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

If any of the following counters have a value other than 0, operator-specific-priority-based inter-frequency handover has taken effect:

- **N.HO.InterFreq.OpDedPri.PrepAttOut**
- **N.HO.InterFreq.OpDedPri.ExecAttOut**
- **N.HO.InterFreq.OpDedPri.ExecSuccOut**

8.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

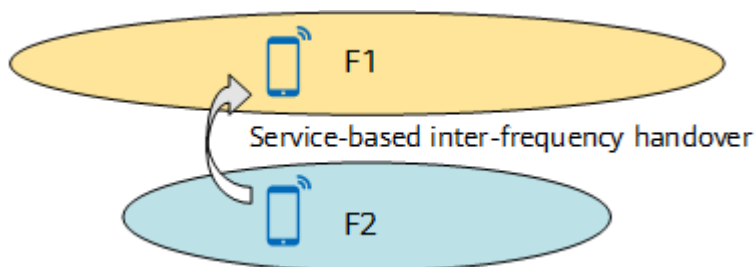
9 Service-based Inter-Frequency Handover

In multi-frequency networking scenarios, service-based inter-frequency handover allows services with different QCI to be carried on different frequencies, achieving flexible networking and service steering, as shown in [Figure 9-1](#).

NOTE

In SA networking, the 5G Core Network (5GC) notifies the gNodeB of the 5QI corresponding to each QoS flow. For detailed description about QCI and 5QI, see *QoS Management*.

Figure 9-1 Service-based inter-frequency handover

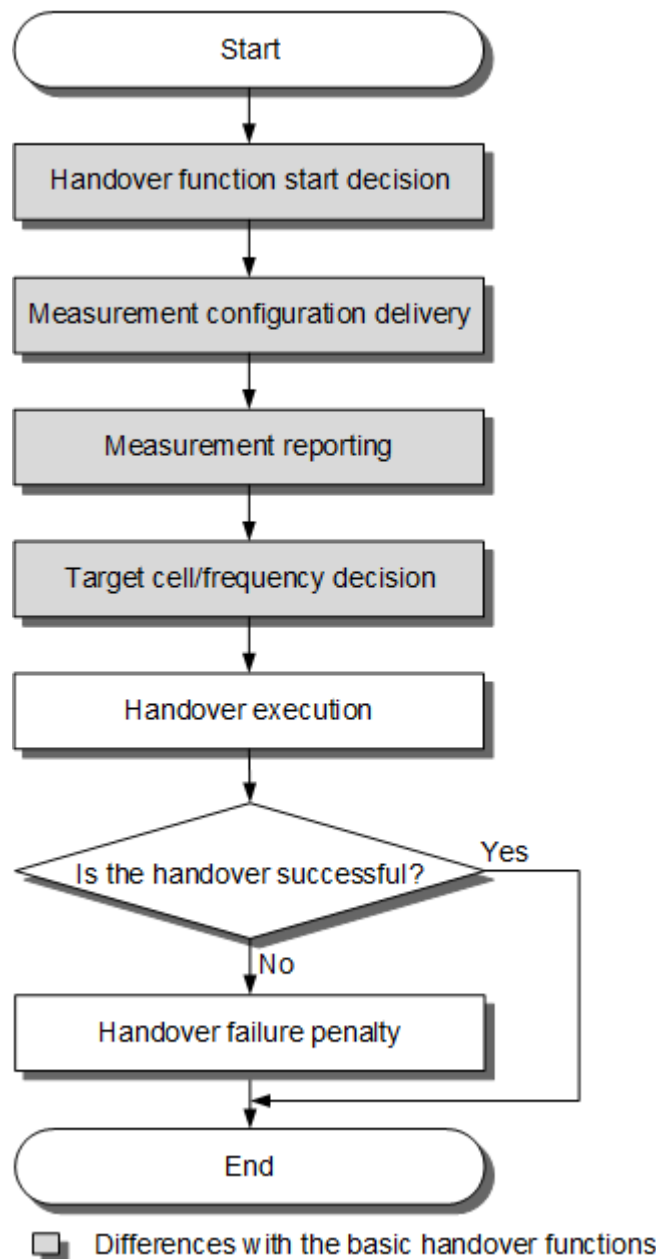


9.1 Principles

Service-based inter-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports only handover but does not support redirection.

With service-based inter-frequency handover, a target frequency group can be configured for each QCI. After service-based inter-frequency handover is enabled, the gNodeB continuously monitors a UE's service status. If the UE's QCI changes, the gNodeB sends the measurement configurations for the specified frequency group to the UE. The UE performs inter-frequency measurements accordingly and then sends a measurement report. After receiving the measurement report from the UE, the gNodeB determines the optimal candidate target cell for handover execution. [Figure 9-2](#) shows the procedure of service-based inter-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 9-2 Procedure of service-based inter-frequency handover



9.1.1 Handover Function Start Decision

If a UE initiates a QCI-specific service that has the highest priority, this function is started when all the following conditions are met:

- The **SERVICE_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected.
- The base station checks the priority of the QoS flow corresponding to the QCI-specific service. That is, the **gNBQciBearer.SrvInterFHoPriJudgeFlag** parameter is set to **TRUE** for the QCI.
- The QCI-specific service is associated with a valid target frequency group. That is, the frequency group for service-based inter-frequency handover is

associated with the QCI through the **gNBOperatorQciParam.gNBSrvInterFHoFreqGroupId** parameter and the frequency group ID is not 255.

- The serving frequency of the UE is not in a target frequency group associated with the QCI-specific service (namely, the frequency group associated with the **gNBOperatorQciParam.gNBSrvInterFHoFreqGroupId** parameter for service-based inter-frequency handover).
- The UE supports inter-frequency measurement and inter-frequency handover. The UE supports the frequency bands containing the non-serving frequencies to be measured.
- In the case that service-based inter-frequency handover enhancement is enabled (by selecting the **SERV_BASED_ENH_HO** option of the **NRCellMobilityConfig.InterFreqHoExtSwitch** parameter), the gNodeB must receive a measurement report indicating that the serving cell meets the entering condition for event A2.

Event A2 represents that the serving cell's signal quality drops below a specific threshold, which is specified by the **NRCellInterFHoMeaGrp.InterFreqServHoA2ThldRsrp** parameter. For details about the definition, entering conditions, and leaving conditions for event A2, see [Table 4-6](#). For the variable configurations for event A2, see [Table 4-8](#).

In particular, for a CA UE whose PCC and SCC frequencies are both the serving frequencies, the following additional factors are taken into account on start of this function:

Whether or Not the PCC Frequency Is in the Target Frequency Group Associated with the QCI-specific Service	Whether or Not the SCC Frequency Is in the Target Frequency Group Associated with the QCI-specific Service	Whether Service-based Handover to SCC Frequencies Is Enabled	Whether to Start Service-based Inter-Frequency Handover
Yes	-	-	Not allowed
No	Yes	Disabled	Not allowed
No	Yes	Enabled	Allowed
No	No	-	Allowed
Note: • "-" in the table indicates that the corresponding factor does not need to be considered. • Service-based handover to SCC frequencies is controlled by the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter.			

 NOTE

For details about the priority configuration for each QCI, see *QoS Management*.

Anti-Ping-Pong Function Between Service-based Inter-Frequency Handover and Voice-Quality-based Inter-Frequency Handover

In low-frequency scenarios, an anti-ping-pong function is introduced to reduce the number of ping-pong handovers when both service-based inter-frequency handover and voice-quality-based inter-frequency handover are enabled in the serving cell.

- The anti-ping-pong function is enabled when the **VOICE_INTER_FREQ_HO_COLLAB_SW** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter is selected. For a UE that is handed over to a cell (the serving cell) through a voice-quality-based inter-frequency handover, the anti-ping-pong function starts a 10s timer when the incoming handover is complete. Initiating a service-based inter-frequency handover for the UE is allowed only when the timer expires. If the UE accesses the serving cell in other ways, relevant processing is not performed.
- The anti-ping-pong function is disabled when the **VOICE_INTER_FREQ_HO_COLLAB_SW** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter is deselected.

For details about voice-quality-based inter-frequency handover, see *VoNR*.

9.1.2 Measurement Configuration Delivery

After service-based inter-frequency handover is started, the gNodeB selects frequencies and delivers related measurement configurations based on frequency selection and event delivery principles as follows:

Frequency Selection Principles

The gNodeB selects the target frequencies for service-based inter-frequency handover and delivers related measurement configurations.

Specifically, the gNodeB selects frequencies in descending order of priorities indicated by the **NRCCellFreqRelation.ConnFreqPriority** parameter until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.

Target frequencies for service-based inter-frequency handover are included in the target frequency group determined by the **gNBSrvInterHoFreqGrp** MO that is associated with the **gNBOperatorQciParam.gNBSrvInterFHoFreqGroupId** parameter.

Event Delivery Principles

The type of a measurement event delivered by the gNodeB is specified by the **NRCCellMobilityConfig.InterFreqMeasEventConfig** parameter.

- If this parameter is set to **EVENT_A4**, the gNodeB delivers measurement configurations related to event A4.
- If this parameter is set to **EVENT_A5**, the gNodeB delivers measurement configurations related to event A5.

Event A4 indicates that the signal quality of a neighboring cell exceeds a specific threshold. Event A5 indicates that the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2. The thresholds for events A4 and A5 are specified by the following parameters:

- Threshold for event A4:
 - If the threshold for event A4 related to RFSP-specific service-based inter-frequency measurements is not customized for a UE, the RSRP threshold for event A4 related to service-based inter-frequency measurements equals ***NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld***.
 - If the threshold for event A4 related to RFSP-specific service-based inter-frequency measurements is customized for a UE, the RSRP threshold for event A4 related to service-based inter-frequency measurements equals ***NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld*** plus ***gNBRfspConfig.ServBasedInterFHoA4RsrpOfs***.

For more information, see [Table 4-10](#).

- Thresholds for event A5:
 - Threshold 1 for event A5 is specified by ***NRCellInterFHoMeaGrp.InterFreqServHoA2ThldRsrp***.
 - Threshold 2 for event A5 is set as follows:
 - If threshold 2 for event A5 related to RFSP-specific service-based inter-frequency measurements is not customized for a UE, RSRP threshold 2 for event A5 related to service-based inter-frequency measurements equals ***NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld***.
 - If threshold 2 for event A5 related to RFSP-specific service-based inter-frequency measurements is customized for a UE, RSRP threshold 2 for event A5 related to service-based inter-frequency measurements equals ***NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld*** plus ***gNBRfspConfig.ServBasedInterFHoA4RsrpOfs***.

For more information, see [Table 4-11](#).

9.1.3 Measurement Reporting

The UE performs measurements based on the event configuration included in the measurement configuration. If neighboring cells that meet the requirements are detected, a measurement report is sent to the gNodeB, which then uses the report to determine the target cells or frequencies.

If no suitable neighboring cell is reported 6s after the measurement of event A4 or A5 has started, the gNodeB releases the measurement configurations, enabling the UE to stop the measurement for this event A4 or A5. In this scenario, if the UE is always in connected mode, the gNodeB no longer sends the same measurement configurations.

9.1.4 Target Cell or Frequency Decision

After the gNodeB receives a measurement report of event A4 or A5, the gNodeB selects the optimal cell from the reported cells as the target cell for handover

execution. Note that whether the gNodeB instructs a UE to perform a handover is also affected by subscription to uplink interference information about inter-frequency neighboring cells and inter-frequency handover collaboration, as detailed in [Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells](#) and [Inter-Frequency Handover Collaboration](#).

Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells

When the function of subscription to uplink interference information about inter-frequency neighboring cells is enabled (by selecting the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch.NCellInfoCtrlSwitch** parameter) for the serving cell, the serving cell obtains uplink interference information about intra- or inter-base-station neighboring cells, based on which the serving cell determines whether to instruct the UE to initiate a handover. Specifically,

- The UE is instructed to perform a handover if the serving cell determines that the target cell is exposed to low uplink interference.
- Otherwise, the UE is not instructed to perform a handover.

NOTE

For details about how the serving cell determines whether a neighboring cell is exposed to low uplink interference, see [10.1.4 Target Cell or Frequency Decision](#).

If service-based inter-frequency handover is enabled for the serving cell and uplink-interference-based inter-frequency handover is enabled for the target cell, you are advised to enable the function of subscription to uplink interference information about inter-frequency neighboring cells for the serving cell, avoiding ping-pong handovers.

Inter-Frequency Handover Collaboration

When the inter-frequency handover collaboration function is enabled (by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter) for the serving cell, the serving cell further determines whether to instruct the UE to initiate a handover as follows:

If the serving cell identifies that the target cell has entered the high load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.

The serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.

 NOTE

When service-based inter-frequency handover is enabled for the serving cell, you are advised to enable the inter-frequency handover collaboration function for the serving cell to avoid ping-pong handovers if any of the following functions is enabled for the target cell:

Inter-frequency MLB by transferring UEs in connected mode (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter)

Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter)

For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see *Mobility Load Balancing in Connected Mode*. For details about high-load beams and multi-frequency beam coordination, see *Multi-Frequency Smart Selection*.

9.2 Network Analysis

9.2.1 Benefits

This function provides a method for handing over UEs based on services and allows flexible networking, thereby helping operators implement service steering.

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

This function results in the following network impacts:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- Networking of the live network and the impact relationship between service-based inter-frequency handover and other handover functions (such as frequency-priority-based inter-frequency handover) must be considered to prevent ping-pong handovers and ensure user experience. For details about networking requirements, see [9.3.4 Networking](#).
- When service-based inter-frequency handover enhancement takes effect, the number of outgoing service-based inter-frequency handovers from the serving cell decreases and the unnecessary measurement overheads decline, resulting in fluctuations in the number of UEs served by a cell and the values of mobility-related performance counters. This is because a service-based inter-frequency handover can be triggered only when the serving cell's signal quality measured by a UE meets the event A2 threshold. A smaller value of the event A2 threshold results in a lower probability of triggering event A2 reports and inter-frequency measurements. A larger value of the event A2 threshold produces the opposite effects.

9.3 Requirements

9.3.1 Licenses

There are no license requirements.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based inter-frequency handover	SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Service-based inter-frequency handover enhancement (controlled by the SERV_BASED_ENH_HO option of the NRCellMobilityConfig.InterFreqHoExtSwitch parameter) requires service-based inter-frequency handover to be enabled.

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	If service-based inter-frequency handover is disabled, intra-RAT ANR will not be triggered by service-based inter-frequency handover because such handover does not take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both UE-number-based connected mode MLB and service-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both spectral-efficiency-based connected mode MLB and service-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and the NRCellMlb.MlbTriggerMode parameter set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	If (1) both radio-resource-based connected mode MLB and service-based inter-frequency handover are enabled, and (2) the target cell rejects a handover initiated by either of the enabled functions, then a handover cannot be initiated by the other enabled function in the source cell until a short period of time elapses.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- When the service-based handover to SCC frequencies function is disabled (by deselecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are not allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers. - When the service-based handover to SCC frequencies function is enabled (by selecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- When the service-based handover to SCC frequencies function is disabled (by deselecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are not allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers. - When the service-based handover to SCC frequencies function is enabled (by selecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_IN TER_BAND_C A_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- When the service-based handover to SCC frequencies function is disabled (by deselecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are not allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers. - When the service-based handover to SCC frequencies function is enabled (by selecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_IN TER_BAND_U L_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- When the service-based handover to SCC frequencies function is disabled (by deselecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are not allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers. - When the service-based handover to SCC frequencies function is enabled (by selecting the SERV_BASED_HO_TO_SCC_FREQ_SW option of the gNBMobilityCommParam.MobilityAlgoSwitch parameter), CA UEs are allowed to transfer to cells on their SCC frequencies through service-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for service-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, service-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to FALSE for the target cell's frequency in the

RAT	Function Name	Function Switch	Reference	Description
				frequency group associated with the roaming UE, service-based inter-frequency handover does not take effect for the roaming UE.
Low-frequency TDD	Uplink-experience-based inter-frequency handover	UL_EXP_BASED_INTER_FREQ_HO_SW option of the NRDUCellRedCap.RedCapUplinkExperienceBasedInterFrequencyHandoverSwitch parameter	<i>RedCap</i>	Uplink-experience-based inter-frequency handover is not triggered for UEs for which service-based inter-frequency handover has taken effect.
Low-frequency TDD	QCI-specific handover priority configuration	gNBQciBearer.QciPriorityForRedCapHo	<i>RedCap</i>	If both QCI-specific handover priority configuration and service-based inter-frequency handover take effect, the QCI priority is preferentially determined by the gNBQciBearer.QciPriorityForRedCapHo parameter. If this parameter is set to 255 , the QCI priority will then be determined by the gNBQciBearer.QciPriorityForHo parameter.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through a service-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a service-based inter-frequency handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through a service-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

The NR cell's frequency relationships ([Table 9-1](#)), neighbor relationships ([Table 4-20](#)), frequency priorities ([Table 9-2](#)), and the NR cell's operator-specific policies ([Table 9-3](#)) must be configured before this function is activated.

[Table 9-4](#) describes the function switches.

[Table 9-5](#) describes other parameters used for function optimization.

Table 9-1 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Frequency Priority for Connected Mode	NRCellFreqRelation.ConnFreqPriority	It is recommended that this parameter be set to a non-zero value for frequencies providing contiguous coverage, with a larger value set for a frequency in a lower frequency band. It is recommended that this parameter be set to 0 for frequencies providing discontinuous coverage, which means that such a frequency will not be selected as the target frequency.

Table 9-2 Parameters used to configure frequency groups for service-based inter-frequency handover

Parameter Name	Parameter ID	Setting Notes
gNodeB Srv-Based Inter-Freq HO Freq Group ID	gNBSrvInterHoFreqGrp.gNBSrvInterFHoFreqGroupId	Set this parameter based on the network plan.
SSB Frequency Index	gNBSrvInterHoFreqGrp.SsbFreqIndex	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	gNBSrvInterHoFreqGrp.SsbDescMethod	Set this parameter based on the network plan.
SSB Frequency Position	gNBSrvInterHoFreqGrp.SsbFreqPos	Set this parameter based on the network plan.

Table 9-3 Parameters related to NR operator-specific QCI configurations

Parameter Name	Parameter ID	Setting Notes
Operator ID	gNBOperatorQciParam.OperatorId	Set this parameter based on the network plan.
QoS Class Identifier	gNBOperatorQciParam.Qci	Set this parameter based on the network plan.
gNodeB Srv-Based Inter-Freq HO Freq Group ID	gNBOperatorQciParam.gNBSrvInterFHoFreqGroupId	Set this parameter based on the network plan.

Table 9-4 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch . <i>InterFreqHoSwitch</i>	The SERVICE_BASED_HO option determines whether to enable service-based inter-frequency handover in measurement-based mode. Set this option based on the network plan.
Inter-Frequency Handover Extend Switch	NRCellMobilityConfig . <i>InterFreqHoExtSwitch</i>	The SERV_BASED_ENH_HO option determines whether to enable service-based inter-frequency handover enhancement. Set this option based on the network plan.
Inter-Frequency Handover Switch	NRCellAlgoSwitch . <i>InterFreqHoSwitch</i>	When service-based inter-frequency handover is enabled, it is recommended that the INTER_FREQ_HO_COOPERATION_SW option be selected if inter-frequency MLB by transferring UEs in connected mode is also enabled.

Table 9-5 Other related parameters

Parameter Name	Parameter ID	Setting Notes
Srv-Based Inter-Freq HO Priority Judge Flag	gNBQciBearer . <i>SrvInterFHoPriJudgeFlag</i>	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-Freq Measurement Event Configuration	NRCellMobilityConfig.InterFreqMeasEventConfig	Use the recommended value.
Inter-freq Handover Measurement Group ID	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
Inter-freq Measurement Event Time to Trigger	NRCellInterFHoMeaGrp.InterFreqA4A5TimeToTrig	Use the recommended value.
Inter-freq Measurement Event Hysteresis	NRCellInterFHoMeaGrp.InterFreqA4A5Hyst	Use the recommended value.
Service-based Inter-Freq HO A2 RSRP Thld	NRCellInterFHoMeaGrp.InterFreqSrvHoA2ThldRsrp	This parameter indicates the RSRP threshold for event A2 related to service-based inter-frequency handover and RSRP threshold 1 for event A5 related to service-based inter-frequency handover. Use the recommended value.
Srv-Based Inter-freq A4 RSRP Threshold	NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld	This parameter indicates the RSRP threshold for event A4 related to service-based inter-frequency handover and RSRP threshold 2 for event A5 related to service-based inter-frequency handover. Use the recommended value.
Service-Based Inter-Freq A4 RSRP Thld Offset	gNBRfspConfig.ServBasedInterF-HoA4RsrpOfs	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeas-GroupId	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	NRCellQciBearer.NsaInterFreqHoMeasGroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.
Neighboring Cell Information Control Switch	NRCellAlgoSwitch.NCellInfoCtrlSwitch	The INTER_FREQ_UL_INT RF_SW option specifies whether to enable subscription to uplink interference information about inter-frequency neighboring cells. Use the recommended value.
Mobility Algorithm Switch	gNBMobilityCommParam.MobilityAlgoSwitch	The SERV_BASED_HO_TO_SCC_FREQ_SW option of this parameter specifies whether to allow CA UEs to be handed over to cells on their SCC frequencies through service-based inter-frequency handover. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	The VOICE_INTER_FREQ_HO_COLLAB_SW option of this parameter specifies whether to enable the anti-ping-pong function between service-based inter-frequency handover and voice-quality-based inter-frequency handover. If both service-based inter-frequency handover and voice-quality-based inter-frequency handover are enabled in the serving cell, it is recommended that this option be selected.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1;
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8580, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N79, SubcarrierSpacing=30KHZ, ConnFreqPriority=2;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=UNLIMITED;
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=1, CellName="B", PhysicalCellId=1,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8580, FrequencyBand=N79,
NrNetworkingOption=UNLIMITED;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=1, CellIndividualOffset=DB0;
```

```
//Configuring a frequency group for service-based inter-frequency handover
ADD GNBSRVINTERHOFREQGRP: gNBSrvInterFHoFreqGroupId=1, SsbFreqIndex=0, SsbFreqPos=8000,
SsbDescMethod=SSB_DESC_TYPE_GSCN;
ADD GNBSRVINTERHOFREQGRP: gNBSrvInterFHoFreqGroupId=2, SsbFreqIndex=0, SsbFreqPos=8580,
SsbDescMethod=SSB_DESC_TYPE_GSCN;
//Configuring operator-specific QCI parameters
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=1, gNBSrvInterFHoFreqGroupId=1;
MOD GNBOPERATORQCIPARAM: OperatorId=1, Qci=9, gNBSrvInterFHoFreqGroupId=2;
//Reconfiguring the service-based inter-frequency handover priority judge flag
MOD GNBQCIBEARER: Qci=1, SrvInterFHoPriJudgeFlag=TRUE;
MOD GNBQCIBEARER: Qci=9, SrvInterFHoPriJudgeFlag=TRUE;
//Activating service-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=SERVICE_BASED_HO-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring a group of measurement parameters for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, SrvInterFreqA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, SrvInterFreqA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=2,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, SrvInterFreqA4RsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=3,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, SrvInterFreqA4RsrpThld=-104;
//Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement for SA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//Optional) Configuring QCI-specific measurement parameters for NSA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, NsaInterFreqHoMeasGroupId=2;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, NsaInterFreqHoMeasGroupId=3;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Enabling service-based inter-frequency handover enhancement
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoExtSwitch=SERV_BASED_ENH_HO-1;
//Setting the RSRP threshold for event A2 related to service-based inter-frequency handover and RSRP
threshold 1 for event A5 related to service-based inter-frequency handover
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, InterFreqServHoA2ThldRsrp=-43;
//Configuring the type of measurement event for unnecessary inter-frequency handovers (event A5 as an
example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A5;
//Recommended when both service-based inter-frequency handover and voice-quality-based inter-
frequency handover are enabled for the serving cell) Enabling the anti-ping-pong function between service-
based inter-frequency handover and voice-quality-based inter-frequency handover for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=VOICE_INTER_FREQ_HO_COLLAB_SW-1;
//Recommended when service-based inter-frequency handover is enabled for the serving cell while inter-
frequency MLB by transferring UEs in connected mode or multi-frequency beam coordination is enabled for
the target cell) Turning on the inter-frequency handover collaboration switch for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Recommended when service-based inter-frequency handover is enabled for the serving cell and uplink-
interference-based inter-frequency handover is enabled for the target cell) Turning on the switch for
subscription to uplink interference information about inter-frequency neighboring cells for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
//Allowing CA UEs to be handed over to cells on their SCC frequencies through service-based inter-
frequency handover
MOD GNBMOBILITYCOMMPARAM: MobilityAlgoSwitch=SERV_BASED_HO_TO_SCC_FREQ_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating service-based inter-frequency handover  
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=SERVICE_BASED_HO-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

If any of the following counters has a value other than 0, service-based inter-frequency handover has taken effect:

- **N.HO.InterFreq.Service.PrepAttOut**
- **N.HO.InterFreq.Service.ExecAttOut**
- **N.HO.InterFreq.Service.ExecSuccOut**

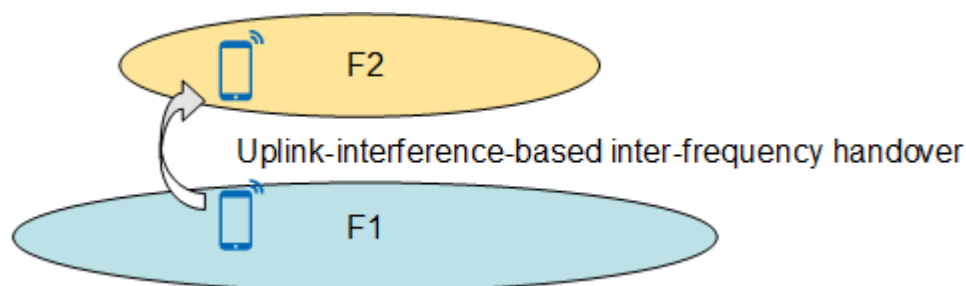
9.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

10 Uplink-Interference-based Inter-Frequency Handover

In multi-frequency networking, if a frequency has uplink interference, the uplink user experience on this frequency will be affected. In this case, uplink-interference-based inter-frequency handovers can be performed to hand over UEs from the serving cell to inter-frequency neighboring cells, ensuring the uplink user experience ([Figure 10-1](#)). Uplink-interference-based inter-frequency handover is supported only in low frequency bands.

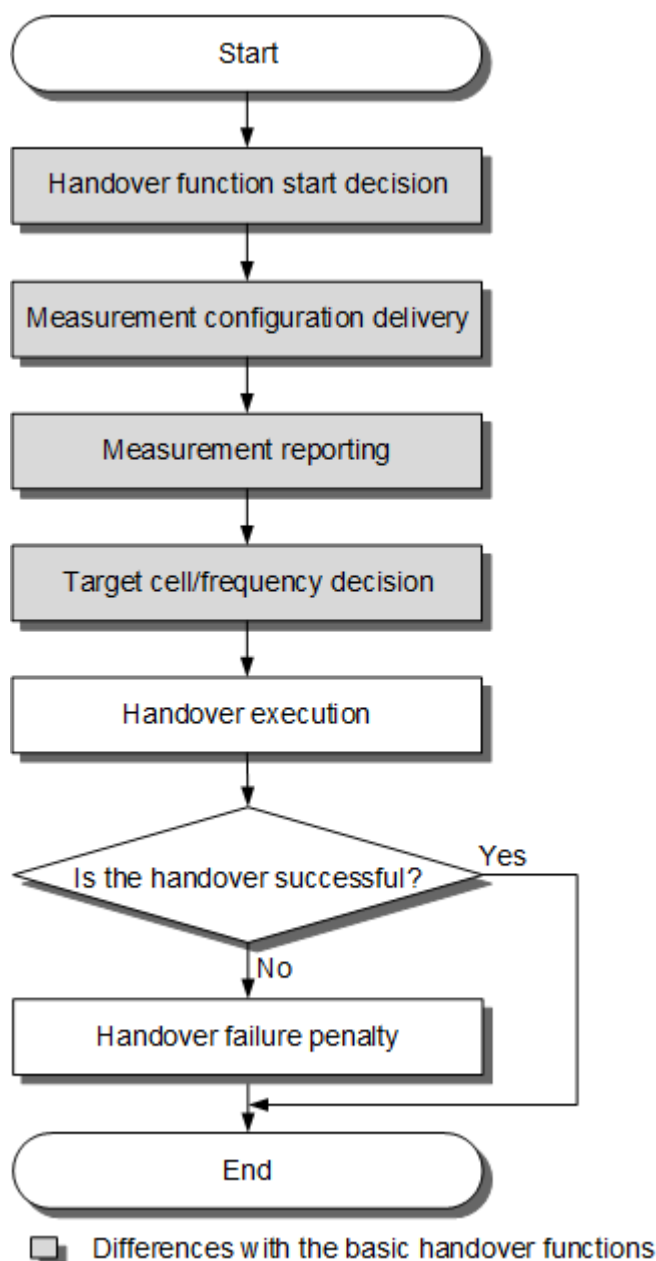
Figure 10-1 Uplink-interference-based inter-frequency handover



10.1 Principles

Uplink-interference-based inter-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports both handover and redirection. [Figure 10-2](#) shows how uplink-interference-based inter-frequency handover works. The gNodeB determines whether to start uplink-interference-based inter-frequency handover based on the strength of uplink interference in the serving cell. Specifically, if the uplink interference strength of the serving cell is greater than a specified threshold, the gNodeB sends inter-frequency measurement configurations to low-SRS-SINR UEs performing uplink large-packet services in the serving cell. The UEs then perform inter-frequency measurements. After receiving measurement reports from the UEs, the gNodeB determines the optimal candidate target cells for handover execution. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 10-2 Procedure for uplink-interference-based inter-frequency handover



10.1.1 Handover Function Start Decision

Uplink-interference-based inter-frequency handover is started when all of the following conditions are met:

- The switch for uplink-interference-based inter-frequency handover is turned on.
 - In SA networking, the **UL_INTRF_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected.
 - In NSA networking, the **UL_INTRF_BASED_HO** and **NSA_UL_INTRF_BASED_HO** options of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter are both selected.

- The strength of uplink interference in the serving cell is greater than the uplink interference threshold (specified by the **NRDUCellIntrfIdent.UlIntrfThld** parameter).

10.1.2 Measurement Configuration Delivery

After uplink-interference-based inter-frequency handover takes effect, the conditions that apply to a UE to which the gNodeB sends measurement configurations vary with the setting of the **NRDUCellIntrfIdent.UlIntrfBasedHoUeSvcType** parameter. Specifically,

- If this parameter is set to **UL_LARGE_PACKET_UE**, the gNodeB sends measurement configurations to a UE that meets all of the following conditions:
 - The UE has a low SRS SINR (less than the threshold specified by **NRDUCellIntrfIdent.SrsSnrThld**).
 - The UE is transmitting large packets in the uplink. This is the case when all of the following conditions are met:
 - Amount of data in the uplink data buffer (indicated in the buffer status report, also known as BSR) sent by the UE to the scheduler + Amount of uplink data already sent by the UE > 4 MB
 - Proportion of TTIs in which the UE is scheduled in the uplink within 1s (Number of TTIs in which the UE is scheduled in the uplink within 1s/Number of TTIs in the uplink in the cell within 1s) ≥ 80%
 - Uplink PRB usage of the UE ≥ 30%
- If this parameter is set to **ALL_UE**, the gNodeB sends measurement configurations to a UE that meets the following condition:
The UE has a low SRS SINR (less than the threshold specified by **NRDUCellIntrfIdent.SrsSnrThld**).

Measurement configurations include the target frequencies for uplink-interference-based inter-frequency handovers and the corresponding event A3 configuration. The details are as follows:

- Target frequencies for uplink-interference-based inter-frequency handovers: The frequencies included in the measurement configurations are those for which the interference-based frequency priority (specified by **NRCellFreqRelation.IntrfFreqPriority**) is set to a value other than 0. Specifically, the gNodeB selects frequencies in descending order of interference-based frequency priorities until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.
- Offset for event A3: The RSRP offset for event A3 is specified by the **NRCellInterFHoMeaGrp.IntrfInterFreqHoA3RsrpOfs** parameter. For details about other variable configurations, see [Table 4-9](#). Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset.

10.1.3 Measurement Reporting

After receiving the measurement configuration from the gNodeB, the UE performs measurements in descending order of interference-based frequency priorities (specified by **NRCellFreqRelation.IntrFReqPriority**). If the UE detects a neighboring cell that meets the specific requirements, it reports the measurement result to the gNodeB and stops measurements for this event A3. The gNodeB then determines the target cells or frequencies based on the measurement report.

If no suitable neighboring cell is reported 6s after the measurement of event A3 has started, the gNodeB releases the measurement configuration, enabling the UE to stop the measurement for this event A3. To prevent the UE from experiencing continued failures in outgoing handovers from the serving cell, cell-level supplementary selection in polling mode is performed. Specifically, the gNodeB periodically (at a fixed interval of 30s) selects a maximum of five low-SRS-SINR UEs performing large packet services in the serving cell's uplink in polling mode and sends measurement configurations to them, enabling them to start measurements for uplink-interference-based inter-frequency handover.

10.1.4 Target Cell or Frequency Decision

After receiving an event A3 measurement report, the gNodeB selects the optimal cell from the reported cells as the target cell for handover execution. In addition, the gNodeB determines whether to instruct the UE to perform a handover or redirection to the target cell based on the information such as the target cell load, uplink interference of the target cell, and bearers of the UE. If the UE supports inter-frequency handover, the gNodeB instructs the UE to initiate a handover attempt to the target cell. If the UE does not support inter-frequency handover, the gNodeB instructs the UE to perform a redirection to the frequency of the target cell. The following assumes that the handover policy is handover.

1. Before the gNodeB determines whether to instruct the UE to perform a handover to the target cell, the serving cell needs to obtain the load or uplink interference information about the target cell in the following ways:
 - When a function of MLB by transferring UEs in connected mode (for example, UE-number-based connected mode MLB) involving exchange of inter-cell load and interference status information is enabled,
 - the gNodeB triggers load information exchange between intra-base-station cells by default, so that the serving cell obtains load and uplink interference information about intra-base-station neighboring cells.
 - if there are candidate inter-base-station neighboring cells, the gNodeB triggers load information exchange between inter-base-station cells so that the serving cell obtains load and uplink interference information about inter-base-station neighboring cells.
 - When the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch.NCellInfoCtrlSwitch** parameter is selected to enable the subscription to uplink interference information about inter-frequency neighboring cells,
 - the serving cell can directly obtain uplink interference information about intra-base-station neighboring cells.

- the serving cell can directly obtain uplink interference information about inter-base-station neighboring cells if there are candidate inter-base-station neighboring cells.
- 2. After the serving cell obtains load or uplink interference information about the target cell, the gNodeB determines whether to instruct the UE to perform a handover to the target cell based on the following conditions.
 - If the serving cell obtains load and uplink interference information about the target cell:
In the case that the RSRP value of the lightly-loaded, low-uplink-interference target cell is greater than the value of **NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld**, the UE will be handed over to the target cell. Otherwise, the UE continues to camp on the serving cell and handover is not performed.
where
 - A neighboring cell is regarded as a low-uplink-interference cell when its uplink interference strength is less than or equal to **NRDUCellIntrfIdent.UlIntrfThld**.
 - A neighboring cell is regarded as a lightly-loaded cell when it does not meet the conditions for entering the MLB state.
 - If the serving cell obtains only uplink interference information about the target cell:
In the case that the RSRP value of the low-uplink-interference target cell is greater than the value of **NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld**, the UE will be handed over to the target cell. Otherwise, the UE continues to camp on the serving cell and handover is not performed.
In this case, a neighboring cell is regarded as a low-uplink-interference cell when its uplink interference strength is less than or equal to **NRDUCellIntrfIdent.UlIntrfThld**.
 - If the serving cell does not obtain uplink interference information about the target cell:
In the case that the RSRP value of the target cell is greater than the value of **NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld**, the UE will be handed over to the target cell. Otherwise, the UE continues to camp on the serving cell and handover is not performed.

Note that whether the gNodeB instructs a UE to perform a handover is also affected by the following functions: inter-frequency handover collaboration, anti-ping-pong function based on the inter-frequency handover waiting time, and anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time. For details about these functions, see [Inter-Frequency Handover Collaboration](#), [Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time](#), and [Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time](#).

Inter-Frequency Handover Collaboration

When the inter-frequency handover collaboration function is enabled (by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the

NRCellAlgoSwitch.InterFreqHoSwitch parameter) for the serving cell, the serving cell further determines whether to instruct the UE to initiate a handover as follows:

If the serving cell identifies that the target cell has entered the high load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.

The serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.

NOTE

When uplink-interference-based inter-frequency handover is enabled for the serving cell, you are advised to enable the inter-frequency handover collaboration function for the serving cell to avoid ping-pong handovers if any of the following functions is enabled for the target cell:

Inter-frequency MLB by transferring UEs in connected mode (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter)

Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter)

For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see *Mobility Load Balancing in Connected Mode*. For details about high-load beams and multi-frequency beam coordination, see *Multi-Frequency Smart Selection*.

Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time

For a UE that is handed over to a cell (the current serving cell) through any handover function, a timer (specified by **NRCellMobilityConfig.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an uplink-interference-based inter-frequency handover before the timer expires.

If the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time preferentially takes effect.

Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time

For a UE with "QCI-specific cell-load-dependent inter-frequency handover" in effect and handed over to a cell (the current serving cell) through any handover function, a timer (specified by **gNBScenarioDiffGrp.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an uplink-interference-based inter-frequency handover before the timer expires.

If the anti-ping-pong function based on the inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific load-dependent inter-frequency handover waiting time takes precedence.

10.2 Network Analysis

10.2.1 Benefits

Uplink-interference-based inter-frequency handovers are performed to ensure uplink user experience when the serving cell has strong uplink interference.

10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

Uplink-interference-based inter-frequency handover results in the following impacts on the network:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- Networking of the existing network and the impact relationship between uplink-interference-based inter-frequency handover and other handover functions (such as frequency-priority-based inter-frequency handover) must be considered to prevent ping-pong handovers and ensure user experience. For details about networking requirements, see [10.3.4 Networking](#).

10.3 Requirements

10.3.1 Licenses

This function is not under license control.

10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

10.3.2.1 Prerequisite Functions

None

10.3.2.2 Mutually Exclusive Functions

None

10.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	In multi-carrier scenarios, ping-pong handovers easily occur between a cell enabled with uplink-interference-based inter-frequency handover and an inter-frequency cell enabled with protection of UEs under weak uplink coverage (controlled by the UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter). As such, you are advised to further select the INTER_FREQ_UL_INTRF_SW option of the NRCellAlgoSwitch.NCellInfoCtrlSwitch parameter for the aforesaid inter-frequency cell to prevent handovers to cells experiencing high interference.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for uplink-interference-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency.
Low-frequency TDD	Uplink symbol power saving	UL_SYMBOL_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	When uplink symbol power saving takes effect, uplink-interference-based inter-frequency handover does not take effect.

10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

10.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types.

Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an uplink-interference-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through an uplink-interference-based inter-frequency handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an uplink-interference-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

The NR cell's frequency relationships (including frequency priorities) and neighbor relationships must be configured before this function is activated. For details about the NR cell's frequency relationship configurations, see [Table 10-1](#). For details about neighbor relationship configurations, see [Table 4-20](#).

[Table 10-3](#) describes the function switches.

[Table 10-2](#) and [Table 10-4](#) describe other parameters used for function optimization.

Table 10-1 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Intrf-based Frequency Priority	NRCellFreqRelation.IntrfFreqPriority	Set this parameter based on the network plan.

Table 10-2 Parameters used to configure frequency groups for uplink-interference-based inter-frequency handover

Parameter Name	Parameter ID	Setting Notes
Inter-freq Handover Measurement Group ID	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
Intrf-based Inter-Freq HO A3 RSRP Offset	NRCellInterFHoMeaGrp.IntrfInterFreqHoA3RsrpOfs	Use the recommended value.
Intrf-based Inter-Freq HO RSRP Thld	NRCellInterFHoMeaGrp.IntrfInterFreqHoRsrpThld	It is good practice to set this parameter to a value greater than the value of NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld to prevent ping-pong with coverage-based inter-frequency handover.
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
NSA UE Inter-freq HO Measurement Group ID	NRCellQciBearer.NsaInterFreqHoMeasGroupId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.

Table 10-3 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	In SA networking, select the UL_INTRF_BASED_HO option to enable uplink-interference-based inter-frequency handover. In NSA networking, select the UL_INTRF_BASED_HO and NSA_UL_INTRF_BASED_HO options to enable uplink-interference-based inter-frequency handover. Set this parameter based on the network plan.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	When uplink-interference-based inter-frequency handover is enabled, it is recommended that the INTER_FREQ_HO_COOPERATION_SW option be selected if inter-frequency MLB by transferring UEs in connected mode is also enabled.

Table 10-4 Other related parameters

Parameter Name	Parameter ID	Setting Notes
Uplink Interference Threshold	NRDUCellIntrfIdent.UlIntrfThld	Use the recommended value.
SRS SINR Threshold	NRDUCellIntrfIdent.SrsSinrThld	Use the recommended value.
Neighboring Cell Information Control Switch	NRCCellAlgoSwitch.NCellInfoCtrlSwitch	The INTER_FREQ_UL_INT RF_SW option specifies whether to enable subscription to uplink interference information about inter-frequency neighboring cells. Use the recommended value.
Inter-Frequency Handover Wait Time	NRCCellMobilityConfig.InterFreqHoWaitTime	Use the recommended value.
UL-Intrf-based Handover UE Service Type	NRDUCellIntrfIdent.UlIntrfBasedHoUeSvcType	Set this parameter to UL_LARGE_PACKET_UE if uplink-interference-based inter-frequency handover is enabled to take effect only for UEs running large-packet services. Set this parameter to ALL_UE if uplink-interference-based inter-frequency handover is enabled to take effect for all UEs. Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDLPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoULPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Wait Time	gNBScenarioDiffGrp.InterFreqHoWaitTime	Set this parameter based on the network plan.

10.4.1.2 Using MML Commands

Before using MML commands, refer to [10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, IntrFreqPriority=16;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=UNLIMITED;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Activating uplink-interference-based inter-frequency handover in SA networking
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=UL_INTRF_BASED_HO-1;
//Activating uplink-interference-based inter-frequency handover in NSA networking
MOD NRCELLALGOSWITCH: NrCellId=0,
InterFreqHoSwitch=UL_INTRF_BASED_HO-1&NSA_UL_INTRF_BASED_HO-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring a group of measurement parameters for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, IntrInterFreqHoA3RsrpOfs=3,
IntrInterFreqHoRsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1, IntrInterFreqHoA3RsrpOfs=3,
IntrInterFreqHoRsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=2, IntrInterFreqHoA3RsrpOfs=3,
IntrInterFreqHoRsrpThld=-104;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=3, IntrInterFreqHoA3RsrpOfs=3,
IntrInterFreqHoRsrpThld=-104;
//(Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement for SA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//(Optional) Configuring QCI-specific measurement parameters for NSA UEs
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, NsaInterFreqHoMeasGroupId=2;
```

```
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, NsaInterFreqHoMeasGroupId=3;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1, InterFreqHoWaitTime=10;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Turning on the switch for subscription to uplink interference information about inter-frequency
neighboring cells
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
//Modifying the interference identification configuration of the NR DU cell
MOD NRDUCELLINTRFIDENT: NrDuCellId=0, UlIntrfThld=-100, SrsSinrThld=-4;
//Recommended when uplink-interference-based inter-frequency handover is enabled for the serving cell
while inter-frequency MLB by transferring UEs in connected mode or multi-frequency beam coordination is
enabled for the target cell) Turning on the inter-frequency handover collaboration switch for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Setting the inter-frequency handover waiting time
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoWaitTime=10;
//Configuring the UE service type in uplink-interference-based inter-frequency handover
MOD NRDUCELLINTRFIDENT: NrDuCellId=0, UlIntrfBasedHoUeSvcType=UL_LARGE_PACKET_UE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating uplink-interference-based inter-frequency handover in SA networking
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=UL_INTRF_BASED_HO-0;
//Deactivating uplink-interference-based inter-frequency handover in NSA networking
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=NSA_UL_INTRF_BASED_HO-0;
```

10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

10.4.2 Activation Verification

If **INTRF_BASED** is displayed for **HoCause** in the external CHR event **PRIVATE_INTRA_RAT_HO_OUT**, uplink-interference-based inter-frequency handover has taken effect.

10.4.3 Network Monitoring

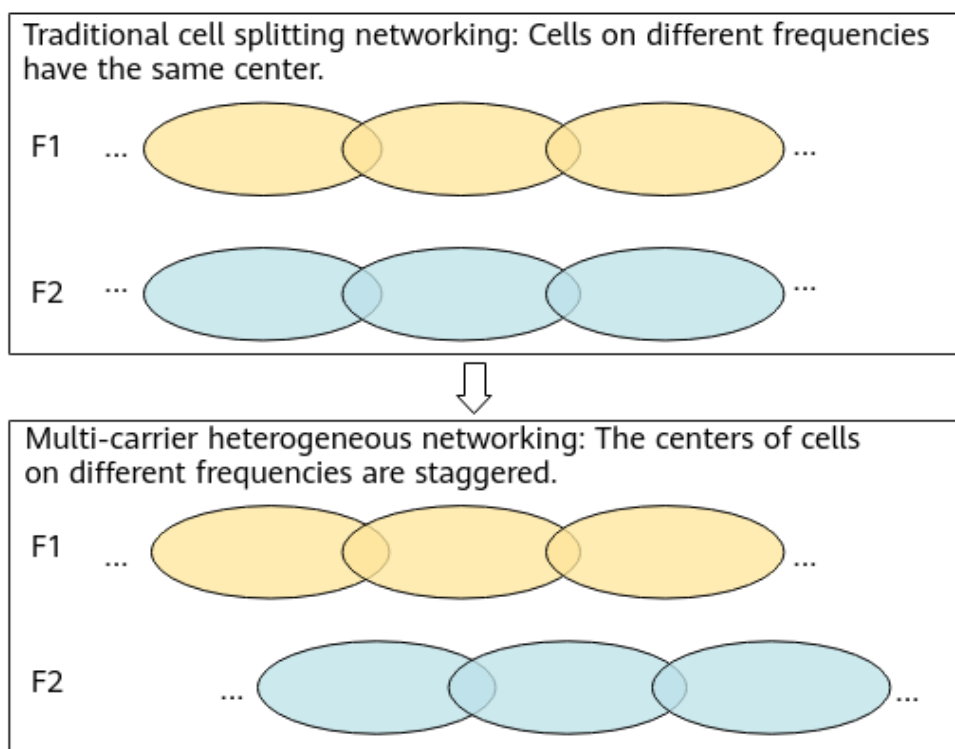
This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

11

SSB SINR-based Inter-Frequency Handover (for LampSite)

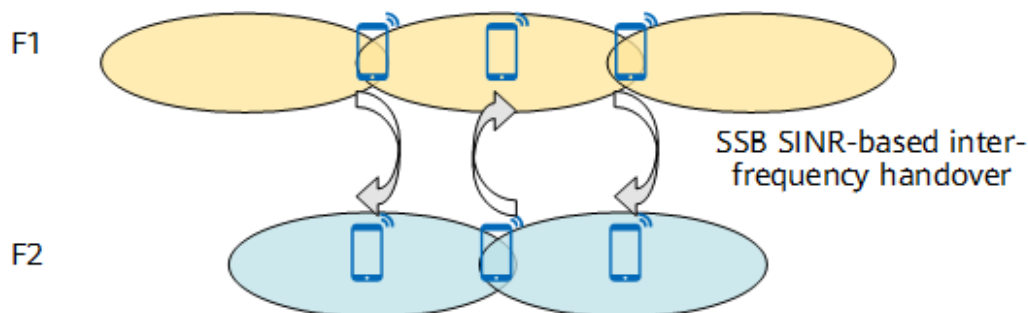
For LampSite base stations intended for high capacity scenarios, cell splitting can be used to split a congested cell into multiple smaller cells for capacity expansion. In traditional cell splitting networking as shown in [Figure 11-1](#), cells on different frequencies have the same cell center and similar cell edges. This means that user experience at the cell edge is poor. Cell splitting heterogeneous networking is recommended to meet high requirements on cell edge user experience. In this networking, the cell centers of cells on different frequencies are staggered. That is, the cell edge area of a cell on a frequency is aligned with the cell center area of a cell on a different frequency.

Figure 11-1 Traditional cell splitting networking and cell splitting heterogeneous networking



In cell splitting heterogeneous networking as shown in [Figure 11-2](#), SSB SINR-based inter-frequency handover can be used to hand over UEs from the edge of a cell on a frequency to the center of another cell on a different frequency. This reduces interference between intra-frequency cells and improves user experience and system capacity. SSB SINR-based inter-frequency handover is supported only by LampSite base stations in low frequency bands.

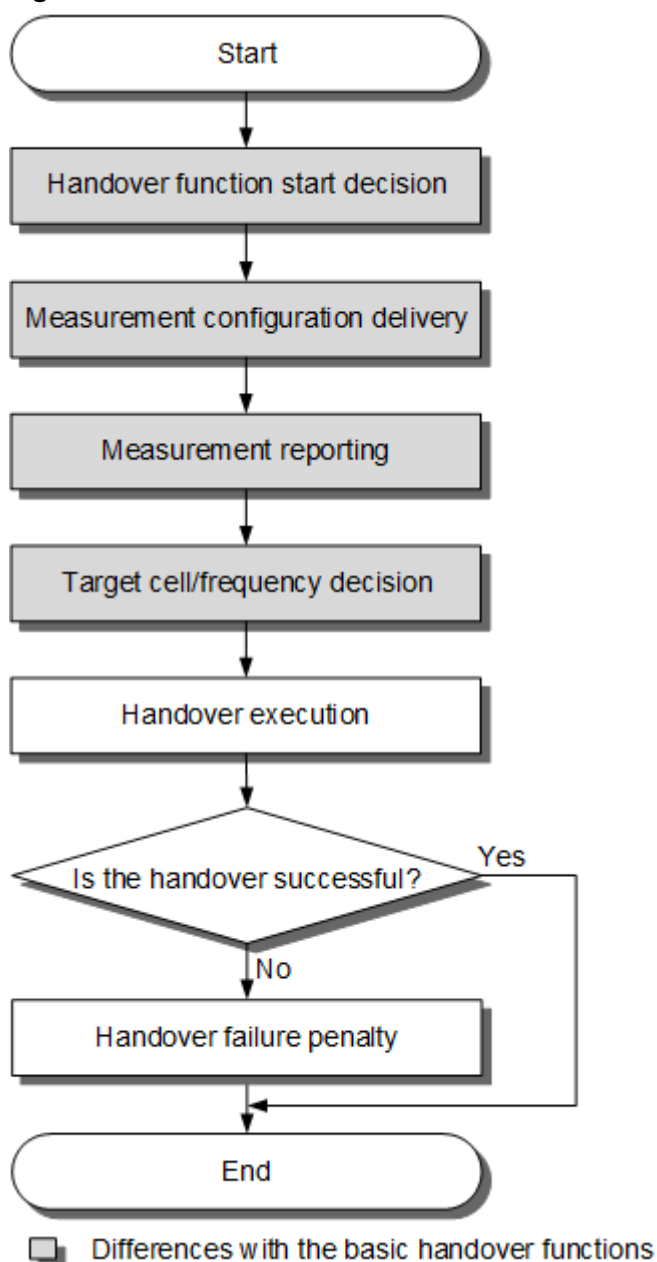
Figure 11-2 SSB SINR-based inter-frequency handover



11.1 Principles

SSB SINR-based inter-frequency handover supports only measurement-based mode but does not support blind mode. The handover policy supports only handover but does not support redirection. [Figure 11-3](#) shows the procedure of SSB SINR-based inter-frequency handover. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 11-3 Procedure of SSB SINR-based inter-frequency handover



11.1.1 Handover Function Start Decision

SSB SINR-based inter-frequency handover is started when all of the following conditions are met:

- The **SSB_SINR_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected.
- The gNodeB receives a measurement report indicating that the measured SINR of the serving cell meets the entering conditions for event A2. Event A2 indicates that the signal quality of the serving cell drops below a specific threshold. Event A2 involves SSB SINR measurements and the SINR threshold for event A2 is specified by the **NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA2Thld** parameter.

For details about the definition, entering conditions, and leaving conditions for event A2, see [Table 4-6](#). For the variable configurations for event A2, see [Table 4-8](#).

11.1.2 Measurement Configuration Delivery

After SSB SINR-based inter-frequency handover is started, the serving cell determines whether to deliver measurement configurations to the UE, based on the configuration of suppression of unnecessary handover for the UE's QCI-specific bearers. This process is referred to as QCI-specific suppression of unnecessary inter-frequency handover. The details are as follows:

- If the UE has a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB does not deliver measurement configurations to the UE.
- If the UE does not have a QCI-specific bearer for which the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter is selected, the gNodeB delivers measurement configurations to the UE.

NOTE

When SSB SINR-based inter-frequency handover is enabled for the serving cell, you are advised to select the **INTER_FREQ_HO_SUPPRESS_SW** option of the **gNBQciBearer.MobilityFunInd** parameter for each QCI-specific bearer associated with the target frequency group to avoid ping-pong handovers if service-based inter-frequency handover (specified by the **SERVICE_BASED_HO** option of the **NRCCellAlgoSwitch.InterFreqHoSwitch** parameter) is enabled for the target cell.

The gNodeB selects frequencies and delivers related measurement configurations based on the following principles:

- The gNodeB selects frequencies based on the following principles:
The gNodeB selects frequencies in descending order of priorities indicated by the **NRCCellFreqRelation.ConnFreqPriority** parameter until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.
- The gNodeB delivers measurement configurations related to events based on the following principles:
The gNodeB sends the measurement configuration to instruct the UE to stop event A2 measurements as well as the measurement configuration to instruct the UE to start event A3 and event A1 measurements.

Event A3 indicates that the signal quality of a neighboring cell is higher than that of the serving cell by a certain offset. Event A1 indicates that the signal quality of the serving cell exceeds a specific threshold. Event A3 involves SSB RSRP measurements, and event A1 involves SSB SINR measurements. For details about the variable configurations for event A3, see [Table 4-9](#). The **NRCCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld** parameter specifies the SINR threshold for event A1. For details about the configuration of other variables for event A1, see [Table 4-7](#).

11.1.3 Measurement Reporting

A UE performs measurements based on the event measurement configuration and reports measurement results.

- If event A3 is reported, the gNodeB determines the target cells or frequencies based on the measurement report.

If no neighboring cell is reported 6s after the event A3 measurement has started, the gNodeB releases the measurement configurations for events A3 and A1 to enable the UE to stop measurements for this event A3 and event A1, and starts the SSB SINR-based inter-frequency handover penalty timer specified by **gNB Mobility Comm Param. SsbSinrInterFHoPunishTimer**.

Before the inter-frequency handover penalty timer expires, if the UE always remains in the RRC_CONNECTED state, the gNodeB no longer sends measurement configurations for event A2. After the inter-frequency handover penalty timer expires, if the UE always remains in the RRC_CONNECTED state, the gNodeB re-sends measurement configurations for event A2, and after SSB SINR-based inter-frequency handover is started, the gNodeB sends measurement configurations for events A3 and A1.

- If event A1 is reported, the gNodeB sends the measurement configuration to instruct the UE to stop event A3 and event A1 measurements and the measurement configuration to instruct the UE to restart event A2 measurements. This means that the UE will stop measurements for this event A3 and event A1 and restart event A2 measurements.

11.1.4 Target Cell or Frequency Decision

After receiving an event A3 measurement report, the gNodeB generates a target cell or frequency list based on the filtering rules and selects the optimal cell from the reported cells as the target cell for handover execution.

where

- In addition to the filtering rules described in [4.1.6 Target Cell or Frequency Decision](#), the filtering rules dedicated to SSB SINR-based inter-frequency handover are also considered, which are described in [Filtering Rules Dedicated to SSB SINR-based Inter-Frequency Handover](#).
- Note that whether the gNodeB instructs a UE to perform a handover is also affected by the following functions: subscription to uplink interference information about inter-frequency neighboring cells, inter-frequency handover collaboration, anti-ping-pong function based on the inter-frequency handover waiting time, and anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time. For details about these functions, see [Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells](#), [Inter-Frequency Handover Collaboration](#), [Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time](#), and [Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time](#).

Filtering Rules Dedicated to SSB SINR-based Inter-Frequency Handover

If SSB SINR-based inter-frequency handover is enabled and measurement-based inter-frequency handover (see [6.1.1 Measurement-based Inter-Frequency](#)

Handover) or measurement-based inter-frequency redirection (see [6.1.2 Measurement-based Inter-Frequency Redirection](#)) in a coverage-related scenario is also enabled, the gNodeB further filters out the following cells to generate a target cell or frequency list.

- If only RSRP is used as the triggering quantity for event A2 related to coverage-based inter-frequency handover, the gNodeB filters out the cells whose RSRP is less than the result of **"NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld – NRCellInterFHoMeaGrp.InterFreqA1A2Hyst"**.
- If RSRP or RSRQ is used as the triggering quantity for event A2 related to coverage-based inter-frequency handover, the gNodeB filters out the following cells:
 - Cells whose RSRP is less than the result of **"NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld – NRCellInterFHoMeaGrp.InterFreqA1A2Hyst"**
 - Cells whose RSRQ is less than the result of **"NRCellInterFHoMeaGrp.CovInterFreqA2RsrqThld – NRCellInterFHoMeaGrp.InterFreqA1A2Hyst"**

The **NRCellMobilityConfig.MeasTrigQuantity** parameter specifies the triggering quantity for event A2 related to coverage-based inter-frequency handover. When both coverage-based inter-frequency handover and SSB SINR-based inter-frequency handover are enabled, the **NRCellMobilityConfig.MeasTrigQuantity** parameter cannot be set to **RSRP_OR_SINR**.

Subscription to Uplink Interference Information About Inter-Frequency Neighboring Cells

When the function of subscription to uplink interference information about inter-frequency neighboring cells is enabled (by selecting the **INTER_FREQ_UL_INTRF_SW** option of the **NRCellAlgoSwitch.NCellInfoCtrlSwitch** parameter) for the serving cell, the serving cell obtains uplink interference information about intra- or inter-base-station neighboring cells, based on which the serving cell determines whether to instruct the UE to initiate a handover. Specifically,

- The UE is instructed to perform a handover if the serving cell determines that the target cell is exposed to low uplink interference.
- Otherwise, the UE is not instructed to perform a handover.

NOTE

For details about how the serving cell determines whether a neighboring cell is exposed to low uplink interference, see [10.1.4 Target Cell or Frequency Decision](#).

If SSB SINR-based inter-frequency handover is enabled for the serving cell and uplink-interference-based inter-frequency handover is enabled for the target cell, you are advised to enable the function of subscription to uplink interference information about inter-frequency neighboring cells for the serving cell, avoiding ping-pong handovers.

Inter-Frequency Handover Collaboration

When the inter-frequency handover collaboration function is enabled (by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the

NRCellAlgoSwitch.InterFreqHoSwitch parameter) for the serving cell, the serving cell further determines whether to instruct the UE to initiate a handover as follows:

If the serving cell identifies that the target cell has entered the high load state through inter-cell load information exchange, the gNodeB does not instruct the UE to perform a handover to the target cell, avoiding ping-pong handovers.

The serving cell filters out the target cell corresponding to a high-load beam based on SSB beam information reported by the UE. That is, the gNodeB does not instruct the UE to perform a handover to the target cell.

NOTE

When SSB SINR-based inter-frequency handover is enabled for the serving cell, you are advised to enable the inter-frequency handover collaboration function for the serving cell to avoid ping-pong handovers if any of the following functions is enabled for the target cell:

Inter-frequency MLB by transferring UEs in connected mode (controlled by the **INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.MlbAlgoSwitch** parameter)

Multi-frequency beam coordination (controlled by the **MULTI_FREQ_BEAM_COORD_SW** option of the **NRDUCellFeatureSw.MultiCarrierSwitch** parameter)

For details about the high-load state and inter-frequency MLB by transferring UEs in connected mode, see *Mobility Load Balancing in Connected Mode*. For details about high-load beams and multi-frequency beam coordination, see *Multi-Frequency Smart Selection*.

Anti-Ping-Pong Function Based on the Inter-Frequency Handover Waiting Time

For a UE that is handed over to a cell (the current serving cell) through any handover function, a timer (specified by **NRCellMobilityConfig.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an SSB SINR-based inter-frequency handover before the timer expires. An unnecessary handover function enables a UE to be handed over only to the neighboring cell with the strongest signal strength indicated by the measurement report.

If the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific cell-load-dependent inter-frequency handover waiting time preferentially takes effect.

Anti-Ping-Pong Function Based on the QCI-specific Cell-Load-Dependent Inter-Frequency Handover Waiting Time

For a UE with "QCI-specific cell-load-dependent inter-frequency handover" in effect and handed over to a cell (the current serving cell) through any handover function, a timer (specified by **gNBScenarioDiffGrp.InterFreqHoWaitTime**) is started when the incoming handover is complete. The UE cannot be handed over back to the original serving cell through an SSB SINR-based inter-frequency handover before the timer expires.

If the anti-ping-pong function based on the inter-frequency handover waiting time is also enabled, the anti-ping-pong function based on the QCI-specific load-dependent inter-frequency handover waiting time takes precedence.

11.2 Network Analysis

11.2.1 Benefits

In cell splitting heterogeneous networking, SSB SINR-based inter-frequency handover reduces interference between intra-frequency cells and improves user experience and system capacity.

11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

This function results in the following network impacts:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- Networking of the live network and the impact relationships between SSB SINR-based inter-frequency handover and other handover functions (such as frequency-priority-based inter-frequency handover) must be considered to prevent ping-pong handovers and ensure user experience. For details about networking requirements, see [11.3.4 Networking](#).

11.3 Requirements

11.3.1 Licenses

There are no license requirements.

11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

11.3.2.1 Prerequisite Functions

None

11.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If coverage-based inter-frequency handover and SSB SINR-based inter-frequency handover are both enabled, the NRCellMobilityConfig.MeasTrigQuantity parameter cannot be set to RSRP_OR_SINR .
Low-frequency TDD	Inter-frequency redirection	REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If inter-frequency redirection and SSB SINR-based inter-frequency handover are both enabled, the NRCellMobilityConfig.MeasTrigQuantity parameter cannot be set to RSRP_OR_SINR .
High-frequency TDD	None	None	None	None

11.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch.AnrSwitch parameter	<i>ANR</i>	When both SSB SINR-based inter-frequency handover and intra-RAT ANR are enabled, intra-RAT ANR does not take effect in SSB SINR-based inter-frequency handover scenarios.

11.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

11.3.4 Networking

During network planning, if an unnecessary handover function needs to be enabled for multiple cells, a UE must be prevented from being handed over back to the source cell through unnecessary handovers of the same or different types. Otherwise, the UE experiences repeated handovers and user experience is poor. Handovers other than coverage-based intra-frequency handovers and coverage-based inter-frequency handovers are called unnecessary handovers. Following are examples:

- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an SSB SINR-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through an SSB SINR-based inter-frequency handover.
- During network planning, avoid the following situation: After a UE is handed over from cell 1 to cell 2 through an SSB SINR-based inter-frequency handover, the UE is handed over from cell 2 back to cell 1 through a frequency-priority-based handover.

11.4 Operation and Maintenance

11.4.1 Data Configuration

11.4.1.1 Data Preparation

The NR cell's frequency relationships and neighbor relationships must be configured before this function is activated. For details about the NR cell's frequency relationship configurations, see [Table 11-1](#). For details about neighbor relationship configurations, see [Table 4-20](#).

[Table 11-2](#) describes the function switches.

[Table 11-3](#) describes the parameters for other configurations of this function.

Table 11-1 Parameters used to configure frequency relationships for NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	NRCellFreqRelation.SsbDescMethod	Use the recommended value.
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Subcarrier Spacing	NRCellFreqRelation.SubcarrierSpacing	Set this parameter based on the network plan.
Frequency Priority for Connected Mode	NRCellFreqRelation.ConnFreqPriority	It is recommended that this parameter be set to a non-zero value for frequencies providing contiguous coverage, with a larger value set for a frequency in a lower frequency band. It is recommended that this parameter be set to 0 for frequencies providing discontinuous coverage, which means that such a frequency will not be selected as the target frequency.

Table 11-2 Inter-frequency handover algorithm switches

Parameter Name	Parameter ID	Setting Notes
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	The SSB_SINR_BASED_HO option determines whether to enable SSB SINR-based inter-frequency handover. Set this parameter based on the network plan.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	When SSB SINR-based inter-frequency handover is enabled, it is recommended that the INTER_FREQ_HO_COOPERATION_SW option be selected if inter-frequency MLB by transferring UEs in connected mode is also enabled.

Table 11-3 Other related parameters

Parameter Name	Parameter ID	Setting Notes
Inter-freq Handover Measurement Group ID	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId	Set this parameter based on the network plan.
SSB SINR Inter-freq HO A1 Threshold	NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA1Thld	Use the recommended value.
SSB SINR Inter-freq HO A2 Threshold	NRCellInterFHoMeaGrp.SsbSinrInterFreqHoA2Thld	Use the recommended value.
Inter-frequency Handover A3 Offset	NRCellInterFHoMeaGrp.InterFreqHoA3Offset	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Inter-freq HO Measurement Group ID	NRCellQciBearer.InterFreqHoMeas-GroupId	Set this parameter based on the network plan.
QoS Class Identifier	gNBQciBearer.Qci	Set this parameter based on the network plan.
QCI Priority for Handover	gNBQciBearer.QciPriorityForHo	Set this parameter based on the network plan.
Mobility Function Indication	gNBQciBearer.MobilityFunInd	The INTER_FREQ_HO_SUPPRESS_SW option specifies whether to enable QCI-specific suppression of unnecessary inter-frequency handover. When SSB SINR-based inter-frequency handover is enabled for the serving cell, you are advised to select this option for each QCI-specific bearer associated with the target frequency group if service-based inter-frequency handover is enabled for the target cell.
SSB SINR Inter-freq HO Punish Timer	gNBMobilityCommParam.SsbSinrInterFHoPunishTimer	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
Neighboring Cell Information Control Switch	NRCellAlgoSwitch.NCellInfoCtrlSwitch	The INTER_FREQ_UL_INTERF_SW option specifies whether to enable subscription to uplink interference information about inter-frequency neighboring cells. Use the recommended value.
Inter-Frequency Handover Wait Time	NRCellMobilityConfig.InterFreqHoWaitTime	Use the recommended value.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.
Inter-Frequency Handover Wait Time	gNBScenarioDiffGrp.InterFreqHoWaitTime	Set this parameter based on the network plan.

11.4.1.2 Using MML Commands

Before using MML commands, refer to [11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Adding a frequency relationship for an NR cell
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1;
//Adding an external cell
ADD NREXTERNALNCELL: Mcc="302", Mnc="220", gNBId=1, CellId=0, CellName="A", PhysicalCellId=0,
Tac=0, SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, FrequencyBand=N77,
NrNetworkingOption=UNLIMITED;
//Adding a relationship with the NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, CellIndividualOffset=DB0;
//Activating SSB SINR-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=SSB_SINR_BASED_HO-1;
```

Optimization Command Examples

TDD configuration examples

```
//Configuring a group of measurement parameters for inter-frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, SsbSinrInterFreqHoA1Thld=16, SsbSinrInterFreqHoA2Thld=10, InterFreqHoA3Offset=2;
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1,
InterFreqA4A5TimeToTrig=320MS, InterFreqA4A5Hyst=2, InterFreqA1A2TimeToTrig=320MS,
InterFreqA1A2Hyst=2, SsbSinrInterFreqHoA1Thld=16, SsbSinrInterFreqHoA2Thld=10, InterFreqHoA3Offset=2;
// (Optional) Modifying the parameter settings of QCI-specific measurement or QCI-specific cell-load-
dependent measurement
MOD NRCELLQCIBEARER: NrCellId=0, Qci=1, InterFreqHoMeasGroupId=0;
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, InterFreqHoMeasGroupId=1;
//Configuring QCI priorities for handovers
MOD GNBQCIBEARER: Qci=1, QciPriorityForHo=1;
MOD GNBQCIBEARER: Qci=9, QciPriorityForHo=7;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=0, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1, InterFreqHoWaitTime=10;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Enabling QCI-specific suppression of unnecessary inter-frequency handover for the QCI-specific bearers
associated with the target frequency group
MOD GNBQCIBEARER: Qci=1, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
MOD GNBQCIBEARER: Qci=9, MobilityFunInd=INTER_FREQ_HO_SUPPRESS_SW-1;
//Modifying the SSB SINR-based inter-frequency handover penalty timer (Infinity is used as an example.)
MOD GNBMOBILITYCOMMPARAM: SsbSinrInterFHOPunishTimer=INFINITY;
// (Recommended when SSB SINR-based inter-frequency handover is enabled for the serving cell while inter-
frequency MLB by transferring UEs in connected mode or multi-frequency beam coordination is enabled for
the target cell) Turning on the inter-frequency handover collaboration switch for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
// (Recommended when SSB SINR-based inter-frequency handover is enabled for the serving cell and uplink-
interference-based inter-frequency handover is enabled for the target cell) Turning on the switch for
subscription to uplink interference information about inter-frequency neighboring cells for the serving cell
MOD NRCELLALGOSWITCH: NrCellId=0, NCellInfoCtrlSwitch=INTER_FREQ_UL_INTRF_SW-1;
//Setting the inter-frequency handover waiting time
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqHoWaitTime=10;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Deactivating SSB SINR-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=SSB_SINR_BASED_HO-0;
```

11.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

11.4.2 Activation Verification

If any of the following counters have a value other than 0, SSB SINR-based inter-frequency handover has taken effect:

- **N.HO.InterFreq.SSB.SINR.PrepAttOut**
- **N.HO.InterFreq.SSB.SINR.ExecAttOut**
- **N.HO.InterFreq.SSB.SINR.ExecSuccOut**

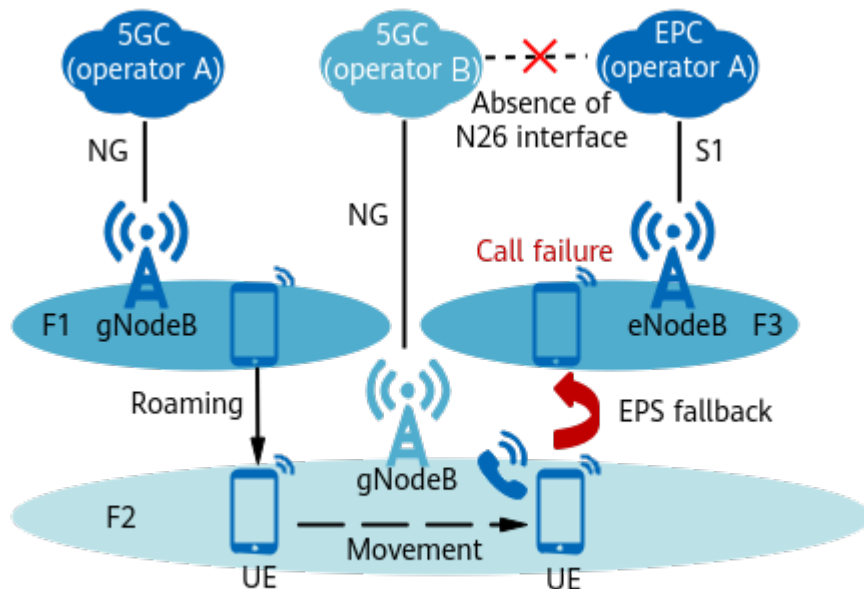
11.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)** KPI.

12 Voice-Capability-based Inter-Frequency Handover Back to the HPLMN

In the scenario shown in [Figure 12-1](#), a VoNR-incapable UE of operator A roams from a 5G cell of operator A to a 5G cell of operator B. Operators A and B are referred to as the home and visited operators, respectively. When the UE is paged in the visited 5G cell, the UE will experience an EPS fallback to a 4G cell of the home operator. In this scenario, if no N26 interface is configured between the home EPC and the visited 5GC, the home EPC cannot obtain the UE context from the visited 5GC over the N26 interface, resulting in a call failure.

Figure 12-1 Call failure for a VoNR-incapable UE of the home operator



Note: The UE in the figure is a VoNR-incapable UE of operator A. Operator A provides the home network and operator B provides the visited network.

The **VOICE_INTER_FREQ_HO_POLICY_SW** option of the **NRCellFreqRelation.AggregationAttribute** parameter can be selected for the visited operator's 5G frequency in the home 5G cell to prohibit VoNR-incapable UEs of the home operator from roaming to the visited 5G cell from the home 5G cell. For details, see [4.1.4.1.2 Frequency Selection for Measurement](#). However,

selecting this option only prohibits VoNR-incapable UEs in RRC_CONNECTED mode of the home operator from roaming to the visited 5G cell from the home 5G cell through handovers. In this situation, VoNR-incapable UEs in RRC_IDLE mode of the home operator can still roam to the visited 5G cell through cell selection or reselection. As such, the preceding call failures cannot be fully avoided.

To further avoid such call failures, VoNR-incapable UEs in RRC_CONNECTED mode of the home operator that have roamed to the visited 5G cell can be proactively handed over from the visited 5G cell back to the home 5G cell before they are paged. To achieve this, Huawei introduces the voice-capability-based inter-frequency handover back to the HPLMN function, which is controlled by the **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator.HplmnAlgoSwitch** parameter. Currently, this function applies only to low-frequency scenarios. Here, the HPLMN is short for home public land mobile network, indicating the subscribed home operator of a UE.

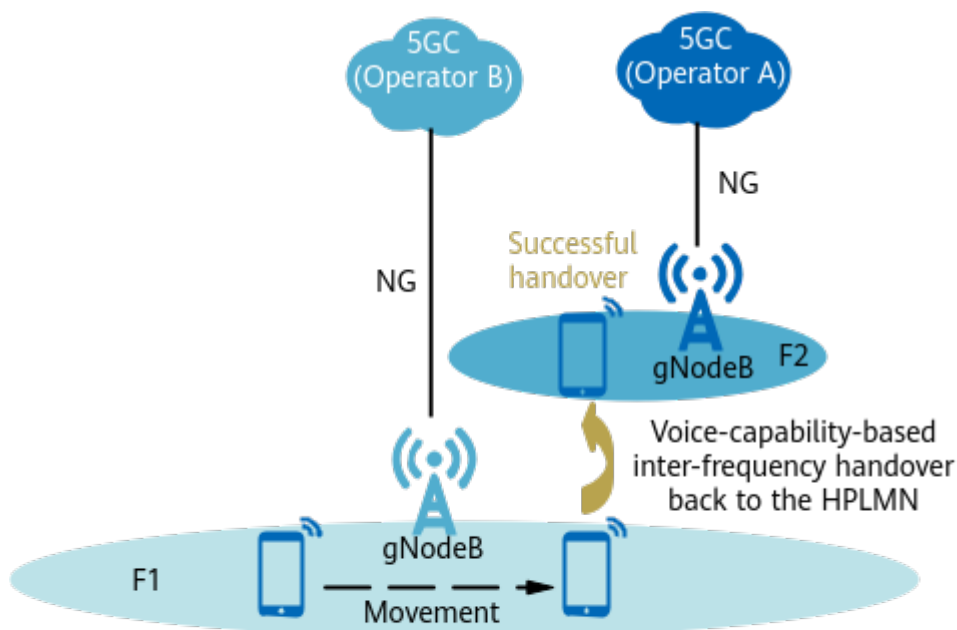
NOTE

For a VoNR-incapable UE in RRC_IDLE mode of the home operator that has roamed to a visited 5G cell, the UE will experience a redirection from the visited 5G cell to the home 4G cell when it initiates a voice call, resulting in a call failure. This occurs because no N26 interface is configured between the home 4G core network and the visited 5GC.

As shown in [Figure 12-2](#), the voice-capability-based inter-frequency handover back to the HPLMN function enables a VoNR-incapable UE of the home operator to be quickly handed over from a visited 5G cell back to its home 5G cell belonging to the HPLMN. It is recommended that this function be enabled on the local network for the home operator of roaming UEs when the coverage of the roaming UEs' home network overlaps with that of the local network. In addition, this function requires that the local operator and the roaming UEs' home operator work in RAN sharing with common carrier mode of multi-operator sharing and the serving cell for the roaming UEs be a shared cell.

To prevent ping-pong handovers when this function is enabled on the local network for the home operator of roaming UEs, it is recommended that the **VOICE_INTER_FREQ_HO_POLICY_SW** option of the **NRCellFreqRelation.AggregationAttribute** parameter be selected for the home 5G cell.

Figure 12-2 Voice-capability-based inter-frequency handover back to the HPLMN

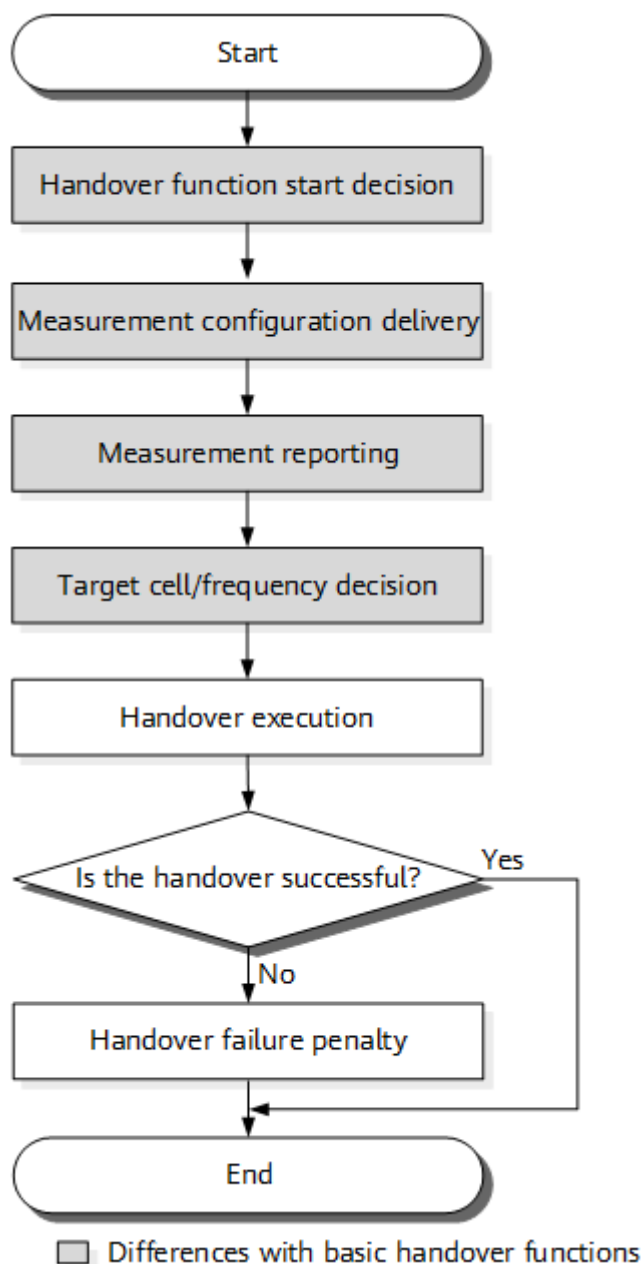


Note: The UE in the figure does not support VoNR. Operator A is the UE's home operator and operator B is the UE's visited operator.

12.1 Principles

Voice-capability-based inter-frequency handover back to the HPLMN supports only measurement-based mode but does not support blind mode. The handover policy supports both handover and redirection. [Figure 12-3](#) shows the procedure of voice-capability-based inter-frequency handover back to the HPLMN. This section describes the differences between this function and the basic handover functions presented in [4 Basic Functions of Mobility Management in Connected Mode](#).

Figure 12-3 Procedure of voice-capability-based inter-frequency handover back to the HPLMN



12.1.1 Handover Function Start Decision

Voice-capability-based inter-frequency handover back to the HPLMN is started when all of the following conditions are met:

- The gNodeB identifies a UE as a roaming UE.
The selected PLMN of the UE is the same as the roaming PLMN configured for the UE's home network on the base station side, and the **gNBOperator.RoamingPlmnFlag** parameter is set to **TRUE** for the roaming PLMN. Here, the selected PLMN is a PLMN selected by the UE in the wireless network.

For example, the PLMN of operator A is "Mcc: 460, Mnc: 00", and the roaming PLMN configured for operator B is "Mcc: 460, Mnc: 50" under the base station of operator A. This example assumes that a UE of operator B accesses the base station of operator A, and **gNBOperator.RoamingPlmnFlag** is set to **TRUE** for the roaming PLMN ("Mcc: 460, Mnc: 50") of operator B under the base station. If the selected PLMN of the UE is "Mcc: 460, Mnc: 50", the base station of operator A identifies the UE as a roaming UE.

- On the local gNodeB, the **VOICE_CAPB_MOB_TO_HPLMN_SW** option of the **gNBOperator.HplmnAlgoSwitch** parameter is selected for the home operator of the roaming UE.
- The roaming UE does not support VoNR.

For details about whether a UE supports VoNR, see section 4.2.13 "IMS Parameters" in 3GPP TS 38.306 V15.5.0.

NOTE

The gNodeB queries the VoNR capability of a UE only in scenarios such as initial access or inter-gNodeB incoming handovers. As such, the following occurs:

- If the VoNR status of a roaming VoNR-capable UE in connected mode is manually changed from enabled to disabled, the visited gNodeB cannot sense the UE's VoNR capability change and retains the UE's VoNR status from the last UE capability query. Consequently, the roaming UE may be mistakenly identified as supporting VoNR and this function cannot take effect, resulting in a call failure in the visited 5G cell.
- If the VoNR status of a roaming VoNR-capable UE in connected mode is manually changed from disabled to enabled, the visited gNodeB cannot sense the UE's VoNR capability change and retains the UE's VoNR status from the last UE capability query. Consequently, the roaming UE may be mistakenly identified as not supporting VoNR and this function takes effect. In this case, the UE is handed over or redirected back to the home 5G cell, failing to acquire user experience of the visited 5G network.

The PLMN IDs of multiple operators are broadcast only in a shared cell working in RAN sharing with common carrier mode in a multi-operator sharing scenario. As such, this function requires that the local operator and home operator of a roaming UE work in RAN sharing with common carrier mode in a multi-operator sharing scenario and that the roaming UE's serving cell be a shared cell. For details about multi-operator sharing, see *Multi-Operator Sharing*.

12.1.2 Measurement Configuration Delivery

After voice-capability-based inter-frequency handover back to the HPLMN is enabled, the gNodeB selects frequencies and delivers related measurement configurations based on the following principles:

- The gNodeB selects frequencies based on the following principles:
The gNodeB selects the frequencies in the frequency list group bound to the roaming UE and with the **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** parameter set to **TRUE** as target frequencies, and further selects these frequencies in descending order of priorities indicated by the **NRCellFreqRelation.ConnFreqPriority** parameter until the number of selected frequencies reaches the maximum. For details about the maximum number of frequencies to be delivered, see [4.1.4.1.2 Frequency Selection for Measurement](#). If multiple frequencies have the same priority, the NR frequency with a smaller *SsbFreqPos* value is preferentially selected.

- The gNodeB delivers measurement configurations related to events based on the following principles:

The gNodeB delivers inter-frequency event A5 measurement configurations to the UE based on the selected target frequencies.

Event A5 indicates that the signal quality of the serving cell becomes lower than threshold 1 and the signal quality of a neighboring cell becomes higher than threshold 2. The variables related to event A5 are set as follows:

Table 12-1 Variable configuration of event A5

Variable	Parameter Name	Parameter ID	Description
Hys	Inter-freq Measurement Event Hysteresis	None	This variable is fixed at 1 dB.
Thresh1	Voice-Capb-based Inter-freq HO Back to HPLMN A5 RSRP Threshold1	None	This variable is fixed at -43 dBm.
Thresh2	Cov Inter-freq Handover A5 RSRP Threshold2	NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld2	Threshold 2 for event A5 related to voice-capability-based inter-frequency handover back to the HPLMN equals the sum of the values of NRCeIIInterFHoMeaGrp.CovInterFreqA5RsrpThld2 and gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs .
	Inter-freq A5 RSRP Threshold2 Offset	gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs	
Ofn	Frequency Offset for Connected Mode	NRCeIIInterFHoMeaGrp.ConnFreqOffset	This variable represents the connected-mode frequency-specific offset of a non-serving frequency and is configured through this parameter.

Variable	Parameter Name	Parameter ID	Description
Ocn	Cell Individual Offset	NRCellRelation. CellIndividualOffset	This variable represents the CIO of a neighboring cell. • If the value is not DB0, this parameter is included in measurement configurations. • If the value is DB0, this parameter is not included in measurement configurations and 0 is used for calculation by default.
TimeToTrig	Inter-freq Measurement Event Time to Trigger	None	This variable is fixed at 320 ms.

12.1.3 Measurement Reporting

The roaming UE performs measurements based on the event configuration included in the measurement configurations. If neighboring cells that meet the requirements are identified, a measurement report is sent to the gNodeB, which then uses the report to determine the target cells or frequencies.

If no qualified neighboring cell is identified within 6s after an event A5 measurement starts, the gNodeB removes the measurement configurations so that the roaming UE stops the event A5 measurement. In this case, the gNodeB delivers event A5 measurement configurations to the roaming UE at an interval of 90s.

NOTE

If a roaming VoNR-incapable UE of the home operator initiates a voice call on the visited 5G network but the UE does not identify the home operator's 5G coverage when measuring the home operator's 5G frequencies, the call fails.

12.1.4 Target Cell or Frequency Decision

Upon receiving an event A5 measurement report, the gNodeB extracts frequency information from the report and determines the target cells.

If the cell's PLMN in the measurement report is the home PLMN of the roaming UE, the gNodeB determines that the cell can be a target cell.

The gNodeB instructs the roaming UE to initiate handover or redirection attempts towards the cells that meet the preceding requirements one by one, which are

sorted in the order that measurement reports are submitted as well as in descending order of RSRP of these cells.

- When the **gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag** parameter is set to **TRUE** for the target cell's frequency in the frequency list group bound to the roaming UE, the following mechanism is applied:
 - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell.
 - In the case that the roaming UE does not support handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency.
- When the **gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag** parameter is set to **FALSE** for the target cell's frequency in the frequency list group bound to the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency.

The gNodeB uses the following procedure to determine whether a cell's PLMN in the measurement report is the home PLMN of a roaming UE:

1. The gNodeB acquires the cell's PLMN (including the MCC and MNC) based on the mapping from the cell's PCI in the measurement report from the neighboring cell list (NCL).
2. The gNodeB compares the cell's PLMN in the measurement report with the home PLMN configured for the roaming UE on the gNodeB side, and determines whether the cell's PLMN is the home PLMN of the roaming UE.

The HPLMN of a roaming UE is specified by the following parameters:

- Operator ID: **gNBRoamingUeHomePlmn.OperatorId**
- Home mobile country code: **gNBRoamingUeHomePlmn.HomeMcc**
- Home mobile network code: **gNBRoamingUeHomePlmn.HomeMnc**

12.2 Network Analysis

12.2.1 Benefits

When the home network coverage for roaming UEs overlaps with the local network coverage, it is recommended that the function be activated on the local network for the home operator of roaming UEs. This enables roaming VoNR-incapable UEs to return to their home network as soon as possible and reduces the number of call failures of roaming VoNR-incapable UEs, improving their network experience.

12.2.2 Impacts

The impacts between this function and other functions are described in [12.3.2.3 Function Impacts](#).

It is recommended that this function be enabled on the local network for a roaming UE's home operator when the home network coverage area of the roaming UE overlaps with the local network coverage area. The network impacts of the function are as follows:

- During a handover, the UE synchronizes with the target cell and initiates random access. This process introduces a delay due to service interruption, affecting data rates.
- Inter-frequency measurements require the use of measurement gaps, leading to decreased throughput.
- The number of outgoing EPS fallback preparation attempts in the coverage area of 5G cells on the local network decreases.

12.3 Requirements

12.3.1 Licenses

There are no license requirements.

12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

12.3.2.1 Prerequisite Functions

None

12.3.2.2 Mutually Exclusive Functions

None

12.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	• COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell • REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for coverage-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: • When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: – In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. – In the case that the roaming UE does not support handovers: If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected, the gNodeB instructs the roaming UE to immediately perform

RAT	Function Name	Function Switch	Reference	Description
				a redirection to the frequency. If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is deselected, coverage-based inter-frequency handover does not take effect for the roaming UE. • When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: – If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency. – If the REDIRECTION option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is deselected, coverage-based inter-frequency handover does not take effect for the roaming UE. In the decision-making phase for coverage-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover

RAT	Function Name	Function Switch	Reference	Description
				back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the COVERAGE_BASED_HO and REDIRECTION options of the NRCellAlgoSwitch.InterFreqHoSwitch parameter are respectively deselected and selected for the serving cell, then the gNodeB will instruct the roaming UE to immediately perform a redirection to the target cell's frequency.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-speed UE outmigration	NRCellMobilityConfig.LowSpeedUeOutgoingPolicy parameter for the serving cell	<i>High Speed Mobility</i>	In the handover decision-making phase for low-speed UE outmigration in a high-speed cell, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the NRCellMobilityConfig.LowSpeedUeOutgoingPolicy parameter is set to MEASUREMENT_BASED for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target cell's frequency in the

RAT	Function Name	Function Switch	Reference	Description
				frequency group associated with the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency. In the handover decision-making phase for low-speed UE outmigration in a high-speed cell, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the NRCellMobilityConfig.LowSpeedUeOutgoing-Policy parameter is set to BLIND_REDIRECTION_BASED for the serving cell, then the gNodeB instructs the roaming UE to immediately perform a blind redirection to the target cell's frequency.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for frequency-priority-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, frequency-priority-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, frequency-priority-based inter-frequency handover does not take effect for the roaming UE.
Low-frequency TDD	Operator-specific-priority-based inter-frequency handover	OP_DED_PRI_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>SA Mobility Management in Connected Mode</i>	If (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the OP_DED_PRI_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then voice-capability-based inter-frequency handover back to the HPLMN takes effect while operator-specific-priority-based inter-frequency handover does not.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Service-based inter-frequency handover	SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for service-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, service-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to FALSE for the target cell's frequency in the

RAT	Function Name	Function Switch	Reference	Description
				frequency group associated with the roaming UE, service-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RFSP-based inter-frequency handover back to the HPLMN	BACK_TO_HPLMN_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.RfspAlgorithmSwitch parameter for the serving cell	<i>Flexible User Steering</i>	In the decision-making phase for RFSP-based inter-frequency handover back to the HPLMN, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the BACK_TO_HPLMN_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.InterFrequencyHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Protection of UEs under weak uplink coverage	UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter for the serving cell	<i>Multi-Frequency Smart Selection</i>	In the handover decision-making phase for protection of UEs under weak uplink coverage, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the UL_WEAK_COV_UE_PROTECT_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink-interference-based inter-frequency handover	UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for uplink-interference-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, the gNodeB instructs the roaming UE to immediately perform a redirection to the frequency. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target

RAT	Function Name	Function Switch	Reference	Description
				cell's frequency in the frequency group associated with the roaming UE, the gNodeB instructs the roaming UE to immediately perform a redirection to the target cell's frequency.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intelligent carrier shutdown	INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter for the serving cell	<i>Energy Conservation and Emission Reduction</i>	In the handover decision-making phase for intelligent carrier shutdown where the specific timer has not expired, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, intelligent carrier shutdown-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
				When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target cell's frequency in the frequency group that is associated with the roaming UE, intelligent carrier shutdown-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast carrier shutdown	- For capacity-layer cells: INTRA_GNB_MULTICAST_W or INTER_GNB_MULTICAST_W option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter FAST_CARRIER_SHUTDOWN_SW option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch parameter - For basic-layer cells: FAST_CARRIER_SHUTTING_ASSURANCE_W option of the NRDUCellAlgoSwitch.PowerSavingExtSwitch	<i>Energy Conservation and Emission Reduction</i>	In the handover decision-making phase for fast carrier shutdown where the specific timer has not expired, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the INTRA_GNB_MULTICAST_W or INTER_GNB_MULTICAST_W option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, fast carrier shutdown-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.R

RAT	Function Name	Function Switch	Reference	Description
		<i>h</i> parameter		<i>oamingUeHplmnInterF-HoFlag</i> parameter is set to FALSE for the target cell's frequency in the frequency group that is associated with the roaming UE, fast carrier shutdown-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SWITCH option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter for the serving cell	<i>CA Experience Improvement</i>	In the decision-making phase for CA-UE-specific frequency-priority-based inter-frequency handover, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, CA-UE-specific frequency-priority-based inter-frequency handover does not take effect for the roaming UE. - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-

RAT	Function Name	Function Switch	Reference	Description
				HoFlag parameter is set to FALSE for the target cell's frequency in the frequency group associated with the roaming UE, CA-UE-specific frequency-priority-based inter-frequency handover does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dedicated cell reselection priority	MO gNBFreqPriorityGroup	SA Mobility Management in Idle Mode Multi-Operator Sharing Flexible User Steering	• In roaming scenarios, if there is a VoNR-incapable UE of the home operator and the VOICE_INTER_FREQ_HO_POLICY_SW option of the NRCellFreqRelation.AggregationAttribute parameter is selected for a visited network's frequency in the home network's cell, the home network's gNodeB will not send the UE an RRCRelease message containing this frequency, and the UE will not reselect this frequency based on dedicated cell reselection priorities, including operator-specific and RFSP-specific ones. • In roaming scenarios, if there is a VoNR-incapable UE of the home operator and the VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgorithmSwitch parameter is selected for the home network on the visited network's gNodeB, the gNodeB will send the UE an RRCRelease message. The message contains the frequencies for which the gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag parameter is set to TRUE in the frequency list group bound to the roaming UE on the gNodeB side. In this case, the UE can reselect one

RAT	Function Name	Function Switch	Reference	Description
				of these frequencies based on dedicated cell reselection priorities, including operator-specific and RFSP-specific ones.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF module fast deep dormancy	FAST_DEEP_DORM_SW option of the RRUEXTINFO.RRUDOROPTFEATSW parameter	<i>Multi-RAT Coordinated Energy Saving</i>	In the handover decision-making phase for RF module fast deep dormancy where the specific timer has not expired, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the INTRA_GNB_MULTI_CARR_SD_SW or INTER_GNB_MULTI_CARR_SD_SW option of the NRDUCellAlgoSwitch.PowerSavingSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. - In the case that the roaming UE does not support handovers, inter-frequency handover for RF module fast deep dormancy does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
				When the gNBFreqPriorityGroup.RoamingUeHplmnInterF-HoFlag parameter is set to FALSE for the target cell's frequency in the frequency group associated with the roaming UE, inter-frequency handover for RF module fast deep dormancy does not take effect for the roaming UE.

12.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

12.3.4 Networking

This function enables a VoNR-incapable UE that has roamed to a 5G cell of the local network to be quickly handed over back to a 5G cell of its home network. As such, this function requires that the coverage of the roaming UE's home network overlap with that of the local network.

The PLMN IDs of multiple operators are broadcast only in a shared cell working in RAN sharing with common carrier mode in a multi-operator sharing scenario. As

such, this function requires that the local operator and home operator of a roaming UE work in RAN sharing with common carrier mode in a multi-operator sharing scenario and that the roaming UE's serving cell be a shared cell. For details about multi-operator sharing, see *Multi-Operator Sharing*.

In the frequency list group associated with a roaming UE, there must be a frequency for which the **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** parameter is set to **TRUE** and this frequency must be a neighboring frequency of the serving frequency.

If the handover policy used by a roaming UE is handover instead of redirection, the N14 interface must be available between the home operator's 5GC and the visited operator's 5GC.

12.4 Operation and Maintenance

12.4.1 Data Configuration

12.4.1.1 Data Preparation

Table 12-2 lists the parameters used for function activation.

Table 12-2 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag	Set this parameter based on the network plan.
Roaming UE HPLMN Inter-Freq Handover Flag	gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag	Set this parameter based on the network plan.
Roaming PLMN Flag	gNBOperator.RoamingPlmnFlag	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
HPLMN Algorithm Switch	<i>gNBOperator.HplmnAlgoSwitch</i>	The VOICE_CAPB_MOB_TO_HPLMN_SW option specifies whether to enable voice-capability-based inter-frequency handover back to the HPLMN. It is recommended that this option be selected on the local network for the home operator of roaming UEs when the coverage of the roaming UEs' home network overlaps with that of the local network. To prevent ping-pong handovers when this option is selected on the local network for the home operator of roaming UEs, it is recommended that the VOICE_INTER_FREQ_HO_POLICY_SW option of the NRCellFreqRelation.AggregationAttribute parameter be selected for the home 5G cell.
Operator ID	<i>gNBRoamingUeHomePlmn.OperatorId</i>	These parameters specify the home PLMN of a roaming UE. Set these parameters based on the network plan.
Home Mobile Country Code	<i>gNBRoamingUeHomePlmn.HomeMcc</i>	
Home Mobile Network Code	<i>gNBRoamingUeHomePlmn.HomeMnc</i>	

Parameter Name	Parameter ID	Setting Notes
Cov Inter-freq Handover A5 RSRP Threshold2	NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2	Threshold 2 for event A5 related to voice-capability-based inter-frequency handover back to the HPLMN equals the sum of the values of NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 and gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs .
Inter-freq A5 RSRP Threshold2 Offset	gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs	
QoS Class Identifier	NRCellQciBearer.Qci	Set this parameter based on the network plan.
Handover Measure Algorithm Switch	NRCellMobilityConfig.HoMeasAlgoSwitch	It is recommended that the HIGH_LOAD_DIFF_HO_POL_SW option be selected in scenarios where UEs are largely stationary, network load is high, and overlapping cell coverage occurs (for example, during concerts).
High Load Differentiated HO UE Num Threshold	NRCellMobilityConfig.HighLoadDiffHoUeNumThld	Set this parameter based on the network plan.
High Load Differentiated HO DL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoDlPrbThld	Set this parameter based on the network plan.
High Load Differentiated HO UL PRB Threshold	NRCellMobilityConfig.HighLoadDiffHoUlPrbThld	Set this parameter based on the network plan.
Scenario Differentiated Group ID	NRCellQciBearer.ScenarioDiffGrpId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Scenario Differentiated Group ID	gNBScenarioDiffGrp.ScenarioDiffGrpId	Set this parameter based on the network plan.
Scenario Type	gNBScenarioDiffGrp.ScenarioType	Set this parameter to HIGH_LOAD when high-load-specific handover is required.
Operator ID	gNBScenarioDiffGrp.OperatorId	Set this parameter based on the network plan.
Inter-freq Handover Measurement Group ID	gNBScenarioDiffGrp.InterFreqHoMeasureGroupId	Set this parameter based on the network plan.

12.4.1.2 Using MML Commands

Before using MML commands, refer to [12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

The following configuration examples assume that the local operator and home operator of a roaming UE work in RAN sharing with common carrier mode in a multi-operator sharing scenario. For details about the configuration example related to this mode, see *Multi-Operator Sharing*.

```
//The configuration examples are based on the following assumptions: The local operator and home operator of a roaming UE work in RAN sharing with common carrier mode in a multi-operator sharing scenario and the roaming UE's serving cell is a shared cell. The operator IDs of the local network and home network of the roaming UE are 0 and 1, respectively. The local network's PLMN is "Mcc: 460, Mnc: 00". The roaming PLMN configured for the home network of the roaming UE is "Mcc: 460, Mnc: 50". The home network's PLMN of the roaming UE (also called the home PLMN of the roaming UE) is "Mcc: 460, Mnc: 03".
//Configuration on the local network: Adding a gNodeB frequency priority group (given an assumption that the NR frequency configured by using the following MML command is the NR frequency of the roaming UE's home network)
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=NR, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN, RoamingUeHplmnFreqFlag=TRUE, RoamingUeHplmnInterFHoFlag=TRUE;
//Configuration on the local network: Adding an operator-specific policy for the NR cell (adding a gNodeB
```

```

frequency priority group for the roaming UE's home network)
ADD NRCELLPOLICY: NrCellId=0, OperatorId=1, gNBFreqPriorityGroupId=0;
//Configuration on the local network: Adding an NR cell frequency relationship (given an assumption that
event A5 is used to trigger handover execution)
ADD NRCELLFREQUENCYRELATION: NrCellId=0, SsbFreqPos=8000, SsbDescMethod=SSB_DESC_TYPE_GSCN,
FrequencyBand=N77, SubcarrierSpacing=30KHZ, ConnFreqPriority=1, InterFreqHoEventType=EVENT_A5;
//Configuration on the local network: Setting RSRP threshold 2 for event A5 related to coverage-based inter-
frequency handovers
MOD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=0, CovInterFreqA5RsrpThld2=-106;
ADD NRCELLINTERFHOMEAGRP: NrCellId=0, InterFreqHoMeasGroupId=1, CovInterFreqA5RsrpThld2=-106;
//Configuration on the local network: Setting the offset of RSRP threshold 2 for inter-frequency event A5
ADD GNBOPERATORQCIPARAM: OperatorId=1, Qci=9, InterFreqA5RsrpThld2Ofs=1;
//Enabling high-load-specific handover
//Turning on the switch for high-load-specific handover and setting the related thresholds
MOD NRCELLMOBILITYCONFIG: NrCellId=0, HoMeasAlgoSwitch=HIGH_LOAD_DIFF_HO_POL_SW-1,
HighLoadDiffHoUeNumThld=100, HighLoadDiffHoDlPrbThld=20, HighLoadDiffHoUlPrbThld=80;
//Configuring a gNodeB scenario differentiation group
ADD GNBSCENARIODIFFGRP: ScenarioDiffGrpId=0, OperatorId=1, ScenarioType=HIGH_LOAD,
InterFreqHoMeasGroupId=1;
//Associating the gNodeB scenario differentiation group with a QCI
MOD NRCELLQCIBEARER: NrCellId=0, Qci=9, ScenarioDiffGrpId=0;
//Configuration on the local network: Adding operator information (enabling the voice-capability-based
inter-frequency handover back to the HPLMN function for the roaming UE's home network)
ADD GNBOPERATOR: OperatorId=0, OperatorName="Operator A", Mcc="460", Mnc="00",
OperatorType=PRIMARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=FALSE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;
ADD GNBOPERATOR: OperatorId=1, OperatorName="Operator B", Mcc="460", Mnc="50",
OperatorType=SECONDARY_OPERATOR, NrNetworkingOption=SA, RoamingPlmnFlag=TRUE,
HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-1;
//Configuration on the local network: Adding home PLMN information for a roaming UE
ADD GNBROAMINGUEHOMEPLMN: OperatorId=1, HomeMcc="460", HomeMnc="03";

```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```

//Deactivating voice-capability-based inter-frequency handover back to the HPLMN
MOD GNBOPERATOR: OperatorId=1, HplmnAlgoSwitch=VOICE_CAPB_MOB_TO_HPLMN_SW-0;

```

12.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

12.4.2 Activation Verification

The following verification procedure assumes that in SA networking, a roaming UE is handed over from the visited NR cell to the home NR cell through the function of voice-capability-based inter-frequency handover back to the HPLMN, and the handover policy is handover.

- Step 1** Enable a VoNR-incapable UE on the home network to roam to a visited 5G network, with services running on the UE. Then, trigger a voice-capability-based inter-frequency handover for the UE to make it return to the HPLMN.

- Step 2** On the MAE-Access for the visited operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **Uu Interface Trace** in the navigation pane of the page. In the displayed **Uu Interface Trace** page, check that the to-be-measured frequency carried in inter-frequency event A5 measurement configurations in the gNodeB-delivered RRCReconfiguration (MC) message is the serving frequency's neighboring NR frequency on the home network, which is configured by running the **ADD GNBREQPRIORITYGROUP** command. In addition, the **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** parameter is set to **TRUE** for the to-be-measured frequency.
- Step 3** On the MAE-Access for the visited operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **NG Interface Trace** in the navigation pane of the page. In the displayed **NG Interface Trace** page, check that the gNodeB receives a HANDOVER COMMAND message from the 5GC.
- Step 4** On the MAE-Access for the visited operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **Uu Interface Trace** in the navigation pane of the page. In the displayed **Uu Interface Trace** page, check that the target cell's frequency carried in the gNodeB-delivered RRCReconfiguration(HO) message is the serving frequency's neighboring NR frequency on the home network, which is configured by running the **ADD GNBREQPRIORITYGROUP** command. In addition, the **gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag** parameter is set to **TRUE** for the target cell's frequency.
- Step 5** On the MAE-Access for the home operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **Uu Interface Trace** in the navigation pane of the page. In the displayed **Uu Interface Trace** page, check that the gNodeB receives an RRCConnectionReconfigurationComplete message from the UE, meaning that voice-capability-based inter-frequency handover back to the HPLMN has taken effect.

----End

The following verification procedure assumes that in SA networking, a roaming UE is handed over from the visited NR cell to the home NR cell through the function of voice-capability-based inter-frequency handover back to the HPLMN, and the handover policy is redirection.

- Step 1** Enable a VoNR-incapable UE on the home network to roam to a visited 5G network, with services running on the UE. Then, trigger a voice-capability-based inter-frequency handover for the UE to make it return to the HPLMN.
- Step 2** On the MAE-Access for the visited operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **Uu Interface Trace** in the navigation pane of the page. In the displayed **Uu Interface Trace** page, check the following settings:
1. The to-be-measured frequency carried in inter-frequency event A5 measurement configurations in the gNodeB-delivered RRCReconfiguration(MC) message is the serving frequency's neighboring NR frequency on the home network, which is configured by running the **ADD GNBREQPRIORITYGROUP** command. In addition, the

gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag parameter is set to **TRUE** for the to-be-measured frequency.

2. The target cell's frequency carried in the gNodeB-delivered RRCRelease message is the serving frequency's neighboring NR frequency on the home network, which is configured by running the **ADD GNBREQPRIORITYGROUP** command. In addition, the **gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag** parameter is set to **FALSE** for the target cell's frequency.

Step 3 On the MAE-Access for the home operator, choose **Monitor > Signaling Trace > Signaling Trace Management**. The **Signaling Trace Management** page is displayed. Choose **Uu Interface Trace** in the navigation pane of the page. In the displayed **Uu Interface Trace** page, check that the gNodeB receives an RRCConnectionSetupRequest message from the UE, meaning that voice-capability-based inter-frequency handover back to the HPLMN has taken effect.

----End

12.4.3 Network Monitoring

When the coverage of the roaming UEs' home network overlaps with that of the local network, activating the function on the local network for the home operator of roaming UEs enables roaming VoNR-incapable UEs to return to their home network as soon as possible. This reduces the number of call failures of roaming VoNR-incapable UEs, improving their network experience. Currently, this function does not involve performance monitoring on the network side.

13 Appendix

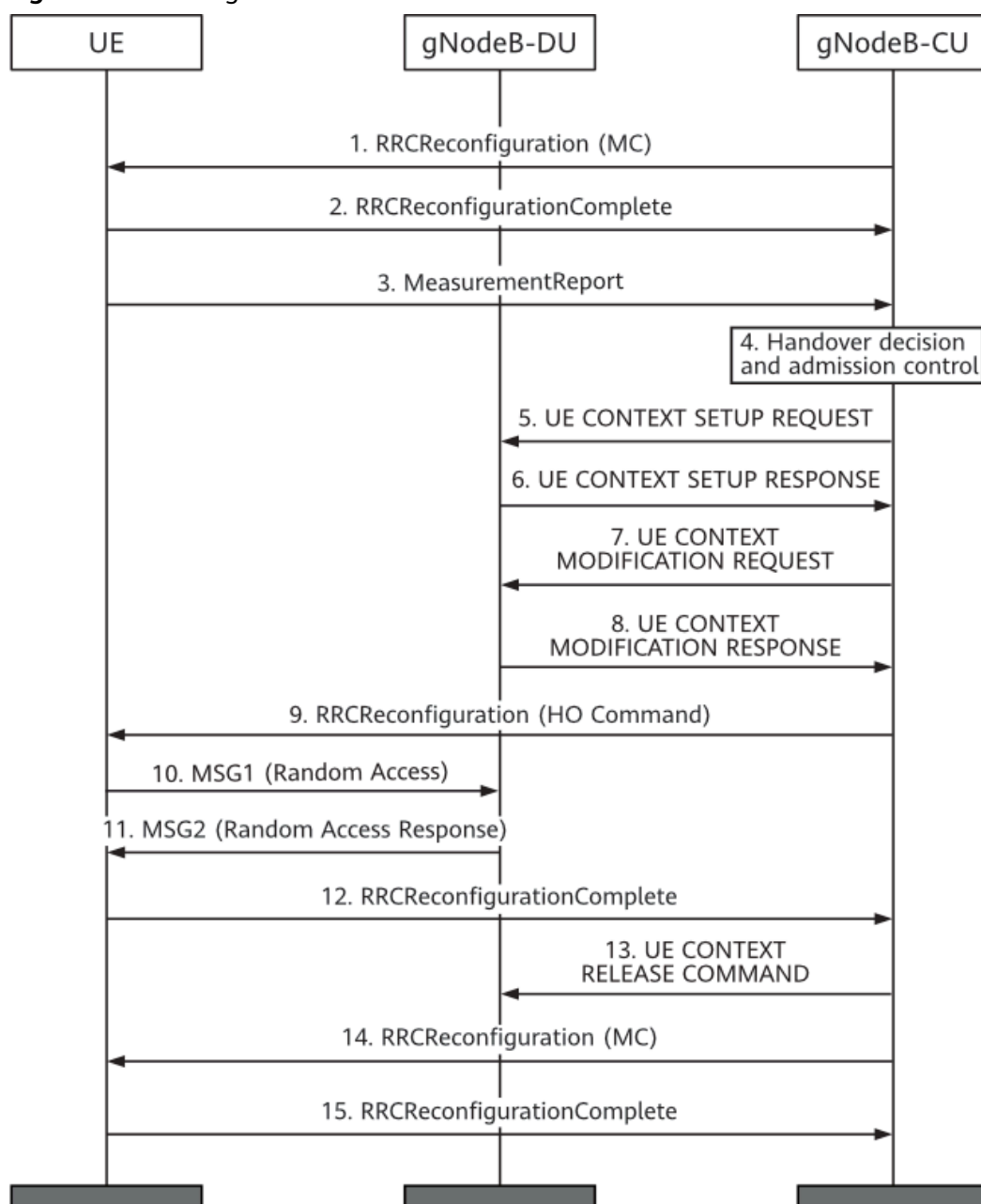
This chapter describes the signaling procedures for mobility management in SA networking. For details about the signaling procedures for mobility management in NSA networking, see *NSA Mobility Management*.

- SA mobility management in idle mode: Cell selection and reselection mainly occur on UEs and do not involve signaling exchange between UEs and the gNodeB.
- Connected-mode mobility in SA networking
 - For details about the signaling procedure for an intra-base-station handover, see [13.1 Intra-Base-Station Handover](#).
 - For details about the signaling procedure for an Xn-based handover, see [13.2 Xn-based Handover](#).
 - For details about the signaling procedure for an NG-based handover, see [13.3 NG-based Handover](#).
- SA mobility management in inactive mode: For details about the signaling procedures for RNA update and inactive state transition, see *SA Mobility Management in Inactive Mode*.

13.1 Intra-Base-Station Handover

An intra-base-station handover is a process where a UE is handed over from one cell to another cell served by the same gNodeB.

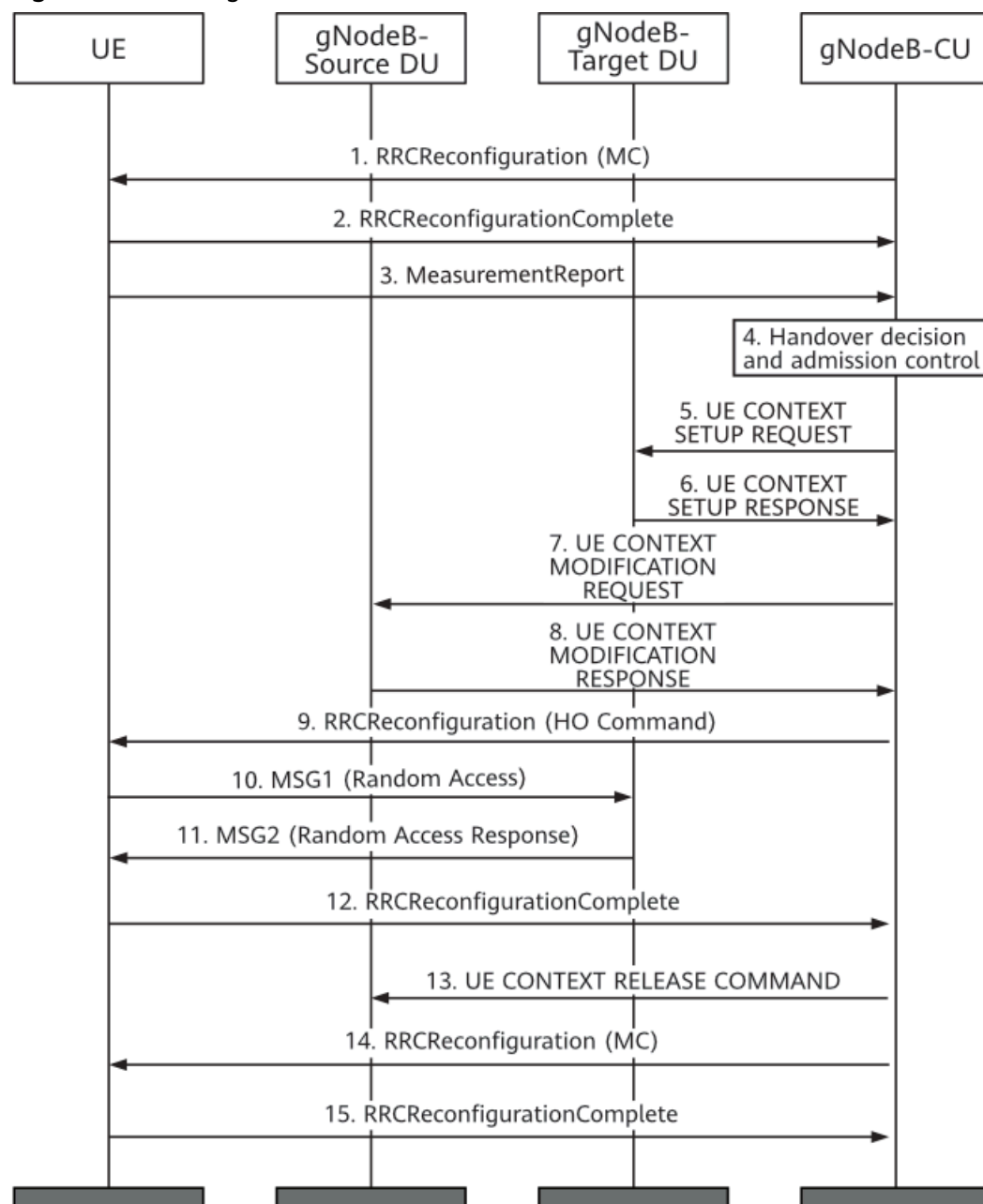
Figure 13-1 Intra-gNodeB-DU handover



1. The gNodeB-CU delivers measurement configurations to the UE through an RRCReconfiguration message.
2. The UE sends an RRCReconfigurationComplete message to the gNodeB-CU.
3. The UE sends a measurement report to the gNodeB-CU.
4. The target cell performs handover decision and admission control.
5. The gNodeB-CU sends a UE CONTEXT SETUP REQUEST message to the gNodeB-DU, requesting new UE resources for the target cell.
6. If the gNodeB-DU successfully allocates resources, it sends a UE CONTEXT SETUP RESPONSE message to the gNodeB-CU.
7. The gNodeB-CU sends a UE CONTEXT MODIFICATION REQUEST message to the gNodeB-DU.

8. The gNodeB-DU sends a UE CONTEXT MODIFICATION RESPONSE message to the gNodeB-CU.
9. The gNodeB-CU sends an RRCReconfiguration message to the UE, instructing the UE to perform a handover.
10. The UE initiates a non-contention-based random access procedure in the target cell by sending an MSG1 to the gNodeB-DU.
11. The gNodeB-DU replies with an MSG2.
12. The UE sends an RRCReconfigurationComplete message to the gNodeB-CU, notifying the gNodeB-CU that the UE has accessed the target cell.
13. The gNodeB-CU sends a UE CONTEXT RELEASE COMMAND message to the gNodeB-DU. The source cell releases the UE that has been handed over.
14. After the UE is handed over to the target cell, the gNodeB-CU sends the UE an RRCReconfiguration message containing the measurement configurations of the target cell.
15. After receiving the new measurement configurations from the gNodeB-CU, the UE returns an RRCReconfigurationComplete message to the gNodeB-CU.

Figure 13-2 Inter-gNodeB-DU handover



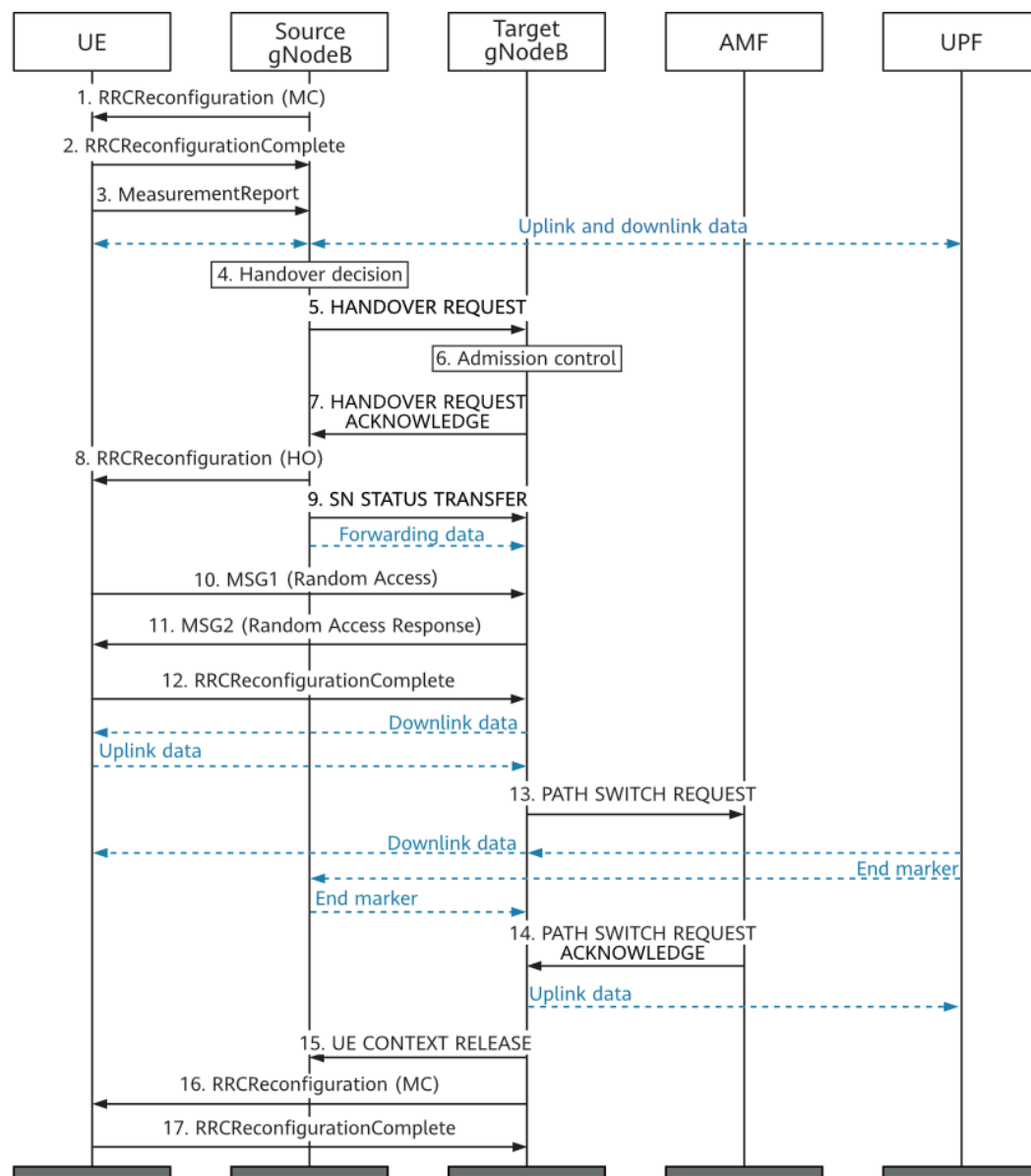
1. The gNodeB-CU delivers measurement configurations to the UE through an RRCReconfiguration message.
2. The UE sends an RRCReconfigurationComplete message to the gNodeB-CU.
3. The UE sends a measurement report to the gNodeB-CU.
4. The target cell performs handover decision and admission control.
5. The gNodeB-CU sends a UE CONTEXT SETUP REQUEST message to the target gNodeB-DU, requesting new UE resources for the target cell.
6. If the target gNodeB-DU successfully allocates resources, it sends a UE CONTEXT SETUP RESPONSE message to the gNodeB-CU.
7. The gNodeB-CU sends a UE CONTEXT MODIFICATION REQUEST message to the source gNodeB-DU.

8. The source gNodeB-DU sends a UE CONTEXT MODIFICATION RESPONSE message to the gNodeB-CU.
9. The gNodeB-CU sends an RRCReconfiguration message to the UE, instructing the UE to perform a handover.
10. The UE initiates a non-contention-based random access procedure in the target cell of the target gNodeB-DU by sending an MSG1 to the target gNodeB-DU.
11. The target gNodeB-DU replies with an MSG2.
12. The UE sends an RRCReconfigurationComplete message to the gNodeB-CU, notifying the gNodeB-CU that the UE has accessed the target cell.
13. The gNodeB-CU sends a UE CONTEXT RELEASE COMMAND message to the source gNodeB-DU. After receiving this message, the source cell served by the source gNodeB-DU releases the UE which has been handed over to the target cell.
14. After the UE is handed over to the target cell, the gNodeB-CU sends the UE an RRCReconfiguration message containing the measurement configurations of the target cell.
15. After receiving the new measurement configurations from the gNodeB-CU, the UE returns an RRCReconfigurationComplete message to the gNodeB-CU.

13.2 Xn-based Handover

During an Xn-based handover, the source cell and the target cell are served by different gNodeBs that belong to the same AMF region, and an Xn interface exists between the source and target gNodeBs.

Figure 13-3 Xn-based handover



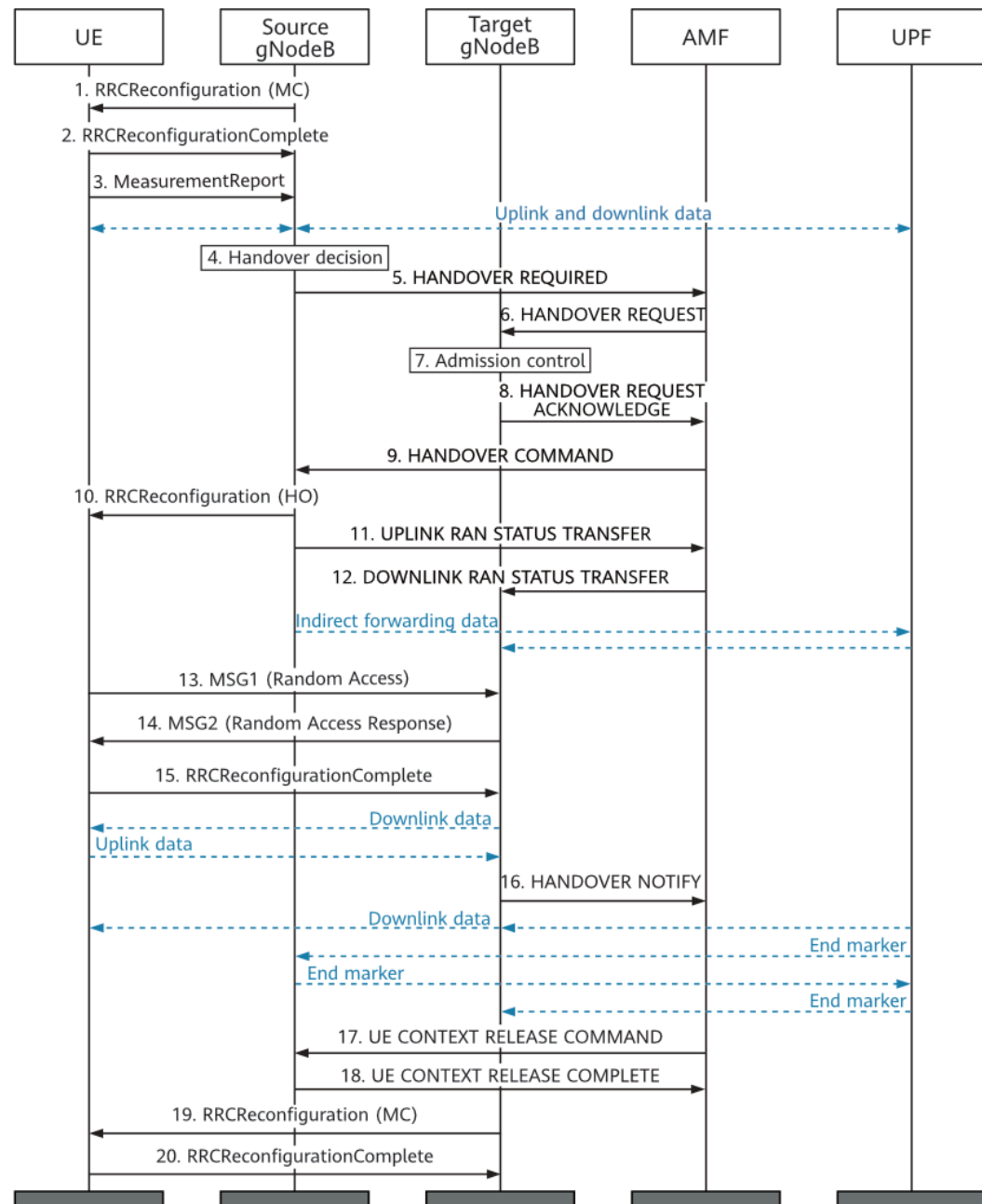
1. The source gNodeB delivers measurement configurations to the UE through an RRCReconfiguration message.
2. The UE sends an RRCReconfigurationComplete message to the source gNodeB.
3. The UE sends a measurement report to the source gNodeB.
4. The source gNodeB performs the handover decision.
5. The source gNodeB sends a HANDOVER REQUEST message to the target gNodeB serving the target cell over the Xn interface to initiate a handover request.
6. After receiving the handover request, the target gNodeB performs admission control.
7. The target gNodeB sends a HANDOVER REQUEST ACKNOWLEDGE message to the source gNodeB, indicating that the incoming handover is allowed.

8. The source gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to perform a handover.
9. The source gNodeB sends a PDCP SN to the target gNodeB through an SN STATUS TRANSFER message.
10. The UE initiates a non-contention-based random access procedure in the target cell of the target gNodeB and sends an MSG1 to the target gNodeB.
11. The target gNodeB replies with an MSG2.
12. The UE sends an RRCReconfigurationComplete message to the target gNodeB, notifying the target gNodeB that the UE has accessed the target cell.
13. The target gNodeB sends a PATH SWITCH REQUEST message to the AMF, notifying the AMF that the serving cell of the UE has changed.
14. The AMF sends a PATH SWITCH REQUEST ACKNOWLEDGE message to the target gNodeB.
15. The target gNodeB sends a UE CONTEXT RELEASE message to the source gNodeB. After receiving this message, the source gNodeB releases the UE which has been handed over to the target cell.
16. After the UE is handed over to the target cell, the target gNodeB sends the UE an RRCReconfiguration message containing the measurement configurations of the target cell.
17. After receiving the new measurement configurations from the target gNodeB, the UE sends an RRCReconfigurationComplete message to the target gNodeB.

13.3 NG-based Handover

During an NG-based handover, the source cell and the target cell are served by different gNodeBs that belong to different AMF regions, or the source gNodeB and target gNodeB belong to the same AMF region but no Xn interface exists between them.

Figure 13-4 NG-based handover



1. The source gNodeB delivers measurement configurations to the UE through an RRCReconfiguration message.
2. The UE sends an RRCReconfigurationComplete message to the source gNodeB.
3. The UE sends a measurement report to the source gNodeB.
4. The source gNodeB performs the handover decision.
5. The source gNodeB sends a HANDOVER REQUIRED message to the AMF over the NG link to initiate a handover request.
6. The AMF sends a HANDOVER REQUEST message to the target gNodeB serving the specified target cell to initiate a handover request.

7. After receiving the handover request, the target gNodeB performs admission control.
8. The target gNodeB sends a HANDOVER REQUEST ACKNOWLEDGE message to the AMF, indicating that the incoming handover is allowed.
9. The AMF sends a HANDOVER COMMAND message to the source gNodeB.
10. The source gNodeB sends an RRCReconfiguration message to the UE, instructing the UE to perform a handover.
11. The source gNodeB sends a PDCP SN to the AMF through an UPLINK RAN STATUS TRANSFER message.
12. The AMF transfers the PDCP SN to the target gNodeB through a DOWNLINK RAN STATUS TRANSFER message.
13. The UE initiates a non-contention-based random access procedure in the target cell of the target gNodeB and sends an MSG1 to the target gNodeB.
14. The target gNodeB replies with an MSG2.
15. The UE sends an RRCReconfigurationComplete message to the target gNodeB, notifying the target gNodeB that the UE has accessed the target cell.
16. The target gNodeB sends a HANDOVER NOTIFY message to the AMF, notifying the AMF that the UE has accessed the target cell.
17. The AMF sends a UE CONTEXT RELEASE COMMAND message to the source gNodeB. After receiving this message, the source gNodeB releases the UE which has been handed over to the target cell.
18. The source gNodeB sends a UE CONTEXT RELEASE COMPLETE message to the AMF.
19. After the UE is handed over to the target cell, the target gNodeB sends the UE an RRCReconfiguration message containing the measurement configurations of the target cell.
20. After receiving the new measurement configurations from the target gNodeB, the UE sends an RRCReconfigurationComplete message to the target gNodeB.

14 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

15 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

16 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

17 Reference Documents

- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.133: "NR; Requirements for support of radio resource management"
- 3GPP TS 38.306: "NR; User Equipment (UE) radio access capabilities"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 23.501: "System architecture for the 5G System (5GS); Stage 2"
- 3GPP TS 38.423: "NG-RAN; Xn application protocol (XnAP)"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- 3GPP TS 23.203: "Policy and charging control architecture"
- *5G Networking and Signaling*
- *EPC-based NSA Basics*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *NSA-SA Selection Based on User Experience*
- *ANR*
- *Flexible User Steering*
- *Mobility Load Balancing in Connected Mode*
- *QoS Management*
- *Multi-Operator Sharing*
- *Carrier Aggregation*
- *NG-flex*
- *Standards Compliance*
- *Energy Conservation and Emission Reduction*
- *High Speed Mobility*
- *VoNR*
- *ViNR*
- *URLLC*
- *Multi-Frequency Smart Selection*

- *Multi-Frequency Smart Aggregation*
- *Virtual Grid-based Multi-Frequency Coordination*
- *Resource Control in NSA Networking*
- *Network Slicing*
- *UE Power Saving*
- *NSA Mobility Management*
- *MRO*
- *SA Mobility Management in Idle Mode*
- *SA Mobility Management in Inactive Mode*
- *NG and Xn Self-Management*
- *CA Experience Improvement*
- *RedCap*
- *Multi-RAT Coordinated Energy Saving*
- *Idle Mode Management in the eRAN feature documentation*
- *Mobility Management in Connected Mode in the eRAN feature documentation*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*